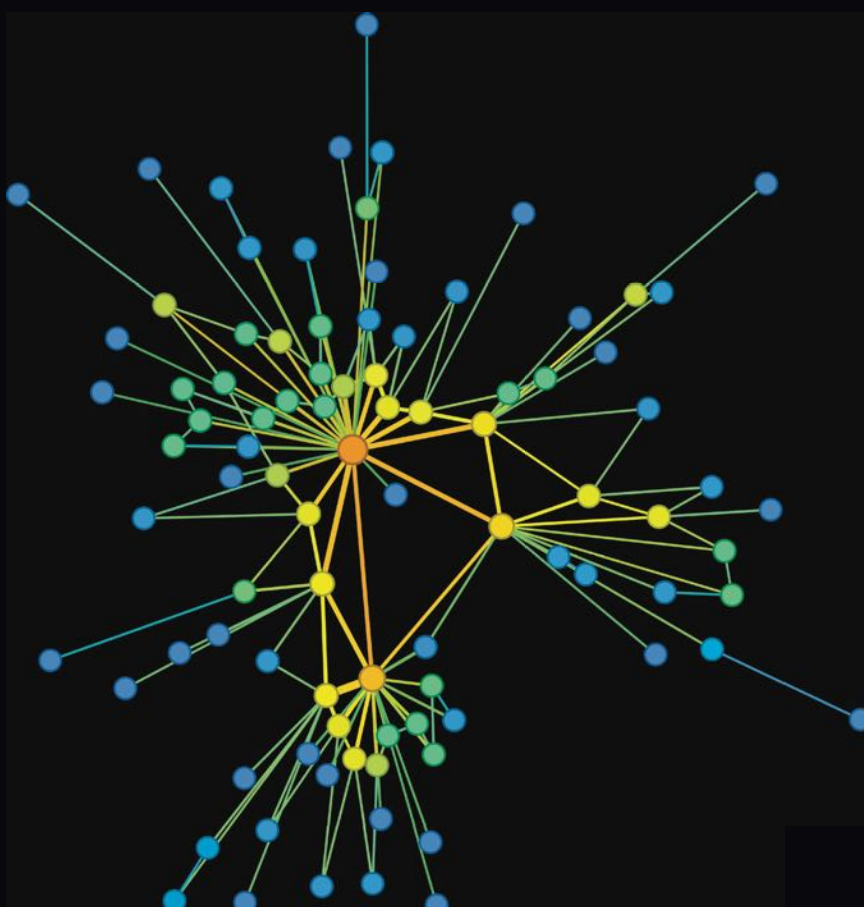
BEHAVIORAL NETWORK SCIENCE

Language, Mind, and Society



THOMAS T. HILLS

Behavioral Network Science

Behavioral Network Science explains how and why structure matters in the behavioral sciences. Exploring open questions in language evolution, child language learning, memory search, age-related cognitive decline, creativity, group problem solving, opinion dynamics, conspiracies, and conflict, readers will learn essential behavioral science theory alongside novel network science applications. This book also contains an introductory guide to network science, demonstrating how to turn data into networks, quantify network structure across scales, and hone one's intuition for how structure arises and evolves. Online R code allows readers to explore the data and reproduce all the visualizations and simulations for themselves, empowering them to make contributions of their own. For data scientists interested in gaining a professional understanding of how the behavioral sciences inform network science, or behavioral scientists interested in learning how to apply network science from the ground up, this book is an essential guide.

Dr. Thomas T. Hills is Professor of Psychology at the University of Warwick, concentrating on how humans represent and navigate information in the mind and society. He directs the Behavioral and Data Science MSc and has held fellowships with the Alan Turing Institute and the Royal Society.

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Part II Language

Additional Resources

For those keen to explore the methods detailed in this book further, all the R code for producing the figures and simulations is available online, separated by chapter. This resource is fully commented to walk you through various techniques, from transforming raw data into insightful networks and then visualizing, analyzing, and modeling these networks. You'll also find all the necessary data to recreate the figures and simulations on your own. For datasets that are frequently updated, links are provided directly to the original researchers' latest versions. This online suite is designed to be a hands-on extension of the book, perfect for readers looking to deepen their practical knowledge of network analysis.

R code available at: https://thomasthills.github.io/BehavioralNetworkScience.html

Introduction
Structure Matters

Why do we see the behaviors that we do in the world? This is a question that has challenged many notable figures, including Darwin, Saussure, Wittgenstein, Lévi-Strauss, Durkheim, and many other past and recent thinkers. The conclusions that they arrived at, time and again, identified how things in the world – from species, to thoughts, to culture – relied on the interconnectivity of the systems in which they arose. Behaviors are a consequence of the environments in which they emerge and the historical trajectories that have brought them about.

The biologist Theodosius Dobzhansky claimed as much when he said "Nothing in biology makes sense except in the light of evolution" (Dobzhansky, 1973: p. 125). The linguist John Firth reiterated it when he said that "You shall know a word by the company it keeps" (Firth, 1957: p. 11). These are structural claims about the origins of life and meaning. Organisms originate from a marriage of ancestry and environment. The meanings of words depend on their relationships with other words. And so on it goes. People aggregate into mobs of like-minded fanatics and beliefs nestle among one another like a huddle of penguins.

One of the tools that has brought this structural vision of life and meaning into focus is network science. By describing the relationships that bind systems together, network science provides a quantitative precision to these earlier ideas. As a ruler measures length, allowing us to compare human height with the Burj Khalifa, network science measures structure, giving us the ability to compare networks of human relationships with the organization of the environments in which they occur, the patterns of activation in our brains and – more enchantingly – with the architecture of our thoughts. When we stop seeing isolated things and start focusing on the relationships between them, we are inviting a structural view of behavior that can be compared across disciplines.

This cross-disciplinary insight is one of the catalysts behind the science of complex systems. Complex systems approaches attempt to identify general principles of systems that apply across domains. By general principles, I mean principles that are abstracted away from the particulars of the environment in which they are found. When someone describes a scurrying procession of driver ants, searching for prey on the forest floor in the Gold Coast of Ghana, alongside the rejoicing voices of a group of improvisational singers in a nearby church, we have something that approximates literature. Understanding how the ants and the singers solve a similar coordination problem is a view from complex systems.

Both the ants and the singers are navigating structured spaces – whether among the forest's fractal underbrush or the melodic and rhythmic conventions of song. And, for both, the quality of their navigation depends on the structure of their communication: who communicates with whom, when, and by what means. The structural features of communication the ants and singers share apply to 1970s jazz fusion as well as they do to a murmur of starlings, the challenges of international politics, or the way people use what they know to learn what they don't. This shared structure across domains invites us to think about behavior using network science and, when we do, we have behavioral network science.

Viewing behavior in this way, phenomena become symptoms of the structural processes that govern them. The goal of the behavioral network scientist is to articulate the structure and processes of this government.

0.1 Structure in the World

We should care about structure because it helps us to understand how and why systems work like they do. That, in turn, helps us to anticipate what they are likely to do next – and what, if anything, we should do about it. That is, for all the grand unified generality of complex systems, behavioral network science shines most brightly when it tells us about specific things. Let me give you three examples.

Structure can tell us what kinds of systems we are dealing with. When Krebs (2002) mapped the network structure of the 9/11 terrorist cells – identifying each individual and the relationships among them – what he observed was peculiar for a social network. The 19 hijackers had sparsely interacted. This is rare among social networks. Our friends' friends tend to become our friends. Close-knit groups form from like-minded individuals. And coordinated groups need coordinated communication. On the face of it, the structure of the terrorist cells lacked these features. But this was by design. Sparsely connected social networks are perfectly sensible for criminal networks. Many criminal networks invoke similar behavioral immune systems, insulating themselves from compromise by limiting what the individuals know about one another and how often they interact. This makes it harder for authorities to identify and compromise them.

Structure also tells us about the kinds of individuals we need to maintain social health. Such social health is critical for winter-over crews at the Amundsen-Scott South Pole Station in the Antarctic. These crews are, for all practical purposes, people trapped together in a cage for nine months. Johnson et al. (2003) pioneered research on the network structure of these teams and found that the social health of the group went hand in hand with its social structure. This structure also depended critically on specific group roles. Certain kinds of people were the linchpins of community health. One of these people was the leader. Not just any leader, however. Successful leaders were recognized for their expressive leadership – a form of leadership focused on social cohesion. This requires knowing what needs to be done but also getting people to care enough about one another to do it with good cheer. Expressive leaders build self-motivated communities. Another kind of person was the 'positive deviant' – that is, the clown. Through their ability to defy social conventions, the clown acts as what network scientists call a broker between various teams (such as nonscientists and scientists), eventually bringing individuals together who would otherwise remain apart.

Structure also helps predict where certain kinds of events will happen. Consider Juárez, a city that lies on the border between the USA and Mexico. During the early 2000s it was the murder capital of the world. What makes a place like Juárez so dangerous? In our effort to explain such things, we often reach for exhausted stereotypes or true but inexact explanations like "drug cartels." The latter is half right, but there are drug cartels in many peaceful places far beyond Juárez. What differs about Juárez is almost entirely structural – it is what it is because of its relationship with other places. According to Wainwright (2016), Juárez is one of the critical gateways for drug trafficking between South and North America. Those who control Juárez control access to a large slice of the North American market. Juárez is therefore a place with what network scientists call high betweenness: to go from one place to another, one must pass through it. During the 1990s, control over Juárez was thrown into question, and the disputes that arose reflected the value of its betweenness.

These three examples demonstrate several things. Foremost, they demonstrate how an understanding of the basic principles of network science can help guide our behavioral intuition. When we see a certain behavior in the world, we can better hypothesize about what gave rise to it if we understand how structure works. In the same way that a knowledge of Latin can help us infer the meaning of a novel word, a knowledge of structure helps us make sense of complex relationships.

What these examples also demonstrate is how behavior can bring network science to life. When we see an abstract principle from network science embodied in issues that we care about and for which we already have some understanding, the principle becomes approachable. And it starts to become something we can recognize at a party.

0.2 Structure and Process

Like a map, structure is a simplification. But when our understanding of structure functions at its best, it helps us think about how structure and behavioral processes interact.

For sparsely connected terrorist cells, we can see how a disconnected structure prevents a certain kind of disruptive process such as getting one individual to rat out the others. But if the process were different – for example, if the police took down criminals like lions single out a stray from the herd – then the structure would be different. Criminals would crowd together like gazelles on the African savanna. Similarly, our mental health benefits from our social relationships, as if it were contagious. Knowing this, it becomes evident why people who act as social connective tissue keep communities healthy. And, finally, Juárez's structural position matters because of where drugs originate and how they move. Legalizing drugs or otherwise changing the conditions by which they are trafficked would change the processes driving the drug market and change Juárez as a result. Structure lives and dies by its processes.

There are countless examples. US Route 66 is a structural bygone that fell into decline not because we changed the physical structure of space, but because we changed the way we moved through it. Similarly, our social networks have changed because online environments make it easier to find people who share our obsessions. And the processes we use to search memory mean the aging mind can both know more and be harder to search at the same time. Behavior is awash with examples like these. Creativity, child learning, opinion dynamics, the beliefs of conspiracy theorists, and much more are a consequence of structure and process. The beauty of structure is that it helps us understand the processes better, and vice versa.

0.3 Behavioral Network Science

Behavioral Network Science is written for those who are interested in gaining a better grasp of the relationship between structure and behavior. It does this by combining behavioral science with network science in a way that allows them to inform one another.

Behavioral science is a developing interdisciplinary field focused on understanding what makes people tick. It encompasses fields such as behavioral ecology, business, behavioral economics, psychological science, informatics, experimental philosophy, and the cognitive sciences more generally. Above all, behavioral scientists are comfortable with theoretical and experimental tests of hypotheses that utilize quantitative approaches (i.e., advanced counting). Network science is a quantitative discipline aimed at describing structure and the processes that influence it.

By combining these, this book aims to provide a serious investigation of theory and practice in behavioral science in a way that highlights why structure matters and how we can use network science to understand it better. In all cases, I have aimed to communicate ideas as if the reader knew little about the behavioral theory or the network methodology. An interested reader should have no problem seeing how it all comes together. That said, the behavioral science chapters focus on open questions in the behavioral sciences and, in most cases, present novel analyses which attempt to integrate and extend what we already know. I could not bring myself to write a book that merely relayed what I already knew; there is excitement in knowing the snow is fresh.

There are of course many excellent books that provide an introduction to network science. I could not have written this book without them. These range from Wasserman and Faust's (1994) classic text *Social network analysis: Methods and applications* (which I read like it was on fire many years ago), to more recent contributions like Newman (2018) *Networks*, Barabási and Posfai's (2016) *Network science*, Menczer, Fortunato, and Davis's (2020) *A first course in network science*, and Borgatti, Everett, and Johnson's (2018) *Analyzing social networks*. All of these books should be read twice. They go more deeply into the technical aspects of network theory than this book and they should be on any card-carrying network scientist's reading list. There are also books dedicated to the history of social networks (Freeman, 2004), the networks of crowds and markets (Easely & Kleinberg, 2012), and networks in neuroscience (Fornito et al., 2016; Sporns, 2016). These books are all excellent and inspiring for anyone interested in network thinking.

However, there are few books that focus on the psychology and cognitive theory of behavioral science as it is informed by network science. And that's a shame, because the two topics go well together. Moreover, if part of the aim of network science is to understand how it matters for people, then it makes sense to take a close look at what we already know about what makes people do what they do. That is something about which behavioral scientists have much to say.

The meat of this book presents a series of chapters that each tackle a new problem in the behavioral sciences, laying out its theoretical and experimental foundations, and then offering an approach to better understanding it using network science. In essence, each of these chapters poses a scientific riddle about behavior and then uses network science to unravel it.

0.4 The Structure of the Book

This book was written to meet readers at their level of interest. It is laid out in four parts. Readers who are interested in enriching their understanding of network science can start at the beginning with **A Brief Guide to Network Science**. This will give readers a short course on how networks are constructed, what we can measure with them, and how we can grow our own. It is designed to be quick and dirty, with an emphasis on practical understanding, and I try to provide brief but memorable real-world examples for many of the concepts. If read while working through the online code, the reader will be making and measuring networks in short order.

Following the brief guide, the book gets down to the business of behavior. Readers with a specific behavioral interest can choose the chapter that most piques their interest in one of the three parts that make up the behavioral section of the book. Part II focuses on issues of **Language**, showing how we can apply network science to understand language, language acquisition, language emergence, and language evolution. Part III, **Mind**, presents a series of chapters investigating mental processes, such as false memories, memory search, cognitive aging, and creativity. Part IV, **Society**, visits a series of topics associated with social dynamics, focusing on social illusions, the wisdom of crowds, opinion dynamics, group conflict, and an alternative pathway for scientific funding inspired by a merit-based universal basic income.

The behavioral science chapters also follow a methodological arc. Earlier chapters focus more on applying and developing metrics for measuring nodes, edges, and networks. Later chapters develop behavioral models that try to explain behavior by generating it, in the same way we might try to write down a recipe for vegan cheesecake. It is not easy, but it can be done. And as better recipes produce better cakes, better models produce better explanations of behavior. The later chapters focus more on getting beliefs into individuals and use agent-based models to let them act out their parts in the social fabric.

0.4.1 The Online Code

Most everything in this book comes with the code and data to reproduce it. The book is written exclusively in R (R Core Team, 2023) using RMarkdown (Allaire et al., 2020). The annotated code necessary to reproduce the analyses and figures is provided on the book's website for each chapter. The primary graphing package is igraph (https://r.igraph.org/reference/index.html). All the additional packages necessary to run the code are designated in the code. Statistically, I've generally gone for the interocular trauma test: the results should hit you between the eyes. In the few cases where this isn't true, more statistical details can be found in the online code. I've dominantly left the statistical minutiae out of the text, as it would have otherwise been too cumbersome. Generally speaking, the statistical tests one might use for networks are not different from other data types. Other books describe the underlying details of these tests better than I can (e.g., Field et al., 2012). Just the same, networks do have their peculiarities and I address these where they arise.

0.4.2 Using This Book in the Classroom

Early drafts of this book have been used as a text for undergraduate and postgraduate coursework. These have been taught to social scientists interested in learning more about network science and mathematical scientists interested in learning more about behavioral science.

Short courses can focus on the first three chapters, to provide a foundation in network science that will rapidly bring readers up to speed. They are especially effective when combined with specific examples cherry-picked from later chapters that demonstrate principles in action.

Longer courses can start from the beginning and explore a few chapters per week alongside the online code.

To give a bird's-eye view of the terrain, Table 0.4 provides each chapter's behavioral theme alongside its methodological focus.

Table 0.4 Each chapter's behavioral theme and methodological focus

Chapter	Theme	Methods
1	Network representation	Kinds of networks and how to build them from data
2	Network metrics	How to measure properties about nodes, communities, and networks as a whole
3	Generative network models	How to grow your own networks
4	Language structure	Degree and degree distributions
5	Child vocabulary learning	Fitting models of network growth to real data
6	How distinctiveness aids learning	Various ways to build edges based on shared features in multilayer networks
7	Comparing early and late talkers	Three approaches to measuring small-world networks
8	Where new words are born into language	Using biological theory to understand node origins
9	Language emergence	Agent-based modeling of community information emergence
10	False memories	Spreading activation on a network
11	Memory search	Comparing animal foraging models with network search algorithms
12	Explaining age-related cognitive decline as enrichment	Using prediction-error models to simulate aging

Table 0.4 (Continued)

Chapter	Theme	Methods
13	Creativity	Producing structural differences from process differences
14	Structural illusions	Degree-attribute and aggregation illusions in network data
15	Group problem solving	Agent-based models of collective search and network annealing
16	Opinion dynamics and perceived normality	Agent-based models of segregation with Bayesian beliefs
17	Conspiracy beliefs and incoherence	Comparing network coherence and pointwise mutual information
18	Game theory and brinkmanship	Agent-based models of cooperation and escalation
19	Merit-based universal basic income	Agent-based models of wealth redistribution and eigenvector centrality

0.5 Acknowledgments

In a book about structure, one cannot help but to consider the network of all influencers. This must start with things like the Great Programmer, ancestors, grandparents, parents, and all the other scientific mentors along the way. This book owes much to them. The subgraph of more recent influencers is, however, easier to list.

This book was made possible by fellowships from the Royal Society and the Alan Turing Institute. By supporting research that bridges the mathematical and social sciences, they helped me to realize the appeal and importance of crossing the yawning divide that sometimes separates these disciplines. The University of Warwick was also instrumental in creating an interdisciplinary atmosphere where the behavioral sciences flourish. It has continued to promote behavioral science initiatives within and across disciplines. This has created a lively community of researchers willing to walk around campus to visit one another for nothing more than a biscuit and some lively discussion.

The pleasure of writing this book was enhanced by the many people who helped along the way. Cynthia Siew, especially, got the ball rolling during a too-short visit to Warwick. Cynthia played a substantial role in the early stages to help shape the vision of this book. Stephen Acerra at Cambridge University Press was also ever optimistic, and his enthusiasm helped the book to reach the finish line. Many others have acted as expert readers and advisers, providing feedback and offering invaluable criticism. Thanks to Adrian Bangerter, Adriano Lamieri, Alessandro Miani, Charlie Pilgrim, Chris Strelluf, Clay Beckner, Daniel Read, Daniel Sgroi, Danyang Hu, Dasol Jeong, Dirk Wulff, Emmanuelle Volle, Ethan Freestone, Eugene Malthouse, Eugenio Proto, Eva Jimenez, Friederike Schlaghecken, Gordon Brown, Halleyson Li, Iain Murdoch, Jens Vogelgesang, Joshua Cox, Kita Sotaro, Li Ying, Lucas Castillo, Mahathi Parvatham, Marcela Ovando-Tellez, Massimo Stella, Michael Hattersley, Michael Vitevitch, Mikhail Spektor, Paolo Turrini, Pascal Wagner, Richard Moore, Rowan Munson, Simon De

Deyne, Yoed Kenett, Yuqi Ye, Yuval Hart, Zoran Kovacevic, and many others. They have all helped to straighten my crooked thinking. Many have also been generous with data and code. Thanks also to the many students at the University of Hohenheim, the University of Lugano, and the University of Warwick, who suffered through early drafts of this work and pointed out the errors that others too kindly assumed I would recognize on my own. The remaining errors are left in for others to enjoy.

Finally, thanks to my closest friend, Sara, who has kept the craft of life properly ruddered, and whose eye for detail has played a critical part in helping shape the final form of this book. And thanks to Fionniain, Kaleia, and Sonora, whose unfailing curiosity and good humor hold the hand of thought open to William James's "wild facts" – where the new discoveries lie.

Part I

A Brief Guide to Network Science

1 Making and Recognizing Networks

1.1 What Is a Network?

Networks are made up of **nodes** and **edges**. Nodes (or **vertices**) often represent people, places, or things. Edges represent relationships or **links** between nodes. If we have a list of nodes (N) and edges (E), then we have a network. Networks are sometimes referred to as **graphs** and indicated by $G(N, E)$.

The network in Figure 1.1 has six nodes and eight edges.

We can imagine there are six people and eight relationships. Node 6 has four friends and one of them is Node 2. But there is no reason why these have to be people and their relationships. Nodes could represent words and edges could represent shared meanings. They could be places and paths between them, symptoms and comorbidity, brands and shared product lines, organisms and shared environments, beliefs and their co-occurrence in the general population. Whenever we have 'things' (in the broadest sense of the term) and relationships, we have nodes and edges. Then we have a network.

1.2 Representing Networks

We can represent the node and edge relationships in two distinct ways: either as an edge list or an adjacency matrix. An **edge list** lists each relationship: there is a row for every edge, listing the names of the vertices in two columns. The list in Table 1.1 corresponds to the eight edges shown in Figure 1.1. V1 and V2 correspond to the two vertices that are connected by an edge.

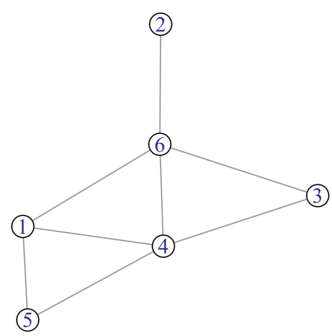

Figure 1.1 A basic network.

Table 1.1 An edgelist for our basic network.

V1	V2
1	4
3	4
1	5
4	5
1	6
2	6
3	6
4	6

Table 1.2 An adjacency matrix for our basic network.

	1	2	3	4	5	6
1	0	0	0	1	1	1
2	0	0	0	0	0	1
3	0	0	0	1	0	1
4	1	0	1	0	1	1
5	1	0	0	1	0	0
6	1	1	1	1	0	0

The first edge in the edge list is between Node 1 and Node 4. This corresponds to an edge we can see in the visualization. The edge list is not in any particular order from top to bottom.

We can also represent a network using an **adjacency matrix**, which is a matrix indicating which nodes are connected by edges. Table 1.2 provides an example.

For every edge in the edge list, there is a nonzero value in the corresponding cell of the adjacency matrix. The value of the edge between node i and node j is represented by A_{ij}. Row 4, column 1 has a value of 1: $A_{4,1} = 1$, indicating the presence of an edge. Note that $A_{1,4}$ also has a 1. Whenever we have an **undirected network** – meaning that edges do not indicate any form of directionality – then the adjacency matrix is symmetric: $A_{ij} = A_{ji}$.

The **diagonal** of the matrix refers to all cells A_{ij} for which $i = j$. In other words, they represent **self-loops**. If no nodes connect to themselves, the diagonal is zero. We can add a self-loop for Node 2 by adding an edge from 2 to itself: $A_{2,2} = 1$, as shown in Table 1.3 and Figure 1.2.

Self-loops are infrequent in behavioral network science. But they are sometimes useful. For example, if edges represent people who share cake with one another, then we might imagine that some people eat some of their own cake. Those people would have a self-loop.

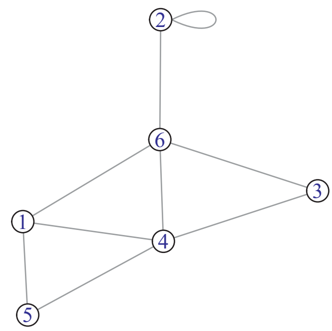

Figure 1.2 A network with a self-loop.

Table 1.3 Adjacency matrix for a network with a self-loop.

	1	2	3	4	5	6
1	0	0	0	1	1	1
2	0	1	0	0	0	1
3	0	0	0	1	0	1
4	1	0	1	0	1	1
5	1	0	0	1	0	0
6	1	1	1	1	0	0

1.3 Four Types of Networks

The network described in Figure 1.1 is the simplest of networks: Edges have values of 0 or 1 and are undirected. There is only one kind of node and there is only one kind of edge. It's all very basic. All other types of networks diverge from such basic networks by varying different properties of edges and nodes.[1]

1.3.1 Weighted Networks

When edges can take values different from 0 or 1, the network is **weighted**. Figure 1.3 shows a weighted network.

We might imagine that the eight relations are each valued according to the intensity of their love. Node 4 gets the most love. But the weights could just as easily be duration, frequency of contact, relative similarity, or number of shared features. Any relation that varies can be represented by a weighted edge.[2]

The corresponding weighted adjacency matrix contains edge weights in each cell, as shown in Table 1.4.

[1] The term **simple network** is reserved for networks with no self-loops and no multi-edges. Thus, pretty much all the networks we discuss here are simple by this definition.

[2] A slight variation is a **signed network**, which can represent positive (love) and negative (hate) relationships, useful for tracking bullying or international alliances.

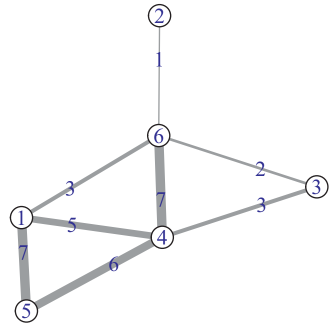

Figure 1.3 A weighted network. Edge thickness is proportional to edge weight.

Table 1.4 Weighted adjacency matrix.

	1	2	3	4	5	6
1	0	0	0	5	7	3
2	0	0	0	0	0	1
3	0	0	0	3	0	2
4	5	0	3	0	6	7
5	7	0	0	6	0	0
6	3	1	2	7	0	0

Table 1.5 Weighted edge list.

V1	V2	weight
1	4	5
3	4	3
1	5	7
4	5	6
1	6	3
2	6	1
3	6	2
4	6	7

We can write an edge list for the same network by adding the weights in a third column, as in Table 1.5.

1.3.2 Directed Networks

When edges point from one node to another, the network is a **directed network**, as shown in Figure 1.4. Now we can imagine our six nodes are sumo wrestlers. Directed edges represent victories. These could go either way, but here we will let the edge point from the loser to the victor. Nodes 6 and 5 are undefeated.

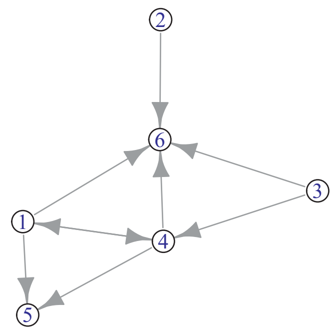

Figure 1.4 A directed network.

Table 1.6 Directed edge list.

V1	V2
1	4
3	4
1	5
4	5
1	6
2	6
3	6
4	6
4	1

Table 1.7 Directed adjacency matrix.

	1	2	3	4	5	6
1	0	0	0	1	1	1
2	0	0	0	0	0	1
3	0	0	0	1	0	1
4	1	0	0	0	1	1
5	0	0	0	0	0	0
6	0	0	0	0	0	0

For a directed network the edge list indicates which node points to which other node: The convention is the first column points to the second column. This creates a corresponding asymmetry in the adjacency matrix. The row node points to the column node. Compare the edge list in Table 1.6 with the adjacency matrix in Table 1.7.

When an edge points in both directions, $A_{ij} = A_{ji}$, we say it is a **reciprocal edge**. Nodes 4 and 1 in our sumo network share reciprocal wins.

Crossing unweighted with weighted and undirected with directed produces four network types. Most of the networks we deal with will be one of the following four

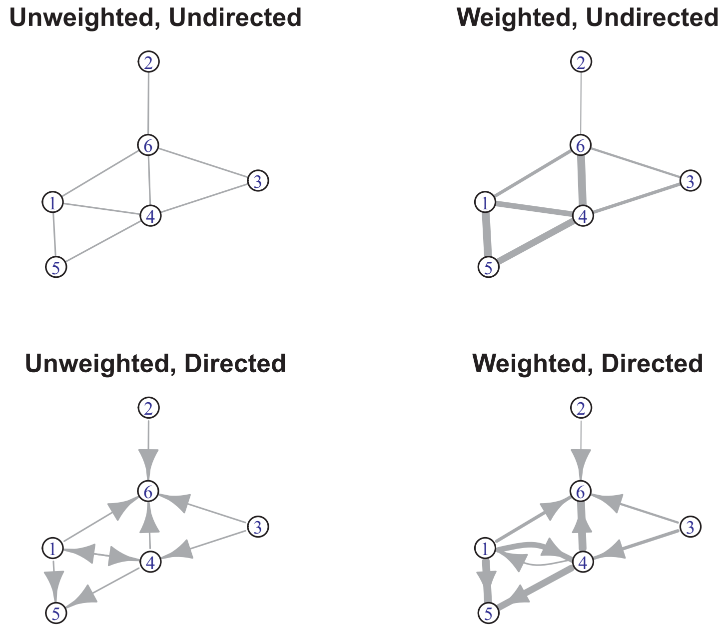

Figure 1.5 Four common network types.

(Figure 1.5): (1) unweighted and undirected, (2) weighted and undirected, (3) unweighted and directed, or (4) weighted and directed.

1.3.3 Sparse Matrices, Edge Lists, and Adjacency Matrices

Data often comes in the form of edge lists or adjacency matrices. For example, if we have people list their friends, we have an edge list. When the output of a similarity measure produces a large matrix of pairwise similarities, we have an adjacency matrix. One can quickly be turned into the other.

A savvy network scientist can quickly infer properties of a network by looking at the adjacency matrix. For example, is the network symmetric? Is it weighted? It is often useful to dummy check the adjacency matrix to make sure it reflects your understanding and to verify that edges are where and how you think they should be.

However, for networks with many nodes, the adjacency matrix can be unreasonably large. A directed network with 1,000 nodes might have 1,000,000 possible edges, with the same number of values in the adjacency matrix. As this grows, so grows the computational resources needed to handle it.

Luckily, most networks are **sparse matrices**, meaning that most of the edge values are 0. The number of nonzero edges is much smaller than the number of possible edges. This means that the edge list will be a much smaller computer memory object than the adjacency matrix. If our network with 1,000,000 possible edges had only one actual edge, we could represent it with one row in an edge list. Most network packages have methods for interacting with large but sparse matrices in fast and accurate ways.

1.4 Thresholding

Our basic network is often the easiest to study, visualize, and interpret. When it captures the relevant information necessary to address a particular problem, it is preferred to a weighted or directed network. When that is the case, it is useful to be able to transform a more complex network into a basic network. There are two traditional routes to achieve this.

1.4.1 Thresholding Weighted Networks

Suppose we have a network representing how often ministers in the UK Parliament vote together. We would have a weighted network with each edge representing the number of shared votes. If we plotted this as a network, every node would be connected to every other node because ministers occasionally vote unanimously. How might we turn this into a useful unweighted and undirected network? One approach is **thresholding**: only keep edges with weights above some threshold value. Here the ministers must share more than a threshold number of votes. Below this threshold we assign them an edge value of 0 and above it we assign a value of 1.

In Figure 1.6 there are two types of nodes (dark and light). Edges between nodes of the same color have larger weights. You cannot see this in the visualization if the threshold is 0; everything is connected. As the threshold increases, the two groups become visible. At too high a threshold, all nodes become isolates.

What is the optimal threshold? That depends on the problem. Best practice usually involves choosing multiple thresholds over a range that captures this transition and presenting the results for all of them. This reduces the worry that results are sensitive to an arbitrary threshold.

1.4.2 Thresholding Directed Networks

Suppose we ask a group of students in a classroom to tell us who their friends are. The resulting friendship network is directed, with arrows pointing toward hopefully reciprocated friendships. How might we turn this into an unweighted and undirected network?

An undirected edge can be defined for the two variations of directed edges: (a) Either at least one directed edge is present, or (b) the edges are reciprocal – each node directs

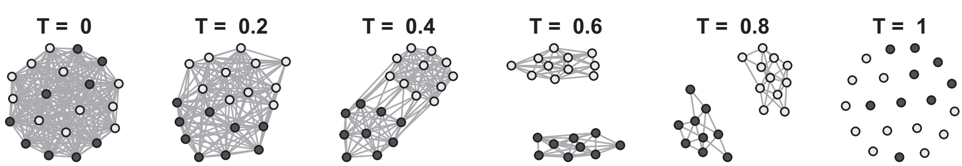

Figure 1.6 Thresholding weighted networks. Dark and light nodes have stronger weights with other members of their group. This is easiest to see when the threshold for edge weights is an intermediate value.

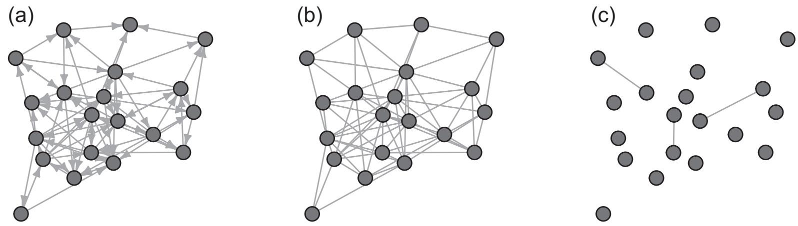

Figure 1.7 Thresholding directed networks. The network in panel (a) shows the full directed network. The panel (b) network transforms this into an undirected network, making an edge wherever at least one node has a directed edge to the other. The network in panel (c) only preserves reciprocal edges.

an edge to the other. Figure 1.7 shows a directed network and the products of applying the at-least-one-edge and reciprocal-edge rules.

Which approach is best will depend on the question. If the researcher is studying airborne disease spread, then the direction of the edge may not matter as long as the edge represents contact. But if the researcher is studying a potential outbreak of tuberculosis – which requires prolonged contact – then reciprocal edges may be more meaningful.

1.5 Attributes of Nodes and Edges

Network representations can be enriched further by assigning different properties to the nodes and edges. The **attributes** can be categorical (e.g., democrat or republican) or continuous (e.g., height in millimeters). If we are trying to make predictions about nodes or edges, we can think of these attributes as additional predictors in our model.

If nodes are people, they can have different political affiliations, social strategies, ages, socioeconomic status, ethnicity, and so on. If nodes are words, they could have different valences (positive or negative meaning), grammatical types (e.g., nouns or verbs), frequencies of use, levels of concreteness, and many other features. Edges can vary along different dimensions as well.

These attributes often form the basis of hypotheses. For example, do individuals of different political affiliations (node property) have different numbers of social contacts? Does the age of a friendship (edge property) influence the well-being of the individuals (nodes) who share that friendship? Figure 1.8 visualizes these relationships.

1.5.1 Node Attributes: Bipartite Networks

Bipartite networks occur when nodes of one type are only associated with nodes of another type. Sometimes one of the node types represents possible features or attributes that can be shared among nodes of the other type.

Bipartite networks have many uses. For example, if people choose behaviors (e.g., movies to watch), a bipartite network can be used as a recommender system to identify which movie to recommend next. Bipartite networks can also be used to determine which

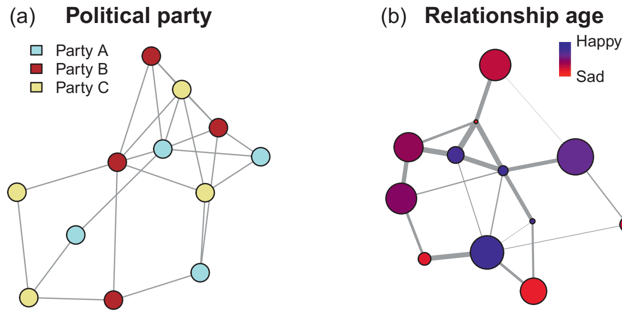

Figure 1.8 Categorical and continuous node attributes. (a) A network showing social contacts among representatives of different political parties. (b) A network of friendships. The edge width in the “Relationship age” network indicates the age of the relationship (an edge weight but also an edge attribute). Node size indicates age of the individual. Color indicates happiness.

individuals or behaviors are similar to one another, by looking at how nodes of one type share nodes of the other type.

Groups of nodes connected to a node in a bipartite network are functionally connected by their common affiliation. These are sometimes called **affiliation networks** or **hypergraphs**. In that sense, bipartite networks can be thought of as networks where edges connect more than two individuals at a time – for example, when individuals are members of clubs.

The two networks on the top of Figure 1.9 are both the same network. The one on the left is in a typical force-directed graph visualization.[3] The representation on the right is a bipartite projection, clearly showing the two node types. The bipartite projection shows how the nodes do not connect with nodes of the same type but only with nodes of the other type. For example, executives may be on multiple company boards, senators can vote for the same bills, people can share similar hobbies, and objects can share similar features.

The adjacency matrix of a bipartite network – often called an **incidence matrix** – assigns different types of nodes to the rows and columns of the matrix (see Table 1.8). Unlike the square monopartite matrix, with only one node type, the bipartite network can be represented as a rectangular adjacency matrix.

Bipartite networks can be **projected** onto one or the other node type. Nodes are then connected with one another if they share a connection with a node of the other type. The bottom two networks in Figure 1.9 are the one-mode (monopartite) projections for the two-mode (bipartite) network shown in the same figure.

1.5.2 Edge Attributes: Multiplex Networks

Edges in the same network can be of different types. Such **multiplex** (or **multilayer**) networks are commonly used to characterize systems in which nodes share different kinds of

[3] **Force-directed graph layouts** are physical simulations of the network that often treat edges as springs and nodes as repulsive electrostatically charged particles. The forces can be tuned to enhance the aesthetic features of the visualization.

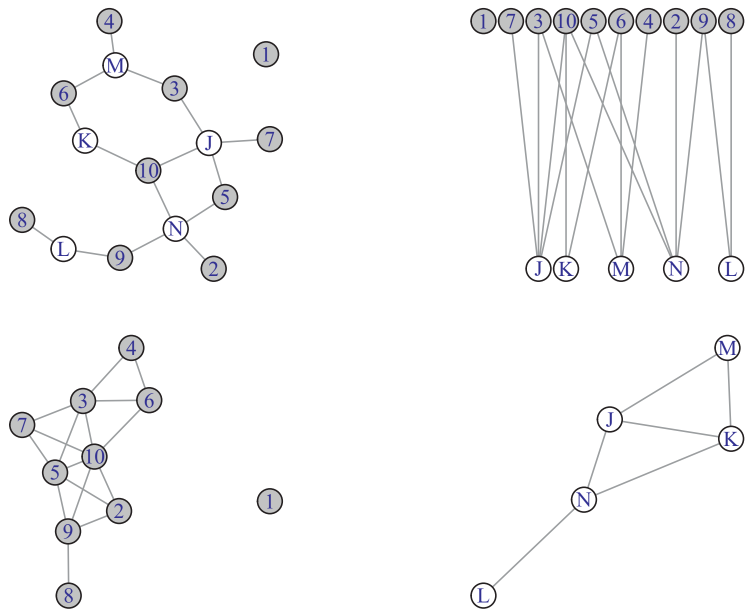

Figure 1.9 The top two networks show a single bipartite network plotted in two different ways. The bottom two networks are the results of projecting the top network onto each of the two node types.

Table 1.8 Bipartite adjacency matrix with two node types.

	J	K	L	M	N
1	0	0	0	0	0
2	0	0	0	0	1
3	1	0	0	1	0
4	0	0	0	1	0
5	1	0	0	0	1
6	0	1	0	1	0
7	1	0	0	0	0
8	0	0	1	0	0
9	0	0	1	0	1
10	1	1	0	0	1

edge relationships. These relationships can then be separated or combined to investigate new networks with novel properties.

For example, in social networks, edges could represent work, family, *and* local community relationships. In language networks, edges could represent phonological (sound) *and* semantic (meaning) relationships. Multiplex networks combine these edge types in different ways to understand their relative contributions and patterns of interaction. Figure 1.10 shows a full network with three different edge types. It also shows the network associated with each edge type separately.

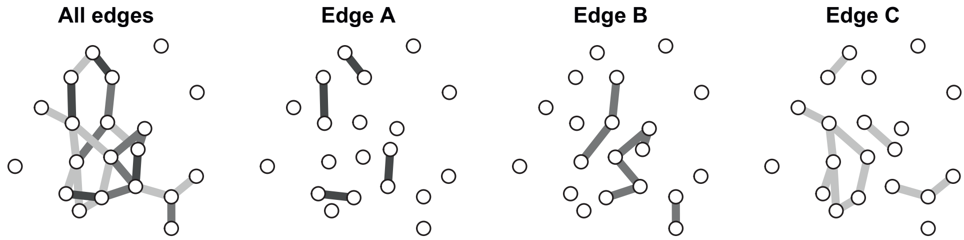

Figure 1.10 Multiplex network in which the combined representation is made up of edges of three different types.

1.6 Turning Data into Networks

Data does not always come to us in the form of an edge list or adjacency matrix. Instead, there are surveys, lists of remembered stimuli, decisions made by individuals at different times, natural language, collections of documents, and so on. All of these can be turned into networks, but there is no single answer as to how to do it.

I will offer several examples here, but many others are provided throughout the book and many more are published each day. In general, however, the workflow is as follows:

First, identify the nodes. Nodes are typically based on theoretical questions: Are we interested in individuals, groups of individuals, specific behaviors, categories of behavior, or something else? This is a research decision, but networks invite us to consider it carefully. For example, we might want to know how disease symptoms progress through a population. We could treat individuals as the nodes and attempt to identify edges between individuals based on symptom similarity. Alternatively, we could treat symptoms as nodes, noting that insomnia is comorbid with depression and possibly has a directed edge, such that one leads to the other. The breadth of possibilities is wide and it is useful to consider them with some creativity.

Second, identify the edges. When the data comes in the form of relationships between nodes of interest, things are often clear. In many cases, though, there are yet more veiled relationships. They might be inferred from the data or we might bring some outside data to bear on the problem. For example, when nodes have collections of features they share with other nodes, we have a bipartite network. We could choose a projection from a bipartite network to form edges. But edges could also be weighted based on the number of shared features (as in bibliometric networks, e.g., Newman, 2018), or on the cosine similarity between vectors of shared neighbors (as in semantic space models, see Chapter 12), or on other measures of similarity or distinctiveness (see Chapter 8). We could have many people evaluate the similarities among pairs (see Chapter 13) or use statistical models to compute similarities among documents or words that appear otherwise unrelated. When order matters, we can use ordered lists to construct networks that would regenerate the lists that produced them (Zemla & Austerweil, 2018). It is often useful to consider different edge possibilities as representing different hypotheses about what matters. Then we can see how well they explain the phenomenon of interest.

1.6.1 Survey Data

If we have a list of items that we care about, such as long-term health risks, we can ask a group of people to evaluate each of these along a common scale. We then have rows of individuals and columns that indicate how they answered each question. **Network psychometrics** approaches have developed numerous tools for asking about the relationships among such variables (e.g., Christensen et al., 2018; Isvoranu et al., 2022). The general principle is that one infers edges based on the correlations or covariance between questions (i.e., nodes) in the pattern of people's responses to them. Edges with more weight represent more similar patterns of responses across participants. Now nodes are not participants but the items they are evaluating. Edges are similarities in evaluation.

One of the key complications of these approaches is that the number of edges grows dramatically with the number of variables. If there were 20 variables (nodes), then we would have 190 edges to estimate. And we may not have enough data to do that well. To deal with this problem, filtering methods can be used to reduce the number of edges. Several of these filtering methods are shown in Figure 1.11. The development of these methods is ongoing. They are also not without controversy. That said, there is no better way to understand the controversy than to see it in action with your own data.

Suppose we have binary responses to survey data as shown in Table 1.9. This data might represent "yes" or "no" responses to a series of five questions each answered by an individual participant.

Table 1.9 A sample of cross-sectional survey data. Rows are participants and columns are binary responses to survey questions.

	A	B	C	D	E
1	0	0	0	0	0
2	0	1	0	1	1
3	0	0	0	0	0
4	1	1	0	0	0
5	0	1	0	1	0

Table 1.10 A sample of cross-sectional survey data. Rows are participants and columns are responses to survey questions with continuous ratings on a scale from 1 to 7.

	A	B	C	D	E	F	G
1	4	4	4	5	5	5	5
2	3	3	5	3	1	5	4
3	6	4	4	6	5	5	5
4	2	2	4	3	3	5	4
5	6	5	3	6	3	4	6

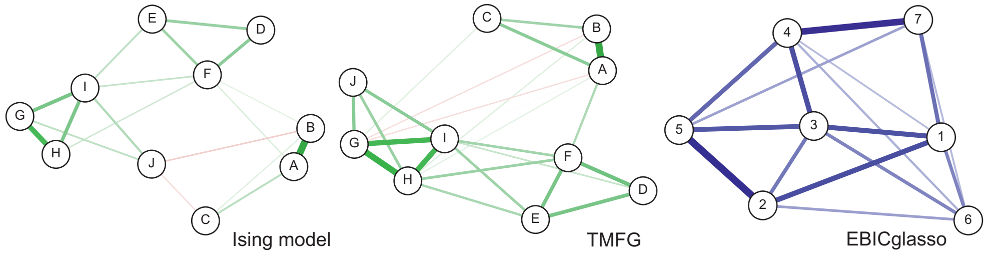

Figure 1.11 Network psychometric graphs based on survey measures. Networks shown are the output of the Ising model and the TMFG information filtering graph for a 10 question survey with 1,700 individuals. Also shown is the output of an EBIC graphical lasso model from a stylized data set representing continuous survey data from 150 participants on 7 survey questions, each rated from 1 to 7.

One method for identifying edges in this data is the *Ising model* (Van Borkulo et al., 2014). This is based on logistic regression, predicting each variable based on the other variables – giving stronger edges to better predictors. It then penalizes edge weights by limiting the total edge weight possible over the entire graph, shrinking smaller edge weights toward zero. The Ising model attempts to identify the best solution by choosing the network that makes the best trade-off between minimizing error and minimizing model complexity. This trade-off is handled using a variation of the Bayesian Information Criterion, which is discussed in more detail in Chapter 6. Figure 1.11 shows the output of this model run on a binary data set with 12 variables and 3,000 individuals.

Information filtering networks offer another approach. This approach computes edges based on the binary equivalent of Pearson's correlations. These edges are then ordered and added to the network one at a time according to a set of rules that minimizes edge overlap (see Massara et al., 2017). This produces the TMFG (triangulated maximally filtered graph) network shown in Figure 1.11. See Christensen et al. (2018) for further details on this approach and the Ising model.

If the data is continuous, such as survey data with rating scales, we have something like that shown in Table 1.10. Edges can be created here using the partial correlation coefficients combined with what is called a *graphical lasso*. The graphical lasso penalizes edges, similar to the Ising model mentioned earlier, simplifying the overall edge structure by keeping only those edges with the strongest weights (Epskamp & Fried, 2018). This produces the EBICglasso network shown in Figure 1.11.

There are many other approaches. Rarely do a few months go by that someone does not come up with a new and insightful way to generate nodes and edges. The possibilities are far from exhausted.

In this chapter, we have gone from thinking about nodes and edges to building a variety of different network types from our data. In the next chapter, we will start evaluating network properties.

2 Network Metrics

A valuable feature of networks is their ability to quantify structural relationships. Much like a chemist, who, having purified their sample, can bring to bear a wide range of tools to examine its qualities, a network scientist, having transformed their data into a network, can leverage a wide range of tools to understand its structure.

Network analysis allows us to do this at three different scales:

- Micro: What are the structural properties of individual nodes and edges?
- Meso: What are the structural properties of local communities or **subgraphs** (i.e., subsets of the network)?
- Macro: What are the structural properties of entire networks?

By tuning our analysis across these three different levels, we can bring different features of the network into focus. For example, nodes have structural features such as how many neighbors they have (degree) and how far away they are from other nodes (shortest path length). Whole networks have the average of their individual nodes' features, but they also have graph-level properties such as overall connectivity (density), the distinctiveness of their communities (modularity), and the likelihood that similar nodes connect with one another (assortativity). Subgraphs – which can be identified using various methods – can be evaluated both at the node and graph level, as well as compared with one another.

The number of possible measurements is vast and there are too many to cover in any single book. There are many excellent resources for learning more than are presented here (e.g., Barabási & Posfai, 2016; Menczer et al., 2020; Newman, 2018). More pragmatically, it is not uncommon to develop one's own measures that are fit to purpose, and this book provides several examples of that as well. Here we will cover some of the most popular off-the-shelf measures as these are the ones researchers are most likely to encounter and use. Table 2.1 provides a quick reference for the nomenclature. These also provide a solid foundation for understanding and developing others. I will present these in a concise format, with brief examples and accompanying online code, noting that we will cover these in more applied cases later.

2.1 Nodes, Edges, and Density

Leo Tolstoy once noted that "all happy families are alike; each unhappy family is unhappy in its own way." If we consider a network as a kind of family, and happy families as being made up of individuals that are all happy with one another, then we can translate our

Table 2.1 Basic network measures.

Variable	Definition
N	Number of nodes
E	Number of edges
ρ	Density
L	Average shortest path length
D	Diameter
C	Clustering coefficient
k	Degree
b	Betweenness centrality
c	Closeness centrality
x	Eigenvector centrality
r	Assortativity
Q	Modularity

happy family into a network with edges between every pair of nodes. Such a network is a **fully connected graph** or a **complete graph**. When edges go missing, the house becomes divided, and the possible ways in which this can happen proliferate.

A graph-level measure of this division is **density**, which is the number of observed edges divided by the number of possible edges. For an undirected network, it is therefore the probability that if you select a random pair of nodes, you will find an edge between them.

If an undirected network is fully connected – all nodes are connected with one another – the density is 1. Groups of nodes that have this property are called a **clique**: they are fully connected and therefore have a density of 1 among themselves.

If there are no connections between nodes, this is an empty graph and the density is 0. Nodes that are completely unconnected are called **isolates** (or **hermits**).

How many possible edges are there for a network of size N? A square adjacency matrix with N nodes has N rows and N columns and is therefore size $N \times N$. If every node can have a directed edge to every other node, including itself, then every cell in the adjacency matrix could have a value of 1, giving us $N \times N = N^2$ possible edges. This powers *Metcalfe's law*: the value within network relationships grows as N^2.

If we don't include self-loops, then we subtract the diagonal, which gives us $N^2 - N = N(N-1)$, which is the possible number of edges in a directed network without self-loops.

An undirected network has only one edge possible between every pair of nodes; recall that it is symmetric and therefore $A_{ij} = A_{ji}$. So it has half as many possible edges as a directed network. The total number possible is then $N(N-1)/2$ for an undirected, unweighted network with no self-loops.

Thus, we can define density, ρ, for an undirected network as:

$$\rho = \frac{2E}{N(N-1)} = \frac{\textit{observed edges}}{\textit{possible edges}}$$

By definition, as the density increases it becomes more likely that nodes in a network will be connected together.

2.1.1 Paths and Path Length

“Six degrees of separation” and “it’s a small world after all” reflect our surprise that people are often more socially near one another than we anticipate. In network terminology, we say that the **path length** is short. The number of edges we would need to traverse to get from one person to the other is small.

The shortest path between nodes i and j is called their **geodesic distance**, and it is the minimum distance in edges needed to travel between them. It can be written as d_{ij}. In a simple network, the distance is the count of edges. If the network is directed, the path must follow the direction of the edges.

The dark edges in panel (a) of Figure 2.1 show that the shortest distance between node 1 and node 5 is 3. There can be more than one shortest path if several paths have the same length. For example, the path here could also travel from $1 \to 3 \to 6 \to 5$.

If two nodes are not connected by a path, the path is infinite.

The **average shortest path length**, L, for a network is the average of the geodesic distances between all pairs of nodes that have a finite path length. Figure 2.2 provides an example with each node labeled with its average shortest path length. Table 2.2 provides the corresponding matrix of path distances for each node to every other node.

The **diameter**, D, of a network is the longest path length we would have to travel to get between the two most distant nodes if we took the shortest path between them. It

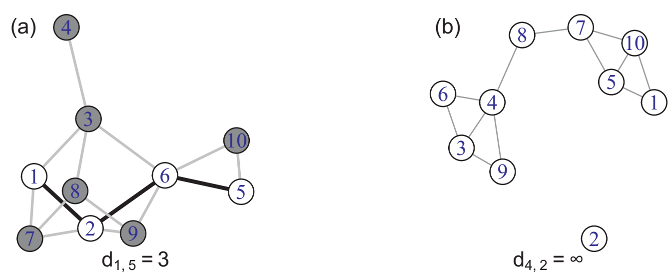

Figure 2.1 Examples of path length.

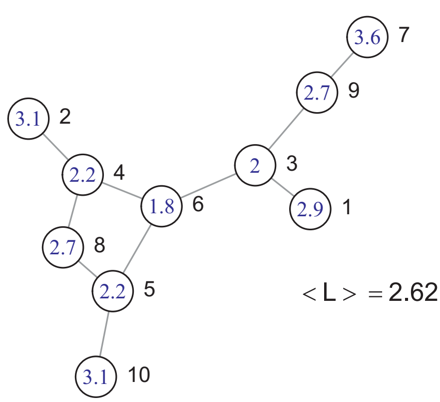

Figure 2.2 The average shortest path length. Each node is labeled with its average shortest path length to all other nodes (inside the circle). The average of these is the average shortest path length for the network. Each node is also labeled to the right with its node ID from Table 2.2.

Table 2.2 The distance table for the graph shown above. Each cell indicates the shortest path between the nodes indicated by the row and column.

	1	2	3	4	5	6	7	8	9	10
1	0	4	1	3	3	2	3	4	2	4
2	4	0	3	1	3	2	5	2	4	4
3	1	3	0	2	2	1	2	3	1	3
4	3	1	2	0	2	1	4	1	3	3
5	3	3	2	2	0	1	4	1	3	1
6	2	2	1	1	1	0	3	2	2	2
7	3	5	2	4	4	3	0	5	1	5
8	4	2	3	1	1	2	5	0	4	2
9	2	4	1	3	3	2	1	4	0	4
10	4	4	3	3	1	2	5	2	4	0

is therefore the longest of the shortest paths. It is analogous to measuring the diameter of a circle.

The path length of a weighted network can be computed in terms of number of edges (as noted earlier) or it can use the weights of edges. How we use the weights depends on what the weights represent. If the weight represents some measure of closeness (such as similarity), then to correspond to distance it should be construed as the inverse of the weight, $1/w$. If the weight indicates a form of distance along the edge, then path length can be computed straightforwardly as the sum of the weights.

Because infinite paths are computationally intractable, the average shortest path length is computed among only those nodes that share a path – that is, they are **reachable**. This can create some curious behavior. Imagine a network that changes its structure by breaking into dyads, like a dance party. When this happens the average shortest path length is 1, but most nodes are not connected at all. The average shortest path length would then become a poor indicator of the connectivity of the network. In general, as nodes become unreachable via paths, average shortest path length becomes a less desirable measure of network structure. Some other measure of connectivity is likely to be better, such as the number of components.

2.2 Components

Various estimates suggest there are on the order of 100 groups of Indigenous peoples who remain uncontacted by the larger non-Indigenous society we might think of as the world community. The majority of these isolated groups live in South America and, because they remain uncontacted, their true numbers are difficult to estimate. In the small world experiment of Stanley Milgram (1967), in which people attempted to send letters via acquaintances to people they did not know, these Indigenous peoples could not have been included in the data because they are, by definition, unreachable. The six degrees of separation we often use to describe the structure of our social environment – and which was

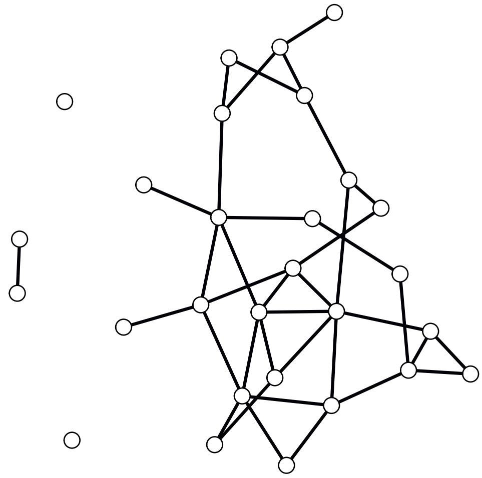

Figure 2.3 A network with 4 components.

in some ways confirmed by Milgram's experiment – does not really apply to everyone. It only applies to those who are reachable. Unreachable nodes are said to be in a different component.

A **component** is a collection of nodes that are all reachable, one to the other. Every node in a component can reach every other node in that component by traversing a path over consecutive edges. Figure 2.3 has four components: the largest or **giant component** with 24 nodes, a smaller **island** with 2 nodes, and two solitary isolates.

When a network is directed, a component is **strongly connected** when all its nodes are reachable by following the direction of path edges. A component is only **weakly connected** if some of its nodes are only reachable by swimming upstream – going the wrong direction along a directed edge.

The number of components in a network can be a useful measure of how disconnected a network is. Many network studies focus on resilience and attack tolerance by deliberately damaging networks to evaluate their resistance to fractionation into multiple components (see Chapter 13). The thresholding described in Chapter 1 is one example of this process for weighted networks.

When focusing on network-level measurements, many network analyses focus only on the giant component. This helps avoid some of the measurement issues for multicomponent networks, such as infinite path lengths.

Components are a form of meso-scale or community measure. Zooming out to the whole network we might see it is made up of multiple components. But we can also zoom in on a single component to see that some communities within it are collections of nodes that are all directly connected with one another: a clique. This should give you a sense of the richness of this meso-scale, of which we have only touched the surface.

2.2.1 Network Size, Edge Probability, and Components

When networks change their size, they can change their structure. They can do this even when the underlying processes in the network are the same. Consider a social constraint, like *Dunbar's number*, which suggests that humans can meaningfully hold relationships with on the order of 100 people. In a network of 100 people all exercising their Dunbarian extroversion, the density would be 1 – everyone knows everyone well. But in a 1,000 person network, the density would be 0.1. We would make an error in inference if we concluded that people were less social in larger networks because the density is lower. One would want to use a node-level measure to evaluate that.

If two different networks of the same size had different densities, the structural inference is less problematic. But when they are different sizes, things can become more confusing.

Consider isolates. The probability that any node remains an isolate is the probability that it does not share an edge with any other node in the network. One might imagine that the number of isolates should increase as the size of a network increases, but the intuition is backward if the probability of an edge remains the same. Suppose the probability of an edge is p. Then the probability of not sharing an edge with a specific other node is $(1-p)$.[1] As there are $N-1$ other nodes, the probability that a node is not connected to

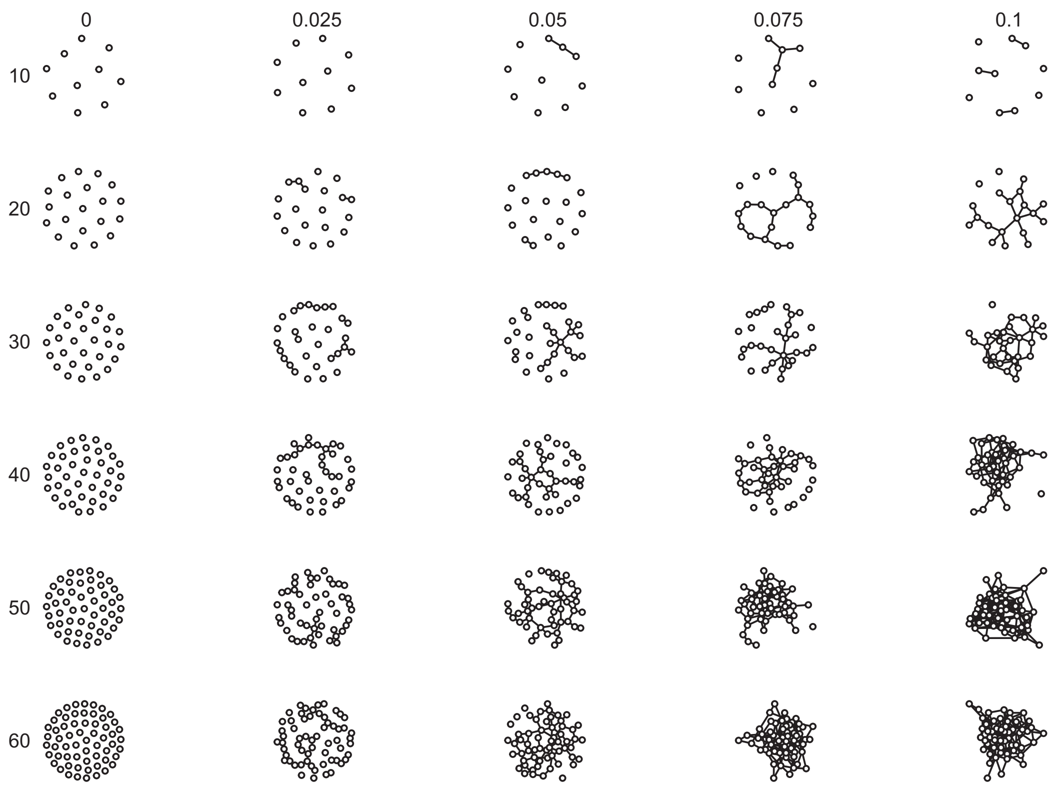

Figure 2.4 The influence of density and number of nodes on components and connectivity. Number of nodes is listed along the rows. Edge probability is listed along the columns.

[1] The network generating probability is p. The density, ρ, is the probability we observe from the network data. As an example, the probability of an edge in a two-node network might be $p = 0.5$, but it will always have an observed value ρ of either 1 or 0.

any other node is $(1-p)^{N-1}$. Because this approaches 0 as N increases, the probability that a node is an isolate shrinks as the size of the network increases. If we assume the probability of forming an edge is fixed then the more nodes there are to connect with, the more likely it is that there will be a connection.

This is science that any party-goer can understand. It is the basis of *the birthday paradox*: even if the probability that two people share a birthday is a mere $1/365$, the number of possible shared birthdays (edges) in a group of 23 people is 253. Therefore, the probability that at least two people will share a birthday in a group of size 23 is $1-(1-1/365)^{(23\times 22)/2}=0.5$. This is another consequence of Metcalfe's Law.

Figure 2.4 shows random networks generated for different edge probabilities and numbers of nodes. As the number of nodes increases, the size of the giant component gets larger and the number of isolates falls. Similarly for the probability of an edge, as the probability increases, giant components get larger and the isolates fall away.

2.3 Centrality

When we are more concerned with the structural properties of nodes than with the network as a whole, we are interested in measures of **centrality**. These vary from local measures that only look at relationships with direct neighbors (e.g., degree) to global measures that involve computing relationships with all other nodes in the network (e.g., betweenness and closeness). Understanding how nodes differ along these measures gives us a sense of the diversity in the network and how nodes differ in importance. Centrality measures therefore give us a node's-eye view of the network. Averaging over them gives us a sense of the average microlevel structure of the network.

There are many centrality measures because there are many different ways of measuring structural relationships.[2] They are each relevant to different structural questions. We cover the more common ones here and more specialized ones later in the book.

2.3.1 Degree

When we say that someone is an extrovert, what we mean is that they have many social relations: their degree is high. An introvert, on the other hand, has a low degree by comparison. Formally, the **degree** of a node, k, is the number of links it has to other nodes.

Hubs are nodes with a larger degree relative to other nodes in a network. Isolates have a degree of 0. In Figure 2.5, every node is labeled with its degree. For most networks, different nodes will have different degrees. A **regular graph** has nodes all of the same degree.

There is a clear relationship between degree and density. To see this, consider the three-node network shown in panel (a) in Figure 2.6. The maximum possible average degree would occur if each node is connected to the other two nodes and therefore has a degree

[2] The Periodic Table of Network Centrality gives a sense of the breadth available here, and this does not cover them all: www.schochastics.net/sna/periodic.html.

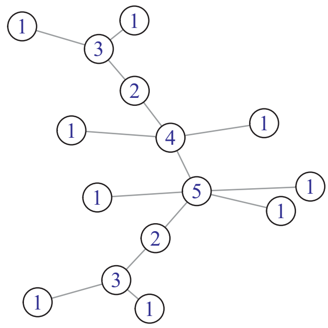

Figure 2.5 Degree. Each node is labeled with its degree.

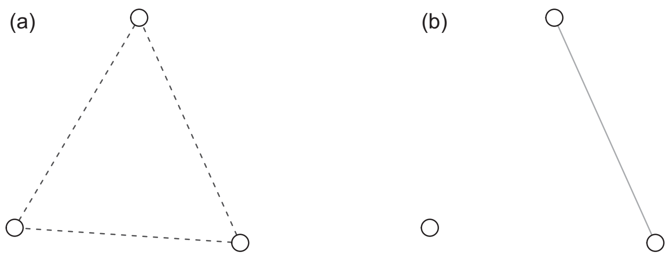

Figure 2.6 Density. Possible versus observed edges in a network with density 1/3.

of 2. In panel (b), the network only has one edge: two nodes have a degree of 1 and one has a degree of 0. The average degree is then $(1 + 1 + 0)/3 = 2/3$. The average degree, $2/3$, divided by the maximum possible degree, 2, is $1/3$. This is equivalent to the density: A three-node network has three possible edges. If it has one edge, then the density is $1/3$.

For undirected networks, the degree of node i is indicated by k_i and is the sum along the row *or* column of the adjacency matrix:

$$k_i = \sum_j A_{ij} = \sum_j A_{ji}$$

In the equation, A_{ij} is the value in the adjacency matrix that corresponds to the edge value between node i and node j. Here, Σ_j means to add up all the values of A_{ij} for all values of j.

Indegree and Outdegree If the network is directed, then each node will have an indegree and an outdegree. **Indegree** is the number of edges directed toward a node. **Outdegree** is the number of edges directed outward from a node. In Figure 2.7 each node is labeled with its indegree and outdegree.

Summing across a row gives the outdegree:

$$k_i^{out} = \sum_j A_{ij}$$

Table 2.3 The adjacency matrix for a directed network showing the row sums (outdegree) and column sums (indegree), which, when summed over all nodes, are equivalent.

	1	2	3	4	5	6	7	8	9	10	Row sums
1	0	1	1	0	1	0	0	0	0	0	3
2	1	0	0	0	0	0	0	0	0	0	1
3	0	1	0	0	0	0	1	0	0	1	3
4	0	1	0	0	1	0	1	1	0	0	4
5	0	0	0	0	0	0	1	0	1	0	2
6	1	1	0	0	0	0	1	1	1	0	5
7	1	1	0	0	0	1	0	0	0	0	3
8	0	0	0	0	0	0	0	0	0	0	0
9	0	0	0	1	0	0	0	0	0	0	1
10	0	0	0	0	1	0	0	0	0	0	1
Column sums	3	5	1	1	3	1	4	2	2	1	23

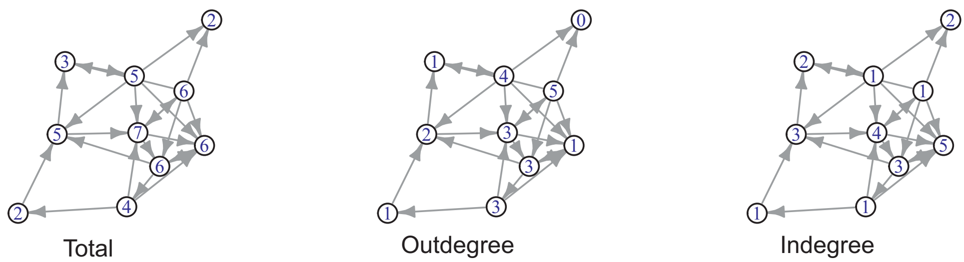

Figure 2.7 Indegree and outdegree alongside total degree. Each node is labeled with its total degree, indegree, or outdegree.

Summing down a column gives the indegree:[3]

$$k_i^{in} = \sum_j A_{ji}$$

Adding indegree and outdegree gives the total degree:

$$k_i^{total} = k_i^{in} + k_i^{out}$$

The average degree for the network is the sum of degrees over nodes divided by the number of nodes:

$$< k >= \frac{1}{N} \sum_{i=1}^{N} k_i$$

The number of out-directed edges equals the number of in-directed edges. Therefore, the average indegree and outdegree are the same, as shown in Table 2.3.

3 A_{ij} is defined here as an edge from i to j. Some texts invert this for mathematical efficiency, assigning A_{ij} as an edge from j to i. It is best to check the source carefully, noting that sources will vary, as do computational tools.

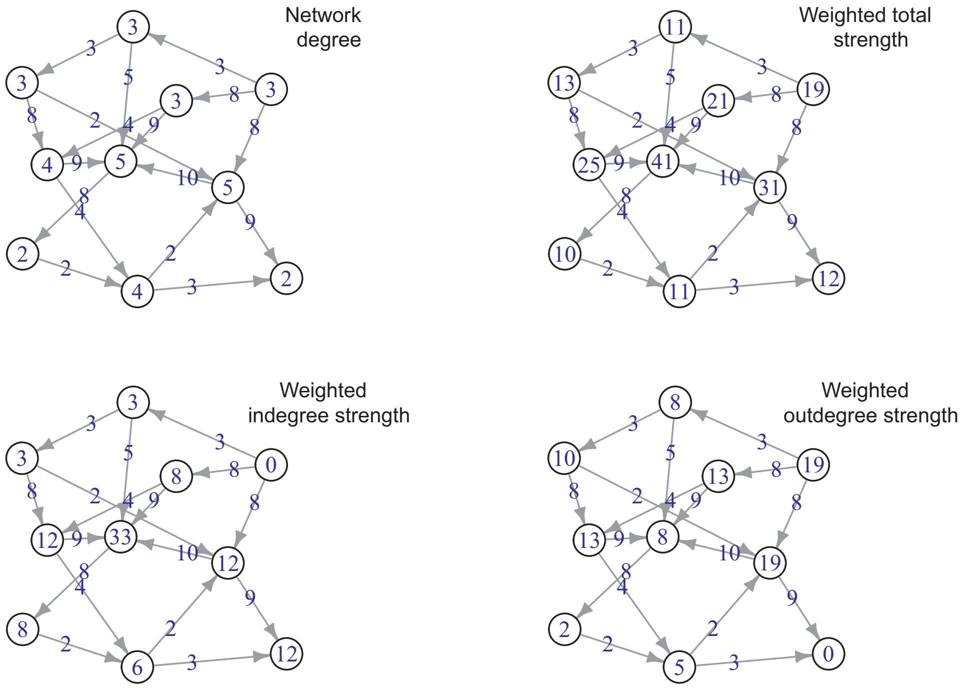

Figure 2.8 Various ways of computing the degree in directed and weighted networks. Nodes are labeled with their relevant degree or strength.

Strength When edges have weights, we want to be able to evaluate the overall strength of a node's relationships with other nodes. We can of course compute the unweighted degree for a weighted network by simply counting edges, not weights. But we can incorporate weights into the degree by summing the relevant weights. This is called **strength**. Networks that are both weighted and directed will have strengths associated with outdegrees and indegrees. See Figure 2.8.

2.3.2 Clustering Coefficient and Transitivity

If all my friends' friends are my friends, then we form a group of people who all know one another. Formally, we are a clique. Less formally, we are a cluster. Clustering is often formally computed using the **clustering coefficient** which measures the extent to which a node's neighbors are themselves neighbors.

The clustering coefficient has two forms. These view clustering from different vantage points. The first is from the perspective of an individual node – it is a node-level measure called the **local clustering coefficient**. This measures the proportion of a node's neighbors that are neighbors of one another. That is, it measures the extent to which friends are friends. See Figure 2.9

If all of a node's neighbors are connected to one another by an edge, then the clustering coefficient is 1. If all are unconnected, the clustering coefficient is 0.

Mathematically, the clustering coefficient for a node is the observed number of edges, e, between its k neighbors over the number of possible edges between its k neighbors. Like

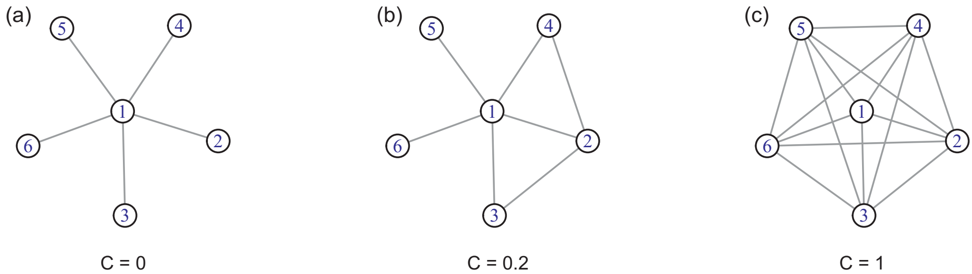

Figure 2.9 Local clustering coefficient. The clustering coefficient for the node (1) in panel (b) is shown below the network.

the network as a whole, the number of possible edges between k nodes is $k(k-1)/2$. Thus, the clustering coefficient is:

$$C_i = \frac{2e}{k_i(k_i - 1)}$$

Once we have the local clustering coefficient for individual nodes, we can average over all nodes to get the average clustering coefficient for the graph.

A second and different way to get a graph-level measure of clustering coefficient is called transitivity.[4] **Transitivity** measures the extent to which friends of friends are friends. However, rather than measure this for individual nodes, transitivity measures this for the graph as a whole. Transitivity does this by identifying all cases where two nodes share a common neighbor and are therefore friends of friends. Then it simply takes the proportion of all these cases where the friends of friends are friends.

In network language, these cases where two nodes share a common neighbor are called a **triplet**. The two nodes at the end of the triplet are friends of a friend. They either share an edge or not. If they share an edge, they make a closed triangle and are called a **transitive triplet** (see Figure 2.10).[5] If they don't share an edge, they are an intransitive triplet and form a 'Λ' shape: what is also called a **2-star**. A 2-star network has a single center node and two nodes that radiate out from it (who could be connected, or not).[6] So every transitive triplet has three 2-stars and every intransitive triplet has one. Transitivity measures the proportion of 2-stars in the network that are transitive (i.e., a triangle).

We can represent this iconically as follows:

$$T = \frac{3\Delta}{\Lambda}$$

[4] The function for computing local and global clustering coefficient in the online code uses igraph and is called 'transitivity'. It is best to ask for the local clustering coefficient or transitivity explicitly, so as not to get confused.

[5] Transitivity comes from the mathematical meaning of the term. If A = B and B = C then A = C, which is a consequence of the transitive property of mathematics.

[6] More generally, a **star network** is a network with a single hub node and all other nodes with degree 1 that radiate out from it.

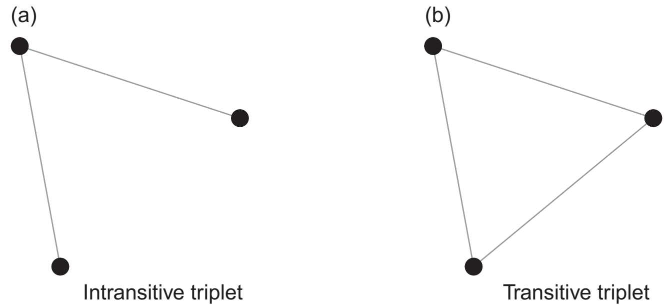

Figure 2.10 Intransitive and transitive triplets. Transitivity is only computed using triplets with either of these two formations. In panel (a), we have one 2-star. In panel (b), the transitive triplet has three 2-stars.

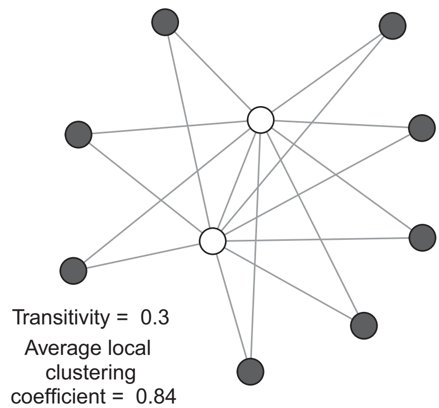

Figure 2.11 The wheel network demonstrates the difference between transitivity and average local clustering coefficient. As the outer nodes increase, the average local clustering coefficient approaches 1 and the transitivity approaches 0.

The total number of 2-stars is designated by Λ. The total number of transitive triplets is Δ. Because each transitive triplet contains three 2-stars, we multiply the number of triplets by three to account for all the 2-stars they make transitive.

Transitivity and the average local clustering coefficient are not the same. They can diverge. The majority of individual nodes may all have neighbors who are connected – making the average local clustering coefficient high – even though the majority of triplets are non-transitive – making the transitivity low. Figure 2.11 demonstrates this. For each new gray node added that connects with the two inner white nodes, the average local clustering coefficient rises toward 1. This is because each new gray node has a local clustering coefficient of 1. But each new gray node also adds many new intransitive 2-stars, causing the transitivity of the network as a whole to fall toward 0.

If people (or nodes) have only one friend, then there is no question about friends of friends – they do not exist. As a consequence, local clustering coefficient and transitivity are undefined for nodes of degree less than two; groups of isolates or dyads (two nodes connected by an edge) have neither an average local clustering coefficient nor a transitivity.

Because graph-level transitivity and average local clustering coefficient are not the same thing, I prefer to use transitivity when referring to the graph-level measure and average clustering coefficient (or clustering coefficient) when referring to the local measure.

2.3.3 Closeness Centrality

If you were looking to live near an airport with the shortest flights to all other cities, you would be looking for an airport with high closeness centrality. If you were hoping to identify a social influencer in a social network whose influence would spread most rapidly to others, you would again be looking for someone with high closeness centrality.

Closeness centrality is defined as 1 over the sum of the shortest path lengths to all other nodes:

$$c_i = \frac{1}{\sum_{j=i}^{N} d_{ij}}$$

Figure 2.12 shows three networks with each node labeled with its closeness centrality. Taking an end node from the network in panel (a), the sum of the distances to the two other nodes is $1 + 2 = 3$. The inverse of this gives the closeness centrality: 0.33. The same logic applies to the middle node, which gives us 0.5.

In the network in panel (b), nodes are again labeled with their closeness centrality. There are two things to notice here. First, closeness centrality goes down with the number of nodes included in the distance summation. Larger components will generally have lower average closeness centrality. Second, an island can contain nodes that have a closeness centrality of 1 even though they are unreachable from the majority of the nodes. Closeness is only calculated for reachable nodes. As a consequence, it is mainly only meaningful for nodes within a single component.

The network in panel (c) is the same as the middle, except here closeness centrality is normalized. The normalized closeness centrality is the same as the inverse of a node's average shortest path length. We can see this somewhat plainly if we consider the gray node near the center. One over its closeness centrality gives us $1/0.58 = 1.72$, which is the number of edges on average needed to travel to each of the other nodes in that component.

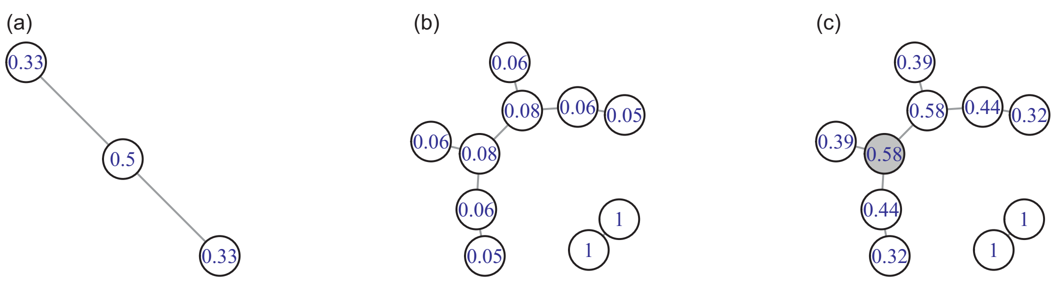

Figure 2.12 Closeness centrality. Each node is labeled with its closeness centrality.

2.3.4 Betweenness Centrality

When a node lies on the shortest path between many other pairs of nodes it has high betweenness centrality. Cities along the Silk Road, such as Xi'an and Constantinople, became what they are today because they had high betweenness centrality along early trade routes. Many port cities have benefited from the same middle-man status in economic trade. This is not always a good thing. Mexico's Juárez is a city with high betweenness in the drug trafficking trade, falling along the border between the USA and Mexico. Its high betweenness made it one of the murder capitals of the world as cartels fought for control of the city.

The **betweenness centrality** for a node is the number of shortest paths between all other nodes that pass through it. Formally, this is computed as:

$$b_i = \sum_{i \neq j \neq k} \frac{\sigma_{jk}(i)}{\sigma_{jk}}$$

Here, $\sigma_{jk}(i)$ represents the number of shortest paths that pass from j to k through i. To account for the fact that sometimes there are multiple shortest routes, we divide by the total number of routes: σ_{jk}. If all the shortest paths between j and k pass through i, this will add 1 to the betweenness centrality. If some don't pass through i, then the increase in betweenness for node i is the fraction of shortest paths that pass through i. Figure 2.13 labels each node with its betweenness centrality.

Betweenness centrality increases with the size of the network. As betweenness is the sum of paths, more nodes means more paths. To resolve this, betweenness is often normalized by dividing by the maximum possible number of shortest paths. If all paths pass through a node, as they do for the central node in Figure 2.13, the normalization scales its betweenness value to 1. Using the same logic, betweenness centrality can also be computed for edges. Wherever many paths converge to cross from one place to another, the betweenness is high.

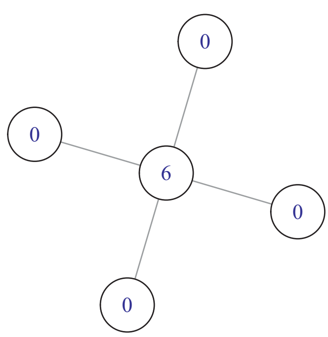

Figure 2.13 Betweenness centrality. Each node is labeled with its betweenness centrality – the number of shortest paths between other nodes that pass through it. Six paths travel through the central node, but none pass through the outer nodes.

2.3.5 Eigenvector Centrality

Eigenvector centrality measures how important a node is based on how important the nodes are that it is connected to. To provide an intuitive example, if a person has only one friend (degree one), they are still likely to be a more important person if their one friend is Paul McCartney than if it is, say, Jim Kooti from southern Oklahoma. Everyone knows Paul. Many fewer know Jim. Eigenvector centrality measures this difference in prestige. To be prestigious, one must receive prestige from others with prestige.

The definition is recursive. It requires that we know how prestigious nodes are before we can compute how prestigious nodes are. Like many chicken-and-egg problems, this one can be solved algorithmically by taking a random guess and then letting the network work out the implications. Assign nodes random prestige values and then allow them to transfer this to their neighbors. Receiving neighbors then take the prestige they receive and send it out to their neighbors. Repeated several times, the relative prestige values will stabilize. Because many people are likely to give prestige to Paul, and few give it to Jim, Paul can in turn give more prestige than Jim. If you're only going to have one friend, pick someone with good eigenvector centrality.

Of course, eigenvector centrality is not specific to prestige. It applies to any adjacency matrix and it is easily solved in practice using matrix algebra. Given a vector of values, $\mathbf{x}$, representing the prestige of every node such that the highest value is 1, we can multiply this by the adjacency matrix, $\mathbf{A}$. This distributes prestige to the neighboring nodes. When this distribution is stable, the product of this multiplication will be the original vector, $\mathbf{x}$, multiplied by a scaling factor, λ. The scaling factor is called the eigenvalue. Formally, that is:

$$\mathbf{Ax} = \lambda\mathbf{x}$$

Software like R will handle this readily, and some example eigenvector centrality measures are shown in the networks in Figure 2.14.

Eigenvector centrality is often most useful for undirected networks. **PageRank** and **Katz centrality** are useful alternatives for directed networks. Each makes slightly

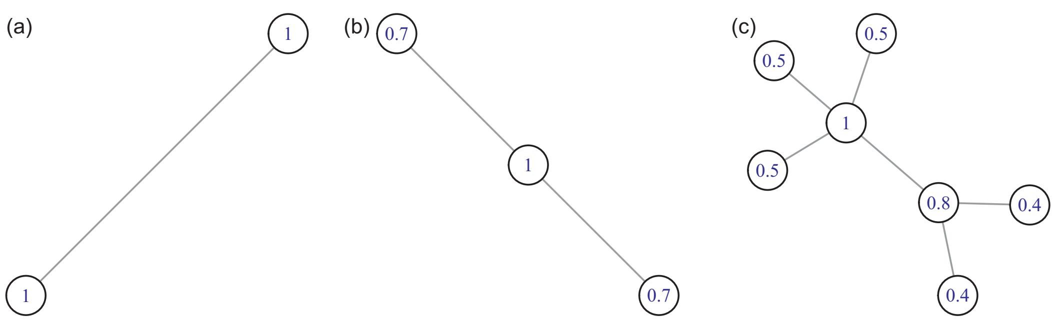

Figure 2.14 Eigenvector Centrality. Each node is labeled with its eigenvector centrality. Note that in the network in panel (c), nodes with the same degree can have higher scores if they connect with nodes of higher eigenvector centrality.

different assumptions about how value is distributed, but all are fairly equivalent in their underlying logic. Chapter 19 discusses these measures in more detail.

2.3.6 Comparison of Centrality Measures

It is useful to see centrality measures side by side. I have created the Mouse to demonstrate these different centrality measures, shown in Figure 2.15. It is modeled after the pocket mouse of the desert Southwest and it gives some personality to these measures, which emphasize the head, body, or tail, for example.

The Mouse is inspired by **Krackhardt's kite network** (Krackhardt, 1990). The Kite is provided here for further comparison (Figure 2.16). Together the Mouse and the Kite should help crystallize these centrality measures and where you might be likely to find them.

There is no reason not to consider developing your own centrality measures. Often a specific behavioral problem will require a measure that is fit to purpose. It is common practice to do this to solve specific problems when you are unable to find an existing measure.

In summary, high degree nodes can be found anywhere. High clustering coefficient nodes are found in communities (small or large). High closeness nodes are found near the center of the network. High betweenness nodes tend to separate different communities. And high eigenvector nodes tend to occur near higher degree nodes in assortative communities, which is discussed next.

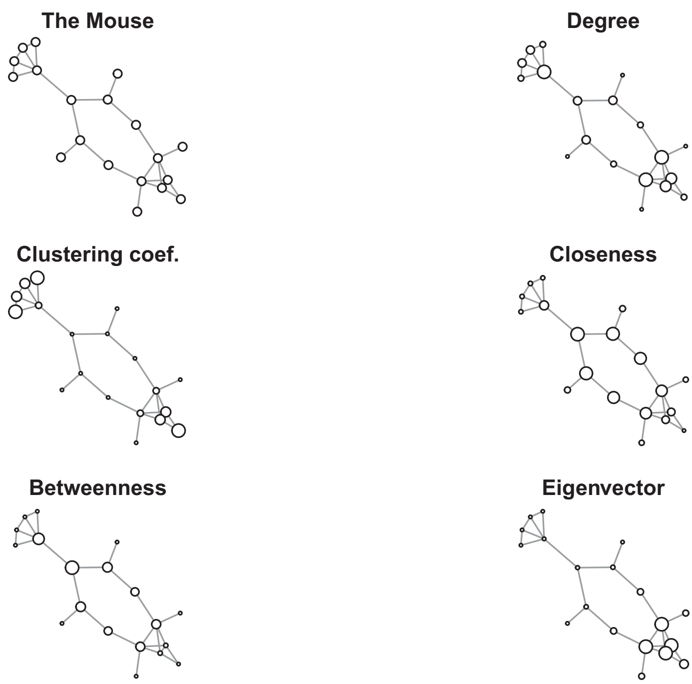

Figure 2.15 Comparison of the Mouse's centrality measures. Node sizes are scaled relative to the maximum and minimum of the centrality measure to highlight the differences.

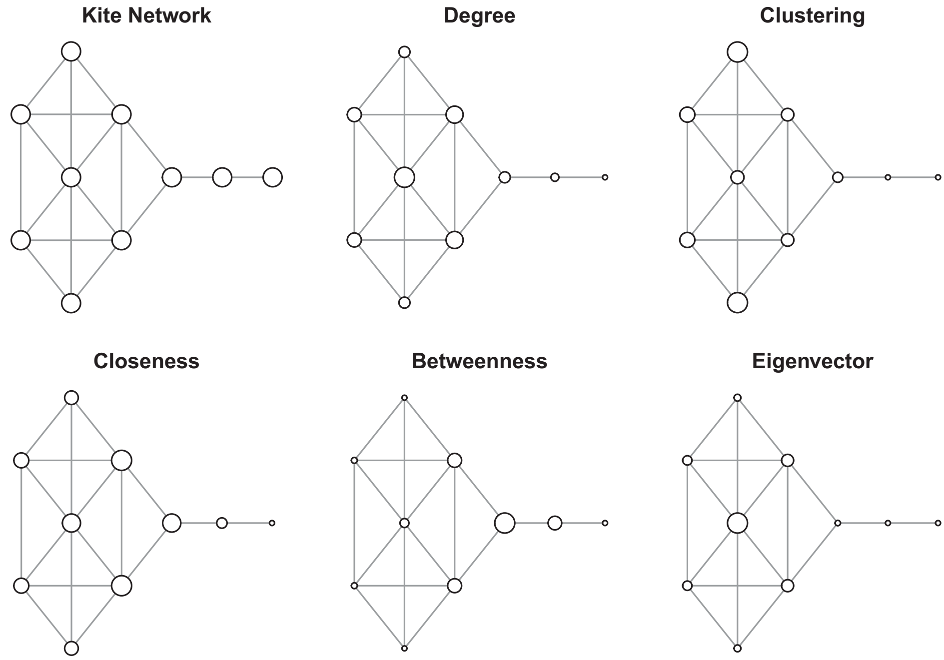

Figure 2.16 Comparison of centrality measures using Krackhardt's Kite. Node sizes are scaled relative to the maximum and minimum of the centrality measure to highlight the differences, as for the Mouse in Figure 2.15.

2.4 Assortativity and Homophily

Assortativity evaluates the degree to which nodes with similar properties connect with each other. In social networks, this is known as *homophily*: "birds of a feather flock together." People tend to associate with others who share similar features. Network analysis allows us to ask this about structural properties: for example, do nodes of a similar degree connect with one another? We can also ask about similarity in other attributes: for example, do individuals connect with others of a similar age or job?

To evaluate assortativity we compute an assortativity coefficient, r, which is the Pearson correlation between pairs of connected nodes in the network with respect to the value in question. To do this, we can generate an edge list from the network, replace the node labels with the value for each node, and take the correlation of the two columns of values.

The assortativity coefficient lies between -1 and 1, indicating the extent of the network's assortativity. When $r > 0$, the network has assortative mixing, or positive assortativity. When $r = 0$ the network is nonassortative. When $r < 0$ the network is disassortative or is negatively assortative – similar nodes are not connected.

Figure 2.17 shows two networks that differ in their assortativity coefficients. It is clear from the visualization that, in panel (a), nodes of similar sizes (indicating degree) are connected with one another. In panel (b), nodes tend to be connected with nodes of different sizes.

Assortativity can also be computed for node attributes. Figure 2.18 shows networks that are assortative or disassortative with respect to node color.

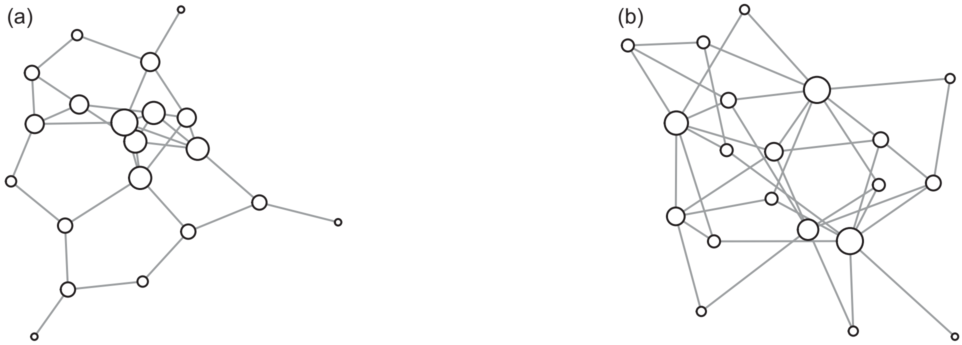

Figure 2.17 Two networks that are assortative or disassortative by degree. The assortativity coefficients are 0.45 (left) and −0.62 (right). Node sizes are scaled to reflect relative degree.

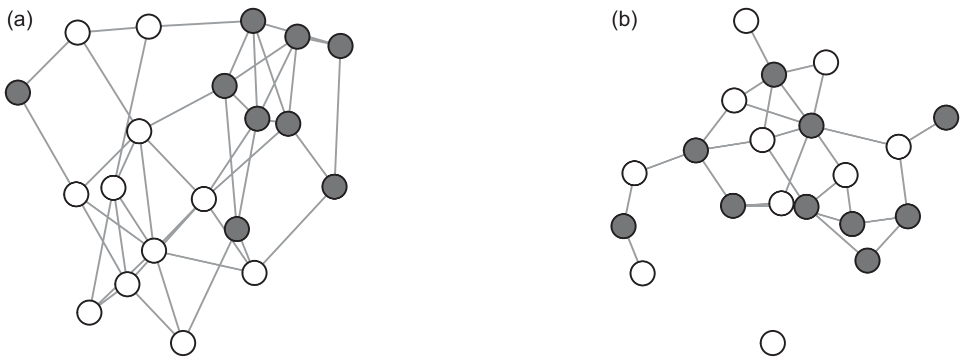

Figure 2.18 An assortative (left) and disassortative (right) network by color. The network in panel (a) has an assortativity coefficient of 0.57. The one in panel (b) is −0.65.

2.5 Community Detection

Networks are excellent for identifying communities. Given a set of people or things with some definition of edges, we can determine which nodes are in which communities as well as how many communities there are. This is powerful as it allows us to sort structured information into groups, identify communities who may be operating together, and predict relationships before they form. Social media platforms that always seem to know which of its many millions of users are our long lost friends are using community detection algorithms to identify them. Other recommender systems, which proffer us appealing movies, clothes, and videos, are also supported by community detection algorithms. With every choice we make, our community alignment becomes more well-defined and the recommendations get harder to refuse.

More formally, the **community detection problem** is one of identifying clusters of nodes that are more well connected to one another than they are to nodes in other communities. Dividing a network into sets of communities is called a **partition**. Identifying the partition that best satisfies the community detection problem is the goal. This is not trivial. The problem is that the number of possible partitions (the Bell number) grows rapidly with the number of nodes: If there are two nodes, there are two partitions: either the nodes are together or they are separate. If there are three nodes, there are five partitions: each node could be in its own community, two could share a community and the third could be on its own, or they could all be in the same community. With twelve nodes, there are more than four million partitions. The number quickly becomes computationally unfriendly.

The solution is to simplify and approximate. Different community detection algorithms make different assumptions to simplify the problem. Some methods are hierarchical clustering methods that assign nodes exclusively to specific groups (e.g., the Girvan-Newman method). Others allow nodes to belong to multiple groups (e.g., the clique percolation algorithm). Like centrality measures, they deserve a book of their own. Here we will cover a few of the more popular algorithms. Before we do that, it is first useful to describe modularity: the measure that allows us to recognize a good partition when we see one.

2.5.1 Modularity

If we saw some people at a party, it would be fairly easy to identify how many groups there were. We identify the groups by placing people who are close together in the same group. If we imagine that people who are near one another share an edge, then we can start to see how we might apply this logic to networks. We identify groups of nodes that share many edges among themselves and few edges with other groups of nodes. We can measure how good our grouping is using modularity. **Modularity**, Q, quantifies the difference between the observed links and the expected links within communities.

The mathematics to quantify this is based on two things. First, we can identify internal links within communities. Second, we can estimate the likelihood that we would see those internal links if edges were distributed at random.

To make this concrete, consider a network of two dyads: four nodes with two edges and each node with degree 1. Let our job be to assign nodes to two communities (let's say black and white), each with two nodes. How many different ways are there to distribute these communities over the network? There are two ways, both shown in Figure 2.19: links can either be within communities or between communities.

If we have to assign communities – so that community members are more likely to be connected – we would do better to assign the same color to connected nodes. Modularity

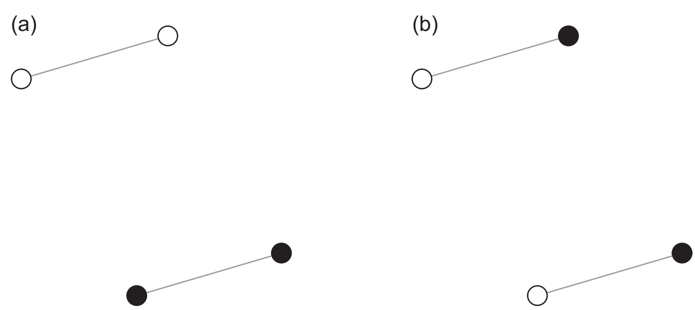

Figure 2.19 Modularity: Two possible network architectures for four nodes with two links, each node with degree 1, and two community assignments (black and white). The modularity for the network in panel (a) is Q = 0.5. For the network in panel (b), Q = −0.5.

tells us the same thing. If we observe links between nodes of the same color, we know there is a $1/2$ chance we would observe this even if the colors were distributed at random. The modularity, Q, is therefore 0.5. If we observe no edges between the same colors, we know there is a $1/2$ chance they could have been between colors. In this case, the modularity, Q, is -0.5.

Here is the equation that formalizes that math for networks of any size:

$$Q = \frac{1}{2E} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{2E} \right] \delta(c_i, c_j)$$

We can walk through this as follows: First, the Kronecker delta function, $\delta(c_i, c_j)$ is 1 if two nodes, i and j, are in the same community: $c_i = c_j$. Otherwise, it is 0. So we are only concerned with cases where i and j are in the same community. Within the square brackets, A_{ij} represents the observed value of the edge between nodes i and j. The product of k_i and k_j divided by two times the number of edges is the probability of an edge between these two nodes observed at random. When considered over all pairs of nodes that are in the same communities, this gives us our observed edges versus our expected edges within communities. Finally, this is all normalized by two times the number of total edges, E. The result, in English, is that modularity is increased whenever two nodes in the same community are connected more than we would expect by chance and reduced whenever they are connected less than we would expect by chance.

Modularity makes the simplifying assumption that a community member's degree values are preserved. To help understand what this means, a three-node network with an edge between two group members and a third node not in the group has a modularity of 0. Because the third node has a degree 0, it can never share an edge with another node by chance. As a result, a set of group members who only have links between them and no links to a set of isolates has modularity 0. This is not always desirable or intuitive. Newman (2018) is a good source for additional thoughts on community detection.

The reason we need modularity is because community detection algorithms often generate many different partitions. Modularity can then be used to identify the partition that best assigns nodes to communities (Fortunato, 2010). Positive Q values that are closer to the maximum value of 1.0 indicate better community structure. In contrast, small or negative Q indicates poor community structure.

In practice, the Q values of large real-world networks such as the Internet and cellular phone networks are relatively high, ranging from 0.42 to 0.78 (Blondel et al., 2008). These values are found by using Q to identify the best partition produced by a community detection algorithm. In the following sections I describe several popular algorithms.

2.5.2 Community Detection Algorithms

2.5.2.1 The Girvan–Newman Method

The **Girvan–Newman Method** (Girvan & Newman, 2002), also called the **edge betweenness method**, is based on the observation that edges connecting separate communities should have high edge betweenness: the shortest paths between members of

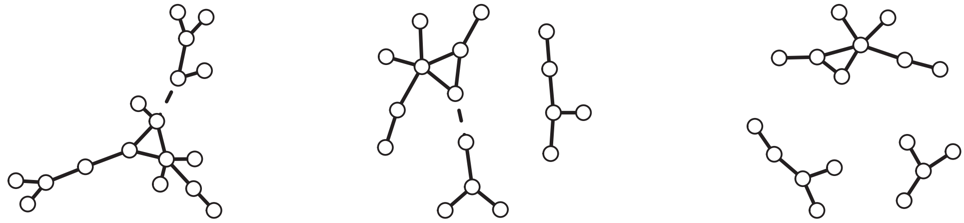

Figure 2.20 A series of partitions created by removing the edge with the highest edge betweenness. The edge to be removed is indicated with a dotted line. The partitions are created from left to right, each time removing the edge with the highest edge betweenness.

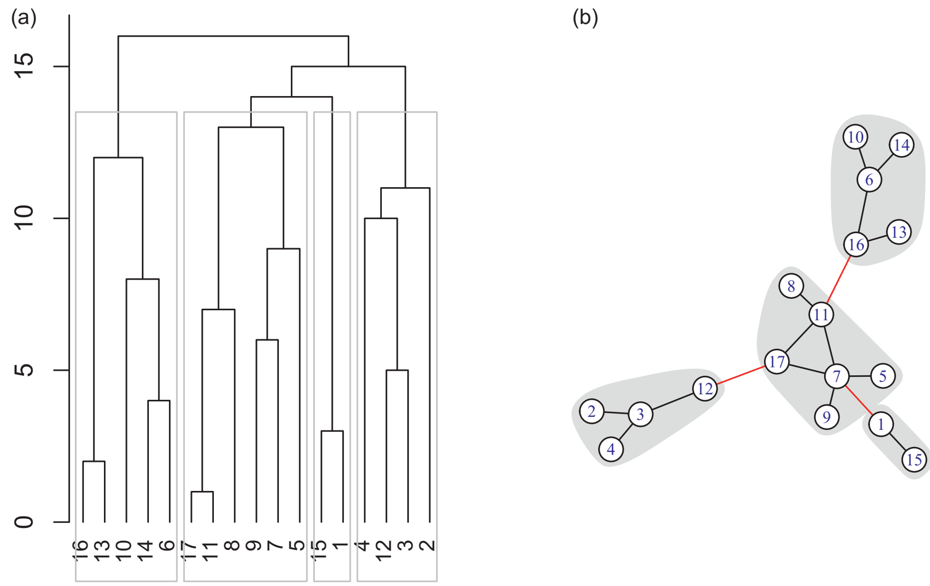

Figure 2.21 The hierarchy of communities identified by the Girvan–Newman method. Any horizontal division along the dendrogram on the left will produce a partition of the network into communities. The y-axis on the dendrogram shows the number of remaining edges for a given partition. The partition with the highest modularity is shown in panel (a) and identified by boxes in panel (b). The modularity of the community shown is 0.51.

different communities will pass through edges with high edge betweenness. This allows us to identify communities by sequentially removing the edge with the highest edge betweenness score. This in turn produces a hierarchy of partitions, which separate the network into smaller and smaller subnetworks until all nodes are isolates. Figure 2.20 shows a few steps in this process.

The hierarchy (or dendrogram) of partitions that is the outcome of this process is shown in Figure 2.21. By cutting the dendrogram horizontally, one identifies a specific partition. The partition in the hierarchy with the highest modularity is used to assign community membership to nodes. The partition with the highest Q is shown in the figure on the right.

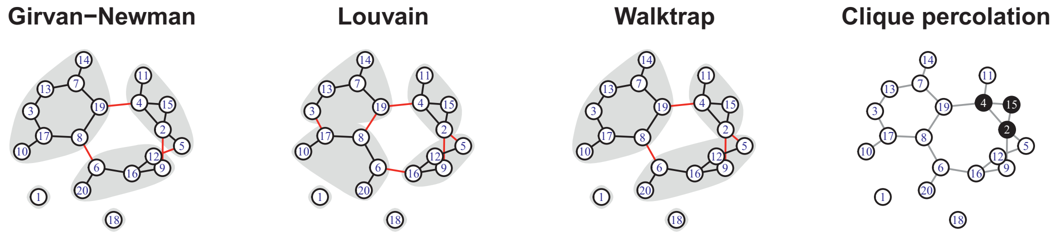

Figure 2.22 Comparison of different community detection algorithms. In the clique-percolation result on the right, a 3-clique is shown in black.

2.5.2.2 Other Methods

Community detection algorithms take a variety of forms. Some examples are shown in Figure 2.22. The **Louvain method** (Blondel et al., 2008) starts by assigning each node to its own individual community. Then communities (initially nodes) are chosen at random and pooled together with another community that best increases the modularity. This is repeated until there is only one community or until the modularity can no longer be increased. The partition with the highest modularity is chosen.

Random walker algorithms, like **Walktrap** (Pons & Latapy, 2005; see also Rosvall & Bergstrom, 2008), assume that random walkers moving between nodes via edges will tend to stay within communities more often than they pass between them. This allows the algorithm to identify a distance between communities using the movement patterns of random walkers. Partitions can then be sequentially identified based on the above distance assumption to produce a hierarchy of partitions. The hierarchy can then be evaluated using modularity.

The **clique percolation algorithm** (Palla et al., 2005) allows individual nodes to belong to multiple communities. It starts by identifying k-cliques, which contain k nodes that all share edges with one another. The smallest is a 3-clique or triangle. Communities are defined as sets of k-clique-adjacent k-cliques. k-cliques are k-clique-adjacent if they share $k - 1$ neighbors. This algorithm often assigns many nodes to isolate communities and preserves only those communities that are engaged in k-cliques of size 3 or more.

Note that some community detection algorithms (e.g., Louvain) apply themselves randomly to the nodes as they search for new partitions. As a consequence, the algorithm may not always return the same partition.

3 Generative Network Models and Network Evolution

Understanding the processes that give rise to networks gives us a better grasp of why we see the networks we do, where we might expect to find them, and how we might expect them to change over time. One way to achieve this is to generate networks of our own. Such simulated networks allow us to build networks based on detailed principles. We can then ask how networks derived from these principles behave. This will give us insight into how our observed networks may be generated by similar principles.

For example, we might hypothesize that the structure of an observed network is based on some assortative property. To evaluate this hypothesis, we can create networks with and without this property and then measure to what extent these networks resemble the observed network. More generally, if we believe a network grows based on a certain principle, we can simulate networks that grow with or without that principle and see how they are like or not like the one we observe.

It is useful to have an understanding of multiple generative network processes by which we can compare the networks we study. Such a toolbox will not only allow us to test our assumptions, but will help us recognize some of the many dimensions along which networks vary.

This chapter focuses on mechanisms for generating networks that are agnostic with respect to the behavioral system under study. They therefore provide useful abstractions which we can use to understand a range of real-world phenomena.

To rapidly develop a conceptual understanding without getting bogged down in unnecessary details, the descriptions here focus on the generative processes necessary to give rise to simple undirected and unweighted networks. Directed and weighted versions of these processes can be derived from this basic framework. I provide some details about how to go about this at the end of the chapter.

3.1 Erdös–Rényi Random Graphs

Suppose your social network was made up of a random sample from all the people in the world. Such a network would be unlike anything you currently experience. Your mother might be from Canada and your father from Jamaica, but they wouldn't know each other and they wouldn't know your grandparents either. We might imagine a dystopian society where everything happens online and all relationships are independently and randomly assigned by the state. This, of course, doesn't make any sense for the typical social network. Most of our new relationships depend on the people we already know. In the real world, networks are like onions. New layers of edges depend on old layers of edges – recall that a friend of my friend is likely to be my friend as well. As a result, edges tend

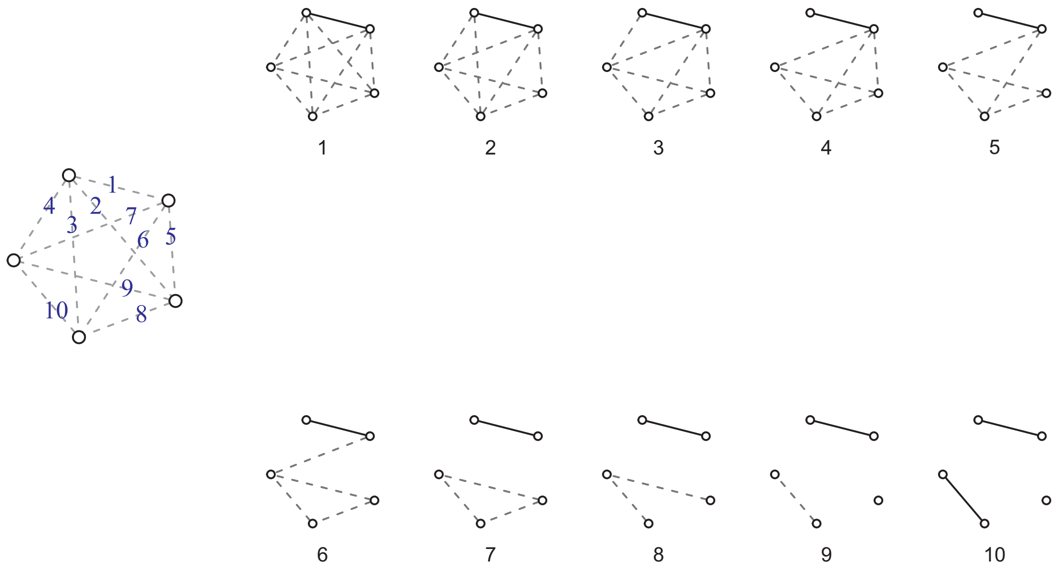

Figure 3.1 The construction of an ER random graph. Starting with the empty graph on the left, we evaluate each possible edge (indicated with dotted lines) and with probability p assign it a value of 1 (solid), or otherwise 0. In this network with five nodes, there are ten edges to evaluate. Each is evaluated in turn in the panels labeled 1 through 10, creating an edge with probability 0.2. Network 10 is one possible output of this random graph generation.

to depend on one another. What would happen if this weren't the case – if all edges were formed independently?

The **Erdös–Rényi (ER) random graph** is an effort to answer this question. First introduced by Erdös and Rényi (1959), it is easy to generate and study and is commonly used as a point of comparison for real-world networks. It is therefore a kind of fruit-fly of the network world. Most network software and the online code for this book will generate these and other network algorithms without complaint, but it is useful to understand the underlying process.

Given a set of nodes, N, an ER random graph places an edge between each pair of nodes with probability p. Each time we generate an ER network, we can get a different answer. But if we do it repeatedly, we will get a distribution of answers that reflect how likely different network structures are if all edges were independently created. See Figure 3.1 for an example.

Generally speaking, comparisons with observed networks involve sampling many hundreds of simulated random graphs and evaluating the observed network with respect to the distribution of simulated networks. Figure 3.2 provides some intuition for the variation in metrics produced by an ER random graph with a fixed edge probability.

A second way of generating ER random graphs uses a fixed number of edges and distributes them randomly. For an undirected network, this can be done by sampling E cells from the adjacency matrix without replacement and adding an edge to each cell sampled. A fixed edge number always produces the same density ($\rho = \frac{2E}{N(N-1)}$). Its distribution of other metrics are also lower in variance because in an ER random graph with probabilistic edges the total number of edges can vary.

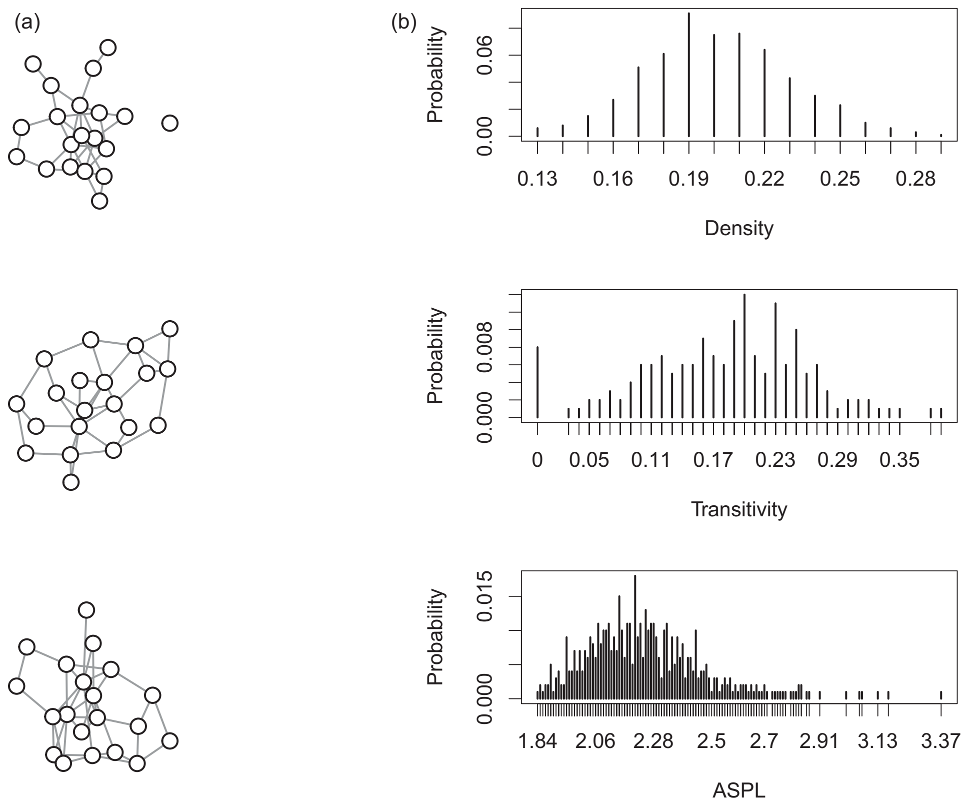

Figure 3.2 Erdös–Rényi random graphs with $N = 20$ and $p = 0.2$. (a) Three networks showing random examples. (b) Histograms of the density, transitivity, and average shortest path length (ASPL) observed for 1,000 random graphs.

3.2 Small-World Networks

If you want to get away from it all, you might travel to one of the world's most remote places, like the Amazonian city of Iquitos in Peru. There are no roads leading to it. If you don't get there by plane from Lima, you'll require a multiday journey by river. If you're in Iquitos, make sure to say "hi" to my old friend Josh, and tell him I sent you.

The term "small world" captures two ideas. The first is that most of the people we know also know each other; we live in local communities of well-connected individuals. The second is that when we meet someone new – even far from home – they often tend to know people we know. In a small world, path lengths between random pairs of nodes are surprisingly short.

To formally capture this notion, Watts & Strogatz (1998) introduced small-world networks. **Small-world networks** have local clustering coefficients that are higher than one would expect from an ER random graph of the same size and density, yet the average path length is relatively similar. In other words, they offer both local clustering and distant connectivity.

There are a number of ways to measure small-world properties, discussed in Chapter 7. Here I will focus on how small-world networks are constructed.

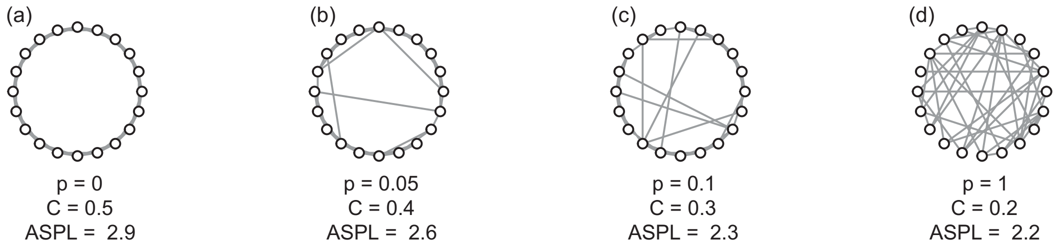

Figure 3.3 The construction of small-world networks. The starting lattice (a) has twenty nodes and each node is connected to its four closest neighbors. Moving from (a) to (b), the probability of a rewiring, p, increases. The clustering coefficient and average shortest path length are shown for the resulting network.

The method described by Watts and Strogatz (1998) involves starting with a regular lattice – all nodes of the same degree – and then rewiring edges with some probability. This process does not alter the network density. However, Watts and Strogatz (1998) showed that the random rewiring reduces the average shortest path length more rapidly than it reduces the clustering coefficient.

Figure 3.3 shows the transition across networks as rewiring probabilities increase. With $p = 0$ the ring maintains its regularity, each node has the same degree, and each is equidistant from all other nodes. With $p = 1$ the network becomes fully disordered. In between, networks are more clustered than a random equivalent even though their path lengths stay relatively short.

3.3 Preferential Attachment

According to the economist Thomas Picketty, author of *Capital in the Twenty-First Century* (2014), rising inequality is an inevitable outcome of capitalism. The rich get richer when the return on capital exceeds economic growth, and they will continue to until an equal and opposite force prevents them. *The Bible* put it more simply in the Gospel of Matthew: "to them that have shall be given."

Social systems with rich-get-richer dynamics tend to be strongly skewed. Some individuals enjoy popularity or wealth that far outstrips that of their nearest neighbor, where most people enjoy little or none of it. Barabási and Albert's (1999) **preferential attachment model** captures this for networks: as new nodes are added to the network, they preferentially attach to nodes that are already well connected. This leads the degree of some nodes to skyrocket, whereas most others fizzle at the smaller end of the scale.

To achieve this, the preferential attachment starts with one node. Then it adds nodes, one at a time. Each new node forms edges with nodes that are already part of the network. The probability of attaching to an existing node in a network of size N is proportional to the degree, k, of the existing node in the following way:

$$P(i) = \frac{k_i^{\alpha} + a}{\sum_{j \in N}^{N} k_j^{\alpha} + a}$$

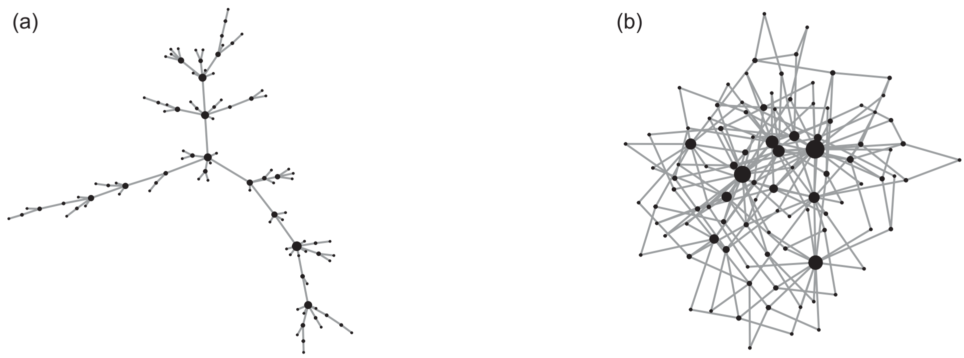

Figure 3.4 Preferential attachment networks. The network adds nodes as it grows, with new nodes attaching to m existing nodes, with a preference for higher-degree nodes. The networks have $m = 1$ (a) and $m = 2$ (b). Node size is proportional to the degree of the node. Both networks have 100 nodes.

To get a feel for this equation, note that if $a = 0$ and $\alpha = 1$, then the probability of forming an edge to node i is directly proportional to its degree, k_i. Higher values of α produce increasingly preferential attachment. At the most extreme, this leads to a star network where one node receives all the edges. Values of α approaching 0 become increasingly insensitive to degree and more random as a result. Figure 3.4 shows two examples of preferential attachment networks generated with $a = 1$ and $\alpha = 1$.

The a is sometimes called the *attractiveness* parameter. It determines the inherent attractiveness of added nodes when their degree is zero (for directed networks) or small (for undirected networks). Tuning the relative values of α and a determines the balance between preferential attachment and inherent attractiveness. The denominator normalizes the probabilities so that across all nodes they sum to 1.

One intuitive instantiation of the preferential attachment model is called the **bag model** (see Figure 3.5). It has $\alpha = 1$ and $a = 1$. The bag model is equivalent to sampling existing nodes from a bag that contains $k_i + 1$ samples for each node, where k_i is the degree of node i. A new node forms an edge to the node drawn from the bag. If multiple edges are formed, multiple nodes are drawn from the bag.

The preferential attachment model has several interesting properties: (1) Early nodes tend to become hubs of increasingly high degree. First movers into the network have a strong advantage over late comers; (2) The degree distribution across nodes is highly skewed. Many nodes will have low degree, whereas a few will have high degree.

This latter property is sometimes referred to as a scale-free distribution (see Chapter 4). Though preferential attachment is sometimes hailed as the quintessential process for generating scale-free networks, there are many generative processes that give rise to scale-free network distributions. For example, forest fire models, preferential acquisition, duplication and divergence, and fitness models can also generate scale-free distributions (all described in the next section).

Another property of preferential attachment models is that if the number of edges that form for each new node is 1, then the graph will not contain any **cycles**. That is, it will not contain paths that are able to end where they originate without crossing over the

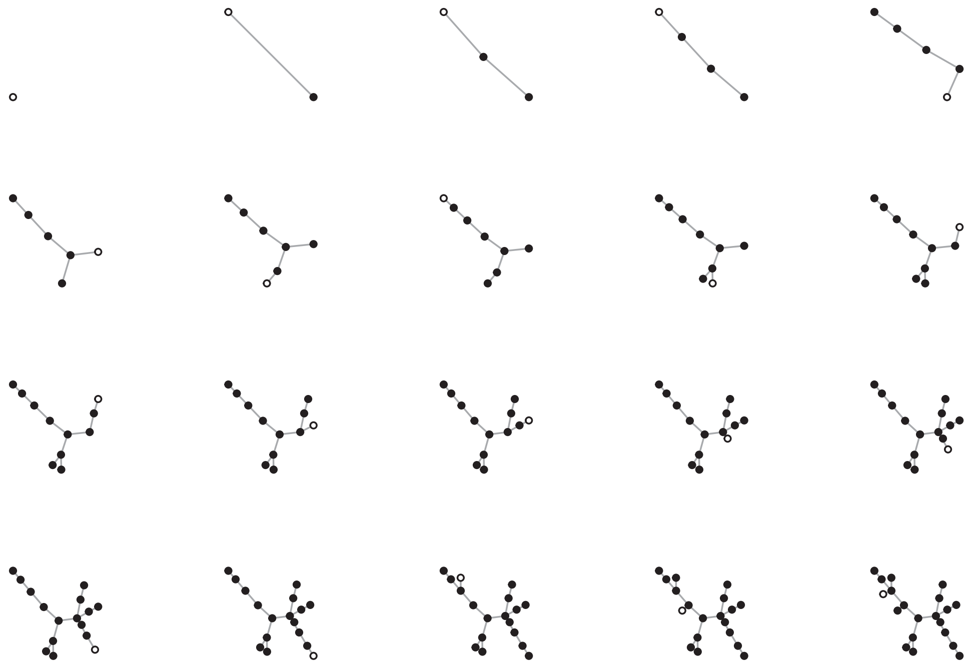

Figure 3.5 The evolution of a preferential attachment "bag" model with $N = 20$. Each frame indicates the addition of the white node to the pre-existing network of black nodes.

same edge twice. Such **acyclic graphs** are peculiar and unlike many social or behavioral systems. However, when they do arise, they may signal certain kinds of behavior such as clandestine networks – where individuals are attempting to minimize contact but maintain connection with the giant component – or minimum spanning trees that may reduce, for example, the cost of maintaining transport networks.

3.4 Fitness Models

In preferential attachment models, early nodes have an advantage. Their degree gains ground early on and that inequality only increases over time. But in some real-world systems, nodes have an advantage because of other qualities unrelated to their degree. For example, they might have an advantage based on attractiveness, and this can happen whenever they enter the network. This dynamic is captured by fitness models.

Fitness models form new edges to existing nodes in proportion to their fitness. The fitness values, η, are randomly assigned to nodes as they are created. This can be combined with Erdös–Rényi random graphs or preferential attachment models.

For Erdös–Rényi random fitness models, edges are formed between nodes chosen at random in proportion to their fitness values. The probability of choosing a node for an edge is:

$$P(i) = \frac{\eta_i}{\sum_{j \in N}^{N} \eta_j}$$

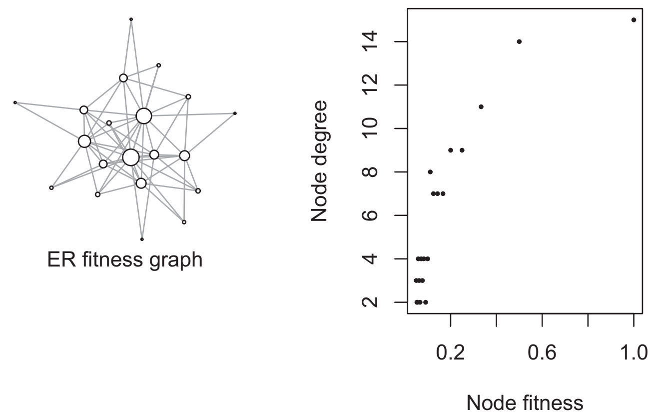

Figure 3.6 Erdös–Rényi fitness-based models. The ER fitness model is based on the scale-free model presented in Goh et al. (2001) with $\alpha = 1$, $N = 20$, and $E = 60$. Node size is proportional to degree.

Two nodes are chosen based on the above probabilities, then an edge is formed between the two nodes. This is repeated for E edges. Figure 3.6 provides an example of an ER fitness network.

Like the preferential attachment network described earlier these ER random graphs can be made to have long-tailed degree distributions by assigning fitness to the nodes in an appropriate way. In Chapter 12, a variation of this approach is used to simulate the structure of conceptual relationships we experience in daily life.

Fitness models can be combined with preferential attachment (e.g., Bianconi & Barabási, 2001), such that the probability of an edge is

$$P(i) = \frac{\eta_i k_i}{\sum_{j \in N}^{N} \eta_j k_j}.$$

The evolution of a fitness-based preferential attachment model is shown in Figure 3.7. Unlike the bag model, nodes in this network can arrive late and still develop high degree.

Fitness networks could have a variety of other implementations, limited largely by one's creativity. For example, attractiveness, a, in the earlier preferential attachment implementation could be a form of fitness and applied independently for each node. Fitness could also be combined with other structural properties besides degree. For example, if fitness were a result of clustering coefficient this could represent a form of group popularity – community fitness.

3.5 Configuration Models

Sometimes we are interested in how the network we observe might be different if only some particular aspect of the world were different. For example, suppose we all had the same number of friends we currently have, but they were replaced with random friends. Hubs are still hubs and hermits are still hermits, but something has changed. What is it?

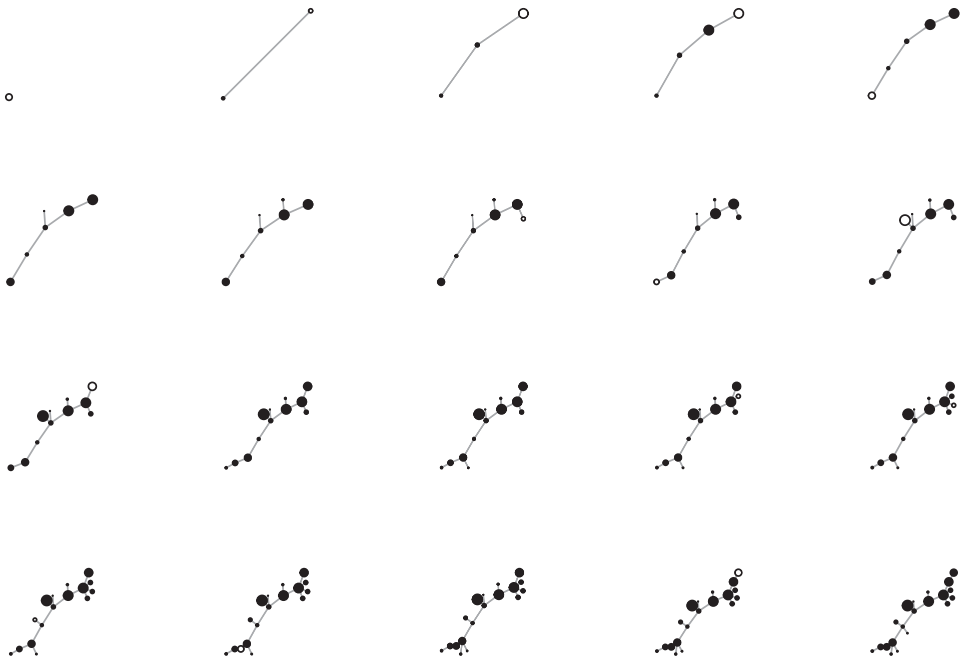

Figure 3.7 The evolution of a fitness-based preferential attachment (Bianconi–Barabási) model with $N = 20$. Fitness is assigned to each node when it originates from a uniform distribution between 0 and 1. The size of the node is proportional to its fitness. Notice that the node with an early headstart in degree development is lower fitness than the nodes that later connect to it. This leads still later nodes to attach preferentially to these higher fitness nodes, not to the early hub.

Configuration models help us understand this by generating networks that preserve features of our original network while randomizing other properties. The random networks will have, for example, the same degree distribution as the observed networks but lack additional structural properties that are not a result of the degree distribution alone (e.g., Milo et al., 2002). Configuration models can be used to evaluate the role of this additional structure because they effectively randomize this structure while preserving degree. It is therefore a useful tool for establishing that something more than degree is responsible for the structure we see.

Configuration models begin with a known network and then cut all edges, turning them into *stubs*. Pairs of stubs are then chosen at random and reattached. This is repeated until all stubs are re-attached (Van Der Hofstad, 2009). See Figure 3.8 for an example.

There are many variations on the configuration model theme. These variations constrain networks to maintain certain network motifs (or subgraph structures), and then generate distributions of networks that preserve these statistical properties. This allows researchers to statistically measure dependencies in the network while controlling for other network features, such as reciprocity, transitivity, and assortativity. For example, to see if reciprocity is greater than we might expect by chance, we can compare the observed network with different randomized variations of itself. This is closely associated with **exponential random graph models**, which are a useful approach for statistically

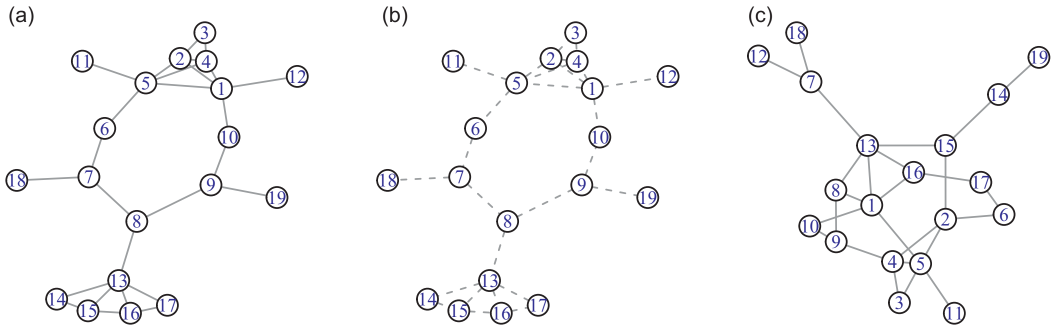

Figure 3.8 Configuration model. Starting with the network in panel (a), all edges are cut and turned into stubs (shown in the middle network). Pairs of stubs are then chosen at random and connected until all pairs are reconnected, producing the network in panel (c). This network has the same number of nodes, density, and degree distribution as the original network.

evaluating network properties relative to controlled variations of your observed network structure (Robins et al., 2007; Snijders, 2001).

3.6 Community Formation

At any local community gathering one can watch as new relationships are formed. These new relationships are rarely made at random. Similar people tend to form edges with one another. Whether by gender, age, or ethnicity, we seem to be drawn to those most like us. How might we capture this process in a network?

Girvan and Newman (2002) – who gave us the Girvan–Newman method for community detection – also present a model for community formation that achieves the desired comingling. Their model for **community formation** starts by assigning each of N nodes to different community memberships. The network is then constructed by forming edges between nodes conditional on whether they belong to the same community: edges are formed with probability p_d if they are from different communities and p_s if they are from the same community, with $p_d < p_s$.

When $p_d = 0$ and $p_s = 1$, communities are completely isolated into cliques of community members. When $p_d = p_s$, there is no differentiation between connections within and between communities. In this latter case, we have an Erdös–Rényi random graph with edges created with probability $p = P_d = P_s$. Figure 3.9 provides an example.

3.7 Forest Fire Models

When we go "down the wormhole" what we are doing in network terms is following a path of relations further and further away from where we started. The network structure left over in our mind tends to look like a forest fire. We started off at one node, and then this led us to activate another, which ignited all the nodes nearby, and so on. This

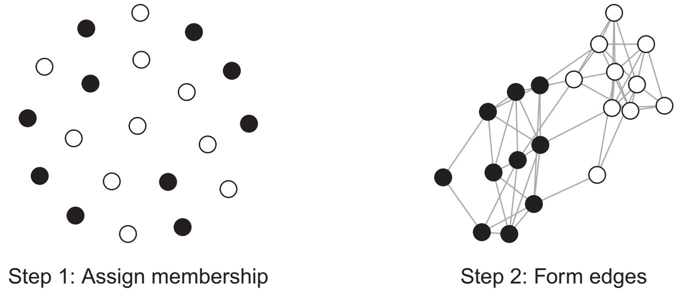

Figure 3.9 Girvan–Newman community formation is achieved in two steps. First, the algorithm assigns nodes to separate communities. Second, edges are formed between nodes, with the probability of an edge dependent on whether the nodes are (p_s) or are not (p_d) in the same community. Here we have $p_s = 0.5$ and $p_d = 0.2$ with $N = 20$ nodes and 2 communities (black and white).

forest fire process mimics many human behaviors. For example, a new relationship leads to introductions to others, and then still others. When we follow citations to scientific research, we create little wildfires of intellectual exploration.[1]

Forest fire models capture this expanding connectivity by allowing a new edge formed to one node to increase the likelihood that edges form to its neighbors as well. Nodes form edges not only to randomly selected nodes, but also with some probability to their neighbors, and then their neighbor's neighbors, and so on.

Numerous models based on forest fire dynamics have been introduced (Vázquez, 2001). These models make different assumptions but roughly follow the variation described here:

Given a pre-existing network of one or more nodes, new nodes are added sequentially. As each node is added, edges are formed to the pre-existing network nodes as follows:

- Step 1. *m ambassador* nodes are chosen at random from the pre-existing network and edges are formed to each of them.
- Step 2. For each ambassador, edges are formed to its neighbors with probability p. Newly attached neighbors become new ambassadors. Step 2 repeats until there are no new ambassadors.

This process allows new nodes to form connections with nodes that are nearby their "entry point" into the network. Occasionally, as Leskovec et al. (2007) describe, this process produces large "conflagrations," in which many new edges are formed by a newly entering node.

In the model shown in Figure 3.10, an approach similar to Vázquez (2001) is used, in which new edges are formed to each ambassador's neighbors, each with probability p.

[1] Many networks might grow like this. Leskovec et al. (2007) found that many real-world networks show an increase in density when the number of nodes added to the network increases. This is a marker of a forest fire process.

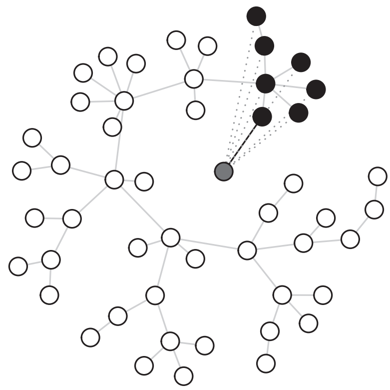

Figure 3.10 A forest fire model based on Vázquez's (2001) recursive search model in which a node attaches to a random node in a network and then follows each edge with probability p, forming a new edge to the adjoining neighbors or otherwise stopping. When all paths are exhausted, the network adds the next node which attaches to a new random target node. Here the new node (gray) is added to a pre-existing preferential attachment network with 50 nodes. The probability of moving down an edge to the next node is $p = 0.94$. The additional new edges that are reached via neighboring nodes are shown as dotted lines.

3.8 Duplication and Divergence

Van Gogh became the artist we now know by making himself the disciple of many of the artists who came before him, copying their works even as he developed his own style. Many novel things arise by a similar process. New genes often start out as copies of existing genes, generated by gene duplication. A protein produced by the duplicated gene maintains protein interactions (edges) held by the gene it duplicated. Over time, further mutations lead to divergence, removing some of the edges and forming new ones as well. New species arise by the same process – the descendants of two siblings slowly diverging over time until one hairy ground-dweller stares up into the feathered face of a distant cousin.

Duplication and divergence models approximate this evolutionary process in network formation (Slanina & Kotrla, 1999; Steyvers & Tenenbaum, 2005; Vázquez, 2003). Starting with a pre-existing network of one or more nodes, the growth process follows these rules:

- Step 1. Identify a node to duplicate. This can be uniformly chosen among existing nodes or chosen proportional to some feature (e.g., degree).
- Step 2. Duplicate the node, giving it the same edges as its parent. This new node is initially **structurally equivalent** to its predecessor as it has the same neighbors.
- Step 3. Modify the edges of the new node to reflect its divergence over time from the original.

An example of this process is shown in Figure 3.11.

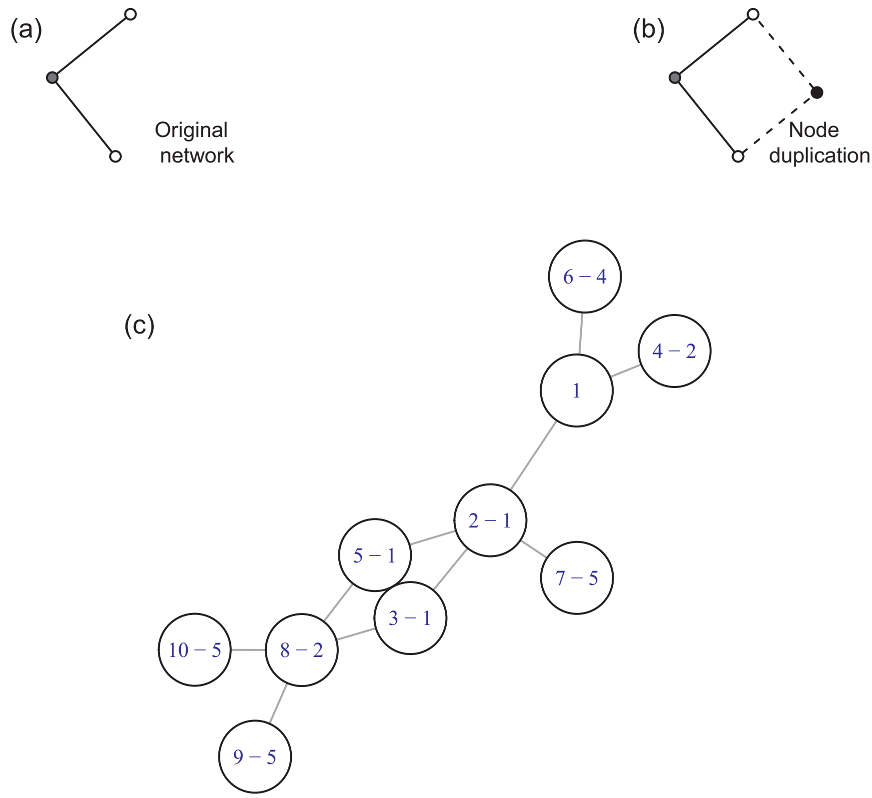

Figure 3.11 Duplication and divergence model. The two networks on the top show the duplication of the gray node (a), which becomes the black node (b). The two dotted edges indicate the duplication of the gray node's two edges. The network (c) shows a duplication and divergence model with $N = 10$ nodes and $p = 0.8$. Each node after node 1 is labeled with "X–Y" where the first number is the order of the node addition – from 1 to 10 – and the second number is the node that was duplicated. For example, the node labeled "10–5" is the 10th node and it is duplicating node 5.

3.9 Preferential Acquisition

The models given earlier all assume the network is generated from thin air. There is no structural information in the environment to inform the network's evolution. This is unlikely for most networks. Social networks exist prior to the formation of online social networks. Spatial networks exist prior to the formation of internet hubs and airport connections. The structure of the world and other forms of information exist prior to our assigning language to it.

Thus, while many of the above networks might be useful for theorizing about the development of networks when other information is absent, we can also consider how networks might develop in response to existing structure.

One network model that does this is called **preferential acquisition** (Hills et al., 2009). It adds nodes to the new developing network with a preference for nodes with certain structural properties in the pre-existing structure.

For example, we might ask which individuals are most likely to join a new social media site. If individuals with high degree in their existing social networks are the first to sign up for the new social media site, the network's development will show a signature pattern of growth that would be different from a network joined first by individuals with lower degree.

This can be formalized by allowing nodes to be acquired in the new network with probability proportional to some developmentally appropriate structural property, d:

$$P(i) = \frac{d_i^{\beta}}{\sum_j^N d_j^{\beta}}$$

Here d represents the preferential feature and β indicates the sensitivity to that feature. When $\beta = 0$ nodes are added at random. For very large β, the network adds the nodes in order of their preferred characteristic. For our social network example, d might be the degree of the individual in their existing social networks prior to joining the social media site.

Figure 3.12 shows preferential acquisition at work. Nodes are selected from a larger pre-existing network (shown in panel (d)) and added to a growing network. The development is shown in three phases, with a preference for attaching higher degree nodes first. As discussed in Chapter 5, this model can be applied to a number of developmental networks to ask which properties best predict network growth.

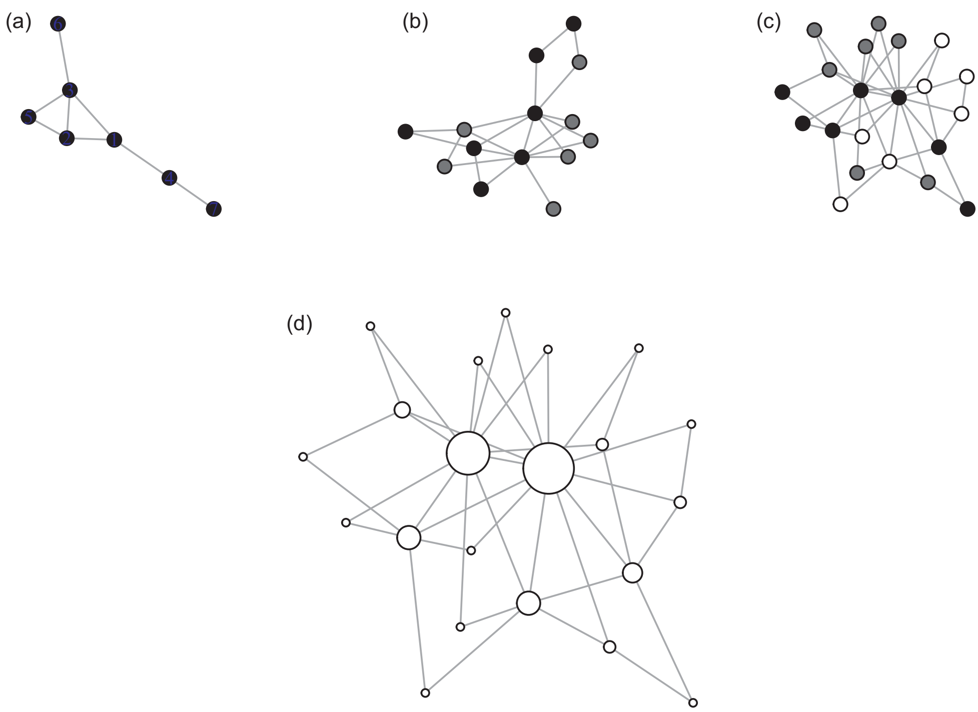

Figure 3.12 Preferential acquisition. The network develops from (a) to (c) along the top. The first nodes to develop in the new network are shown in black. These are followed by the addition of the gray nodes. The final additions are the white nodes shown in panel (c). The network in panel (d) shows the network from which the above networks are preferentially acquired, with node size indicating node degree. Nodes with higher degree are preferentially acquired earlier in development.

3.10 Directed and Weighted Graphs

Each of these models can be generated as directed or weighted networks. The basic logic is a follows: (1) sample from possible directed edges or replace undirected edges with directed edges, and (2) sample weighted edges from predefined distributions. Some possible variations of these approaches are described here, but these are not exhaustive:

- *ER random graph*: Directed edges are randomly assigned between pairs of nodes with the specified probability or edge count. Weighted edges can be drawn from a prespecified distribution, for example, from the distribution of an observed network.
- *Small-world networks*: Directed and/or weighted edges can be specified at the lattice phase.
- *Preferential attachment*: In the original Barabási–Albert preferential attachment model, edges are formed directed toward the preexisting node and indegree is used to evaluate preferential edge formation.
- *Fitness models*: Nodes can have indegree and outdegree fitness. To form an edge, one node is sampled proportional to its indegree fitness across all nodes, and one node is sampled proportional to its outdegree fitness. Weighted edges can be sampled from a predefined distribution or as a function of a fitness value.
- *Configuration models*: As these models recreate an existing structure, they preserve the (un)directed and (un)weighted structure of the original network. Weighted configuration models require additional assumptions to coordinate weights across stubs.
- *Community formation*: As edges are probabilistically evaluated for all pairs of nodes, the probabilities can be defined to produce unreciprocated or reciprocated directed edges. Weighted edges can be sampled from different distributions depending on whether edges are in the same or different communities.
- *Forest fire models*: Edge formation can be directed toward nodes in the preexisting network, emulating a citation process. Weighted edges can be sampled from a predefined distribution or sampled in proportion to some structural property during edge formation, such as the number of new neighbors that an individual node provides access to.
- *Duplication and divergence*: Directed and/or weighted edges can be assigned at random and replicated in the duplication process. Divergence could alter directions or weights.
- *Preferential acquisition*: As this process provides various development pathways for arriving at a pre-existing structure, the directed or weighted values of edges is predefined in the existing structure. These values can then be used to derive the node sampling procedure.

Metrics computed on directed and weighted networks can sometimes be difficult to interpret with respect to a specific problem. For example, there are multiple methods for computing weighted clustering coefficients (e.g., Barrat et al., 2004; Onnela et al., 2005; Opsahl & Panzarasa, 2009). Determining which is most appropriate for a given system and interpreting what it means in relation to the behavioral processes can be a valuable but challenging exercise.

3.11 A Taxonomy of Basic Network Architectures

Basic network architectures provide useful intuition pumps. If we imagine how a process might work on a simplified network, it can help us to generalize to more complicated networks with many more nodes and edges. Figure 3.13 provides a few example networks for comparison. Which of these networks is best insulated from the spread of infection or opinion? Which of these networks is best for spreading good news or most resilient to losing nodes or edges? Which nodes should be vaccinated first against disease or misinformation? Which networks will have the most behavioral diversity or innovation? Which networks would be most coordinated? We will tackle these and many other questions in the chapters to come.

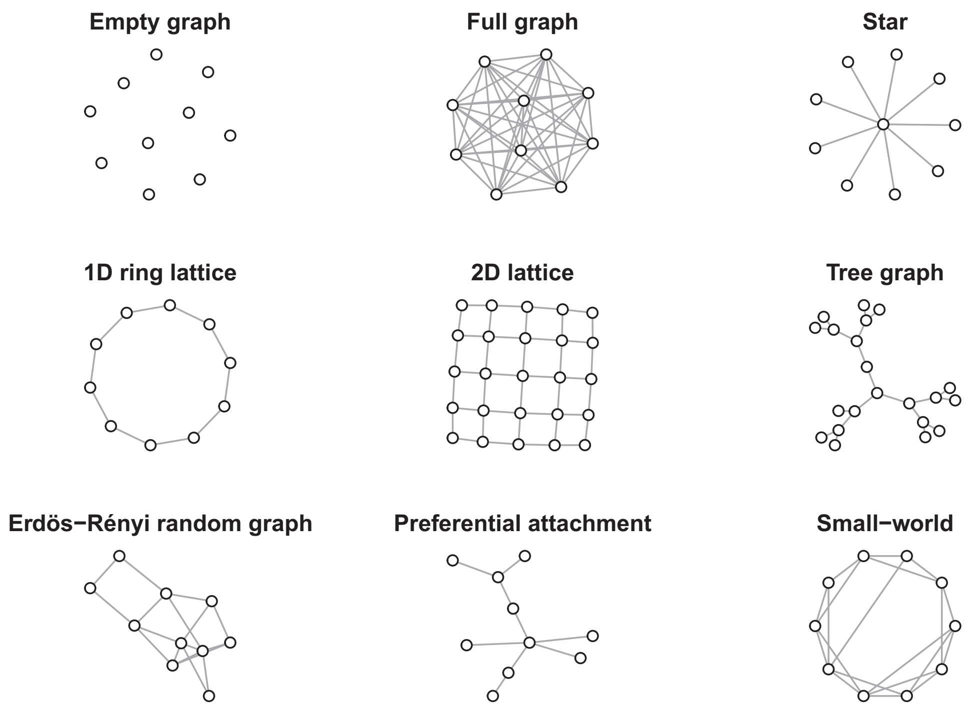

Figure 3.13 A menagerie of some basic network types.

Part II

Language

4 Zipf's Law of Meaning
The Degree Distribution of the Mind

4.1 The Problem

Degree is the simplest of the node-level measures, but its simplicity hides its power. Here we will apply degree to the problem of mental structure. Specifically, what is the structure of meaning in the mind? Are some things more "meaningful" than others? George Kingsley Zipf offered an inroad to this problem when he observed that word frequencies in natural language tend to follow a scale-free distribution. In a scale-free distribution a few things tend to be very large, whereas most tend to be small in comparison, like earthquakes, city populations, and wealth. It has also been suggested that this scale-free distribution applies to relationships *between* words – to their meanings. A few words share meanings with many words while most words share meanings with few. This chapter will explain the theory underlying Zipf's law of meaning and also show how we can explore this idea with the most basic node-level measure: degree.

4.2 Degree

Degree is the count of a node's neighbors. We need not know who those neighbors are, only that they exist. It is the answer to the question "How many friends do you have?" or, more generally, "How many things is X connected to?" Though simple, some of the most interesting work in behavioral network science uses little more than the creative application of degree to address compelling questions.

Most interesting networks are composed of nodes of different degree. By evaluating the *degree distribution* of a network's nodes, we can examine factors like the relative inequality of nodes, how nodes change over time, and how, in directed networks, nodes operate as sources or sinks in resource and information flows.

4.3 Degree Distributions in Art History

Clever uses of degree are excellent sources of scientific innovation. Schich et al. (2014) used degree to understand how patterns of human mobility inform cultural history. To do this, they examined the mobility patterns of more than 150,000 prominent historical artists, such as Liang Kai, Pablo Picasso, and Andy Warhol. Treating places as

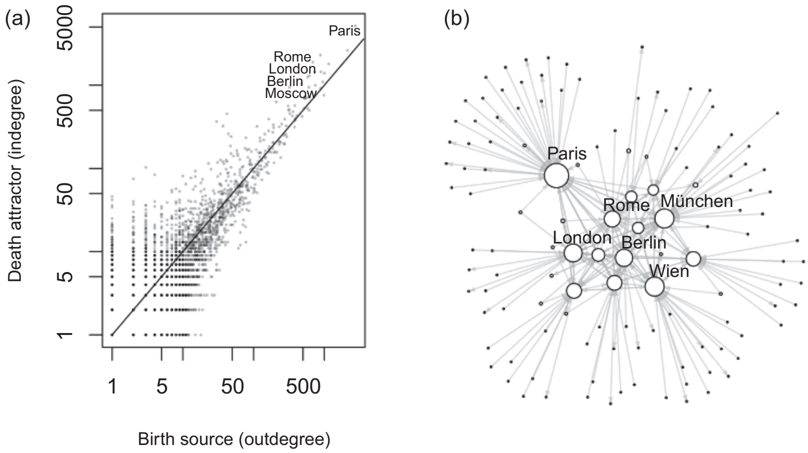

Figure 4.1 Indegree and outdegree as birth sources and death attractors for 112,276 notable historical artists, c. 480bc to 2010ad. There are $N = 46{,}189$ nodes (i.e., places) in the network and 112,276 edges. (a) Dots represent places. Places above the line are cultural centers, where artists tended to be attracted. An artist is more likely to die there than be born there. (b) A representative subset of the network, showing 13 places with a total degree (in + out) of more than 500 plus a sampling of 117 additional nodes that are connected to them. Data is from the General Artist Lexicon (Beyer et al., 2016). The figure is after Schich et al., 2014.

nodes, directed edges between places represented an artists movement from birth to death. These birth-to-death edges allowed Schich et al. (2014) to characterize art history's "birth sources" and "death attractors" by the relative out- and in-degree of nodes.

Death attractors are cultural centers of attraction where individuals die more frequently than they are born; they have a high birth–death imbalance. Schich et al. (2014) used this information to quantify the magnitude and historical rise and fall of cultural centers such as Rome, Paris, Dresden, and Hollywood.

Schich et al. (2014) also observed that as artists numbers increased in the nineteenth century, cultural centers became more attractive. This latter observation suggests a cultural scaling of social proof, in which groups become more attractive the larger their size. This rich-get-richer growth of cultural centers should sound familiar in relation to preferential attachment (discussed in Chapter 3). Figure 4.1 uses the original data from Schich et al. (2014) to produce one of the central figures in that work.

Schich et al. (2014) provides a fascinating example of the innovative use of degree distributions to address cultural history. However, degree in many network analyses is more commonly dealt with by focusing on degree distributions, with one of the more prominent patterns being a power law.

4.4 Power Laws and Zipf's Law

Many real-world phenomena have the property that large events occur much less frequently than small events. Small earthquakes occur frequently, large earthquakes less frequently. The same rule also applies to human behavior. We take many fewer long

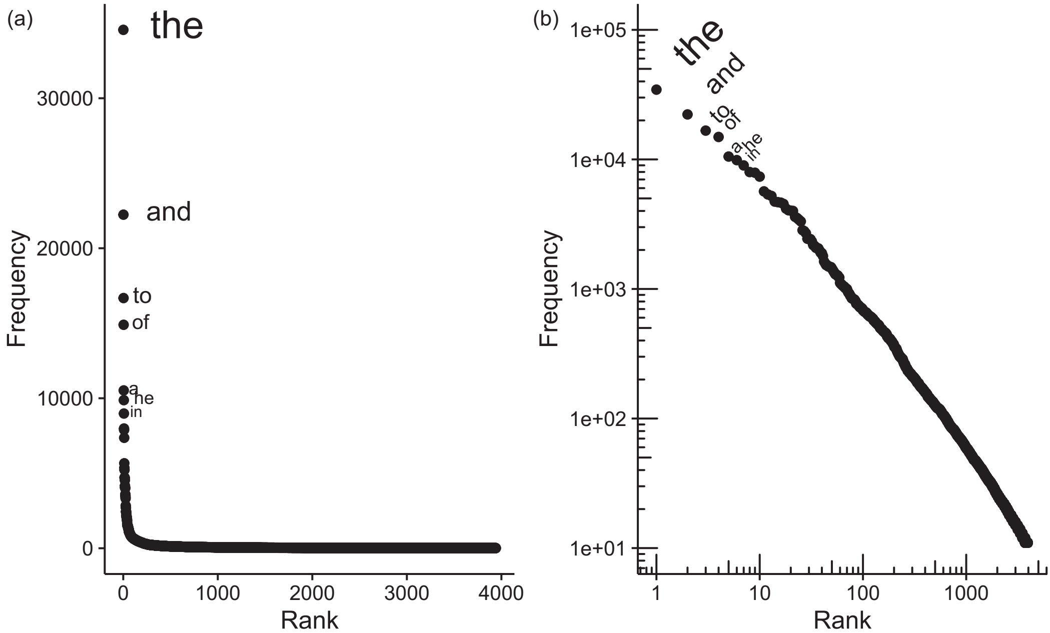

Figure 4.2 Zipf's Law plotted for the 527,667 words in Leo Tolstoy's *War and Peace*, of which there are 3,941 unique words that occur more than 10 times each. In panel (a), the plot is shown without log-log axes. Panel (b) shows the log–log plot. The slope of regression line fit to the log–log plot is −1.13. Each dot represents the frequency and rank of the word. The first 7 words are labeled for comparison.

holidays than we do short holidays. There are many more small communities than large cities. A few people's incomes are substantially larger than the majority of others. This last observation was made by Vilfredo Pareto, and is famously associated with the Pareto principle or 80–20 rule (e.g., 80% of the wealth is owned by 20% of the people).

George Kingsley Zipf (1932) investigated the structure of language using a similar approach. By plotting the frequency of a word against its relative frequency ranking (with 1 being the most frequent), Zipf observed that the shape of the distribution was a line on a log–log plot. Zipf also observed that the line tended to have a slope of −1. This pattern of inverse proportionality with rank is commonly called *Zipf's law* and applies to a wide range of phenomenon.

Figure 4.2 shows the rank–frequency distribution for Leo Tolstoy's *War and Peace*, with and without log axes. The most frequent word, "the," occurs 34,559 times and is ranked 1. Rank 2 is "and" with 22,238 occurrences, and so on. Without log axes, the few large items and the many small items create a plot in which most of the data sticks to the axes. With log axes, the linear relationship among the data becomes clear.

The linear relationship on log–log axes goes by many names, each of which offers insight into the properties of these kinds of distributions.

Heavy-, *long-* or *fat-tailed distributions*: The tail is considered heavy relative to a normal distribution. That is, the data is skewed such that rare events occur more often than one would predict if one were using a normal distribution to make the prediction. Such unexpectedly frequent rare events are the basis of the *black swan* idea made famous by Taleb (2007) and Mandelbrot & Hudson (2010).

Power law distributions: This name comes from the mathematical property that the frequency of a word, n, is proportional to its rank, r, to a specified power:

$$n = ar^{-\gamma}$$

This takes the familiar equation of a line, $y = mx + b$, after taking the log of both sides:

$$log(n) = log(a) - \gamma log(r)$$

Here we can see that a is the intercept, or the frequency of the most frequent word. When $r = 1$ then $n = a$.

Scale-free distributions: Scale-free distributions maintain their relative statistical properties at any scale. Suppose, for example, that we ask how many fewer people have an income of £10 million per year than £1 million per year, and we found that the answer is the same if we asked about £100,000 per year versus £10,000 per year. If we then plotted the frequency of various incomes on a log–log scale, the result would be a line, showing a constant relative frequency distribution.

Human height is normally distributed. What if it were power law distributed with the same average height? If it were, most people would be very short, perhaps a few centimeters high. But we would often encounter people much taller and occasionally someone as tall as a skyscraper, or taller. Unlike a normal distribution, where sizes extremely different from the mean are unlikely, the heavy tail of a power law would guarantee the presence of extremely large individuals.

Many different phenomena produce power law distributions (Newman, 2005). Several of the generative network growth algorithms described in Chapter 3 can produce them: forest fire models, preferential attachment, duplication and divergence, and preferential acquisition. So do phenomena based on other rich-get-richer processes, where an individual can leverage their size to grow still larger. Fractals have a similar scale-free property and also produce power law distributions as shown in Figure 4.3.

Power laws and Zipf's law are often used synonymously. This is because Zipf and many others since have observed power laws in many different behavioral phenomenon, including communication, wealth distributions, city sizes, wars, book sales, music sales, the frequencies of first names, scientific productivity, citation counts, animal movement patterns (e.g., Lévy flights), and many other phenomenon (Gureckis & Goldstone, 2009; Hills, Kalff, et al., 2013; Moreno-Sánchez et al., 2016; Newman, 2005; Reed, 2001). This pattern becomes even more remarkable because it occurs in phenomena unrelated to human behavior as well, such as the size of earthquakes, moon craters, and solar flares (Newman, 2005). This has led to a great deal of research trying to understand the processes that generate power law distributions and when and where we might find them.

The apparent ubiquity of power laws has not gone without controversy. The existence of "true" power laws in networks remains controversial (Broido & Clauset, 2019; Holme, 2019). Real phenomenon are rarely, perhaps never, scale-free at every scale. There are physical limits on how large things can be. For example, no war could kill more earthlings than the population of the earth, even if the power law distribution fit

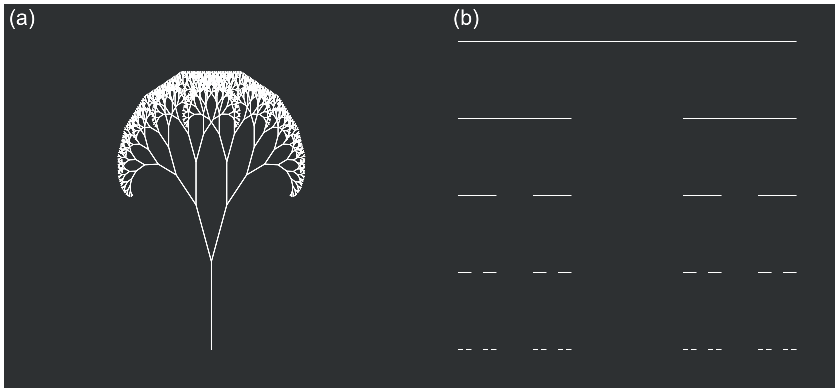

Figure 4.3 Fractals exhibiting scale-free and power law distributions of edge lengths. A recursive tree (a) where each branch leads to two new lines of reduced size. A Cantor Set (b) where each line produces two new lines of 1/3 the size as we move downward. Note that as we move up in size, the number of larger lines is always half the number of lines one size smaller.

to the war data indicates a nonzero probability that such an event could occur. However, whether or not something is a "true" power law is less important for most research than understanding the implications of the observed distributions, in terms of what they mean for understanding the processes that give rise to the behavior. In practice, many researchers are satisfied to refer to near-linear patterns on a log–log plot as a power law or Zipf distribution.

4.5 Zipf's Law of Meaning

In 1937, B. F. Skinner asked if there might also be a Zipf's law of meaning, if speech were "selected on a semantic basis" (p. 71, Skinner, 1937; see also Zipf, 1949). By this he meant that if we counted word's semantic relationships, some words would have very many and some very few, and this relationship would follow that predicted by Zipf for word frequency. For example, there might be a thousand words that share a semantic relationship with *color* – lemon, red, crimson, cherry, wine, blue, and so on – but many fewer words sharing a semantic relationship with *razor* – sharp, cutting, metal, and so forth. Does the frequency of meanings follow Zipf's law? Skinner did not have a direct mechanism to quantify the meanings of words, so he proposed the use of free associations, which is what we will use here.

4.6 Free Associations

Free associations ask people to produce the first word that comes to mind when they are presented with a word like "cat." In this example, "cat" is called the cue word. Given the cue "cat," many people will produce the associative response "dog." A specific cue word can produce many different associates.

Studies of free associates have been collected for over a century (e.g., Jung, 1918). Most modern free association data sets are *normed* – collected from many individuals per cue word – for many thousands of cues (De Deyne et al., 2019; Nelson et al., 2004). This provides a powerful representation of human semantics and is used in a wide variety of domains, from child development, to memory, to marketing. Some contemporary data sets also contain many thousands of cue-response pairs from single individuals (Morais et al., 2013; Wulff et al., 2022).

We can use free-association norms to produce networks. If nodes are words and edges are directed edges from cues to their associates, then the outcome is a free association network. If associations are a valid proxy for shared meaning, as suggested by Skinner, then we can address Zipf's law of meaning by investigating degree distributions in the free association networks. The prediction is that the degree of a word's meanings when plotted against its rank will produce a line on a log–log plot, consistent with a Zipfian distribution of word meanings.

4.6.1 Do Free Association Networks Provide Evidence for Zipf's Law of Meaning?

Steyvers and Tenenbaum (2005) used the University of South Florida free association norms, containing approximately 5,000 cue words, as well as Roget's Thesaurus and WordNet, a hand-coded network of word relations. They found Zipf-like relations for word meaning across each of these data sets.

Here we can use the Small World of Words data set from De Deyne et al. (2019). Cues form directed edges to their responses. We will keep all cue-response edges produced by more than one person. The resulting network contains $N = 21{,}555$ nodes and $E = 144{,}695$ directed edges.[1] Figure 4.4 visualizes the network for a subset of the highest indegree nodes – words that are more often produced as responses. What we can see is, first, that there are words of many types, covering a broad range of everyday concepts like "church," "fun," "hot," and "sex." Second, though many words are missing from this visualization, words tend to cluster together with words of similar meaning. "School" is near "teacher," "hot" is near "cold," and "food" is near "fruit" and "water."

If we follow Skinner in plotting the probability of degree against its rank, we can visually evaluate the linearity of degree on a log–log plot. Figure 4.5 shows this relationship for total degree, indegree, and outdegree. For comparison, it also visualizes the University of South Florida free association norms.

Two observations become immediately clear. First, the two sets of free association norms show similar relationships. This is reassuring. It helps establish that the relationships we see are not an artifact of a particular data set, but rather something that might be true of free associations in general. Second, on visual inspection, total degree is roughly linear over several orders of magnitude. This is consistent with Zipf's law of meaning.

[1] Keeping all cue-response pairs is useful for capturing rarer associations. However, the trade-off is introducing more random associations, misspellings, and so on. Here we would find the power law relationships are unchanged by relaxing the more conservative requirement for two associations, but the semantics became less meaningful. The curious can investigate this further with the online code.

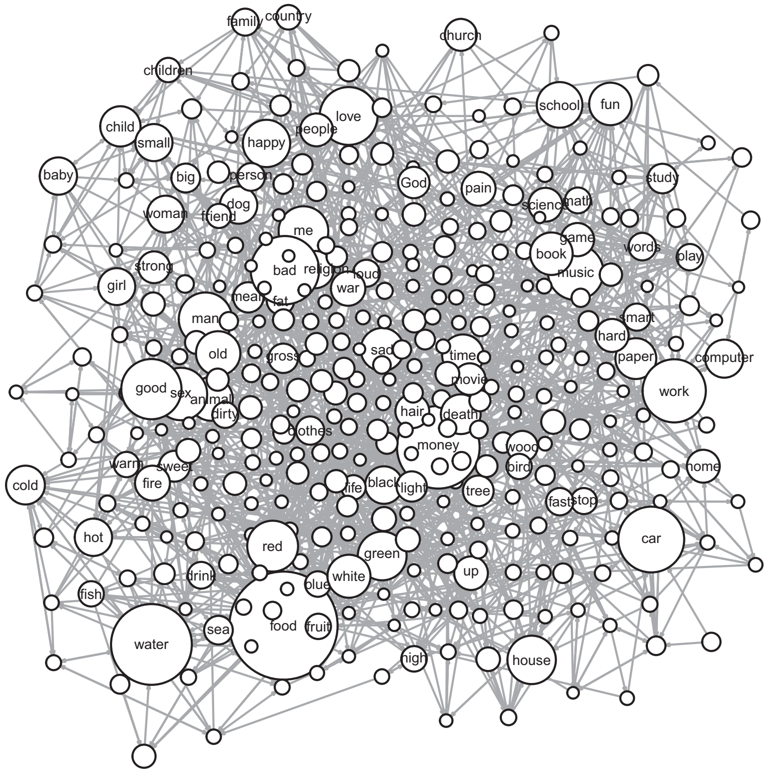

Figure 4.4 The 355 highest degree nodes and their 1,797 directed edges from the Small World of Words free association norms ($k > 60$). The node size is proportional to the indegree in the full network. "Food," "money," "water," "bad," and "car" are the five highest degree nodes in full network.

However, total degree is made up of two parts: it is the sum of a node's indegree and outdegree. The indegree and outdegree distributions are less clearly linear, especially for outdegree. Importantly, only cues chosen by the experimenters can have outdegree. For the Small World of Words, the experimenters chose $n = 12{,}217$ cues – representing about half of all nodes in the network – so only these words can have nonzero outdegree. Their outdegree is also limited by the number of associates participants were allowed to produce. Indegree is a measure of how many different cue words produce a word as an associate, giving us some indication of how often a word comes to mind in association with a broad range of cue words. To level the playing field, in what follows we will limit our analyses to only those words with outdegree greater than zero (indicating that they are cues).

4.6.2 Fitting Power Laws

When comparing networks, it is often useful to fit statistical models to the data. This fits the observed distribution to an expected distribution based on a specified mathematical relationship. Though numerous methods have been described for doing this, there are many challenges (Clauset et al., 2009; e.g., Edwards et al., 2007; Pilgrim & Hills, 2021).

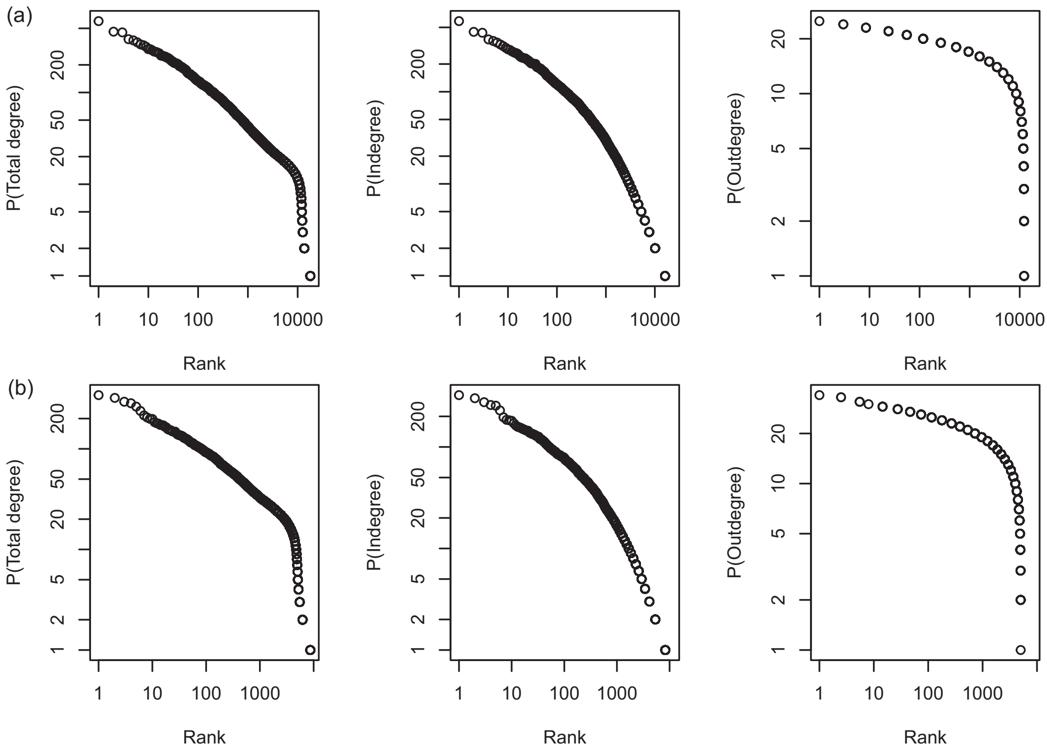

Figure 4.5 (a) The rank–frequency plots for the Small World of Words (total degree, indegree, and outdegree). (b) The same relationships for the University of South Florida free association norms.

One common challenge to fitting a power law distribution is that data in the tail of the distribution is extremely rare. Rare events are often under-sampled. If we want to accurately estimate the frequency of an event that only occurs one in a million times, then we will need millions of samples to do so. Undersampling rare events will lead to an underestimation of the tail of the distribution. This is problematic as these rare events might be exactly the events we most care about, such as large earthquakes and financial crashes. They are also in the area of the distribution most needed to support a power law. Therefore, accurate detection of power laws often requires large samples.

Limitations aside, here we will use a maximum likelihood approach (Clauset et al., 2009). The data is plotted using the complimentary cumulative distribution function. This tells us the probability that a node chosen at random from the network has a degree greater than or equal to some value, $P(i \geq x)$. The details of this method (see e.g., Clauset et al., 2009; Gillespie, 2015) offer valuable tools for comparing power laws with other distributions. We will not focus on this here, but this is useful when attempting to ascertain which functional form best fits the observed data. In Chapter 5 we will discuss model fitting in more detail.

Figure 4.6 shows that both the Small World of Words and the University of South Florida free association networks have similar degree distributions for their cues. Over approximately one order of magnitude, the distributions fit a power law, both with slopes near 3, similar to that found in Steyvers and Tenenbaum (2005). The fitting method also

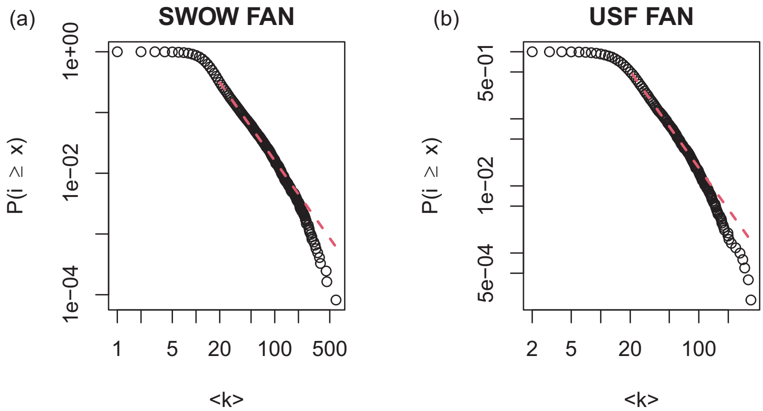

Figure 4.6 (a): Total degree distribution for the Small World of Words for the $n = 12{,}217$ words with outdegree greater than 0. (b): Total degree distribution for the University of South Florida Free Association Norms for the $n = 5{,}018$ words with outdegree greater than 0. The dotted lines indicate power laws fit to the data.

identifies a lower bound, below which the data is not fit. For both, this is approximately 20. Finally, the statistics in the online code also suggest that neither of the distributions are statistically different from a power law – though clearly one should quibble about the tail.

This nicely supports Zipf's law of meaning. The similarity of the two distributions is also reassuring. It suggests that the patterns we observe are not an artifact of a particular time, place, or sampling method.[2]

4.6.3 Shaking off the Tyranny of Power Laws

As we will see in future chapters, the presence of a particular network pattern can hint at the underlying data-generating process. The distribution is a step in that direction, helping us answer the question "How did this particular network come about?" However, the presence of a distribution (e.g., a power law distribution) does not tell us what the underlying process is. It only narrows down the possibilities. It is certainly tempting to infer the process from the distribution (as is often done for preferential attachment and Lévy flights), but many different processes can generate the same distribution (Newman, 2005). Gaining a clear understanding of the behavior and its associated research literature is often critical for narrowing down the possibilities to plausible accounts.

Ideally, we should also test our hypotheses beyond identifying the distribution. In the case of the free association norms, the observation that the degree distribution of free associations does not fit a power law continuously over the range of our data is suggestive that processes may differ depending on whether we are focusing on high or low degree nodes. Indeed, this observation has already been used to suggest that language is made

[2] Free associations for SWOW also exist in Dutch, providing an easy data set to investigate for comparison.

up of two different lexicons: a core lexicon – used by everyone – and a more specialized lexicon dedicated to areas of specialized knowledge (Ferrer-i-Cancho & Sole, 2001).

Moreover, indegree and outdegree are not linear in the same way as total degree, suggesting they may have processes of their own, potentially influenced by the way the data is collected. However, indegree and outdegree are not independent processes – they are positively correlated (Pearson's product moment correlation, $r = 0.27$, $p < 0.001$). Perhaps this is no surprise. If our minds often drift to food, sex, money, and love, then much that we do should be associated with getting them.

4.7 What Processes Generate Power Law Structure?

What kinds of processes might generate the power law distributions we see in free association norms? One possibility is that semantic relationships are an *accretional process*. Astronomical objects, like stars, black holes, solar systems, and planets are accretional objects: they grow by attracting matter, and as they attract more matter their capacity to attract additional matter grows in turn.

Accretion is similar to a familiar process in biology called the *Yule process*. In the Yule process, if each member of a population has a fixed probability of giving birth to a new member, then populations will grow in proportion to their size. It is easily extended to speciation processes and was developed to explain why the distribution of groups within biological taxonomies (e.g., species within a genus) follow a power law distribution (Yule, 1925). The Yule process is a rich-get-richer process and gives rise to power law distributions. It is similar in concept to some of the generative network distributions described in Chapter 3, and it is similar again to the language emergence model we will see in Chapter 9.

The growth of associations could be an accretional process. The more associations that a word has, the more likely it is to be used in novel contexts where it can form new associations. Occasionally these novel associations go on to form new words of their own (e.g., "crowdfunding" and "staycation").

Figure 4.7 allows us to make some straightforward comparisons with three models. Providing them alongside one another makes it easy to make a direct visual comparison. These are an Erdös–Rényi random graph, preferential attachment, and a forest fire model.

The figures demonstrate several important features. First, not all graphs produce a power law distribution. The Erdös–Rényi random graph is clearly not a line on a log–log plot. It is good to have models that fail to resemble the data as they place limits on the range of alternative hypotheses.

Second, the preferential attachment and forest fire models produce a clear linear relationship over a large portion of their distribution. They mimic the pattern seen in the Small World of Words over at least some region of their range.

Third, none of these models reflect the exact pattern we see in the free association norms. That should not be too surprising. The models presented here have suffered under limited tuning, and there is much that could be tuned. We are not yet trying to fit our models to the data. Yet, even if we had a perfect off-the-shelf model, a good critic should argue that the explanation is incomplete. Models are made better when they are tested

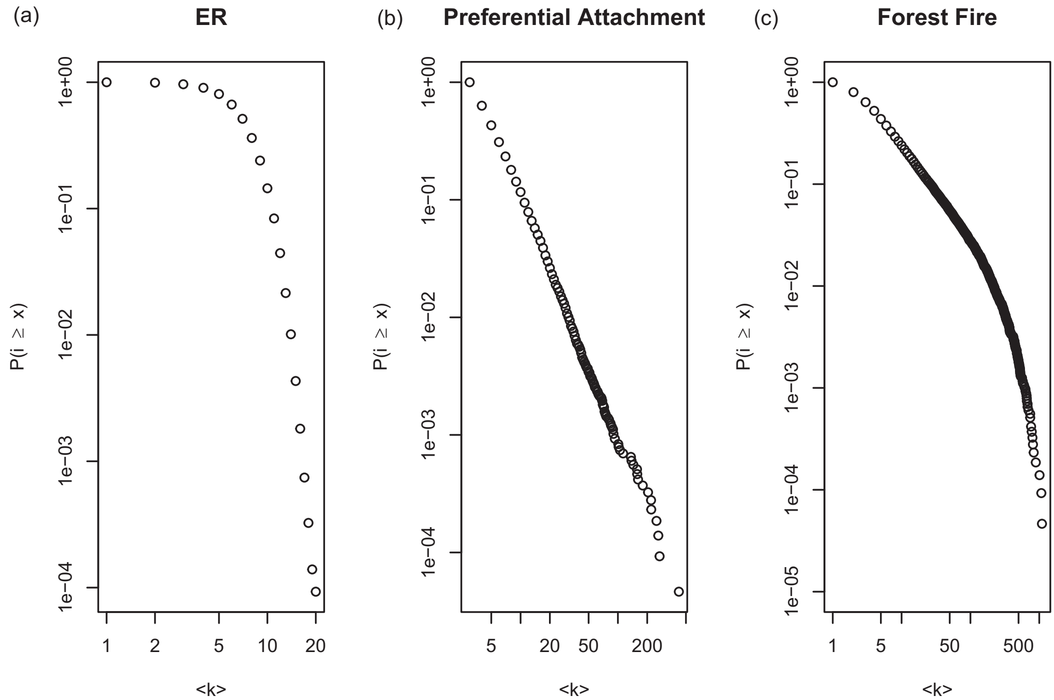

Figure 4.7 The degree distributions of three models: (a) Erdös–Rényi random graph, (b) preferential attachment, and (c) the forest fire model. Each of the graphs has a node count and density near that of the Small World of Words.

against other models. The assumptions of the models would need to be tested in rigorous experiments. Further, identifying these assumptions requires that we formulate the model descriptions in ways that are behaviorally plausible, helping us generate hypotheses and new questions about the behavior in question.

These models are just a few of many that might generate power law distributed degree. As we saw in Chapter 3, there are others that could do the same. Yet, the results of even this simple analysis suggest that an accretional process like preferential attachment or a forest fire model would be worth further investigation. Perhaps the mind grows like a forest fire when conditions are right.

4.8 Summary

Degree is a powerful measure and, though simple, it holds tremendous potential for understanding how behavioral systems work as they do. In this chapter, we explored how degree could tell us about the structure of meaning in the mind. By looking at degree distributions, we found a scale-free structure in free association networks. This helps us consider how the mind works – it is consistent with a rich-get-richer process in the construction of meaning. However, we also found that fitting a distribution to a power law would not be enough to decide on a specific data-generating process. At least two of the models we explored generated power laws over some part of their region, and many other models would have done the same. In the end, we have learned something about the structure, but not yet formally compared models that would give rise to it. We will explore this more in Chapter 5.

4.9 Open Questions

4.9.1 The Pervasiveness of Linguistic Laws in Behavior

Zipf's law of frequency and meaning are but a few of many linguistic laws. For example, the Brevity law recognizes the statistical tendency for more frequently used words to become shorter. Herdan–Heaps' law describes the growth in number of word types as the length of a text increases. Many of these laws apply not only to human language, but also to information contexts across biology, ranging from molecules, to organisms, to ecologies (Semple et al., 2022). For example, Zipf's law of meaning has also been observed by mapping dolphin whistles to the contexts in which they are produced (Ferrer-i-Cancho & McCowan, 2009). What gives rise to these various relationships? Numerous researchers have tackled this problem by assuming that communicators and receivers are attempting to balance various forces, such as communicator effort and receiver comprehensibility (Ferrer-i-Cancho, 2016). These factors can themselves be influenced by numerous other relationships, such as the physical act of producing words (Torre et al., 2019) as well as their semantic relationships (Manin, 2008). Indeed, a model equivalent to random walks over semantic networks has shown how Zipf's law of frequency and Zipf's meaning-frequency relationship may arise from a similar source (Ferrer-i-Cancho & Vitevitch, 2018). What kinds of rules might govern this logic across behavior, biological levels of organization, or information environments more generally? Are there general rules that might lend themselves to all of these various laws, which all tend to involve issues with the efficiency of communication (Semple et al., 2022)?

4.9.2 Language Markets and the Evolution of Scaling Laws

Zipf's law and other scaling laws are ubiquitous in behavioral systems (Kello et al., 2010). However, the scaling relationships often change over time. For example, Pilgrim et al. (2021) have shown that the exponent associated with Zipf's law has changed over the past several hundred years, with a flattening of the curve, consistent with other measures of rising diversity in language. Similarly, Gureckis & Goldstone (2009) found that baby name frequencies fit approximately a power law distribution and that the trend since the 1950s shows a similar flattening of the curve, associated with an increasing diversity of baby names.

Rising diversity in language and naming tells us that the way we communicate is changing and this is influencing what we say. Possibly we are seeing a bias toward more efficient communication on the side of the receiver (Hills & Adelman, 2015; Pilgrim et al., 2021). If this is, for example, a consequence of competition among communicators (who must therefore "pay more" to acquire receivers), then might we predict this pattern in competitive communication markets more generally? How exactly this might work remains an open question.

5 Network Learning
Growing a Lexicon by Degrees

5.1 The Problem

The way networks grow and change over time is called network evolution. Numerous off-the-shelf algorithms have been developed to study network evolution. These can give us insight into the way systems grow and change over time. However, what off-the-shelf algorithms often lack is knowledge of the behavioral details surrounding a specific problem. Here we will develop a simple case we will revisit over the next few chapters: How do children learn words from exposure to a sea of language? One possibility is that the words children learn first influence the words they learn next. Another possibility is that the structure of language itself facilitates the learning of some words over others. Indeed, we know that adults speak differently to children in ways that facilitate language learning, with semantically informative words tending to appear more often around words that children learn earliest. This invites the following question: To what extent does the semantic structure of language predict word learning? This chapter will provide a general framework for building and competing models against one another, with a specific application to the network evolution of child vocabularies.

5.2 Network Evolution in Child Language

Children start learning words in the first few years of life. Most children start to produce their first words around their first birthday. What factors make some words easier to learn than others? For example, do children learn words chosen at random from the language they hear, or do they follow some pattern based on word properties?

Consider word frequency. An often proposed model of word learning is that children learn the words they hear most. We can quickly sharpen our intuition here following Zipf's law. The most frequent word in the English language is "the." But most children will say hundreds of other words before they say the word "the." This is easy to confirm using some of the data sets described later in the chapter. Frequency is a valid predictor under some assumptions, just not generally (Goodman et al., 2008).

What other factors are likely to contribute to word learning? Some possibilities include how words sound (phonology), what they mean (semantics), and what the referent's features are. Phonology, semantics, and features can each be used as the basis to form edges between the word-nodes in a lexical network. This then allows us to examine each as a potential hypothesis for how children's lexical networks grow.

Here we will examine a general process for exploring these alternatives. We will simplify the problem by focusing on free associations. An early finding is that words with higher indegree in free association networks are learned earlier. That is, words that are more often produced as associates in response to cue words are more likely to be learned earlier. We can examine this relationship using the Small World of Words free association norms introduced in Chapter 4, which allows us to produce a free association network. Words are nodes. Edges are directed from cues to the associates they produce. So, if a person says "cat" in response to "dog," then a directed edge points from "dog" to "cat." The indegree then gives us a measure of how often a word is produced as an associate across all cues.

We will use two sources to determine at what age words are learned: (1) The WordBank data indicating at what ages children can say various words listed on a parent survey of child vocabulary called the Child Development Inventory (CDI) (Frank et al., 2021), and (2) the age of acquisition norms from Kuperman et al. (2012), which provides adult ratings for 30,000 words. Age of acquisition in the Kuperman et al. norms and WordBank data are correlated for the words we use here ($r = 0.49$), which are limited to those also found in the Small World of Words. Figure 5.1 shows some example developmental networks for the Kuperman et al. norms and the full network at 30 months for the WordBank data.

Figure 5.2 shows the relationship between a word's indegree in the free association network and the age at which the word is learned. Though both have a negative correlation (see the online code), the pattern in the WordBank data is not easy to see. It is clear, however, when looking at the Kuperman et al. norms. Nodes with high indegree in the free association network are more likely to be learned at younger ages.

Recall from Chapter 4 that free association networks have long-tailed distributions. What models of language acquisition might explain early nodes with high indegree and long-tailed distributions?

5.3 Models of Lexical Acquisition

Formal mathematical models are valuable for at least three reasons. First, they put our intuitions to the test. We can evaluate to what extent a proposed explanation generates the observed data. Second, we can compare explanations to see which best explain the data. And third, and perhaps most important, they force us to state our hypotheses with the clarity and precision necessary to test them. As many scientists attest, an explanation is only as good as our ability to test it. We would like to avoid Wolfgang Pauli's cutting criticism that something is not even good enough to be wrong. An explicit model helps us avoid that.

In the case of lexical networks, models represent the theory and their behavior represents the predictions that are a consequence of that theory. Here, we will explore theories of network evolution that are capable of making predictions about the order in which words are learned.

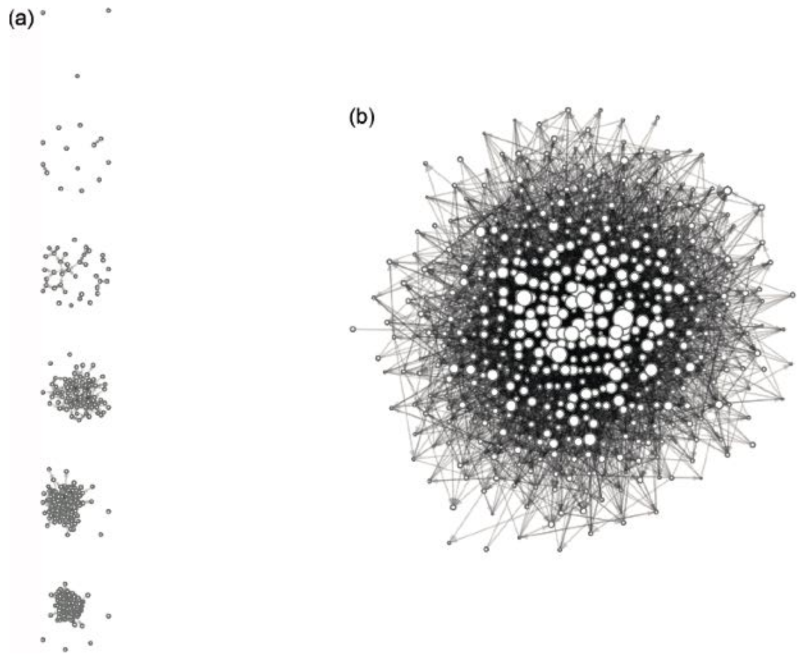

Figure 5.1 Children's free association networks. Edges are from the Small World of Words free association norms, representing relationships between cues and their associates. (a) The growing lexicon for words learned at different ages in the Kuperman et al. norms up to 28, 32, 36, 40, 44, and 48 months. (b) The WordBank network of 527 words learned by 30 months. Node sizes indicate relative indegree.

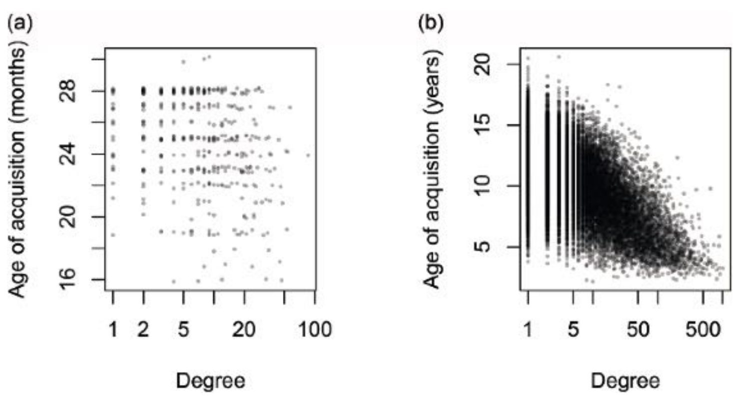

Figure 5.2 Word indegree in the Small World of Words free association norms is plotted against the word's age of acquisition. (a) Age of acquisition is estimated from WordBank, based on the first month at which more than 50% of the children in the database can produce the word – 527 words are shown. (b) Age of acquisition is from the mean age estimated by adults in Kuperman et al.'s age of acquisition norms – 19,912 words are shown. Degree is plotted on a log-scale. Jitter is added to age of acquisition in the left panel to make points more visible.

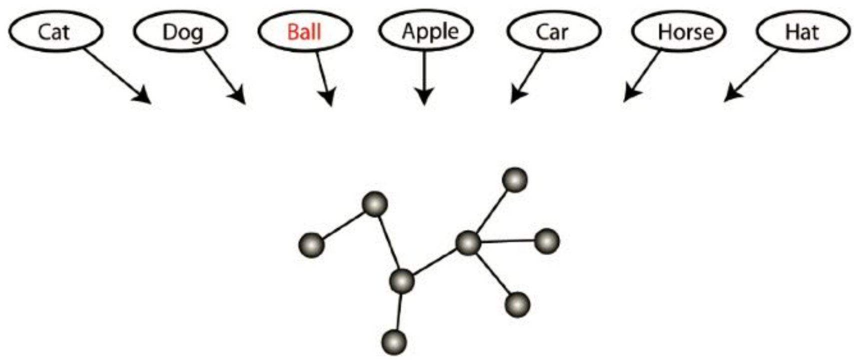

Figure 5.3 The problem our models are trying to answer: Given a known lexical network (gray nodes), and a set of possible words to learn next, what model best predicts the order of word learning observed for children?

Each model needs to answer a fairly simple question: Given a network of known words and a set of unknown words, with what probability should each word be learned next (see Figure 5.3)? Ideally, the best model will assign the highest probability to the words that children actually learn. If our models assign probabilities to all words that could be learned at each step in the network evolution, then we can compare models using the maximum likelihood method as described later in the chapter. First, let us take a close look at the models.

5.3.1 Preferential Attachment

Preferential attachment predicts that new nodes attach preferentially to words that are already well connected. This satisfies two of the requirements of early word learning: it generates long-tailed indegree distributions, and words that are learned earlier have higher indegree than words that are learned later.

Preferential attachment is unconcerned with the language environment beyond the words that children already know. In this sense, preferential attachment is self-determined – new words are chosen based only on how they interact with the existing structure. Steyvers and Tenenbaum (2005) proposed a similar model for lexical growth after observing the power law described in Chapter 4. They provided a proof-of-principle model using duplication and divergence, which they showed could generate the observed power law structure. Here we will compare preferential attachment with several alternatives described in Hills et al. (2009).

5.3.2 Preferential Acquisition

Preferential acquisition is an alternative to self-determined growth models. It is instead driven entirely by the environment. Networks that grow according to preferential acquisition add nodes with the highest indegree in their language environment first. This model is developed based on the observation that nodes of high indegree in the learning environment are already likely to be salient because they appear often in combination with other words.

An example of the importance of this salience is Quine (1960)'s *gavagai problem*. Suppose a language learner is looking at a field and someone points and says "gavagai." Determining what gavagai refers to is challenging because the possibilities are numerous. Hearing a word once in a complex environment is underdetermined. Does "gavagai" mean "field," does it mean "field in open sunlight," does it mean "wheat," or does it mean something else? It's hard to say. But words that co-occur with many associates are likely to occur across environments, and this can help to resolve their meaning from the changing background. If the learner hears the word "gavagai" in context A and later in context B, the learner can surmise that the term refers to something common to both contexts. As it turns out, words that have many associates also occur in more diverse contexts and are learned earlier (Hills et al., 2010).

With preferential acquisition, the emphasis is on the structure of the environment. As noted, language spoken to children is different from language spoken to adults, and it is different in precisely the ways that would facilitate early language acquisition (Hills, 2012). For example, child-directed language tends to pack meaningful associates around words that children are likely to learn earliest. Words like "ball" appear with words like "round," "kick," and "catch" much more often when they are spoken to children than when spoken to adults.

Note that while preferential attachment takes the extreme position that the environment is unimportant, preferential acquisition takes the alternate but extreme view that what a child knows is unimportant. If a child learns one word first, for preferential acquisition this doesn't influence what they learn next. For preferential attachment it does. The next model, the lure of the associates, is a middle ground between these two.

5.3.3 The Lure of the Associates

With the lure of the associates, known words facilitate the learning of unknown words. Once a child knows a set of words, these words invite unknown words into the child's lexicon. The first word in the lure of the associates is learned according to its indegree in the learning environment (i.e., preferential acquisition), as we already know the data favors such a preference. However, after that words are invited in by the words the child already knows. Thus, both what the child knows and the structure of the language environment are important.

To give an example of the lure of the associates, imagine that a child first learns the words "dog" and "cat." Future words are then *lured* in by their relationship to "dog" and "cat," increasing the likelihood of learning words like "pet," "soft," and the phrase "be nice."

Unlike preferential acquisition, the lure of the associates is not based entirely on prominent words in the language environment. Nor is it entirely independent of the language environment: the environment determines which words are related to one another. It is the combination of the structure of the environment and what the child already knows that determines what will be learned next. This represents the subgraph that includes all known words with the addition of each possible new word in turn. The child favors learning new words with higher indegree when included in the subgraph of known words.

5.4 Model Comparisons

Figure 5.4 visualizes the predictions of the three different models (following Hills et al., 2009). In each case, the properties of the network provide a means to quantify the likelihood that a word will be learned. Those properties are a measure of the indegree of the word that will be learned (in the case of preferential acquisition or lure of the associates) or the indegree of the word that a new word will connect to (preferential attachment). In the case of preferential acquisition and the lure of the associates, the indegree is determined differently because the models assume that different things in the environment are important. Preferential acquisition assumes that all relationships among words are important. The lure of the associates assumes that the only relationships that matter for an unknown word are the relationships it has with known words.

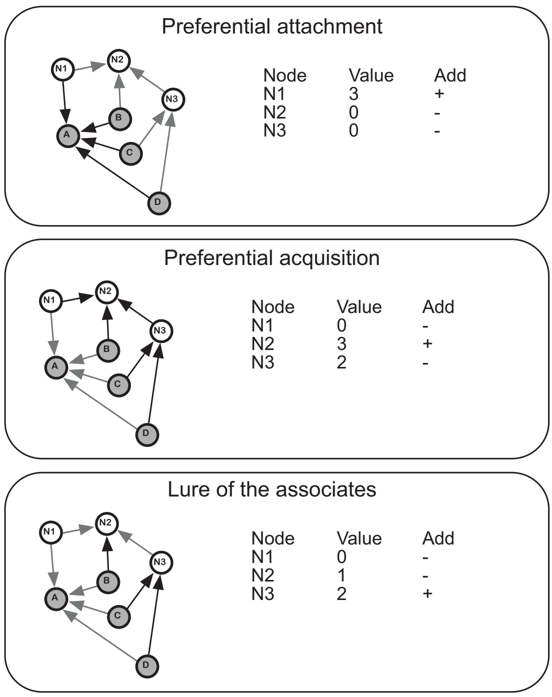

Figure 5.4 The three growth models. Each of the networks are the same, but the models assign different values to the unknown words. The gray nodes are in the "known" network (known words: A–D); white nodes are words to be learned (N1, N2, and N3). Black lines indicate links relevant to the growth models, and gray lines indicate unimportant links. The "Add" column in each illustration indicates which node is favored most for learning by the growth model in question; this is determined by the relative growth values of the possible new nodes. The growth values computed in this example are based on indegree for a directed network – arrow direction is important. In the preferential attachment model, the value of a new node is the average indegree of the known nodes it would attach to (e.g., N1 is attached to A, which has an indegree of 3). In the preferential-acquisition model, the value of a new node is its indegree in the learning environment – that is, the full network (e.g., N2 has an indegree of 3, which includes one link from a known node and two links from unknown nodes). In the lure of the associates model, the value of a node is its indegree with respect to known words (e.g., N3 has a value of 2, based on its two connections from known nodes).

These different measures can be thought of as *learning values*. Each model provides learning values for all unknown words. This allows the models to be compared against one another. One easy way to achieve that is by using the *maximum likelihood estimator* combined with an estimator of model prediction error.

5.4.1 Maximum Likelihood Estimation and Model Comparison

Maximum likelihood estimation is a way to identify the parameter estimates that minimize model error. If we assign a probability to learning each word based on the learning value of that word from its corresponding learning model, then parameters of the model can be tuned to maximize the probabilities in a way that best matches the observed pattern of the data.

To make this easy to understand, imagine a hypothetical language environment in which a number of words are already known but there are only two eligible words that could be learned: "dog" and "telephone." Using the preferential acquisition model we might find that "dog" has a learning value of 4 and "telephone" has a learning value of 2, based on its indegree in the learning environment. We can then assign a probability to each word. Other models will assign different learning values and produce different probabilities. We can produce these probabilities from the learning value by inserting the learning values into the softmax function.

The *softmax function* is a common choice rule that allows us to assign probabilities to different outcomes (or choices) (Daw et al., 2006; Pleskac, 2015). The probability of an outcome is a function of its properties divided by the sum of the same function applied to all possible events. Division by all possible events means that the probabilities sum to 1.

For word learning, the probability of learning word i is a function of its learning value divided by the sum of the same function applied to all possible words, as follows:

$$P(w_i) = \frac{e^{\beta d_i}}{\Sigma_j e^{\beta d_j}}$$

The term on top, $e^{\beta d_i}$, is a function of the learning value, d, for word i. The *sensitivity parameter*, β, determines how important d is. If β is equal to 1, then the learning function for an individual word is e^{d_i}, which means a word's probability of being learned grows exponentially with its learning value. The role of β is to tune the exponential influence of the learning value. If β is 0, then there is no influence of the learning value: $e^0 = 1$. Therefore, all words are learned with equal likelihood. As β grows larger, small advantages in the learning value lead to increasingly larger gains in the relative probability.[1]

Whenever there are different items in a set to choose from, the softmax function can be used to assign probabilities to the different items. In the present context, the goal is to establish which model assigns the probabilities to word learning that most closely reflect the words that children actually learn. The softmax function is sufficiently general that it can also be used to assign probabilities to different choices in judgment and decision

[1] Changing β is similar to changing the base of the exponent. For example, if we let $\beta = ln(2)$, then $e^{ln(2)} = 2$, which means the new learning function is 2^{d_i}.

making tasks (Pleskac, 2015) or to different memories in memory search tasks (Hills et al., 2012). We will see this again in future chapters.

Combining our three different models with the softmax function, we are able to assign probabilities to all the potential words a child could learn for each model. Whenever models assign probabilities to different outcomes, then we can employ the maximum likelihood method to compare the relative accuracy of the different models. We do this by summing up the *log-likelihood* across each of the learned words.

The log-likelihood is basically a penalty for assigning low probabilities to observed data. This is summed across all predicted data to give an estimate of a model's error. Using our probability from the softmax function, we can write the log-likelihood summation as follows:

$$\hat{L} = \sum_{i}^{n} log(P(w_i))$$

The log-likelihood is the log of the model's probability assignment for the observed data. This means that the penalty for a poor prediction gets increasingly large as the probability shrinks. A probability near 1 will have a log value near zero – no penalty. But a probability near 0 will have a log value near negative infinity. Models that assign low probabilities to observed data are less likely to reflect an accurate model of that data. Thus, the maximum likelihood method identifies the model and the parameter values that maximize the $\hat{L}$, making it as close to zero as possible.

There is one caveat which will become more relevant in later chapters (e.g., Chapter 11). Models with more parameters can more accurately capture many different kinds of phenomenon. A more complicated model is a bit like a Swiss Army knife – it can do many things. But you would not want to claim that the best model of a corkscrew is a Swiss Army knife, even though a Swiss Army knife can do what a corkscrew does. We should prefer a model that is as complicated as it needs to be to explain the data, but not more complicated. The maximum likelihood method combined with information criterion can achieve this by assigning an additional penalty for model complexity based on the number of parameters included in the model.

There are two common ways to penalize model complexity: the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC). Both of these transform the log-likelihood into the *negative log-likelihood*, therefore making it positive – it now becomes a value to be minimized. Both models also add a penalty that is a function of the number of parameters.

The AIC penalizes the negative log-likelihood with 2 times the number of parameters, k:

$$\mathrm{AIC} = 2k - 2\hat{L}$$

The BIC penalizes the negative log-likelihood with k times the log of the number of data points, n:

$$\mathrm{BIC} = k \ln n - 2\hat{L}$$

What's the difference between AIC and BIC? The debate continues and its resolution depends on assumptions that may be impossible to verify (e.g., Vrieze, 2012). If we have models with the same numbers of parameters, the AIC and BIC will prefer the same models. When the number of parameters is different, the BIC tends to prefer models with fewer parameters as it penalizes additional parameters more heavily. Practically speaking, it often does not matter, but when it does researchers should report both, carefully consider why and how it matters, and justify their decisions accordingly. For more information about model comparison, Lewandowsky and Farrell (2010) remains a valuable and thorough resource.

5.4.2 Model Competition

Here we use BIC to compare the three models described earlier. Using the Kuperman et al. age of acquisition norms, we focus on the 308 words people report learning before the age of four. These are added to the network from earliest learned to latest learned, computing the probability of adding each word according to each model using the softmax function. These probabilities are then incorporated into the log-likelihood function. The optimal value of β is found for each model using maximum likelihood estimation; this is then used to compute the BIC for each model.

There are numerous optimization algorithms for identifying the β that minimizes the BIC for each model. Here we use a grid search, examining a series of β values over a small region of interest containing the minimum BIC. This allows us to visually compare each model, which can be seen in Figure 5.5.

The results of the model comparison favors preferential acquisition. It has the lowest BIC at $\beta \approx 0.04$. This suggests the best explanation among the alternatives explored is that children attend to the structure of the semantic language environment and choose words with high indegree in that environment. This result was first observed in Hills et al. (2009) using a smaller data set taken from observations of children; here we used the adult-rated age of acquisition norms and found something similar. The utility of

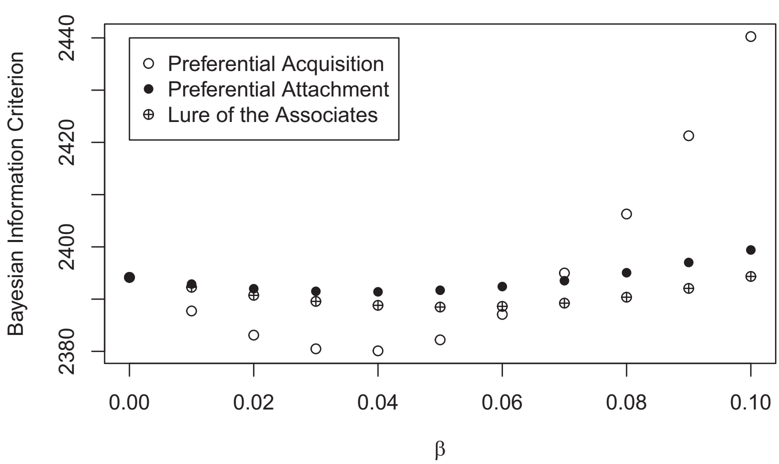

Figure 5.5 Comparing three models of child-language acquisition using Bayesian Information Criterion (BIC). The model that minimizes the BIC is the most favored model. Each model computes a probability for learning each word in turn. In this case, the model computes probabilities for the first 75% (n = 231) of the 308 possible words from Kuperman et al.'s age of acquisition norms. The 308 words are those reported to be produced on average before the age of four. See the online code for further details.

preferential acquisition has since been widely replicated. Fourtassi et al. (2020) have described preferential acquisition as an external pathway, as opposed to the internally driven pathway of preferential attachment. They also found support for preferential acquisition in multiple different languages using the WordBank data described earlier (see also Frank et al., 2017).

Numerous other researchers have explored these and other novel approaches to network growth models in language acquisition (Beckage & Colunga, 2019; Bilson et al., 2015; Ciaglia et al., 2023; Luef, 2022a; Sailor, 2013; Siew & Vitevitch, 2020; Sizemore et al., 2018; Stella et al., 2017, 2018), including the verification of semantic sensitivity in young children (e.g., Arias-Trejo & Plunkett, 2013; Plunkett et al., 2022). Models using semantics tend to favor externally driven models, like preferential acquisition. Some models also explore multiplex networks (using a variety of different edge types) and other language features, such as phonology – and, in these cases, they find interesting variations that paint a more detailed picture of early word learning as a combination of different influences.

5.5 Summary

This chapter described how to develop models of network evolution and then compete these models using quantitative methods for formal comparison. This involved model development, maximum likelihood estimation of the model parameters, and then Bayesian information criterion to determine which models best fit the data. The softmax choice rule was used to compute the probabilities for the addition of each observed node to the network based on the node-level metrics. This represents a general framework that could be applied to many different problems. Thus, while we have compared only three models here, the approach can incorporate a wide range of edge hypotheses and models of network change, including network decay. It could also be used to determine node choice to describe other phenomenology, such as which words might be forgotten on a list, or which items people might select in a judgment and decision making task offering multiple items to choose from. In the present case, the result of the model comparison suggested that children are attuned to the semantic structure of their learning environment and favor words that are of high indegree in that environment. In Chapter 6, we will explore how we might build edges from other network properties, such as the features of individual items, showing how network science offers a diversity of approaches for discriminating among various hypotheses.

5.6 Open Questions

5.6.1 Graph Learning

The capacity to learn a cognitive representation from a pattern of experienced relationships falls into a broader class of problems called *graph learning*. The key idea behind graph learning is that we encounter information in a serial format, much like the way we encounter words, webpages, or places along a path, one event following the next. Such serial input can be described by a network of transitions between nodes represented by edges. How this network is encoded into a cognitive format is the focus of graph learning.

Some of the earliest work in this area focused on children's abilities to parse words from a continuous speech stream. This is a core problem in language recognition because spoken language does not always add pauses between words. For example, in the phrase "I scream for ice cream" there is no audible distinction between "I scream" and "ice cream." However, via a process of *statistical learning*, Saffran et al. (1996) showed that children as young as eight months could detect word boundaries based on their *transition probabilities*. When children experienced a series of syllables like BIDAKUPADOTI, they could detect the boundary between the words BIDAKU and PADOTI if the transition probability within-word syllables (e.g., BI-DA) was 1.0 and between word syllables (e.g., KU-PA) was 0.33. This is effectively a node inference problem that requires recognizing paths among subfeatures of nodes (their syllables).

More recently, this work has been expanded to include a range of experimental and theoretical approaches to presenting and learning from graph representations (Lynn & Bassett, 2020). In these experiments, participants experience a series of nodes representing a route through the network, with each node presented in turn based on its transition probability from the previous node. The time it takes participants to detect presented nodes is a measure of their ability to encode the network. People in these experiments are influenced not only by transitional probabilities, but also network modularity and node centrality measures such as degree and betweenness (e.g., Kahn et al., 2018). This suggests that people encode more than neighbor-to-neighbor transitions. They encode edges between nodes that do not directly co-occur.

Models such as preferential acquisition – as described earlier – are special cases of graph learning in which individuals might be assumed to experience the network in full and then make inferences based on that experience. The details of the acquisition process could of course be otherwise, arising from the process of experiencing language in a serial format.

How might this work? Theories of the mind as a prediction engine, Bayesian brain, and free-energy maximizer all originate from a way of thinking of cognition as a product of analysis-by-synthesis (Clark, 2013). That is, the mind does not simply accumulate sensory information as a collection of edges, but models and predicts its environment in an effort to anticipate what it will experience next. This leads inherently to the many well-supported prediction-error models, like the Rescorla–Wagner model described in Chapter 12. How might models of graph learning be applied to the language data described here? Would such models predict models like preferential acquisition or help reveal pieces of the language acquisition process that are missing?

5.6.2 Semantic Maturation

One thing that is missing from the models described here is the observation that children understand words before they produce them. The pattern of comprehension is not a direct reflection of the pattern of production and this also differs between typical and late talkers. Jiménez and Hills (2022b) modeled these differences in semantic maturation using a two stage network model in which words first joined a comprehension network and then were promoted to production after a period of maturation. This involved parameters for attention and maturation thresholds, with the latter differing between typical and late talkers for nouns and verbs. Unlike the maximum likelihood method described earlier, the complexity of these models required Approximate Bayesian Computation, which is another approach to evaluating model fit. Still, the notion of semantic maturation is an important one because it invites us to ask what structural properties of language might facilitate the advance from comprehension to production. There are many possibilities, and most of them have theoretical backing. For example, how does phonetics matter (Storkel, 2001)? Jiménez and Hills' (2022b) models were based on structure derived directly from child-directed language, and we can ask what structural properties in language might matter as well.

6 What Is Distinctive

Exploring Edge Types in Multilayer Networks

6.1 The Problem

When nodes share features we can combine those features in many possible ways. One standard way is to base relationships on shared features. But there are other possibilities. Here we will apply a number of approaches to investigate the concept of distinctiveness. Distinctiveness is how easy it is to discriminate one thing from another thing. In an important sense distinctiveness is therefore a hypothesis about how the mind works. We say two things are distinctive because a mind can distinguish them. But what makes something distinctive? In this chapter, I will introduce some of the theory behind distinctiveness and then demonstrate how we can use network science to investigate distinctiveness in children's abilities to learn words. This takes a multilayer network approach, in which we will examine many different edge types constructed of various combinations of shared and unshared features. By examining these edge types we will discover how best to combine features and which feature combinations best predict early word learning.

6.2 What Is Distinctiveness?

Distinctiveness is the basis of our ability to tell one thing from another. Psychologically, distinctiveness has a number of interesting properties. For example, making something distinctive makes it more memorable (Murdock, 1960). One of Murdock's main contributions in this literature was noting that, first, distinctiveness is a kind of post hoc explanation based on something that's difficult to quantify, and, second, that we can quantify it. He did so by summing up the distances between various objects along different dimensions. As I will show, there are a variety of ways to do this, and they each represent different hypotheses about how distinctiveness works in the mind.

Some of the earliest work on distinctiveness and memory was done by Hedwig Von Restorff (1933). Von Restorff studied distinctiveness using what is called an *isolation paradigm*. One item in a group differs from the others in one or more ways. For example, she asked people to remember a list such as 8, 14, 3, 17, 9, XLY, 15, and 3. When compared with another list, such as TOZ, DUQ, HOL, COS, QXK, XLY, DRF, TXP, people were much more likely to remember the XLY in the first list than in the second.

Why? It is not easy to tell. Distinctiveness is a subjective measure in these experiments; it is a descriptive term we assign to things that stand out. Psychologists say these things are *salient*. A number of theories have proposed that salience works because it captures attention, and that in turn leads us to think more about it (Hunt, 1995).

When people think more about the underlying meaning of something they create a more distinct impression of it in memory. This *levels of processing theory* argues that the benefit of thinking more deeply about something is that "richer encodings are also more distinctive and unique" (Lockhart et al., 1976). As Hunt (1995) put it, "The idea essentially is that the more one knows about something, the less like other things it becomes" (p. 106).[1]

The two ways described above for thinking about distinctiveness are somewhat circular: distinctiveness leads to additional processing, and additional processing leads to distinctiveness. This does not make for tidy theory. But, luckily, von Restorff helped to tackle this problem.

Von Restorff (1933) and later Hunt (1995) showed that relative distinctiveness matters. They did this by having people remember either one isolated item in a list of nine similar items, or one isolated item in a list of nine dissimilar items. If you imagine a group of short stout people alongside one skinny giant versus a group of people of wildly different shapes and sizes, then you get the idea. What Von Restorff and Hunt found was that people remembered the isolated item in the similar list 80% of the time, but the same item in the dissimilar list only 40% of the time. This is called the *isolation effect* (or the *Von Restorff effect*) and it shows us that distinctiveness is a relative measure between an item and the other members of a set. It is easy to be distinctive among the identical, but quite a different matter among the irregular.

The isolation effect is not driven entirely by surprise. The perceptual distinctiveness at the time of presentation matters little. Even if the isolate were the second item in a list – where it will not drive surprise – it was still easier to remember: "Differences among items affect memory even if those differences are not operationally salient at the time of perception" (Hunt, 1995, p. 110).

What matters is the distinctiveness of memories. For von Restorff – coming as she did from a Gestalt background – things that are similar to one another are "agglutinated" in memory. They lose their differences. This agglutination can also be learned, for example, when we are taught which items belong to which categories: items within categories become harder to discriminate, whereas items in different categories become easier to discriminate (Goldstone, 1994).

Things that are distinctive relative to what we already know are novel and invite further exploration. Humans and other animals seek out and explore novelty. This is supported by the *novel object recognition task*, which uses animal's innate tendencies for novel exploration to evaluate memory (Ennaceur & Delacour, 1988). If an item is familiar, animals explore it less. Neuroscientists have also found that novel items excite dopaminergic neurons, which have been related to reward and learning. Some have interpreted this as fueling a *novelty bonus* in cognitive processing, which drives attention toward unfamiliar objects (Kakade & Dayan, 2002). Thus, in addition to the retroactive value of

[1] It is interesting then that older people, when compared with younger people, tend to say that things are less similar to one another (Wulff, Hills, et al., 2022). See Chapter 13 for more on aging.

distinctiveness in memory, distinctiveness also drives additional investigation and more learning as well.

However, none of this tells us exactly what makes something distinctive. Perhaps like art, we can recognize distinctiveness when we see it. But can we predict it in advance? And how would we do it with network science? If we could characterize distinctiveness in a network, we could use that knowledge to move away from the tautologies noted by Murdock (1960) and Hunt (1995).

6.3 Distinctiveness Measures

When evaluating edges in a network, it is tempting to choose a single measure and work from there. The problem is that there is more than one way to go about determining edges. Instead of assuming one measure is the correct measure, it is almost always better to compare measures to determine which one best addresses our question. This comparative analysis also enriches theory by identifying differences between various hypotheses. This also gives us the opportunity to learn how much something matters by directly comparing it with similar but alternative approaches.[2]

For any form of network construction, the approaches we take to building edges represent hypotheses about what matters. In the case of distinctiveness, we build edges out of features. But if we only have one approach to doing this, we may conclude that the features going into the measure do not matter, when what really does not matter is our particular hypothesis about how they work. Indeed, in Hills et al. (2009) we found that word networks that used features to construct edges were not good predictors of word learning in children. But the problem was not the features, the problem was how we defined edges. We were looking at shared features, not feature distinctiveness.

To provide a concrete though simplified example, suppose we want to establish the relationships between three painters: Manet, Monet, and Van Gogh (see Figure 6.1). If we present people with each painter and ask them to tell us the features that come to mind, then we can use the features they produce as nodes in a bipartite network, as shown in Figure 6.2. If we were doing this in an experiment, we might present people with many painters. For simplicity of understanding, imagine that the features produced are "French," "Landscape painter," "Portrait painter," "Cut off his ear," and "Impressionist."[3] If we ask 100 people, we can take the proportion that mention each feature as the edge weight between the two types of nodes. The adjacency matrix in Table 6.1 provides some example data to show the relationships between the two node types: painters and features.

With these three nodes and five features, we can now produce a variety of different networks that characterize the relationships between the three painters. There are two standard approaches to achieving this based on similarity, and three more based on distinctiveness.

[2] The measures and analysis of word learning in children follows that presented in Engelthaler & Hills (2017).

[3] For the curious: Manet often painted people and, though well respected among Impressionists, his status as an impressionist is disputed – he is more associated with modernist realism. Monet's early paintings share Manet's style, but he is most well known for his impressionistic paintings. He gave the movement its name with his hazy but colorful *Impression, Sunrise*.

Table 6.1 A bipartite adjacency matrix linking two node types: Painters and features.

	French	Landscape	Portrait	Ear	Impressionist
Manet	1	0.1	1.0	0	0.5
Monet	1	1.0	0.0	0	1.0
Van Gogh	0	1.0	0.1	1	1.0

Figure 6.1 Some illustrative paintings: *Boating* by Edouard Manet (a), *Impression, Sunrise* by Claude Monet (b), and *Sunflowers* by Vincent van Gogh (c)

Source: a) The Metropolitan Museum of Art, New York, b) The Metropolitan Museum of Art, New York, c) The Metropolitan Museum of Art, New York

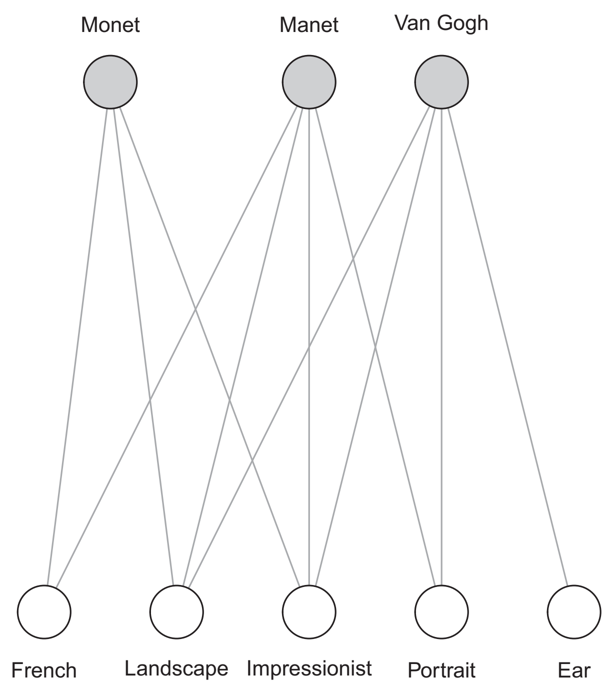

Figure 6.2 Artists and their features: A simplified bipartite feature network with two node types.

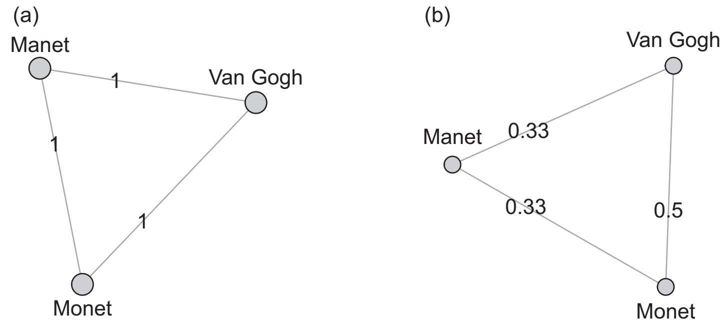

Figure 6.3 Standard bipartite projections based on shared features. These are converted into distances by taking the inverse. (a) Projection based on at least one shared feature. (b) A weighted projection based on the count of shared (nonzero) features.

6.3.1 Standard Bipartite Projections: Unweighted and Weighted

The most common way of basing edges on attributes is to form an edge between two nodes if they share at least one attribute. This is a standard bipartite projection onto one node type, as shown in Figure 6.3. For social networks, the intuition is that people who, for example, belong to the same group are likely to share social contact. Here our focus is on conceptual distinctiveness, so we are interested in shared features not group membership. In the painter example, this would connect Manet and Monet and Van Gogh in a triad with equal edge weights of value 1. This is a measure of similarity, but we can compute distance by taking 1 over the similarity. In this case the values are still 1.

A weighted projection gives edge values equivalent to the number of shared features (sometimes called *multiplicity*). This would give the edge between Monet and Manet a value of 3, the edge between Manet and Van Gogh a value of 3, and the edge between Monet and Van Gogh a value of 2. We will call the *inverse multiplicity* the distance computed by taking 1 over the edge weight. This gives $1/3$, $1/3$, and $1/2$, respectively.

These are far from complete. They throw away information about feature weights: Exactly how many people would agree that Manet is an impressionist? They also throw away information about unshared features: How important is Van Gogh's ear?

I will provide three more methods focusing on distinctiveness that take into account this information in different ways. Thinking in terms of distinctiveness helps us to consider what nonshared features mean.[4]

6.3.2 Nonshared Feature Distance and Symmetric Difference

Nonshared feature distance is the sum of all features that are not shared between two nodes. In set theory this is called the *symmetric difference* between two sets. It is computed as

[4] These three measures are not exhaustive. There are many more sophisticated measures. For example, the SIMPLE model of memory takes into account such things as Shepard's universal law of generality (Brown et al., 2007a). Murdock (1960) also considered the log values of feature dimensions to account for the Weber–Fechner law.

$$d = (n_1 + n_2) - (2n_s)$$

where n_1 is the number of features for the first node, n_2 is the number of features for the second node, and n_s is the number of shared features. If two nodes each have 5 distinct features, but no shared features, then they will have a symmetric difference of 10. If two of their 5 features are shared, they will have a symmetric difference of 6.

Symmetric difference focuses only on dissimilarity. Adding shared features does not change the symmetric difference. Also, features are not weighted. If a feature is rarely mentioned, it is as important to symmetric difference as if everyone mentioned it. A feature is simply present or absent.[5]

6.3.3 Jaccard Distance and Jaccard Similarity

The *Jaccard index*, or *Jaccard similarity*, is the ratio of shared features over total features available for comparison. The Jaccard similarity is always a value between 0 and 1. When all the features are shared, the Jaccard similarity has a value of 1. When all the features are unique, the Jaccard similarity is 0.

The Jaccard distance is one minus the Jaccard similarity. Formally, and using the same notation as used earlier for nonshared feature distance, the Jaccard similarity, J_s, is

$$J_s = \frac{n_s}{n_1 + n_2 - n_s}.$$

The Jaccard distance, J_d, is therefore

$$J_d = 1 - \frac{n_s}{n_1 + n_2 - n_s}.$$

The Jaccard distance is influenced by the number of shared features. But it is also influenced by the number of nonshared features. As it is the ratio of nonshared features to total features, adding more shared features reduces the distance. Adding more nonshared features increases it.

6.3.4 The Manhattan Distance

The *Manhattan distance*, sometimes called the city block distance, is a measure that can be intuitively thought of as the distance it would take to walk between two points along city blocks. To get your intuition right, you'll need to imagine grid-planned city blocks, like you might find in Chicago or, not surprisingly, Manhattan. If we have only two attributes, then this maps onto a two-dimensional grid-plan city nicely. We can scale it to higher dimensions easily enough by summing the distances for each and every attribute. Formally, we can write the Manhattan distance between two nodes as

$$d = \sum_{i}^{n} |p_i - q_i|$$

[5] One solution to this is to use a threshold for feature inclusion – for example, if fewer than $x\%$ of people report a feature, the item is considered not to have that feature.

where n is the total number of features, and p and q represent the proportion of individuals who listed each feature, i, for each node, respectively. This is the method closest to that described by Murdock (1960).

If each feature is either present or absent, then the Manhattan distance is the same as the symmetric distance. It counts up the number of nonshared features. But if features are weighted, then the Manhattan distance will consider the weights. That is, Manhattan distance quantifies how salient the differences are between two nodes along each dimension, where salience can, for example, be measured by how many people do or do not list each feature.

6.3.5 Specificity and Raw Distinctiveness

A further measure that is not a network measure is based on how rare or specific an attribute is. Rarer attributes are more distinctive. *Specificity* can be computed as 1 over the number of objects that share that feature. By summing over feature specificity for all features, we gain a measure of an object's distinctiveness without considering how near or far it is to individual other objects. This is formulated as

$$x = \sum_{i}^{n} \frac{1}{w_i},$$

where x is the raw distinctiveness, n is the number of features, and w_i is the number of objects that have feature i.

To gain some intuition for raw distinctiveness, note that if many objects have a feature, then w_i is larger and therefore that feature will add little to an object's raw distinctiveness. However, if the feature is specific to the object, then it will add a value of 1 to the object's distinctiveness. An object with 10 features that are specific to that object will have a raw distinctiveness of 10 if that object has no other features.

6.3.6 Summary of Measures

Figure 6.4 provides a visual depiction of the three distinctiveness measures used to compute edge weights.

The Manhattan distance and the symmetric difference focus more on unshared features. The Jaccard distance and the shared feature projections focus more on shared features. Raw distinctiveness is not used for computing edge weights, but is a node attribute that can be used as a variable to evaluate its relative influence on behavior.

6.4 Applying Distinctiveness to Child Language Acquisition

Now that we have these measures, we can apply them to the word learning problem introduced in the Chapter 5. However, instead of modeling network growth, we will use the network measures of distinctiveness to predict age of acquisition – a problem with which we are growing increasingly familiar. To enrich the context, we will first ask why features should matter at all in child language learning.

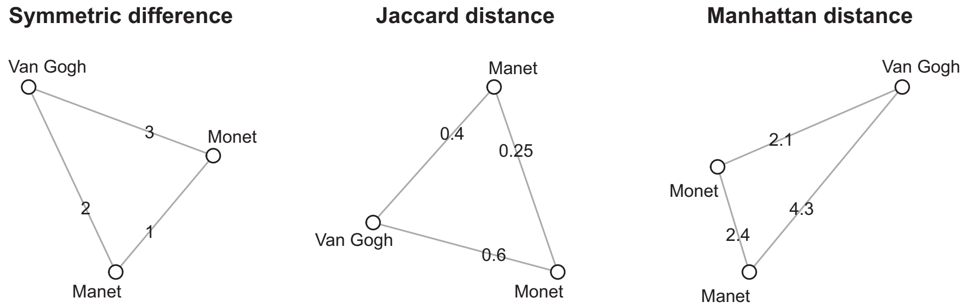

Figure 6.4 Network connectivity measures for distinctiveness. The symmetric difference asks how many features are not shared by a pair of nodes. The Jaccard distance is the number of nonshared features divided by the proportion that could be shared. The Manhattan distance is the sum of the absolute value of the difference in feature values across all features. It is the only metric that takes into account feature weights in the bipartite network.

When people learn new concepts, part of that learning involves learning the boundaries of individual concepts. To know what something is requires knowing how it is or is not like other things. The notion of things being more or less similar to other things should immediately bring to mind a network approach.

Understanding the boundaries of concept meanings often goes under the title of *generalization* (Shepard, 1987). A typical question might be: How does one stimuli generalize to another stimuli? Because few events are ever experienced in exactly the same way, what matters for knowing whether one thing is the same thing as another?

If a dog learns that a raised hand with the palm upwards in the kitchen means sit, will it also sit if the palm is downwards, or if the command is given at the park? Dog owners will be familiar with the confusion. My dog, Samson, sits nicely in the kitchen, but always seems incredulous at the park. Samson's behavior tells me that he recognizes the raised hand as a sit command, but that a variety of other things matter, many of which are not apparent to me. To know what a dog thinks a sit command is, we also have to know what it thinks is not a sit command. When we take the difference, then we start to understand the dog's representation of the sit command.

The idea that what something means is tied up with how it is like or unlike other things is a common assumption of structural linguistics. Saussure put it like this: "language is a system of interdependent terms in which the value of each term results solely from the simultaneous presence of the others" (Saussure, 1916/2011, p. 114). A word like "hot" gains some aspects of its meaning from its association with other terms such as "sun," "burn," "fire," and "ouch." These relationships should invite network thinking.

The question of what kinds of relationships matter for the von Restorff effect, for dogs learning commands, and for word meaning is partly a question about what features to include in the network and how to turn those features into meaningful edges that make valid predictions about learning. If a mind attends to some features and less to others, then objects that share less salient features will be perceived as less similar. If a mind attends more to similarities and less to differences, then Jaccard distance may be a better predictor than nonshared features. If we are trying to understand child cognition or Artificial Intelligence or the cognition of some other species or alien, each of these approaches represents a test of what kind of structure matters.

Children learning words provides a good environment for evaluating feature distinctiveness. A child might learn that the word "spoon" goes with the particular plastic blue spoon they often use. But what exactly does "spoon" refer to in this case? Many children will use a word like "spoon" to refer to forks or other utensils, just as they might use a word like "dog" to refer to horses or other moderately sized mammals. This is a problem of *overextension*. But *overextension* is just a specific part of a broader problem of *indeterminancy* (Hills, 2012). This is Quine's gavagai problem again (described in Chapter 5) which notes that word learning environments are rarely sufficiently constrained to map words onto the precise concepts they refer to.

A useful conceptual lever and partial answer to the problem is the observation that young children often have a *shape bias* (Landau et al., 1988). Shape bias means that when children first learn the name of a solid object, shape is a prominent feature. Material and texture are less salient. Not all soft things with fur are dogs, but dog shaped things are still dogs even if they are made of wood.

A typical test for shape bias involves teaching a child that a novel object has a novel name, like "Dax." The dax is a funny shaped thingamajig with a certain set of features, like color, shape, material, or texture. Then the child is asked which of a set of other objects is also a dax, with some of those objects sharing a similar shape and the others having different shapes but sharing other similarities with the original dax, such as texture or color. Figure 6.5 shows a visual depiction of the problem.

If we were building networks out of information intended to reflect what a child attends to, then features such as shape are a useful hypothesis to incorporate into edge construction. Because we have an immense amount of data on the order in which children learn

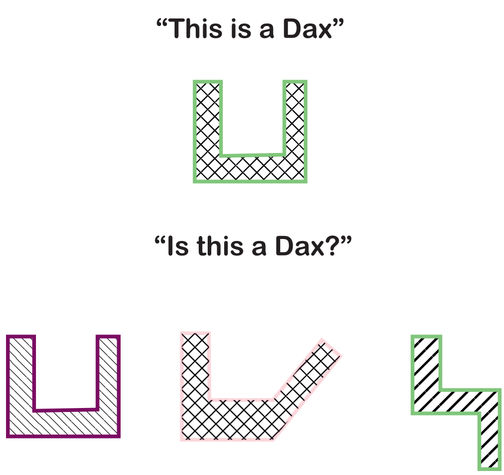

Figure 6.5 Typical shape bias paradigm. A child is taught that the word "Dax" is associated with the top object. Then they are asked if other items below are a "Dax." In the original study, test objects were presented one by one. Color is represented by the border. Texture is represented by the cross-hatching. The figure is roughly based on Landau et al., 1988.

words and we also have a wealth of data about the features associated with each of those words, we can ask which features best predict the order in which children learn words.

To do this, we will build networks using the various similarity and distinctiveness measures described here and then ask how degree centrality based on these measures is correlated with age of acquisition. This will tell us which of the various approaches to edge formation best captures what children might be attending to during word learning.

6.4.1 The Data

6.4.1.1 The Feature Norms

Feature norms represent a collection of words and the features that people say are associated with those words. Here we will use the McRae et al. (2005) feature production norms. This is a collection of 541 nouns. Each noun has features collected from approximately 725 people. There are 2,526 unique features, which are labelled and categorized into 10 different feature types:

- visual form and surface
- visual motion
- visual color
- taxonomic
- encyclopedic
- function
- sound
- tactile
- taste
- smell

Each visual form and surface feature is further labelled as "external component," "external surface property," "material," "internal component," "internal surface property," "systemic property," and "entity behavior." See Table 6.2 for examples of nouns and associated features. The multiplex nature of the network built out of this variety of feature types means that we can investigate the full multilayer network – including all feature types – as well as individual feature types (Stella, Citraro, et al., 2022). This gives us additional leverage in establishing what properties make something distinctive.

Table 6.2 A sample of the McRae Feature Norms. There are 541 nouns (rows) and 2,526 unique features (columns).

	a_musical_instrument	has_keys	requires_air	associated_with_polkas	has_buttons
accordion	0.93	0.57	0.37	0.3	0.27
piano	0.73	0.90	0.00	0.0	0.00
balloon	0.00	0.00	0.50	0.0	0.00

Table 6.3 Correlations between age of acquisition and the different degree and distinctiveness measures. There are n=492 words and each measure is computed as described earlier. CI indicates the 95% confidence interval.

	Model	r	CI
6	Manhattan distance	−0.3400	[−0.42, −0.26]
5	Symmetric difference	−0.3200	[−0.4, −0.24]
3	Raw distinctiveness	−0.2300	[−0.32, −0.15]
4	Jaccard distance	−0.1100	[−0.2, −0.022]
1	Simple degree	−0.0180	[−0.11, 0.071]
2	Inverse multiplicity	−0.0021	[−0.09, 0.086]

6.4.1.2 Age of Acquisition

We will use the Kuperman et al. (2012) age of acquisition norms, which are a collection of about 30,000 words rated by adults for the age at which they were learned. Of these words, 492 of them are present in the McRae feature norms, providing a data set upon which to compare distinctiveness measures.

6.4.2 Comparing Distinctiveness Measures

Table 6.3 shows the correlations between each word's strength (weighted total degree) in the weighted distinctiveness networks and its age of acquisition. Distinctiveness computed from Manhattan distance is the best predictor of age of acquisition. This confirms that the additional information contained in the weighted feature values is informative. The simple bipartite projection (simple degree) is effectively no different from random guessing when predicting age of acquisition.

One interpretation of these results is that children may attend not only to the presence or absence of a feature, but also to its relative intensity. When comparing two objects, they may compare these two intensities across the available features. This interpretation is a prediction of the Manhattan distance model – it becomes a hypothesis that can then be tested in future experimental studies.

6.5 Testing Hypotheses by Deconstructing the Multiplex Network

The feature set we use to compute the various distinctiveness measures contains multiple feature types. Our original edges consisted of these many different features types, constructing relational measures over many different kinds of properties. We can deconstruct this multiplex network into its components to evaluate more specific hypotheses about distinctiveness. Specifically, we can recompute the Manhattan distance for each feature type (e.g., "visual_form_and_surface") to evaluate its relative contribution to explaining the variance in early word learning. To do this, we subset the feature norms to only contain the feature type of interest. We then recompute Manhattan distance for a network

Table 6.4 Correlations between age of acquisition and individual feature types. Each correlation is limited to the *n* words that have that feature. CI indicates the 95% confidence interval.

Feature type	r	CI	n
visual-form_and_surface	−0.27	[−0.35, −0.19]	486
encyclopedic	−0.17	[−0.25, −0.08]	458
function	−0.13	[−0.22, −0.04]	418
taxonomic	0.06	[−0.04, 0.16]	396
tactile	−0.13	[−0.27, 0.03]	165
visual-colour	−0.11	[−0.22, 0.01]	264
taste	−0.13	[−0.39, 0.15]	51
visual-motion	−0.07	[−0.22, 0.09]	165
smell	−0.02	[−0.47, 0.44]	19
sound	−0.01	[−0.21, 0.20]	93

composed only of this feature type. For example, we can take all the features in the "smell" category and ask, among those words with this feature, to what extent do smell features predict age of acquisition.

Table 6.4 shows the correlations. Visual-form_and_surface is the strongest predictor. It is also the feature type reported for the highest number of words, which may further indicate its salience. Together, these suggest that distinctiveness based on visual-form_and_surface properties may help young learners to discriminate object categories.

Encyclopedic and function are two other feature types with confidence intervals nonoverlapping with 0. Encyclopedic refers to features like "lives in Africa" (zebra) and "like a potato" (for potato). Function refers to features like "used for travel" (an airplane feature) and "used for holding boats still" (anchor). These offer other novel hypotheses for thinking about distinctiveness.

Finally, we can unpack visual-form_and_surface properties into their various components. This further limits the analysis to words that have each specific feature type. Among these words, to what extent does the distinctiveness associated with that feature type correlate with age of acquisition?

The results suggest that more distinctive feature types within visual-form_and_surface are *material*, *external component*, and *external surface property* (Table 6.5). "Material" contains features like "made of flour," "made of gold," and "made of brick." Material has been shown to be important for nonsolid objects (e.g., Colunga & Smith, 2005), but it is also possible that materials like "flour" and "brick" are associated with distinctive shapes, like bread and buildings (which we can see in the feature norms data). "External component" contains features like "has a tail," "has 4 wheels," and "has a propeller," which are all likely to be salient at least partly because of their shape. "External surface property" is also associated with shape ("is round") but also size, length, and texture. At the end of our analysis, we find shape is a prominent player in distinctiveness, but it exists alongside other properties as well. In the open questions I discuss still further properties, such as phonology and semantics, which can be combined in various ways and competed against one another in a more complete analysis.

Table 6.5 Correlations between age of acquisition and individual feature types within visual form and surface. n indicates the number of words with this feature type on which the analysis is based.

Feature type	r	CI	n
Made of (material)	−0.25	[−0.37, −0.13]	240
External component	−0.15	[−0.25, −0.05]	377
External surface property	−0.12	[−0.22, −0.01]	346
Internal component	−0.18	[−0.37, 0.02]	113
Internal surface property	−0.1	[−0.51, 0.34]	29

One can also ask if an object is distinctive because it is distinct from other nearby objects or because it belongs to a community of distinctive items. Local communities of items that are similar to one another could be distinctive relative to the majority of items in the network. Our current analysis does not answer this. The above analyses measure distinctiveness from all other objects in the network. However, networks make this local community analysis straightforward. Engelthaler and Hills (2017) addressed this using community detection analysis (Citraro et al., 2023; see also Siew, 2021). The analysis with age of acquisition was then repeated over a range of smaller and smaller community sizes, computing distinctiveness only within these reduced communities. The distinctiveness effects persisted. This suggests that feature distinctiveness is not only a property of global comparisons, but also occurs in comparison with more similar objects in local subgraphs.

6.6 Summary

In this chapter we took a multiplex network approach to the question of distinctiveness. A number of different methods for computing edge weights based on word features were compared. Strength was then computed for each node to evaluate its relative distinctiveness in the network, and this was correlated with age of acquisition for words that children learn when young. The best correlation was found with Manhattan distance, which supports the notion that nonshared features are important and that feature intensity is important as well. These are two things that would be lost in a bipartite projection or standard shared-feature approach.

However, our approach went still further. Given that features belong to certain feature categories (e.g., visual versus tactile), this allowed us to reduce the edge computations to only include features of a certain type. This confirmed that features like shape and material could be important contributors to distinctiveness in child learning as they best correlated with age of acquisition. The distinctiveness and multiplex network approaches taken here are fairly general. Distinctiveness – or, more generally, measures that take into account shared and unshared features – can be used with any bipartite network of objects and features. When those features – and, subsequently, edges – can be categorized into subsets, we can investigate which feature sets provide the best predictors.

6.7 Open Questions

6.7.1 Optimizing Edge Distinctiveness

Manhattan distance is a special case of the more general Minkowski distance:

$$d = \left(\sum_{i}^{n} |p_i - q_i|^p \right)^{\frac{1}{p}} .$$

When $p = 1$, this is the Manhattan distance. When $p = 2$ this is the Euclidean distance. When $p = \infty$, we have the Chebyshev distance. In our word-feature example, the Chebyshev distance is equivalent to the maximal difference along any feature. So if one feature has the largest distance, this is the relative distinctiveness between the two items. What's interesting about these various formulations is their ability to facilitate hypothesis generation. Suppose cognitive systems always noticed the single largest distance in features; this would be akin to Chebyshev distance. Suppose they sum all features distances equivalently; this is the Manhattan distance. Might a different value of p be more appropriate for predicting age of acquisition? What would this mean for the way the mind evaluates feature differences?

6.7.2 Comparing Features with Semantics and Other Edge Types

In Chapter 5 we used the maximum likelihood method combined with Bayesian information criterion to compare growth models for semantics. The same could be done here to compare growth models for feature types based on distinctiveness. Stella et al. (2017) is an excellent resource in this respect. They compared shared features with associations, phonological similarity, and word co-occurrence, and examined the impact of these relationships over time, showing that the relative contributions of these edge types changed across development. They did not, however, examine feature distinctiveness. In many cases, full multilayer networks make the best predictions (e.g., Levy et al., 2021), which is what we also find here. All the features combined work better than any subset of features. Could the edges be improved further by weighting feature types, or by taking into account distinctiveness, or amplifying differences still further using a method similar to the softmax function presented in Chapter 5?

7 The Small-World Spectrum
Using Small Worlds to Compare Networks

7.1 The Problem

Many networks of practical significance are small-world networks. Small-world networks are characterized as having a high clustering coefficient and short average path length relative to a random network of the same size and density. Social networks, neural connectivity, symptoms associated with disease, business networks, transport networks, and so on are small-world networks because they have clustered local communities laced with nodes that bridge to otherwise distant regions of the network. The extent to which different networks achieve this relative balance creates a small-world spectrum, ranging from highly clustered to highly random. This spectrum has fueled a great deal of research into understanding how best to measure small-world networks. One of the useful properties of these measurements is that they compare the structure of an observed network relative to different versions of itself, thereby providing 'control' networks against which to benchmark a measurement. This is useful for comparing networks of different sizes and density, allowing us to home in on what matters and why. In this chapter, I will discuss small-world networks and three different ways to quantify them, and then use these measures to answer a simple question: Are late-talking children just slow versions of early talkers, or are they learning something different? After using the small-world measures to demonstrate the difference, I offer three different explanations as to why – based on function, formation, and emulation – which represent three different approaches to explaining how network structure comes about.

7.2 Living in Small Worlds

In *The WEIRDest people in the world*, Joseph Henrich (2020) tells the story of the Ilahita. They are a clan-based community in the Sepik region of New Guinea. Most communities in this region rarely grew in size beyond 300 people. When they grew larger than this, they tended to break up into smaller isolated groups, often instigated by social disagreements. The Ilahita were an exception. They had a population closer to 2,500, made up of 39 clans. By overcoming this size limit, they wielded considerable power.

According to Henrich, the innovation that allowed the Ilahita to grow so large was the consequence of a cultural mistake. The Ilahita introduced this mistake when they purposefully acquired the ritual practices of what they saw as a more powerful neighboring tribe. These practices, called the Tambaran, created interdependencies in the communities that practiced them. For example, the Tambaran prohibited individuals from eating their own pigs, so they had to be shared. However, when the Ilahita took up the Tambaran, they introduced 'copying errors' that further increased the number of interdependencies in the community. These copying errors included taking brothers out of their own clans and putting them in neighboring Ilahita clans. They also created more powerful nonlocal gods shared across the clans. These relationships (both social and spiritual) brought all the individuals in the Ilahita closer to one another (shortening the average path length) even as they maintained relationships within local communities (preserving the local clustering coefficient). That is, the Ilahita overcame a population size problem by inadvertently introducing cultural practices that allowed them to grow while preserving their *small-world* structure.

Small-world networks are defined as networks with a high average clustering coefficient alongside a short average path length. For social networks, this tends to mean that individuals organize into groups but the groups remain well connected. Individuals in an Ilahita clan – often containing fewer than 100 people – might all know one another personally. Yet the introduction of the modified Tambaran practices mean that they also have many indirect relationships with people in other clans. The basic idea is that the Tambaran strengthened social connectivity, making the Ilahita more resilient to social divisions that might fracture them.

Our real-world familiarity with small worlds and their frequent occurrence raise many interesting questions: What are the typical statistical properties of small world networks (Newman & Watts, 1999; Watts, 1999)? What do small-worlds mean for business and creativity (Kogut & Walker, 2001; Uzzi et al., 2007; Uzzi & Spiro, 2005), neuropathology (Stam et al., 2007), and disease epidemics (Moore & Newman, 2000)? How do people search small worlds (Kleinberg, 2000)? Are there distinguishable subclasses within the small-world regime (Amaral et al., 2000). And how can we best measure them (Neal, 2017)? This is but a taste of this work. One could write books about it, and there are many excellent examples to choose from (Buchanan, 2003; Watts, 2004).

One of the landmark experiments in the small-world literature is the research of psychologist Stanley Milgram. Milgram was interested in answering the following question: "Given any two people in the world, person X and person Z, how many intermediate acquaintance links are needed before X and Z are connected" (p. 62, Milgram, 1967). To answer this, Milgram assigned people in Omaha, Nebraska, Wichita, Kansas, and Boston, Massachusetts with the task of sending a letter to a target individual in Boston about which they only knew things such as their name, occupation, and address. However, the letter senders could not send the letter directly. They were only allowed to send the letter to people they knew on a first-name basis. So, if they did not know the target individual, they needed to send the letter to someone who might know someone who did, for example. Receivers of letters in this letter chain were instructed to do the same – unless, of course, they knew the target person, in which case they could send it directly.

Even though only about one-third of these letters ever reached their targets, among those that did the average path length involved about five people (Milgram, 1967; Travers & Milgram, 1977). A fascinating large-scale attempt to replicate this phenomenon by email produced similar estimates of our global connectivity (Dodds et al., 2003).

Though Milgram's work was groundbreaking, accounts of human interconnectedness were already in the air. The Hungarian writer Frigyes Karinthy wrote a short story in 1929 about a parlor game that anticipated Milgram's experiment in detail – including predicting five links (Karinthy, n.d.). He even proposed that the same logic could be applied to objects: "How can I link, with three, four, or at most five links of the chain, trivial, everyday things of life" (p. 3).

The playwright John Guare entitled his 1990 play *Six degrees of separation*, which has made famous the notion that everyone is connected with everyone else on the planet through a chain of six or so people. Each new person offers so many new connections. As Guare's character Ouisa notes in the final lines of the play, "every person is a new door, opening up into other worlds."

It is often entertaining as an icebreaker to have people unfamiliar with one another attempt to find shared connections, social or otherwise. In my experience, the results are quite dramatic. With a vested effort in looking for similarities with others, they are not hard to find.

You can experiment with this connectivity in other domains. For example, the online "Oracle of Bacon" allows people to examine actor networks, with edges formed via costarring in movies. For example, Morgan Freeman was in *The Sum of All Fears* with Ron Rifkin. Rifkin was in *JFK* with Kevin Bacon. According to the Oracle, Bacon numbers greater than four are rare. Similarly, most quantitative scientists will at some point attempt to compute their Erdös Number: the path length through shared publications to Paul Erdös. Paul Erdös wrote about 1,500 articles, with numerous collaborators. The simple idea is that in real social networks people are often closer to one another than they expect.

7.3 The Small-World Spectrum

Watts and Strogatz (1998) offered a description of how to identify small-world networks. Small-world networks are "highly clustered, like regular lattices, yet have small characteristic path lengths, like random graphs" (p. 440). They proposed two measures on which to base a definition: Small-world networks have high clustering coefficient and short average path length. High and short relative to what? Relative to random graphs of the same size and density. In other words, a world is "small" if it is more clustered than one would expect compared to a random network with the same number of nodes and edges.

Watts and Strogatz (1998) proposed a mechanism for generating small-world networks, which lie between two extremes. In one extreme, networks have a regular structure where all individuals are in small, well-connected, groups. In the other extreme, networks are completely random, and there is no local structure. Watts and Strogatz showed that we can move from one extreme to the other by tuning a single parameter. If we start with a regular lattice, in which all individuals are connected only with their close neighbors, then we can move this network toward the other extreme (a random graph) by randomly rewiring connections between neighbors. If we let the probability of randomly rewiring each edge be p, then for $p = 0$ we have a regular lattice and for $p = 1$ we have a fully random network. For intermediate values of p, such as $p = 0.1$, we have a hybrid between regular and random.

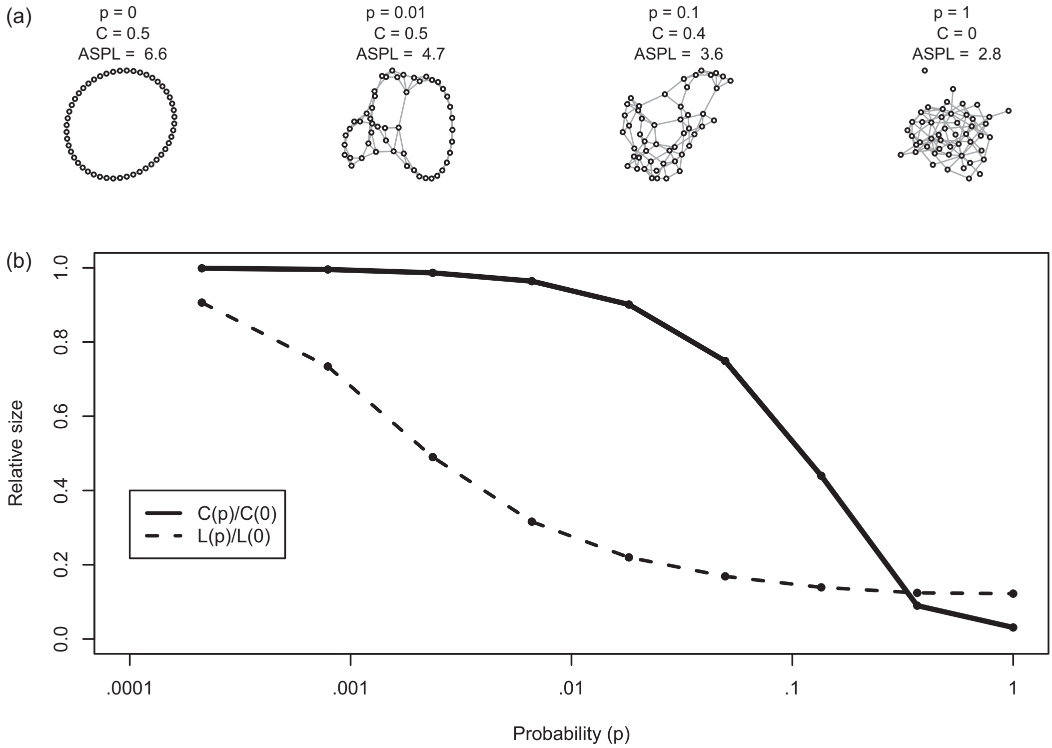

Figure 7.1 Watts–Strogatz small-world model. Starting with a lattice, where each node is connected to local neighbors, each edge is randomly rewired with probability p. For $p = 0$, the network is the regular lattice, which has the highest clustering coefficient with well connected local communities. When $p = 1$, the network is equivalent to an Erdös–Rényi random graph. (a) The network has $N = 50$ nodes and each node has degree 4, giving a density of 0.04. The probability of rewiring edges is shown above each network, along with its clustering coefficient and ASPL. (b) The relative size of the clustering coefficient and ASPL across a range of rewiring probabilities, p. There are 100 simulations for each value of p, with networks of size $N = 400$ each with degree 4. Each measurement is normalized relative to the regular lattice with $p = 0$. Note the x-axis is on a log-scale.

Figure 7.1 presents examples of rewired networks ranging from ordered to random. It also provides a comparison of the clustering coefficient and average shortest path length (ASPL) for various values of p. There is a region over p where the clustering coefficient stays high even as the ASPL quickly falls. It is this region where Watts and Strogatz identify small-world networks.

This region where the clustering coefficient is relatively high but the ASPL is relatively low invites us to think of small-world networks as sitting along a spectrum. This also invites us to ask how networks might be more or less small world.

7.4 Small-World Metrics

How can we evaluate the extent to which a network is a small world? The intuition behind Watts and Strogatz's proposal is that small-world networks lie somewhere between perfectly ordered lattices and random networks.

Since most networks vary somewhere along the small-world spectrum, it becomes useful to be able to quantify the intensity of small-worldness that a network has. A standard approach to achieving this is to compare a network with alternative versions of itself. That is, how much higher is the clustering coefficient, and how much shorter is the path length, for a real-world network than for a random network or lattice networks of the same size and density? Several measures have been proposed to quantify this comparison. I cover three here.

7.4.1 The Small-World Index

Humphries and Gurney's (2008) small-worldness measure, which we will call the *small-world index*, following popular usage, takes the clustering coefficient, C, and the average shortest path length, L, for an observed network (obs) and compares it with the same values computed for an Erdös–Rényi random graph (ran) of the same size and density. More specifically, one computes the ratio of these two quantities as follows:

$$SWI = \frac{\frac{C_{obs}}{C_{ran}}}{\frac{L_{obs}}{L_{ran}}}$$

The intuition behind this measure is that the random network provides a benchmark against which to compare the observed network. A network with an small-world index value greater than 1 is a small-world network. As noted earlier, that is true for many networks. Luckily, the small-world index also provides a relative measure: the more clustered the observed network relative to a random network with the same path length, the higher the small-world index. One generally computes many random networks and takes their average values.

With this measure, the clustering coefficients could be the same and the path lengths differ – though still producing a small-world index greater than 1. This highlights potential problems with the small-world index but also demonstrates how it captures the two senses in which a world is small: local communities and short path lengths. Making communities tighter and connecting to otherwise distant nodes both increase small-world structure.

Note also that the value is uncapped. At least in principle, a very large network could have a much higher clustering coefficient than a random network of the same size. So, the small-world index could be much greater than 1. As Muldoon et al. (2016) and Telesford et al. (2011) show, the small-world index is highly sensitive to network size and density. This makes it more challenging to compare networks. The next measure we discuss deals with this problem nicely.

7.4.2 The Small-World Propensity

An alternative measure proposed by Muldoon et al. (2016) is called the *small-world propensity*, ϕ. The small-world propensity compares a network with both a lattice (lat) and an Erdös–Rényi random graph of the same size and density. The lattice and random network provide upper and lower bounds, respectively, for how organized or disorganized the network could be given its size and density. Like the small-world index, the small-world propensity uses both C and L.

$$\phi = 1 - \sqrt{\frac{\Delta^2 C + \Delta^2 L}{2}}$$

Where

$$\Delta C = \frac{C_{lat} - C_{obs}}{C_{lat} - C_{ran}}$$

and

$$\Delta L = \frac{L_{obs} - L_{ran}}{L_{lat} - L_{ran}}$$

Muldoon et al. (2016) propose that the small-world propensity is a more informative measure than the small-world index for several reasons. First, the small-world propensity ranges between 0 and 1, unlike the small-world index which is uncapped.[1] The range of the small-world propensity is also well conserved whatever the density of the networks, whereas the range of the small-world index is reduced as the network increases in density. Finally, the small-world propensity allows us to investigate the relative contribution of the clustering coefficient and path length using *deviation* measures for each: ΔC and ΔL. These measures were used by Muldoon et al. (2016) to show that the nematode (*C. elegans*) neural network was of questionable small-world status compared with other brain networks because it had a very low clustering coefficient relative to a lattice (high ΔC) with a path length similar to a random network (low ΔL).

7.4.3 Telesford's Small-World Measure

Telesford's small-world measure takes seriously the property that networks can be more ordered or more random and that small-world networks lie somewhere in between. It does this by indicating to which end of the small-world spectrum a network tends to lie. Formally, this involves comparing an observed network's path length with the path length of a random network but its clustering coefficient with the clustering coefficient of a lattice, as follows:

$$\omega = \frac{L_{ran}}{L_{obs}} - \frac{C_{obs}}{C_{lat}}$$

The value of Telesford's measure, ω, ranges between -1 and 1. If the path length is close to that of a random network, $L_{ran} \approx L$, and the clustering coefficient similar to that of a lattice, $C \approx C_{lat}$, then the network has small-world properties and ω is closer to 0. Unlike the small-world index and small-world propensity, Telesford's metric is more small-world as $\omega \approx 0$. Positive values indicate a more disordered network and negative values indicate a more ordered network, with small-world networks lying in between. Neal (2017) usefully suggests taking the absolute value of the difference from 1, $\omega' = 1 - |\omega|$, to provide easier comparison with other indices. But here I leave the measure in its original form to milk it for its informativeness.

[1] ΔC and ΔL are bounded between 0 and 1 to ensure this.

These three measures offer slightly different perspectives on how we might quantify small-world networks.[2] In the next section we will put these measures to work evaluating differences between the vocabularies of early and late talkers.

7.5 Comparing Small-World Lexicons in Early and Late Talkers

Children start producing words at different ages. Some children are *late talkers*. Technically, late talkers are in the bottom percentiles of vocabulary size for their age and many go on to have lifelong learning difficulties (Moyle et al., 2007; Thal et al., 1997). Do late talkers learn different words than other children? Or, alternatively, are they just slower – showing the same pattern of vocabulary learning just on a longer time scale?

If we only count words, it is impossible to answer this question. Many late talkers will eventually know as many words as their earlier counterparts. However, if late and early talkers differ in how they acquire words based on their structure in the language they hear, then we can compare the structural properties of children's vocabularies (Beckage et al., 2011; Jiménez & Hills, 2022b).[3]

To answer these question more definitively I will create a comparison group of early talkers by choosing a mirrored sample in the distribution, reflecting the same percentile of children who are late talkers. We will take late talkers to be children with vocabulary sizes in the bottom 20% of distribution at a given age (following Jiménez and Hills, 2022b). Early talkers will be children in the top 20%.

The vocabularies will be restricted to a subset of nouns and their associations in the Small World of Words. This will avoid potential problems with word class differences (e.g., nouns, verbs, and pronouns) (Hills, 2012; Jiménez & Hills, 2022b). All directed edges are transformed into undirected edges. Figure 7.2 shows the network of nouns included in the analysis.

Limiting children's vocabularies to these words, we can split the children into groups based on the relative sizes of their vocabularies. Figure 7.3 shows the distribution of early and late talkers in the sample of children available in WordBank (Frank et al., 2021). Wordbank represents a large collection of child vocabulary data from many researchers using the Child Development Inventory. The data consists of surveys filled out by adults that indicate the age of their child and what words the child can understand or produce from a selected set of words.

Ideally, we would like to choose a sample of early and late talkers who have roughly the same vocabulary size. This avoids the issue that different size networks have different properties associated with their network metrics which can be unrelated to the variable of

[2] Neal (2017) provides an informative account of small-world metrics including additional variations.

[3] There are many other ways to approach this, with each offering their own nuances and advancing new questions. For example, we could investigate the node-level properties of words children learn using the network structure of the language they hear. Or we could model network evolution for the two groups separately and compare parameters. There are many interesting examples from which to derive inspiration (Beckage et al., 2011; Beckage & Colunga, 2019; Bilson et al., 2015; Jiménez & Hills, 2022b; Sizemore et al., 2018; Stella et al., 2017).

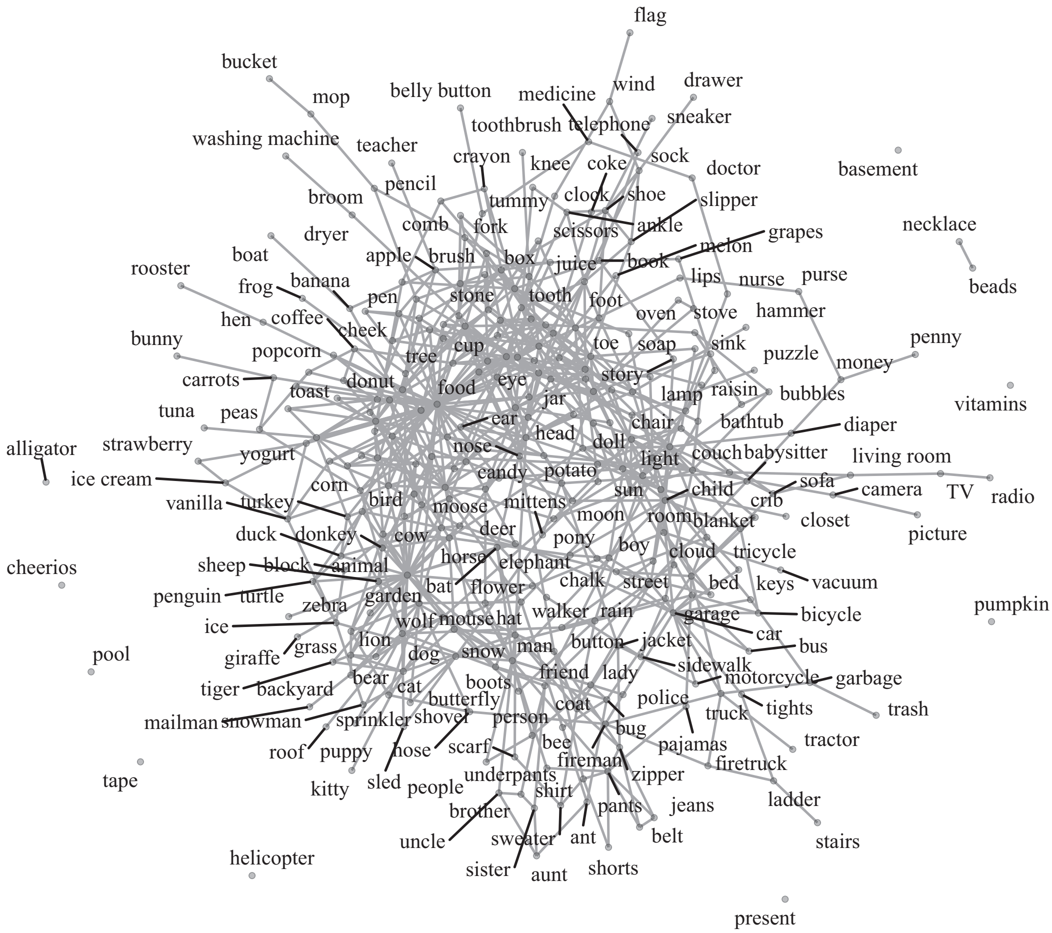

Figure 7.2 Network of 302 nouns in the Child Development Inventory with edges based on the Small World of Words free association norms. Dark lines indicate node-label assignments.

interest. Consider Figure 3.2, where the size of the largest component can vary dramatically with different network sizes even when the density is the same. This will naturally influence ASPL. Effects like these can swamp other differences (e.g., in the learning strategies between late and early talkers). To avoid this, we will take a well-populated slice of vocabulary sizes, allowing us to compare early and late talkers with similar vocabulary sizes. Taking each child identified as an early or late talker and focusing on those with a vocabulary between 150 and 200 words (see dotted lines in Figure 7.3), we can compute for each child their small-world index, small-world propensity, and Telesford's small-world measurement.

We will compare each child with random networks and lattice networks matched to their vocabulary size and density. To do this, we produce 100 Erdös–Rényi random graphs per child. We then compute the average clustering coefficient and shortest path length for each child's simulated networks and average across these to create the random graph values for comparison.

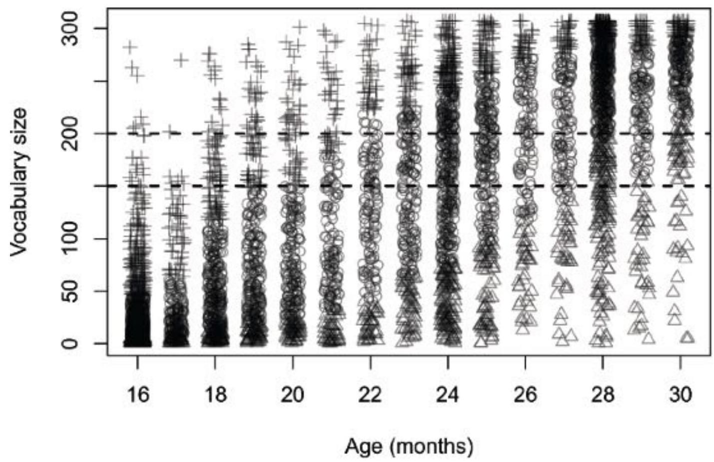

Figure 7.3 Vocabulary size at various ages. Early talkers are near the top at each age (crosses). Late talkers are near the bottom (triangles). In this case, these represent the top and bottom 20% of the distributions at each month. The dotted lines indicate the range within which we will compare early and late talkers with a similar vocabulary size. There are 134 early talkers and 102 late talkers within this range.

The lattice comparisons follow a similar logic. We produce a one-dimensional ring lattice. We give each node in the lattice network a degree equivalent to the average degree in the child's network rounded up to the nearest integer.[4] Rounding up creates a lattice with a higher density than our observed network, but it lets us start with a perfectly structured regular lattice. We then randomly remove edges until we have a lattice network with the same density as the child's network. Because the random removal can differ each time, we do this 100 times for each child, as we did for the random graphs.

The three panels in Figure 7.4 show comparisons for small-world index, small-world propensity, and Telesford's measure. The measures all agree. Late talkers are more small-world for all measures. Recall that Telesford's small-world measure is more small-world the closer the measure is to 0. Though I am not focusing on the statistical analysis here, a statistical analysis supports what is clear to the naked eye.

Figure 7.5 looks more closely at the details. The clustering coefficient is higher for late talkers than early talkers. The ΔC value supports this, showing a smaller difference between late talkers and their associated lattice matrices. Error bars generally overlap for ASPLs, suggesting that the differences in the small-world measures are primarily in the clustering coefficient.

The foregoing observation looks at early and late talkers, but prior work has focused on differences between late and typical talkers, where "typical" means all those who are not late. Nonetheless, this observation fits nicely with recent work by Jiménez & Hills (2022b) showing that nouns are more small-world for late talkers, whereas verbs are more small-world for typical talkers. The original work to look at the relationship between

[4] In some cases, the average degree is less than 1, which gives a lattice clustering coefficient of 0. This makes the Telesford metric undefined. To avoid this, the lattice degree is always set to at least 2 and then random edges are removed as described.

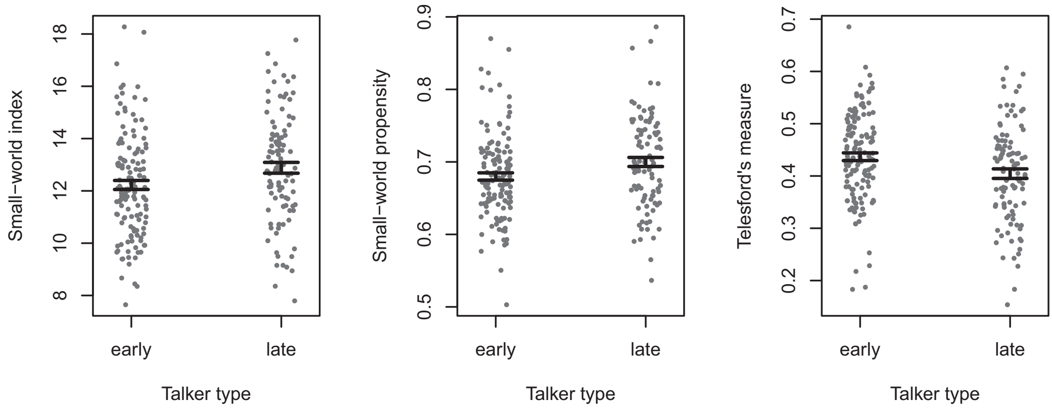

Figure 7.4 Small-world measures for early and late talkers. Telesford's measure is closer to 0 when the measure is more small-world, making this consistent with the small-world index and small-world propensity. Error bars are standard error of the mean.

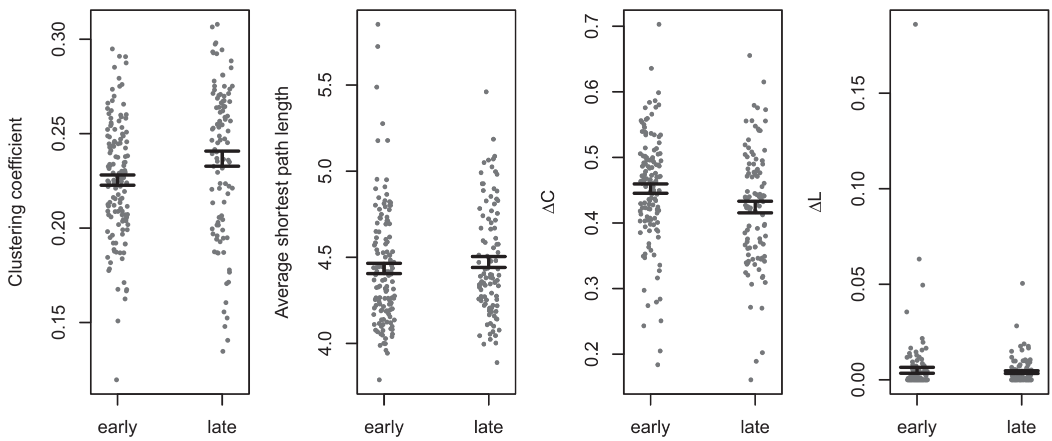

Figure 7.5 Comparison of various measures that contribute to the small-world indices.

late and typical talkers (Beckage et al., 2011) combined nouns and verbs and found that late talkers had less small-world structure. It can only be an irony of network science that more recent approaches find that structure within structure is driving differences in different directions.

7.6 Why Small-World Structure? Function, Formation, and Emulation

There are three general hypotheses for why we see small-world structure. I'll call these function, formation, and emulation. Much like Tinbergen's four questions for explaining the origins of biological phenomenology – which attempt to answer life's questions via evolutionary adaptation, phylogeny, physiology, and development (Tinbergen, 1963) – function, formation, and emulation focus on similar pathways to origination for network structure. That is, what is the network designed or adapted to achieve (function), how did the network develop (formation), and what does the network represent (emulation)?

7.6.1 Function Hypotheses

Function hypotheses argue that small worlds add functional value because of what they facilitate. For example, Sporns et al. (2000) demonstrated that the neural connectivity of macaque and cat brains had small-world structure and that this facilitated their ability to match different patterns of input stimuli. They also showed that these structures were capable of *functional integration and segregation*, meaning that local processing could handle different perceptual or computational tasks, but that these could also be integrated globally.

In a similar fashion, Bassett and Bullmore (2006) proposed that small-world structure in the brain is a compromise between two goals. First, the brain needs segregation or modular structures. These also need to minimize their cost by reducing path length, which is anatomically costly in terms of axons and dendrites. Second, this need is matched by the need for the different modules of the brain to speak with one another. The need for segregation leads to local communities; the need for integration leads to interconnectivity. This may in turn allow the brain to respond quickly and proportionally to changes in the environment. This whole-network synchronization is a property of small-world networks that is not available to lattice networks without long-distance connections (Barahona & Pecora, 2002).

Function hypotheses are often proposed when small-world networks are observed. From Hilgetag and Goulas (2016, p. 2361): "The small-world organization is deemed essential for healthy brain function, as alterations of small-world features are observed in patient groups with Alzheimer's disease [. . .], autism (Barttfeld et al. 2011) or schizophrenia spectrum diseases [. . .]."

Function hypotheses also exist for social communities. For example, small-world networks influence the rate at which local communities can organize and coordinate with other communities. Local communities can facilitate shared beliefs and practices that bind a community together. Good evidence suggests that when like-minded individuals come together they increase one another's confidence in their beliefs, even if the amount of outside evidence they have for those beliefs is unchanged (Hills, 2019b). On the other hand, a lack of contact with outside communities means that they cannot share their ideas or benefit from the ideas of other communities. These effects and their consequences are discussed in more detail in Chapter 15.

Function hypotheses have also been proposed for language. According to Beckage et al. (2011):

> Mature semantic networks possess small-world properties [. . .]. These properties allow for high amounts of local structure combined with global access. Within these semantic networks, there is considerable local structure in the form of clusters of words that are highly interconnected to each other by semantic relatedness, which may be related to category representations [. . .]. However, even with these dense clusters, some words act as bridges and possibly as hubs, connecting semantically distant clusters to each other. This global access is understood as providing easy transitions from one cluster to another and is believed to support online language processing and comprehension. (p. 1)

7.6.2 Formation Hypotheses

Formation hypotheses argue that small-world networks exists because of constraints that exist during the development of the network. The proposal made by Bassett and Bullmore (2006) that local communities exist to minimize anatomical costs for brains is one example. Costs are a formative constraint.

Steyvers and Tenenbaum (2005) propose formative constraints on language evolution with their model of language evolution based on duplication and divergence. As they describe it, the model is a functional hypothesis: new words arise in language around well-connected concepts because these concepts (and presumably the users of those concepts) benefit most from duplication and divergence. Words that have many associations are more likely to give rise to new words with similar connections that then diverge in meaning. That is functional because it serves the purpose of the user.

However, duplication and divergence also have formative constraints. Words may only originate from pre-existing words. For example, one of the words with the most meanings in the English language is *take*. It also has numerous senses: *claim*, *win*, *grasp*, *assume* (as in "take an oath"), *want* (as in "I'll take a bagel"), *suffer* (as in "take a direct hit"), and so on. Correspondingly, it also has many variants in its patterns of usage, such *take-out*, *take-away*, *take hold*, *take heart*, and *take it easy*. Indeed, even the phrase "to have" is proposed to have originated from the word *take*. It comes from the original Proto-Indo-European "kap," which meant "to seize," and is still around in the word *capture*.

We can take *take* as a clear example of a formative constraint of duplication and divergence. That is, if duplication and divergence limits word births to frequently used words (just as they limit new genes to old genes), then this creates a historical path dependence on language change. This creates many confusions for users (especially second-language learners), who do not know how to take "it easy" or "an oath." This formation constraint in language also trickles through to its orthography: consider the pronunciation of words like *magician*, *magic*, *trough*, *plough*, and *through*. The subtleties of semantics may often be better facilitated by novel words, unrelated to past forms.

7.6.3 Emulation Hypotheses

Emulation hypotheses suggests that a network is designed to efficiently mimic or model some useful pattern in the world. Imagine a person who sees every facial expression as completely unique, such that no two expressions are similar – they are all different. Given that a face is never exactly identical in light, shade, angle, size, and context to when we have seen it before, the ability to generalize across faces requires the ability to recognize relationships between images. A person challenged by this, such as Shereshevski – described in Alexander Luria's (1987) book *The mind of a mnemonist* – has trouble recognizing the same individual in different situations. Their network for recognizing similar faces has no local structure. Everything is an isolate.

But we can also imagine the opposite failure, when too many relationships are recognized, such that all objects are seen as roughly equally connected. This is a problem of overgeneralization. It is as if all nodes are connected strongly to one another, so again there is no local structure.

An ideal perceptual and cognitive system recognizes the structural similarities and differences in the world. It neither overgeneralizes nor undergeneralizes. That is, if reality has a small-world structure, then a healthy mind is the one that is capable of *emulating* that structure. Systems that emulate structures may have that structure not because the structure itself is adaptive, or because constraints led to a structure of that exact type, but because the structure already existed and was simply reproduced (which may of course occur for other reasons).

In sum, function hypotheses are about the functional value of the structure. Formation hypotheses focus on the constraints that influence how the structure is formed. Emulation hypotheses hold that the network merely reflects an apparent structure already available in the world.

Preferential acquisition is an example of an emulation hypothesis: a network is formed to reflect structure that already exists in the world. Preferential attachment is a formation based hypothesis: the network has the structure it does because of how it developed. The earlier hypotheses focusing on how brain and language networks facilitate function are function-based hypotheses.

These hypotheses are not always mutually exclusive. For example, a system could evolve to guide itself toward a particular pattern of network formation because the end result produces adaptive value by emulating structure and decision constraints in the world. This is the basic premise behind Herbert A. Simon's (1955) *bounded rationality*, which holds that the mind is constrained to learn and decide in the way it does because it is adapted to its limitations and the structure of its environment (see also Haselton et al., 2009; Hertwig & Pedersen, 2016).

7.6.4 Applying Function, Formation, and Emulation to Early and Late Talkers

We can apply function, formation, and emulation hypotheses to early and late talkers. Each offers different accounts as to the role of small-world networks. By the function account, late talkers' networks are more small-world because they are solving a different problem. This may be because of individual differences in processing information about the world. In Chapter 14 we will see differences in semantic lexicons for creative and noncreative individuals, which may provide benefits to both groups: one that sees the world as more interconnected and possibly overlooks small differences, and one that sees the world as less interconnected and is more sensitive to fine distinctions.

A formation hypothesis might propose that early and late talkers impose different constraints on word learning, either over- or undergeneralizing in ways that alter the semantic structure of the lexicon they learn. A hypothesis about the way information is encoded was suggested by Beckage et al. (2011) in what they called the *oddball hypothesis*. According to this hypothesis, late talkers may require a greater semantic difference to recognize new word-referent mappings. Children with this oddball constraint would tend to learn words that are less well connected and thereby create a more sparse network.[5]

[5] This hypothesis would be more appropriate for verbs, as Beckage et al.'s (2011) results appear to be driven in part by the inclusion of verbs in their analysis.

Finally, if we take a healthy mind as emulating some property of the environment, then it may be that early and late talkers experience different language environments. Recent work by Jiménez & Hills (2022a) finds that the language experienced by early and typical talkers (in child-directed speech) is structured differently. That structure reflects the structure of the lexicons they learn. This is not evidence that the environment is being emulated; it may be that the caregivers adapt their speech to their children's lexicons. However, if the environments are different, some children may reflect this in their lexical networks.

7.7 Summary

Small-world networks are everywhere, so it is useful to be able to measure them in a variety of ways, to tease out the potential differences that may influence their origination and function. This chapter explored the concept of small-world networks, several ways to quantify them along a small-world spectrum, and how they might be used to understand the differences between two populations with similar network sizes (early and late talkers). The observation of differences between groups gives rise to speculation about the cause of the differences. The function, formation, and emulation hypotheses offer different ways to answer that question and in turn generate new questions of their own.

7.8 Open Questions

7.8.1 Modeling Late Talkers

Model assumptions, such as methods of edge construction, matter with respect to predicting word learning and these can matter differently at different times during development (Beckage & Colunga, 2019; Engelthaler & Hills, 2017; Hills & Siew, 2018; Stella et al., 2017). They also matter with respect to what kinds of children we are looking at. Most prior model explorations have focused on typical children. Shifting focus to include late talkers – for which there is ample and growing data – opens up a host of modeling questions. For example, to what extent does the language environment matter? In a large-scale study of differences in language experience among young children, Hart & Risley (1995) found that language experienced by children before the age of three could differ by tens of millions of words. The idiosyncratic nature of language environments also appears to extend to systematic differences in language structure experienced by late and typical talkers (Jiménez & Hills, 2022a). These differences could be a parent's reaction to a child's language ability. In some cases (e.g., late bloomers), children in one environment reach the language ability of children in other environments. That does not rule out environmental effects, but it does support individual differences.

We can also ask about models based on a common environment. Is late talker lexical development better explained by models such as preferential acquisition or the lure of the associates? Do the parameters best fit to the models provide insights into what aspects of their language environment matter most? For example, are late talkers more or less sensitive to semantics, phonology, or perceptual object features than typical talkers?

Even using the same features, how does the edge construction method matter? Recently, free associations were collected from adults with children-oriented responses in mind and these have already been shown to be useful in improving language models (Cox & Haebig, 2023). Might they be helpful here as well? Would distinctiveness help tease apart differences? How do word classes matter, such as nouns and verbs (Horvath et al., 2022; Kueser et al., 2023)? What kinds of network approaches might best explain the differences between comprehension and production in late and typical talkers (Jiménez & Hills, 2022b)? And how might late talkers differ from typical talkers in the way they fill knowledge gaps (Sizemore et al., 2018)? Hills & Siew (2018) discuss these and many other questions for language acquisition.

7.8.2 Building a Better Small-World Metric

The diversity of small-world measures described here should provide some liberation for the industrious. They offer many ways to tease out differences in apparent small-world structure. Taking these measures out of the box and using them without reading the instructions might work often enough, but it is no substitute for taking them apart, tweaking them, and trying to improve them. One place where the small-world propensity and Telesford's measure could be improved is in generating the appropriate lattice. It is not entirely clear what that should be and it is likely to depend on the system. However, suppose we wanted to generate a lattice that preserved the degree distribution of the observed system but was nonetheless maximally clustered? Then the C_{lat} and L_{lat} measures could be derived from this maximally clustered version of the observed network. One way to approach this is to use an adaptive network search, as described by Mason and Watts (2012) and Barkoczi and Galesic (2016) and discussed in Chapter 18, Section 18.6. This repeatedly rewires edges from a starting network – in this case, the observed network – taking only degree-preserving rewirings that increase or decrease the network along some chosen metric. In this case, the metric could be the clustering coefficient. Repeating this rewiring many times, one arrives at a network that maximizes the clustering coefficient given the degree constraints of the observed network. Might this be a useful basis for comparison? Alternatively, one may consider other structural constraints of the kind one finds in discussions of exponential random graphs (Robins et al., 2007).

8 The Birthplace of New Words
Identifying Node Origins

8.1 The Problem

Words, like biological species, are born and then, someday, they die. According to Pagel (2009), the half-life of a word is roughly 2,000 years, meaning that in that interval about half of all words are replaced with an unrelated (non-cognate) word. Where do the new words come from? There are numerous dimensions along which new words could vary from old words, so it may not be easy to see how to enter this problem. However, extending our small-worlds metaphor and the observation of language clusters, we can tell a story that mirrors biological theories about the origins of species. Language has urban centers with well-populated and well-connected meanings (e.g., around words like "food" and "color"). It also has rural fringes, where words live more isolated lives as hermits with limited connections to other words (like "twang" and "ohm"). Are new words more likely to be born in urban centers or in the rural fringes?

8.2 The Birth of New Words

Horace (65 BC–8 BC), in the *Ars Poetica*, paints the following picture of language: "As the forests shed their leaves, as the year declines, and the oldest fall, so perish those former generations of words, while the latest, like infants, are born and thrive."

This process of birth and death in words has been the subject of inquiry for millenia. Algeo (1980) notes that Plato discussed the topic, as did Horace, more than 2,000 years ago.

Perhaps the topic maintains its enduring popularity because of its memorable stories. When talking about words, a good anecdote is never far away. Algeo (1980) tells a particularly juicy one about the origin of the word "cowabunga," made famous by surfers. Did surfers pick up the word on the beaches of Polynesia? Did they adapt it from some other surf-related word? Apparently, neither. According to Algeo, "cowabunga" originated with the *Howdy Doody Show* on 1950s TV. It is a word in the made-up lexicon of character Chief Thunderthud. Perhaps it was intended to sound like a familiar Algonquin word, like "wigwam" or "tomahawk." Surfers learned the word not on the waves, but as children in front of the TV. The word "hobbit" has a similar origin, created by J. R. R. Tolkien, with its own detailed etymology from the languages of Middle Earth.

Algeo (1980) studied the etymological origins of 1,000 words and isolated them into five categories:

- *loan words*: words from other languages, like “macho.”
- *word shortening*: such as “car” from “carriage.”
- *composites*: the combination of two words, like “speedboat.” This includes compounding and affixes.
- *blends*: the combination of two-part words likes “Brexit” (Britain + exit), “brunch” (breakfast + lunch) and “bromance” (bro and romance).
- *shifts*: when a word takes on a new meaning, such as “cool” and “bad.”

Among the 1,000 words Algeo (1980) studied, the majority were composites (~64%). This, he proposes, is the most common origin for new words, which includes the addition of affixes, such as in “final-ize” and “googl-ing.” Coming in a distant second (~14%) are word shifts, followed by shortenings, loan words, and blending, in that order.

The problem with these processes is that they do not yet provide a structural understanding of how words originate. At the simplest, we might ask if words are more likely to originate in language’s urban centers or in its wild frontiers.

We can get nearer to this question if we make a few additional assumptions. For example, we might propose that it is more likely that a composite word will form out of other high-frequency words. If that were true, this would most likely put composites in higher-density areas of the lexicon. Shifts, we might propose, are more likely to occur among lower-frequency words, because less frequent words tend to be less stable. High-frequency words tend to maintain their meanings over time better than low-frequency words (Li & Siew, 2022; Pagel, 2009). Following Zipf – who noted that the most frequent words also tend to be the shortest – we might argue that word shortening should occur mostly as words move from low frequency to high frequency, such as for “car” (carriage) and “flu” (influenza). These might all be the case and have surely been proposed elsewhere. We can attempt to add additional precision with a formal model.

8.2.1 Duplication and Divergence

Recall that Steyvers and Tenenbaum (2005) proposed the duplication and divergence model for language learning and word origins (described in Chapters 3, 5, and 7). They proposed this to explain the observation that the free association norms have a scale-free distribution that is also a small-world network. As a model of word origins, this could be a good fit.

With respect to word origins, the duplication and divergence model makes an assumption “that the probability of differentiating a particular node [i.e., birthing a new word] … is proportional to its current complexity – how many connections it has [its degree]” (Steyvers & Tenenbaum, 2005, p. 57). This is satisfyingly specific and it allows us to move toward a quantitative proposal for where new words should originate. They should originate near lexical hubs.

If the words were identical in connectivity, they would have *structural equivalence* (Everett, 1985; Sailer, 1978); nodes that are structurally equivalent in a network are substitutable without change to the network – they share the same adjacent neighbors.

A perfect duplication of a node would produce a child node that is structurally equivalent to its parent node. In language, structural equivalence would be akin to high similarity. This is also similar to the idea proposed by Steyvers and Tenenbaum (2005) that a new word is needed to distinguish a variation on an existing word. These words would then tend to be found with similar associations: "bee" and "wasp" are both likely to occur "outdoors" at "picnics," "buzzing" and "stinging" with their fascinating "hymenopteran" armatures. As a result, "bee" and "wasp" would share high structural equivalence.

To further elaborate on the quantification of our biological metaphor, we will call for reinforcements.

8.3 Biological Speciation and the Origins of Diversity

Species appear. *Homo neanderthalensis* and modern *Homo sapiens* appeared some 500,000 years ago. And species go extinct, like *Homo neanderthalensis* did some 40,000 years ago and *Homo sapiens* will inevitably do at some point in the future.

Because species must originate from species that came before, the standard model of speciation is that individuals of an existing species become separated from one another by geographic isolation. A river forms, land masses separate, an immigrant colonizes an uninhabited island, or high-altitude species fly to a neighboring mountain top (what are sometimes called sky islands). This division leads to divergent evolution of the two separated populations. Their differences eventually become sufficiently large that they can no longer reproduce. Their reproductive isolation warrants the designation of a new species.

The divergence of a species from its past form is called phyletic evolution. We see the same process in words, as old words take on new meanings. The English word "have" diverged in a phyletic way from its putative Proto-Indo-European ancestor "kap." meaning "to take." Phonetic shifts softened the *k* to *h* and the *p* to *f*, as has happened for many other words. If, as Deutscher (2005) claims, language is a "reef of dead metaphors," then most words originate by a process of duplication and divergence similar to the style of phyletic evolution we observe in "have" (from the Proto-Indo-European "kap").

But, just as often, new word forms are neologisms, arising not from gradual variation of their forebears, but with entirely new meanings. "Cowabunga" and "hobbit" are examples.

Theories of biological speciation go further in terms of postulating where new species originate. Specifically, there are two fascinating theories of speciation which I take as useful metaphors for language origination because they provide precise topological coordinates for word origins. These are *peripatric speciation* and *centrifugal speciation* (see Figure 8.1).

Peripatric speciation (or centripetal speciation) was originally proposed by Ernst Mayr (1954). It is based on the idea that isolated populations, which become separated from their parent population, undergo genetic mutations that allow them to diverge from their parent population. In other words, it is smaller populations on the fringes of larger populations that are the basis of new species. Because populations on the fringes are small, the origin of an adaptive genetic variant can rapidly lead it to become dominant in its population. Smaller populations are more susceptible to rapid genetic change even

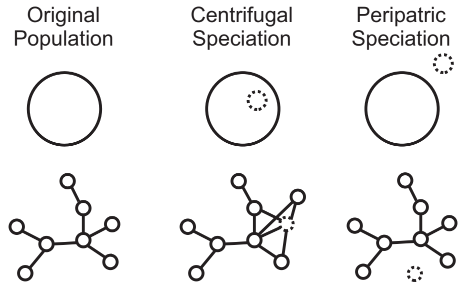

Figure 8.1 Word birth as centrifugal speciation and peripatric speciation. In centrifugal speciation, new words are born near the center of the distribution, in the urban centers of language. In peripatric speciation, new words form in low-density regions of the semantic space, with few or no connections to other words. Larger circles near the top represent dense population centers and the smaller dotted circles represent novel species. The networks below apply the biological metaphor to a lexical network.

when selection is weak. This allows them to more quickly move through the adaptive landscape to find new robust and flourishing forms.

Centrifugal speciation, on the other hand, was proposed by William Brown Jr (1957). Its logic runs the other way, and is just as conceptually convincing. Because population centers of species have such a large size, they are likely to contain greater genetic variation, and as result they are more likely to produce novel genetic adaptations. As populations expand and contract, they leave these novel variants in the periphery, where they are maintained. Meanwhile, the central population continues to churn out new forms even as many of them rapidly die through competition.

Peripatric speciation and centrifugal speciation differ in where they propose novel forms originate:

- Peripatric speciation suggests that they originate in the fringes, among smaller populations.
- Centrifugal speciation proposes that they originate in population centers, among larger and denser populations.

The duplication and divergence model seems to me most like centrifugal speciation. Both suggest that new forms originate where density is highest, either in terms of population or semantic associates. Peripatric speciation suggests that novelty arises at the fringes, in places with lower density and presumably greater opportunity for novel adaptations to take advantage of untapped though potentially scarce resources.

The analogy is not a complete one-to-one mapping. No analogy is. Nonetheless, these two theories of speciation do throw up a concrete vision for where novelty might arise, in language's city centers (centrifugal) or in its uncharted wilderness (peripatric).

8.4 Where Are Words Born in the 1800s?

The methods we will use to address the question of word origins are a marriage of multiple constraints. They offer a starting point for future improvement. Ideally, we would use a distributional semantic model to derive a semantic space from language data available during the precise time period when a word originated (Jones et al., 2006; Landauer & Dumais, 1997; Li et al., 2019). This approach is challenging for many reasons, not the least because word origins can be difficult to identify precisely. It also takes time for a word to develop associations. How much time should elapse before we evaluate a word's associations? Moreover, a word's associations reveal themselves over time. Across what time period should associations be evaluated? These questions can all be addressed, but here we will take a different approach.

The method we will take uses the Small World of Words free association norms (De Deyne et al., 2019), a resource with which we are now somewhat familiar. Though the Small World of Words is unlikely to represent the changing semantic space that existed in the 1800s, it is likely to carry the structure of hub and fringe we need to evaluate a word's urban or rural character. One central measure will therefore be degree, which we will measure as indegree because we are using the Small World of Words, as discussed in Chapter 4. Frequency of words in use and their degree in free association networks are correlated (Deese, 1959; Pagel, 2009; Steyvers & Tenenbaum, 2005). Because high-frequency words also tend to be more stable (Pagel et al., 2007), the nougatty center of language communities are also likely to be more stable.

We will use the Google Ngram frequencies from 1800 and 1900 to evaluate word change. This obviously does not represent all novel words available to us. It is also a specific period in time. English borrowed many loan words from French during the 1600s, and if we looked during that time period we would likely see a different pattern. The particular period chosen here (1800–1900) is a time where we have sufficient data in the Google ngrams and which also occurs before the apparent introduction of artifacts associated with changing data choices in the production of the Google Ngrams (Pechenick et al., 2015). Because it contains many words, it also gives us a large overlap with the words in the Small World of Words free association norms.

Here, we simply take all words that are present in 1800 and all words that are present in 1900 and ask which are *old* words, present in 1800, and which are *new* words, not present in 1800 but present in 1900. We will also require that the words be in at least three books in the year 1900 to avoid misspellings and other nonsense words. We will then take the intersection of these old and new words with the Small World of Words, allowing us to investigate their structural properties in the free association network. This also means we are limiting our investigation to a subset of all possible word origins – excluding things like proper names, abbreviations, made up words, misspellings, numbers, foreign words – which do not show up in the Small World of Words. Nonetheless, this exploration is useful as it allows us to see how we might go about addressing a fairly abstract question by using theory that sets out quantifiable alternatives. This simplifies the question in a way that allows us to provide a preliminary answer – a kind of scientific baby step – which has the potential to be challenged in the future. Many of the assumptions made here are fairly strong, but they are testable, as I note in the Open Questions section at the end of the chapter.

8.4.1 A Lesson from Stop Words

Even among the Small World of Words, not all words are equal. Many of the oldest and more central words in language are likely to belong to what are called *stop words*. In natural language processing, stop words are words that are typically removed because the words generally carry limited semantic information. That is, they are mostly *grammaticalizations*. Grammaticalizations are the colorful elders of language, who have evolved from relative semantic autonomy to take on what Miellet has called "grammatical character." This includes words like *have* (described earlier), forms of *to be*, pronouns, articles like *the* and *a*, contractions like *won't* and *you're*, question words like *what* and *who*, and so on. Many belong to what are called *closed-class words*. Closed-class words are the group of articles, pronouns, or the like where one simply cannot introduce a new word just for fun. To change one's pronoun, one must choose from a fixed list – which is an oppressive feature of language, to be sure. Nonetheless, it is a quality of closed-class words. As a result, stop words tend to represent areas of language where new words are particularly unlikely. If stop words are central, they make everything look rural in comparison and, to compare like with like, it makes sense to remove them.

Yet, there are new stop words appearing after the 1800s. Though I expected there to be no stop words among the new words between 1800 and 1900, my error turns out to be informative. There are in fact eight contractions that arise in the 1800s, such as *couldn't*, *didn't*, *doesn't*, and *isn't*. These are most similar to the blends described by Algeo (1980): the combination of two words, with some foreshortening of the phoneme and associated spelling. Contractions existed before 1800, but they were rare and came into their own during the 1800s. Thus, it is nice to see that our method works in principle by capturing important emerging lexical novelty. Nonetheless, to level the playing field, the analysis described here removes stop words. With the online code, it is easy to put them back in.

8.4.2 Predictions from Duplication and Divergence

Taking those words which appear for the first time between 1800 and 1900, we can now ask what region of the semantic space they originated in. Our comparison group will be the old words that were there in 1800 and still present in 1900. To avoid the problem that new words might begin their lives with few associates, we will investigate degree alongside other measures of centrality. In particular, we will investigate betweenness, closeness, clustering coefficient, and eigenvector centrality. Recall that a node's betweenness is the measure of the number of shortest paths between pairs of nodes that pass through that node. A node's closeness is a measure of how distant it is from other nodes, measured as 1 over the sum of the shortest path distances to all other nodes. A node's clustering coefficient is a measure of what proportion of its neighbors are connected with one another. And a node's eigenvector centrality measures the extent to which it is well connected with other well-connected nodes (see Chapter 3 for more details). Here, we use directed edges for all except clustering coefficient, but the results lead to the same inferences if we use undirected edges.

If words originate by centrifugal speciation, we expect them to generally be toward the higher part of the degree distribution. Low-degree words should be older words that have fallen from fame. We can extend our predictions to the other metrics as well based on the properties of high-degree words.

- New words will be higher in betweenness as hubs tend to bridge between communities.
- New words will have lower clustering coefficients as hubs tend to share edges with many nodes that are unconnected with one another.
- New words should have higher closeness as more central words are closer to the other nodes in the network.
- New words should have higher eigenvector centrality as well-connected words in central communities will be connected with other well-connected words.

8.5 Letting the Data Speak

There are 16,313 words of overlap between the ngrams and the English Small World of Words after removing 131 stop words from a standard list (the "snowball" list), with 14,720 *old* words that were present in 1800 and still present in 1900 and 1,539 *new* words that had zero frequency in 1800 and positive frequency in 1900. Tables 8.1, 8.2, 8.3, and 8.4 show the highest and lowest words ranked along each of the network metrics for old and new words.

New words with the largest growth are largely scientific words, like *biological*, *chromosomes*, and *appendicitis*. This is likely due to the significance of the scientific findings in the 1800s associated with these concepts, though it may also be an early indication of the infiltration of scientific texts into the corpus. Words like *movie*, *scary*, and *airplane* have numerous associations now and have become commonplace, so much so that it is not at all strange to think about *scary movies* and *scary airplanes*. *Container* and *global* connect a number of different regions of the semantic space, having the highest betweenness among new words. *Differentiate* and *enzyme* are also closer to other words than the rest of the new words. And, finally, we see some new words with a high clustering

Table 8.1 The top-ranked new words entering the lexicon in the 1800s along each measure: growth = frequency change, k = indegree, b = betweenness, c = closeness, C = clustering coefficient, and x = eigenvector centrality. For C, many words tie for the highest clustering coefficient with a value of 1, so those shown are a random sample from the top words.

Growth	k	b	c	C	x
biological	movie	container	differentiate	ampere	baseball
chromosomes	scary	global	enzyme	betterment	movie
appendicitis	airplane	airplane	teaspoon	coastal	airplane
voltage	television	environmental	satsuma	reactor	container
electrode	baseball	movie	asexual	bubonic	homework
hydrate	bike	racism	silhouette	mauve	environmental

Table 8.2 The top-ranked old words in the lexicon in the 1800s along each measure. For C, many words tie for the highest clustering coefficient with a value of 1, so those shown are a random sample from the top words.

Growth	k	b	c	C	x
one	food	car	became	looked	love
may	money	fun	especially	observed	water
said	water	love	etc.	thy	food
can	bad	animal	followed	duties	green
upon	work	school	believed	scarcely	man
time	car	food	doctrine	created	red

Table 8.3 The lowest-ranked new words entering the lexicon in the 1800s ranked along each measure. For C, many words tie for the lowest clustering coefficient with a value of 0.

Growth	k	b	c	C	x
unplug	leprechaun	sweatband	leprechaun	slapstick	proofreading
uptight	mishear	unfreeze	mishear	smooches	matriarch
weatherman	multiracial	unplug	multiracial	sweatband	debutante
whirly	slapstick	weatherman	slapstick	unfreeze	soupy
wiggly	unfreeze	whirly	unfreeze	unplug	hypnotic
wriggly	weatherman	wriggly	weatherman	whirly	reincarnation

Table 8.4 The lowest-ranked old words in the lexicon in the 1800s ranked along each measure. For C, many words tie for the lowest clustering coefficient with a value of 0.

Growth	k	b	c	C	x
molt	espresso	gauche	espresso	carport	soggy
ruffian	frankfurters	decypher	frankfurters	gauche	entranced
sew	horrendous	blueish	horrendous	decypher	throb
fake	flipped	molt	flipped	blueish	perfumed
cafe	whorish	ruffian	whorish	molt	pollute
nitrous	lefty	nitrous	lefty	nitrous	baptism

coefficient, which may be due to their low degree and rather specific meanings, such as *ampere*, *betterment*, and *reactor*.

One thing to note is that in order to avoid spelling mistakes, OCR errors, and misdated books, only words that occur in at least three books are included. Without this constraint, the list of lowest-ranked words contains things like *yeti*, *gigolo*, and *jujitsu*. These tell us more about data quality than word origins. *Yeti* does not properly come into English as a loan word from Tibetan/Sherpa until the early 1900s. A look into the books where it occurs in the 1800s finds that it is a misspelling of *yet*, likely caused by automated text methods misreading a smudge on the page as an additional *i*. *Gigolo* is a surname in a book by William Cooke Taylor in 1842. *Jujitsu* appears to be a misdated book. As these are all low-frequency words arising due to errors, we can easily remove them by increasing the frequency required to be included in our analysis. When book number is set to a minimum of three (as shown), our list looks more plausible. The least frequent words by growth, such as *unplug* and *budgerigar* (a kind of bird) appear to genuinely arise in the 1800s. Still, we might enhance our conservatism by setting our threshold for low-frequency words higher.

Thoughts about the costs of Type I and Type II errors should spring to mind – that is, false positives and false negatives, respectively. Such issues are of concern whenever dealing with data sets sufficiently large that they cannot be verified line by line. A general feeling among proponents of large data sets is that the benefits of large sample sizes often outstrip the costs in data quality, assuming those costs are well understood. Best practice is to vary one's criteria for data inclusion and show that the results of interest are unaffected. Additionally, one can provide the code and data and invite readers to explore it themselves.

8.5.1 Isolates and the Degree Distribution of New and Old Words

Are new words relatively more likely to be isolates than old words? We can address this by looking at the proportion of words with degree zero (hermits) in the set of old and new words. Figure 8.2 shows that about 5.3% of new words are isolates, whereas 2.5% of old words are isolates. This suggests that at least some new words are dedicated to charting new regions of the semantic space. Moreover, this is more likely than one would expect based on the base rates for isolates among old words – the 95% confidence intervals for old words gives us a good indication of what we should expect if the new words were sampled from the same distribution as the old words.

The density plot confirms that new words tend to be lower in degree than old words across the board. New words pile up on the left-hand side of the density plot, at the lower end of the degree distribution. Thus, not only are new words more likely to be unconnected semantic isolates, when they are connected they tend to be less well connected than old words. This is consistent with the idea that new words might duplicate old words but lack some of the same associates, thereby leaving them with lower degree. What we do not see is a build up of new words near the higher-degree-valued old words. That is, if duplicates keep some constant fraction of their parent's neighbors, then the distribution we observe reflects a more even sampling from the old word distribution. It is not, as we might expect based on duplication and divergence, a top-heavy distribution. If it were, we would have to make the additional assumption that most new words tend to lose all but a few of their parent's associates, pushing the degree distribution leftward.

The cross-over of the new and old words could correspond to two regimes of word development – what might be called a lexical *core-periphery structure*. That is, it may be similar to Ferrer-i-Cancho and Sole (2001)'s observation of two kinds of words in the Zipfian frequency distribution: one representing a lexical kernel composed of commonly used language (the core) and another representing more specialized low-frequency

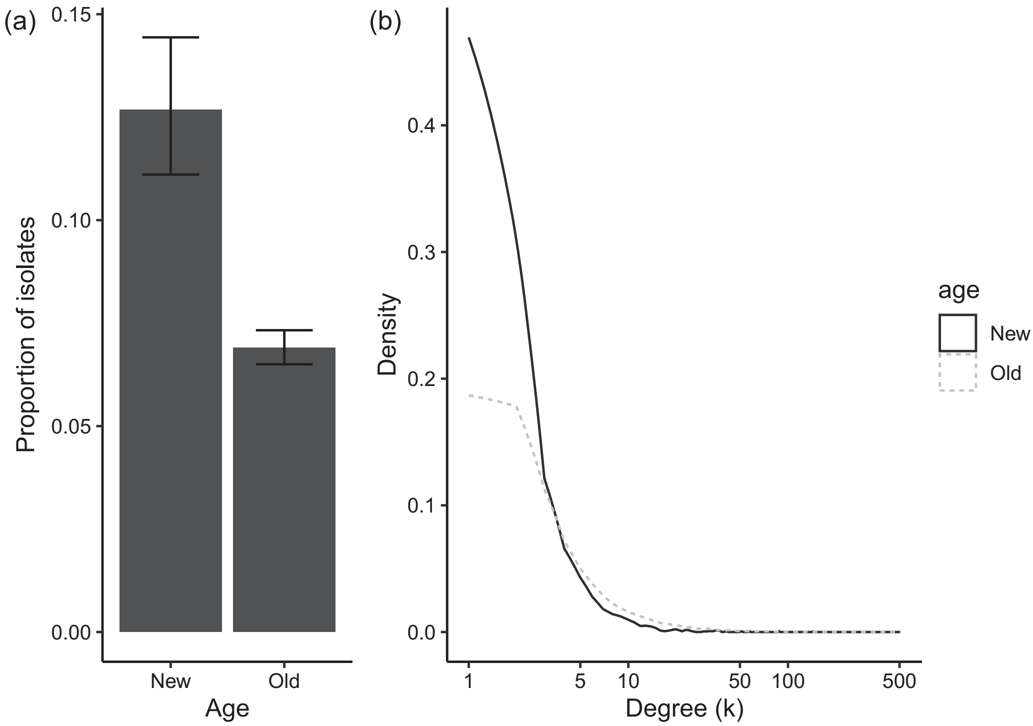

Figure 8.2 (a) The proportion of isolates in the free association network among new words that appeared between 1800 and 1900 and the old words that were present in 1800. Error bars are 95% confidence intervals. (b) The density plot for indegree for old and new words.

language (the periphery). If we identify a lexical core–periphery boundary at the interaction of our old and new word degree distributions, $k < 4$, approximately 84% of new words are in the periphery (including isolates), whereas only 62% of old words are in the periphery. This supports a prominent role for peripatric origins among new words, colonizing new regions of the semantic space. The other 16% of new words occur closer to the center, supporting centrifugal processes.

The density distribution further tells us that we need to control for degree in any comparisons between old and new words. Anything correlated with degree (e.g., betweenness or closeness) is likely to be differently associated with old and new words, not necessarily because of factors influencing their birthplace.

8.5.2 New Words Are Born Near Language's Urban Centers

What are the relative centrality metrics of old versus new words? Figure 8.3 provides a visualization for the relative metrics for betweenness, closeness, clustering coefficient, and eigenvector centrality. These are plotted across the degree range and the lines represent the fitted models for linear mixed effects models with random intercepts for degree.

When controlling for degree in this way, new words appear to be more central: for new words, betweenness is lower; they are not brokers between communities. Closeness is

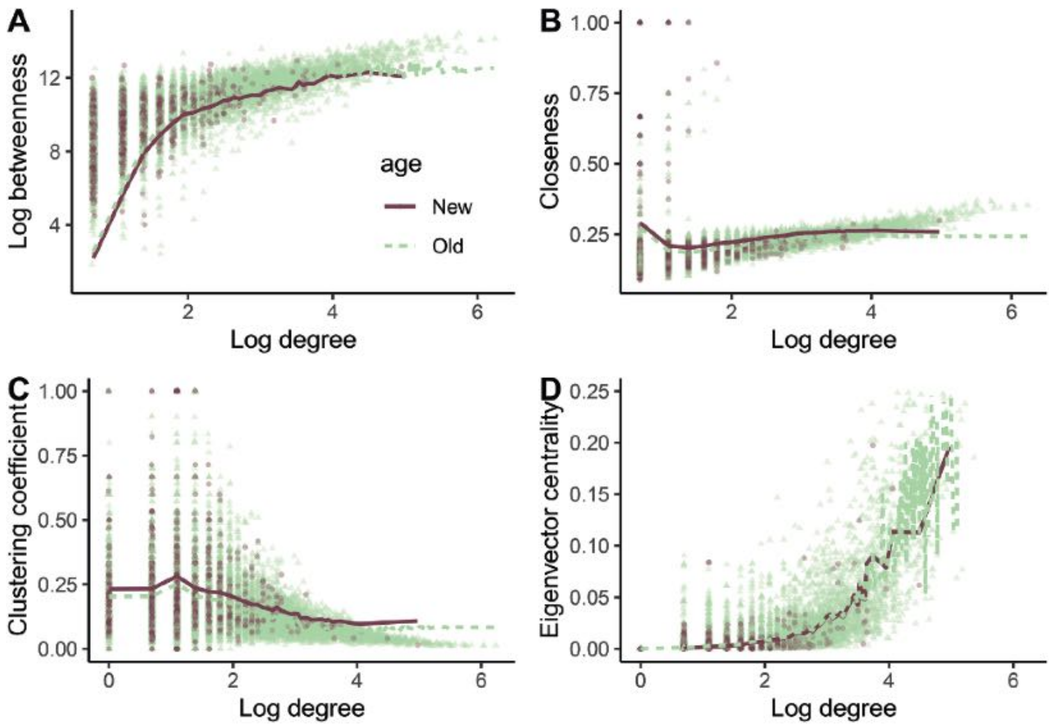

Figure 8.3 Comparisons of different centrality measures across the range of degree values for old and new words. Lines represent linear mixed effect models fit using random intercepts for degree.

higher; they nestle nearer the center of communities than old words. Clustering coefficient is higher; compared to old words of the same degree, their neighbors are more likely to be connected with one another. And eigenvector centrality is higher (subtly), suggesting they are associated with higher degree nodes. These are all statistically warranted (see the code) and they enrich and complicate our understanding. New words, despite showing a proportional increase in isolates relative to old words, appear to be closer to lexical centers than old words.

This is a fascinating result as it suggests a balance between the words that chart out unknown regions of the semantic space (isolates) and words that add additional semantic value to semantic centers that are already well understood.

This balance suggests two different ways in which selection works on word origins. In uncharted regions, old words have had trouble gaining a foothold. If we want to say something new in these regions, we will need new terminology. If that terminology catches on, it will have found a new niche in the ecology of language (e.g., Altmann et al., 2011). Near more dense regions of the semantic space, language is used more frequently in part because there is more here to say. There is more opportunity for productive variation to provide fodder for subtler forces of selection. These forces in turn create the more nuanced and central word forms that appear near lexical centers. The balance of these two forces – each marked by the hypotheses of Mayr (1954) and Brown Jr (1957) – offer an explanation for the peripatric and centrifugal word speciation we observe in the data.

8.6 Summary

Here we focused on node-level properties to differentiate among old and new words originating in the 1800s. Viewed through a lens of biological speciation formulated to operationalize where in a semantic space we might expect to observe the birth of new words, we find that most new words arise in the suburbs of language. These words have fewer associations than we would expect relative to those observed in old words. New words were also proportionally more likely to be isolates than old words. However, our centrality measures provide a more nuanced story. At any given value of degree, the metrics of betweenness, closeness, clustering coefficient, and eigenvector centrality reveal that new words show structural patterns that suggest they are born nearer to community centers. The evidence supports a role for new words at both the core and periphery of language. In sum, this chapter operationalizes a fairly abstract question about word birth in relation to network origins, bolsters this with theory from biological speciation, and then uses several network measures to demonstrate how these offer different lenses through which to answer the question.

8.7 Open Questions

8.7.1 Why Do Words Die?

What causes words to die? *Dancercise* is a colorful blend that had a brief moment in the lexical limelight in the 1980s but it is seldom heard now. Looking further back, there are many gems that have gone out of style or never fully penetrated the English language, such as *crapulence*, *yulehole*, and *hogamadog*, which mean in turn the feeling of overeating and drinking, the hole on your belt you need after eating too much, and the big snowball that forms the base of the snowman. Why didn't these words persist? The analyses used here to focus on word birth could just as easily be focused on word death. In principle, words near the center of language potentially suffer under more competition. But, at the same time, words near the periphery of language may suffer under more limited usage – referring as they must to more rare phenomenology, at least at their birth. Though we all may experience a crapulence, a yulehole requires a belt. How might we characterize these different selective forces and their influence on word longevity?

8.7.2 What Is the Life-History of a Word?

Life-history theory deals with the growth and reproductive strategies of organisms over the course of their lifetime. Stearns (1976) provides an excellent overview of this idea in the biological sciences. We can adapt this metaphor to language. Word frequencies and associations change over time (Li et al., 2019; Li, Hills, et al., 2020; Petersen, Tenenbaum, Havlin, & Stanley, 2012). Words born in one semantic region may migrate to another. Words born near the periphery of language may eventually gather new associations or fall out of use. How can we best characterize this change to evaluate the life histories of words as they transition from birth to death? Do words have specific personalities or patterns of usage in the same way that narratives may be selected from a specific set of plot arcs (Reagan et al., 2016). Do these lifestyles predict longevity or movement away from or toward the urban centers of language? One way to approach this is to construct networks from different periods of time and ask how words originate, fall out of, and move around in this lexical space from one time point to another. The Macroscope (http://macroscope.intelligence-media.com/) provides a means to explore this online for various words (Li et al., 2019). A more systematic treatment would attempt to identify general patterns across words and word classes (e.g., nouns, verbs, etc.).

9 Agent-Based Models of Language Emergence

Structure Favors the Orangutan

9.1 The Problem

For any form of communication to make it beyond the category of talking to oneself, at least two individuals must share a common lexicon. How might shared lexicons originate? Standard explorations of language often look in well-connected social groups such as chimpanzees, frequently numbering in the tens of individuals. But we might ask if language perhaps began in a more humble arrangement, involving social groups of just a few individuals, such as found in the orangutan? Agent-based models combined with network science offer a way to study this problem. By treating nodes as agents with strict rule-based behavior and edges as opportunities for interaction, agent-based models provide frameworks for studying how behavior and connectivity interact to create emergent phenomenon, such as the evolution of cooperation and cultural change. Here we will explore an agent-based model of the naming game to address how structure influences the emergence of shared lexicons.

9.2 Language Games

In Christiansen and Chater's (2022), *The language game*, language is described as the outcome of the human capacity to play charades. Languages, they explain, are never monolithic rule-based outcomes of a central designer. They are emergent outcomes of an ever changing set of intentions and aspirations that lead communicators to improvise meaning construction on the spot. This turns language into a game. The goal of the game is to communicate. This in part requires guessing what is in other people's minds. Good communicators, writers, speakers, and so on play the game better because, as Pinker notes, they make better guesses about what other people are thinking (Pinker, 2015). This notion of a game underlying shared meaning pays homage to Ludwig Wittgenstein's description of language as the construction of meaning through usage, in what he called "language games."

Language games should remind us of game theory. Decisions are made by two or more players and the outcome depends on what everyone decides. A language game involves one player signalling to another player, who in turn chooses a response. Luc Steels (2011) describes numerous experiments where artificial agents engaged in language games can develop a shared lexicon. Might this be the way languages begin?

How languages originated is a diverse and contentious topic. To start, we must answer the question of what exactly language is. Many animal species have complicated communication systems: honeybees dance to signal the location of resources; ants use chemical pheromones to communicate with one another; bats use acoustic, chemical, and visual communication; birds learn song dialects from their local community; meerkats recombine sounds to produce different meanings; and dogs and primates can learn words and symbol systems referring to hundreds of objects and events. But are these true languages? Some have argued that these are all language in a broad sense (Hauser et al., 2002); they are all signal systems designed to intentionally communicate concepts. However, in a more narrow sense, they lack a computational feature common to human language – *recursion* – which allows open-ended expression of unending variety. Recursion is the embedding of signals within signals, such as "The cheese that the mouse that the cat killed ate was in the house that Jack built." Whether or not other animals can or cannot do recursion, and whether or not recursion is the defining characteristic of human language, is a source of unending debate and amendment, a bit like the recursion it often centers around (Corballis, 2007; Pinker & Jackendoff, 2005). Given that we are unlikely to add much to that debate here, we will focus on the simpler problem of how creatures might come to score points in the language game at all.

Niko Tinbergen, a renowned early ethologist, tells a hypothetical story about how the dog's snarl came to indicate threat (Tinbergen, 1952). At first, the dog pulls back its lip when eating to avoid biting itself. It does the same before it bites other dogs. Thus, the raised lip is a predictive signal and when other dogs learn to associate it with the oncoming bite, they can escape before the bite occurs. As this association develops, a dog can use it to communicate its intention to bite. It can simply raise its lip to threaten other dogs, making the bite unnecessary. Cheap talk, perhaps, but the work of language has begun.

The evolution of the snarl as a form of communication requires two things: (1) a reliable signal that arises in a specific context, and (2) an association between the signal and some intention – an object or event – which is commonly understood by both the sender and the receiver of the signal.

The search for commonly understood signals that map onto intentions is the basis of Wittgenstein's language game. It is a coordination game, where each individual must anticipate the understanding of the other. But if this coordination requires that two individuals already understand the intention, then where does the common understanding begin?

It is the question of common understanding that has given rise to theories of language evolution that begin with gestures (Kendon, 2017). With pointing or moving one's hand toward one's mouth, the action signals the intention in a form that is similar to the intended meaning. It is a kind of distilled theater. The outcome is acted out, much like the dog's snarl acts out the sequence before the bite. Similar theories, such as what Aitchison calls the *ding-dong theory* (Aitchison, 2001), arise based on the onomatopoiec sound-similarity between a word and the thing it intends to represent, such as "choo-choo," "woof," and "crash." These cases give us a hint at some possible routes to language's birth.

An assumption underlying these theories is that the communicators are paying attention to the right things (e.g., Poss et al., 2006). Some prominent theories of language development forego the prerequisite of common conceptual understanding and focus instead on shared attention (Tomasello, 2016). Shared attention is a core component of human language learning and children who are better at it learn larger lexicons (Yu et al., 2019). But shared attention appears to require the right evolutionary prerequisites. Hare and Woods (2021), in their book *Survival of the friendliest*, describe how friendliness between individuals allows communicators to develop shared attention. One of their premier examples is the multigenerational domestication of wild Russian foxes. Foxes were chosen for breeding based on their friendliness toward the breeders. Over many years, this produced foxes that were not just friendly, but that could also pay attention to humans and learn from them. These domesticated foxes could use communication from humans to solve problems. Wild foxes were so skittish and distracted that they lacked this ability. Hare and Woods argue that this friendliness gave rise to the productive communication systems that are behind human success.

This all points toward language originating among individuals who can share attention and some degree of interpersonal curiosity around a common set of utilitarian concepts. We see a bare bones version of this in dictionaries of bird song: many birds have a song and an alarm call, satisfying needs for reproductive contests, territorial disputes, and collective vigilance. More advanced forms of instrumental communication arise when humans without a common language interact, forming pidgins around collective goals (like trade), which then become creoles when their children learn these pidgins as a first language. These in turn become more nuanced and morphologically complex as grammaticalization moves concrete words toward more abstract syntactic roles (see *have* in Chapters 7 and 8). This pattern – from shared attention to common concepts to semantic evolution – arises repeatedly.

9.3 Social Structure and Language Emergence

How might social structure influence this process? At first glance, we might expect social organisms with large group sizes, like chimpanzees, to have some advantage. Sociality seems to be an obvious prerequisite for language, and therefore more sociality should be better. Indeed, many studies of language have focused on chimpanzees. Their social network sizes can be 20 or more individuals.

On the other hand, recent comparative reviews have argued that comparisons across hominid species including *Pongo* (orangutans) and *Gorilla* might provide a more complete picture. These different species all share degrees of communicative competence but differ in important genetic and ecological factors (Lameira & Call, 2020). One of these factors is social group size. Orangutans, for example, often live in semisocial groups with dispersed territories, predominantly made up of two individuals: mother and child, who sometimes but infrequently interact with other individuals within their range. Might these differences in social structure matter?

For humans, the answer is a tentative yes. The *linguistic niche hypothesis* is based on the observation that smaller communities tend to maintain more morphological complexity, such as irregular verbs (Lupyan & Dale, 2010; Nettle, 2012). Experimental investigations in the lab have found varying results. For example, Raviv et al. (2019)

asked people to name novel objects that their partners then had to identify. Using various population sizes with fully connected networks, results showed that larger groups produced more systematic languages more quickly than smaller groups (but see Raviv et al., 2020). Investigations of individual differences in people's social network structures have also found structural influences on people's ability to understand and predict what others will say (Lev-Ari, 2019, 2022).

There is still a great deal of work to do in organizing this research. Simulations can be useful here as they help us to identify differences that matter. When combined with empirical research, they can help refine theory and provide demonstrations of phenomenology that would be impossible to create in the real world. One frequently used simulation to understand language emergence is the naming game.

9.4 The Naming Game: A Simple Model of Language Emergence

The naming game is an *agent-based model*. Agent-based models endow entities (i.e., agents) with certain qualities and then allow these agents to interact with their environment to examine how qualities and environmental structure influence emergent outcomes. Agent qualities can include things like beliefs, behaviors, identities, states, group membership, and perceptual or cognitive faculties that allow agents to interact with and learn from their environment. Their environment can consist of analogies to real physical space, which might contain resources of various kinds, or it could simply involve other agents.

Often, agent-based models focus on only one property at a time, allowing researchers to sharpen their intuition around specific assumptions. That is, simpler is better. The point is not to model reality but to understand the implications of a specific set of assumptions about reality. In the case of language emergence, the assumption might be that languages are more likely to emerge when groups are large. Either this is always true or details matter. If details matter, then agent-based models can help to identify those details.

In many cases, agent-based models reveal properties of complex systems that may not be readily apparent by simply thinking about them. For example, Thomas Schelling's segregation model (see Chapter 16) demonstrates that communities will naturally segregate even if most people are willing to be in perfectly mixed neighborhoods. Modeling games of conflict on networks reveals how cooperation can persist even when it would fail to among unstructured populations (see Chapter 18). Agent-based models can also be used to study phenomenon that spread through social systems, such as swarming and flocking, the dynamics of financial markets, the spread of disease, and opinion dynamics (Chapters 15 and 16). There are many excellent books focused on this approach and the richness of its contributions (Boyd & Richerson, 1988a; Epstein, 2014; Epstein & Axtell, 1996; Resnick, 1997).

The naming game is an agent-based model developed to explore how lexicons might emerge in communities based on pairwise interactions among agents (Baronchelli et al., 2006; Steels, 1995). The basic rules of the naming game are as follows:

1. An agent is randomly chosen to be either the speaker or the hearer. Then one of their neighbors is chosen as the hearer or speaker, respectively. This order matters, so we

will call it speaker-first when the speaker is chosen first and hearer-first when the hearer is chosen first.

2. After agent selection, the speaker decides on an object in the world to name. In the minimal version of this game, there is only one object.
3. The speaker then examines their memory for names for that object: If the speaker's memory contains one or more names for that object, they choose one at random and produce it for the hearer. If the speaker's memory for that object is empty, they invent a new word, add it to their memory, and then produce it.
4. The hearer then examines the names it has in memory for that object: If the produced word is in their memory, this communication event is a success. In the case of success, both agents remove all other object-related words for that object from memory, leaving only the successful word. If the hearer does not have the word in memory, the event is a failure. In the case of failure, the hearer adds the word the speaker produced to their memory for that object.

This process is then repeated many times, with new randomly chosen speakers and hearers each time.

The simplicity of this agent-based model is its strength. Whatever happens in this system will be a consequence of the rules that are imposed on it. Because the system is sufficiently simple, how these rules work can be easily teased apart. Once these basic rules and consequences are understood, additional features can be added.

For the naming game, the only way for agents to decide on a collective language is through dyadic interactions. Fittingly for an agent-based model, there is no top-down control. In each pairwise interaction, one agent produces a name for the object, and the other agent either recognizes the name or not. The lexicon that develops is an emergent process of simple, rule-based pairwise interactions among the population's many agents.

As described here, it always takes at least two interactions to achieve a success. The first interaction plants the word as a seed in the memory of the hearer. On the second interaction either the hearer or the speaker in the initial interaction repeats the word to the other. In the minimal version with only one object and a network of only two nodes, success will always occur on the second iteration.

Thinking about a star network can help develop our intuition. If we allow speakers to be chosen first in a star network with n nodes, the central node will often be the hearer. This is because the central node is chosen as the speaker with probably $p = 1/n$, which shrinks with increasing n. If the central node is not the speaker, then they will be the hearer, because a noncentral node's neighbor will always be the central node.

In a speaker-first world, the central node in a star network will hear many different names for an object from different noncentral speaker nodes and the population is likely to remain in a state of perpetual failure for a long time. However, in a hearer-first world, the opposite will happen. Most of the time hearers will be noncentral nodes and the central node will be the speaker. In this case, the lexicon will rapidly stabilize. In general, high-degree nodes are more likely to be hearers when speakers are chosen first; they are more likely to be speakers when hearers are chosen first. Thus, we can already see one implication of this model: if hubs tend to be speakers, communities will coordinate on shared lexicons more quickly than if speakers are chosen at random.

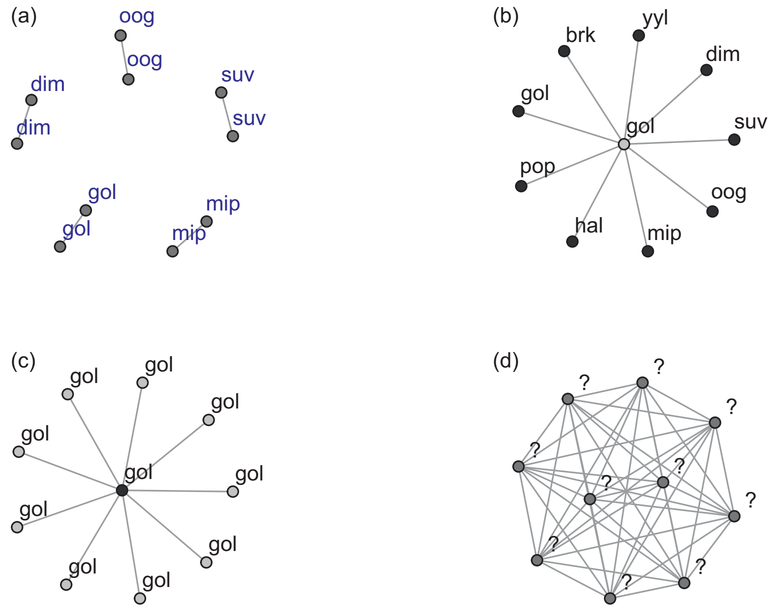

Figure 9.1 The naming game in different environments. (a) With five dyads, each pair will agree on a common word after two iterations. But each dyad will speak a different word from all the others. (b) With a star network, if the speaker is chosen first at random (speaker-first), the hub will spend more time as a hearer (shown in gray), which will inhibit coordination across the network. (c) If the hearer is chosen first (hearer-first), then the hub node will spend more time as a speaker (shown in black), and coordination will happen rapidly. (d) A full network. You might try to take a guess as to where this group will fall relative to the preceding three.

Running the minimal naming game on the four networks shown in Figure 9.1 helps to test our intuitions. Each run of the simulation is a pass through the four steps listed earlier. Figure 9.2 shows the average of 100 simulations of the minimal naming game with 100 iterations of hearer and speaker interactions for each simulation. Because the agents are chosen randomly and sequentially, it takes time for the simulation to reach each agent. The dyads agree most quickly with one another but fail to converge as a community because they are unable to share what they learn. The hearer-first star network also rapidly stabilizes. Here the hub node acts a coordinator. When the speaker is chosen first, numerous word types compete for dominance among the cacophony of speakers and the network is no better than groups of dyads at the end of the 100 iterations. Finally, with a full network, word types initially proliferate to about what is seen in the dyadic network, but this rapidly settles down to that seen in the hearer-first star network.

Numerous network approaches have been taken to studying the naming game. These have revealed influences of network size and degree, memory capacity, average shortest path length, the capacity of local clustering to insulate the system against consensus, and the ability to engineer consensus by seeding clusters with hubs that share a common lexicon (e.g., Dall'Asta et al., 2006; Liu et al., 2009; Lu et al., 2009). In the next section, we will focus on making language harder to learn.

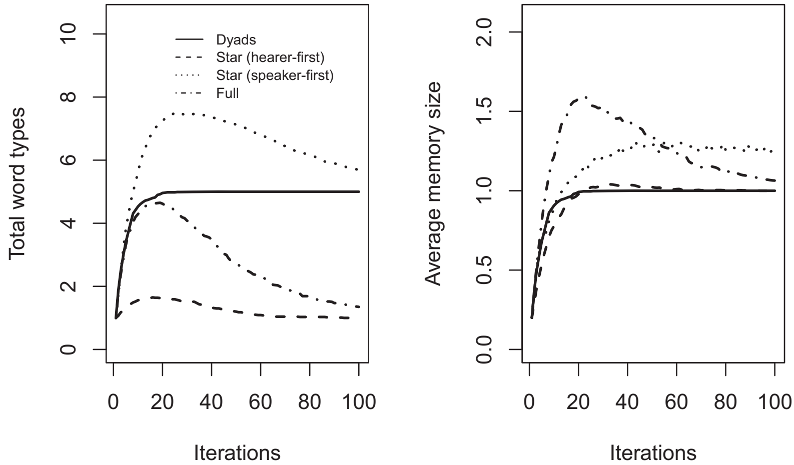

Figure 9.2 The outcome of the minimal naming game simulated on the networks shown in Figure 9.1. In the minimal naming game, agents (i.e., nodes) have only one object to name. Either a speaker (speaker-first) or hearer (hearer-first) is chosen first and then a neighbor is chosen at random to be the hearer or speaker, respectively. The networks all contain $n = 10$ agents. Total word types is the number of unique words known for the object in the population. Average memory size shows the number of words that the average agent has in memory. Both the dyads and the hearer-first network rapidly settle on a shared lexicon among connected nodes. This full network eventually settles on a community-wide lexicon and is faster than the speaker-first star network, which struggles to settle on a shared lexicon.

9.5 The Innovate-and-Propagate Model: Investigating Language Difficulty and Population Size

As described earlier, the naming game is a model of one-trial learning. Once an agent hears a name, they are immediately able to produce it. To reach adult vocabulary sizes of more than 50,000 words, most people learn on the order of 10 words per day up until adulthood (McMurray, 2007). One-trial learning would be an asset in achieving that goal. However, there is also good evidence that for young children, their first words are not learned in a single trial but go through a period of semantic maturation (Jiménez & Hills, 2022b). This likely requires repeated exposures to the same word to shape its conceptual boundaries. Repeated exposure is almost certainly the norm for individuals that do not yet have any common language and may not yet even know that words can refer to objects.

Various studies have investigated the influence of word difficulty on language acquisition and emergence. For example, McMurray (2007) showed that parallel learning of words of different difficulty would generate the characteristic vocabulary explosion seen in early language learners. The primary requirement for this explosion was that difficulty of learning (number of required exposures) be normally distributed.

To explore the interaction of word difficulty and population sizes, Reali et al. (2018) develop a variation of the naming game called the *innovate-and-propagate model* which allows communities to develop multiword lexicons. The model adds several novel features: (a) words are produced in proportion to their past usage and are represented in memory as tokens, each reflecting an instance of past experience; (b) tokens can be forgotten: every token has a fixed probability of being lost from memory; and (c) there is a fixed probability of inventing and producing novel words that are not in the agent's memory.

Agents produce words by sampling from their entire memory pool. An agent's memory pool contains a token for every time a word was used, either heard or produced, in the agent's past experience. Different words will have different numbers of tokens, t_i, indicating the frequency with which they have been encountered in the past. The probability of sampling a particular word, w_i, is then

$$P(w_i) = \frac{t_i}{M+1},$$

where M is the sum of tokens in memory, $M = \sum_i t_i$. This is basically like reaching into memory and sampling a past word usage and repeating it. The addition of 1 in the numerator reflects the probability of inventing a novel word, which is

$$P(w_{new}) = \frac{1}{M+1}.$$

The sum of the probabilities across all known words and novel inventions is 1.

9.5.1 Memory as a Chinese Restaurant Process

The generation of the memory pool reflects a modification of what is called a *Chinese restaurant process* (Pitman, 2002). This is a useful tool for thinking about behavior in settings where the membership of categories can grow over time and novel categories can emerge (e.g., Gershman & Blei, 2012; Kemp et al., 2010). It is called a Chinese restaurant process because it is analogous to customers entering a restaurant with large round tables. Individuals entering the restaurant either choose to sit at occupied tables in proportion to their present population or they choose a new table. The number of people at a table is analogous to the number of tokens for each word type. An individual choosing to sit at a new table is like the invention of a novel word type.

Like many other behavioral processes, this is a rich-get-richer process. As a nod toward the generality of the Chinese restaurant process, Reali et al. (2018) call words *conventions*.

Another aspect of memory and word learning is the problem of forgetting. An approach to forgetting readily adapted to the Chinese restaurant process is Poisson forgetting: allow every token in a speaker's memory to be forgotten with a fixed probability, p. Poisson processes are often used to model how often things occur when events are independent and occur with a fixed small probability per opportunity, such as traffic accidents and cancer incidence.

When applied to memory, this means that an agent will forget approximately $M \times p$ tokens each time forgetting occurs. We will let forgetting occur each time an agent is chosen as a speaker. Forgetting produces an upper limit on an individual's lexicon size:

at some point the number of forgotten tokens will balance out the number of new tokens added through experience with others.

Using the innovate-and-propagate model, Reali et al. (2018) were able to investigate the impact of population size on the ability to learn easy or difficult conventions. Easy conventions require one exposure before they can be produced. Difficult conventions require two. Population size was modeled by artificially generating networks of different size but which all had a structure similar to that found in real-world mobile phone data found in Schläpfer et al. (2014): each network had a clustering coefficient $C \approx 0.25$ and degree scaled such that $k \approx n^{\beta-1}$, with $\beta = 1.677$. Simulating the innovate-and-propagate model on real-world networks was capable of producing results consistent with the language niche hypothesis: larger networks learned fewer hard words than smaller networks.

9.6 The Orangutan Simulation: How Does Social Structure Influence Language Emergence?

The above work sets the stage for addressing the interaction between social structure and language emergence. Reali et al. (2018) do not investigate different social structures as their focus is on the language niche hypothesis. But we can use their framework to study the impact of social structure when language learning is hard, as we might expect it to be when there is no precedent for language in the community. As noted earlier, chimpanzees and orangutans have different social structures. Orangutans exhibit a more semisolitary social strategy (Roth et al., 2020). Chimpanzees live in larger, more well-connected communities (e.g., Tomasello, 1994). Both orangutans and chimpanzees show intentional food-directed communication and gesturing (Bard, 1992), satisfying the basic requirements for shared attention and communicative goals set out earlier.

To model these differences in social structure, we will build networks that consist of multiple subgroups. By varying the extent to which these communities are divided from one another, we can vary the sparseness of the overall community structure. In a nutshell, we are fracturing a fully connected chimp network into orangutan subgroups. To do this, I will use the Girvan–Newman framework introduced in Chapter 3. This produces edges with different probabilities within and between subgroups. To vary group isolation, the probability of an edge between groups will be lowered relative to that within groups by increasing the clustering ratio, r. The clustering ratio, r, is the relative difference between the probability of an edge within or between groups: higher values indicate greater likelihoods of within-group edges. Mathematically, the probability of an edge within a group is $p_g = r/(r+1)$ and between groups is $p_b = 1 - p_g$, such that $r = p_g/p_b$. Using clustering ratios of 1, 10, and 20 produces networks similar to those shown in Figure 9.3.

We simulate the innovate-and-propagate model on three clustering ratios (1, 10, and 20) by generating a new network for each simulation and running each simulation for 200 iterations of randomly chosen speaker–hearer pairs. Each clustering ratio is simulated for 200 networks. Words are either learned after 1 exposure (easy) or 3 exposures (hard). Following Reali et al. (2018), we use a speaker-first implementation and tokens are lost from memory with Poisson forgetting, $p_{forgetting} = 0.20$.

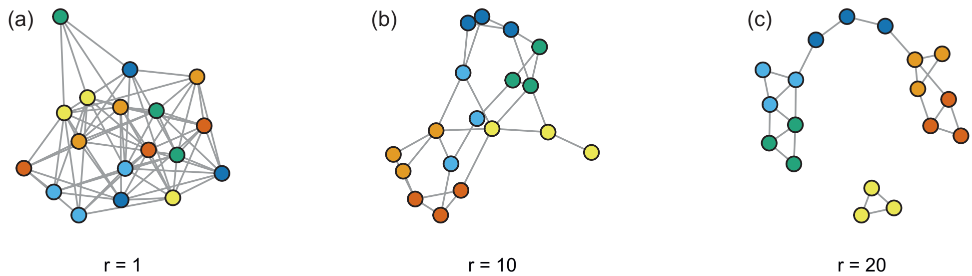

Figure 9.3 The three Girvan–Newman networks used to explore learning difficulty and social structure. By analogy, chimpanzees are shown in panel (a) and orangutans in panel (c). Each network is produced by assigning $N = 18$ nodes to 6 groups, with each group having 3 members. Edges are added between all pairs of nodes with probabilities proportional to their within-group and between-group clustering ratios, shown beneath each network.

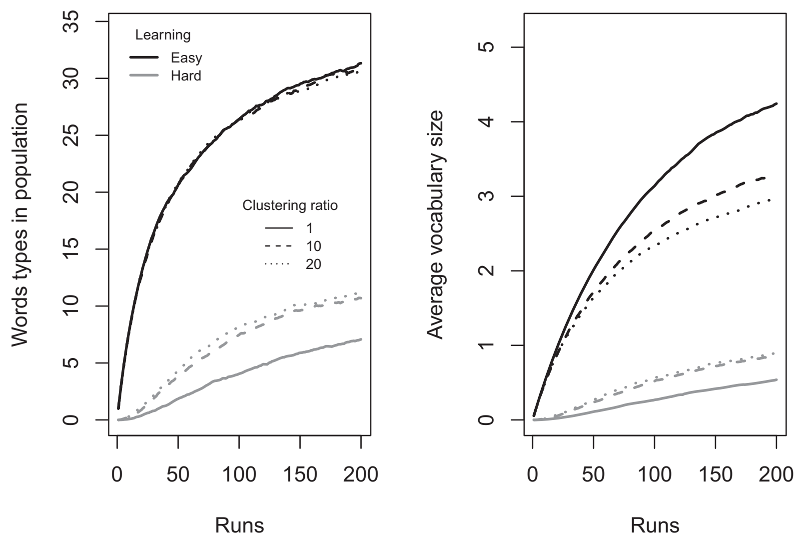

Figure 9.4 The innovate-and-propagate model with various clustering ratios and word difficulty. Simulations are run 200 times for each clustering ratio and each word difficulty. Networks are all the same size, $N = 18$, but have different clustering ratios of 1, 10, and 20. Easy words are learned on the first exposure and hard words are learned after 3 exposures. Forgetting is set to 0.2 for all networks.

Figure 9.4 shows the results of the simulations and reveals a critical interaction between structure and difficulty. When learning is easy, structure is negligible in terms of the total number of word types in use. After 200 iterations, the three networks are not noticeably different. When learning is hard, structure makes a dramatic difference. For hard learning, groups with a higher clustering ratio and therefore smaller local communities learn more words faster and maintain a larger number of words in use than do less clustered communities. When learning is hard, smaller communities provide the repeated exposures needed to help individuals learn before they forget what they know.

The average vocabulary size also tells us something important. When learning is easy and the population contains a lexicon of approximately 30 words, most individuals know only a fraction of these words. Individual vocabularies tend to be larger when the network is more fully connected. However, when learning is hard this trend is reversed. Now individuals in small clustered communities have larger individual lexicons. Many individuals know nothing, but a few manage to get a lexicon off the ground.

The results fit with hominid research on language. For example, the call repertoires in orangutans are likely to be more complex than those in chimpanzees and other species, especially when considering the research time invested in these various species (Lameira, 2022). Moreover, population density also influences the complexity and consistency of communication *within* orangutans: more dense populations of orangutans have less complex, less reproducible, and more transient alarm calls than smaller populations (Lameira et al., 2022). Thus, our model confirms that even when communities are the same size, more sparse and clustered communities are more able to solve the problem of language emergence when learning is hard.

9.7 Summary

How languages emerge in the wild is a complex problem that has generated equally wild hypotheses. Indeed, the challenge of providing evidence for these hypotheses led the Linguistic Society of Paris to ban papers on language origins in 1866 (Aitchison, 2001); as we cannot rerun history, we are left with experimental studies of humans and other animals and theoretical work that attempts to bridge the gap between the two. Though there has been some focus on group size, network science allows us to ask about structure as well. Here we used an agent-based model developed around a recent variation of the naming game. This provides a framework for examining social structure alongside a theoretical motivation based on the stylized community structures of chimpanzees (fully connected) and orangutans (sparse and clustered). The results of the model support empirical observations of the orangutan lexicon, which is more complex when communities are small and less complex when communities are large. Moreover, the model shows that when communities are large, hard language learning is difficult to get off the ground. This demonstrates that a model like innovate-and-propagate, which also supports the complexity and diversity of human languages, can be adapted to language emergence. In doing so, it sharpens our intuition by showing us that more social communities may not be the best place to find language emergence. When learning is hard, lexical emergence and maintenance requires repeat exposure sufficient to overcome forgetting, which is something more clustered communities provide.

9.8 Open Questions

9.8.1 Individual Differences in Language Ability

Lev-Ari (2022) developed elegant studies of individual social networks to help understand how structure matters to an individual's language abilities. We can use simulations to examine the conditions that might give rise to these individual differences. For example, which individuals are most likely to have larger vocabularies in structured language communities? Does betweenness help or hinder individual language development? Which individuals have more specialized lexicons within a larger community? Can we develop experiments to test this in real-world networks? We can also explore how individual faculties and structure interact. For example, might individuals with more limited memory capacities compensate for these limitations by engaging in more locally clustered social communities, which provide repeated language exposure? There are potential real-world implications for this: for example, when older individuals with memory limitations are moved from their small home community into assisted-living facilities with a large language community and new specialized vocabulary, this may then amplify their apparent memory limitations.

9.8.2 Social-Diffusion Chains and Iterated Learning

Iterated learning experiments – sometimes called social-diffusion chains – involve having individuals learn and then teach information to a new individual in the chain, who then repeats this for another individual, and so on. The focus of these experiments is what happens to the information over time. Kirby et al. (2008) showed that iterated learning leads to language simplification, with words being lost and simplified over time. This also happens when stories are listened to and then retold: the number of words taken to retell the story shrinks over time, but nonetheless preserves key information about emotional content (e.g., He et al., 2023). Similarly, when people hear factual information about specific topics, such as nuclear power and antibacterial agents, the total amount of information shared shrinks over time, but specific elements of the information are preserved, and sometimes added to, such as information about risk (Moussaid et al., 2015). Iterated-learning experiments reveal something about memory limitations, which simplify information, and embellishments, that may enhance information content. For example, a study by Jagiello and Hills (2018) showed that information passed through a series of individuals became more persuasive over time than the original information, even though the content became shorter. Are iterated-learning experiments like these the equivalent of learning on large, fully connected networks, in which it is rare that we hear the same message twice? If so, these experiments reflect memory at its limit, where only the easiest concepts can be encoded. Might an antidote to these information limitations be provided by differently structuring information environments, either provided by the way information is shared (its internal structure) or the communities that share it?

Part III

Mind

10 False Memories
Spreading Activation in Memory Networks

10.1 The Problem

What is memory? Scientists have proposed a wide variety of spatial metaphors to understand it. These range from 2D wax tablets to 3D physical spaces that we can figuratively walk around inside. If memory has such a spatial structure, then it suggests a simple rule: items in memory can be near or far from one another. Anything with a near-and-far structure lends itself to a network representation. Such spatial structure also lends itself to being in the wrong place at the wrong time: remembering things that never happened and forgetting things that did. In this chapter, we will explore how structure facilitates memories. We will also look at how structure impacts false memories using a model of spreading activation.

10.2 Structure in Memory

In what is frequently cited as one of the earliest experimental psychological studies, Hermann Ebbinghaus (1885) gave himself the task of memorizing lists of nonsense. By posing himself the problem of learning and later recalling sequences of nonsense syllables (e.g., XYF, TLX, etc.), Ebbinghaus carefully charted the rate of his forgetting over time. He found that forgetting was fastest immediately after learning, and then slowed as time progressed. The pattern has since been called the *forgetting curve*. This curve may as well be a law. Forgetting is always fastest immediately after learning and it slows with time (Brown et al., 2007b; Rubin et al., 1999; Wixted & Ebbesen, 1991).

The forgetting curve is already evidence of structure in memory. It tells us there is temporal structure, specifically a *recency effect*, with better recall for things we have learned recently. However, it also tells us that older memories are less likely to be forgotten than younger memories. This is called *Jost's law*: if two memories are held in memory with equal strength, the newer memory is more at risk of being lost. There is a corresponding *Ribot's law*, associated with traumatic brain injury: brain damage tends to effect recent memories more than long-held memories (Wixted, 2004). On short lists of items learned in the laboratory, we have analogues to both laws: memory is better for the first and last items on the list, corresponding to *recency* and *primacy effects*, respectively (Oberauer, 2003).

Temporal structure can be juxtaposed against semantic structure (what things mean) and phonological structure (what things sound like). Indeed, they are related. One potential explanation for Jost's law and Ribot's law is that memories strengthen over time through their connections with other memories. Ebbinghaus chose nonsense syllables precisely to avoid this problem: unlike common words, nonsense syllables lack associations with information already in memory, and thus they may avoid the structural influences of existing memories. According to Bartlett (1932), Ebbinghaus did this on purpose because he recognized the influence of semantic structure – the memory-enhancing effect of associative meanings.

This effect is driven home by a classic study by Chase and Ericsson (1981). A single individual, called SF, was challenged with remembering lists of numbers. He did this repeatedly over many weeks, spending a total of 250 hours hearing and then trying to recall lists. Initially, SF started with a respectable but typical *number span* of 7 digits. This more or less follows the standard psychological theory that our short term memory can hold 7 plus or minus 2 items (Miller, 1956). However, after many hours of training, SF was able to remember up to 80 digits.

According to Chase and Ericsson (1981), this achievement was the result of SF developing a series of strategic mnemonic devices. SF related number sequences to narratives involving ages, dates, and running times. As a sports enthusiast, SF already had an elaborate and well-structured memory for sports-related figures. By connecting new information to what he already knew, SF turned a long list of unrelated numbers into a shorter list of related numbers. This *chunking* allowed SF to remember more by using structure to organize content in his mind. This is a well-known strategy among memory experts (Maguire et al., 2003). One can experience a simple example by trying to remember the following sequence: MYC-IAW-HOF-BIN-ASA. This list becomes easier to remember if you move each dash one letter to the left, at which point it connects with information you already know.

Connecting new information to existing information often goes by the term *elaborative encoding* (Craik & Tulving, 1975), or sometimes just *elaboration*. Numerous studies have shown that having people engage with the meaning of new information increases the likelihood they will recall it later. For example, people remember things better when they categorize them (Pollio & Gerow, 1968), they remember information better when they organize it during note taking (Jansen et al., 2017), and they have trouble forgetting things that have strong emotional connections (Schacter, 1999).

Chase and Ericsson (1981) demonstrated the importance of elaboration by interfering with it. Specifically, in additional studies with SF, they prevented him from chunking information by giving him other things to do (a secondary task), such as repeating a letter of the alphabet he was told after each number. By interfering with SF's ability to elaborate, SF's memory became no different from when he began the study. Without the ability to elaborate, SF was unremarkably normal.

The importance of existing structure becomes even more clear when considering novices and experts. This was cleverly demonstrated by De Groot's (1978) chess experiments. What De Groot (1978) found was that chess experts were better at remembering chess board positions, but only if the boards were arranged in ways one might actually experience during a chess game (see also Gobet & Simon, 1996). When the chess boards were arranged in patterns atypical of chess games, experts were no better than novices. Like Chase and Ericsson's SF, the ability of chess players to remember chess positions required having a well-developed memory structure with which to elaborate the new information.

10.2.1 Priming

Most contemporary theories of memory treat the mind like an online search engine: memory is cued with a set of key words and then it returns a list of related targets. The *cue* is the pattern of stimulation that activates memory retrieval. What is retrieved is a function of the relationship between the cue and the structure of memory itself.

Because of the relationship between a cue and what memory returns, one can measure the distance between things in memory through priming experiments. In a typical priming experiment, the question is how does experiencing one word (the cue) affect the ability to process a second word (the target). One way to evaluate this is with a *lexical decision task*: a target word is shown and the participant must say whether or not the target is a word or a nonword. The experimenter records the time it takes to make this decision. By providing the cue word alongside the target word, one can detect priming when the cue word decreases the response time to the target word (Hutchison et al., 2013).

In a classic experiment, Meyer and Schvaneveldt (1971) showed people pairs of letter strings simultaneously (BREAD-BUTTER, NURSE-DOCTOR, DOCTOR-BREAD). Participants were asked to respond "yes" if both strings were words, or "no" if one or more of them was a nonword. Priming was demonstrated when people responded more quickly to associated words (BREAD-BUTTER) than to nonassociated words (BREAD-DOCTOR).

Activating one thing in memory prepares other nearby things in memory for recognition. That's useful if you need to anticipate the world. Perhaps less useful is that experiencing one thing can create spurious memories for things that never occurred.

10.3 False Memories

False memories are memories of experiences that one has never had. One of the earliest reported studies of false memories was done by Frederic Bartlett (1932). Bartlett studied memory by asking people to remember and then later recall unfamiliar folk stories, such as *The war of the ghosts* and *The son who tried to outwit his father*. At the time these stories were told, they contained many unfamiliar elements and plot features. Quoting Bartlett (1920, p. 37), "In *The War of the Ghosts*, for example, two young Indians are seal hunting, when they are accosted by warriors from a canoe, who ask them to help in a fight which is about to take place. One of the Indians agrees, and goes with them. In the fight he hears somebody say: 'That young Indian has been hit,' but he feels no hurt. He merely remarks casually: 'Oh, they are ghosts.' He goes back home, tells his friends, lights a fire, and the next morning at sunrise falls down: something black came from his mouth. He was dead."

When people were told the story and later asked to recall it, they tended to distort the story in ways that Bartlett interpreted as culturally familiar. For example, canoes became boats, paddling became rowing, ghosts were forgotten, and the stories tended to acquire morals. The longer people held the story in memory before recalling it, the more the stories became distorted (see also Bergman & Roediger, 1999).

False memories can also be directly implanted. The hot-air-balloon-ride study and the lost-in-the-mall study showed that people could easily be confused into remembering details about events in their childhood that never took place (Scoboria et al., 2017). For example, in a study by Garry and Wade (2005), participants were shown either doctored pictures of a hot air balloon ride they never experienced or heard a narrative story about the same event. Information and photographs were provided from family members or other people who knew the participant well, which helped embed the novel memory among a set of believable and familiar details. Over a series of repeated interviews, participants progressively "remembered" more details about these nonevents, including emotions and visual images.

The *Deese–Roediger–McDermott (DRM) false memory paradigm* is a powerful example of how false memories can be implanted in a laboratory setting (Deese, 1959; Roediger & McDermott, 1995). The experiment involves hearing a list of related words. To get a sense of how it works, read the following list of words once, from beginning to end:

- dream
- bed
- soft
- pillow
- sheet
- rest
- covers
- wake
- alarm
- unconscious
- dark
- morning

Now look away from this list, find a pen and paper, and write down all the words you can remember.

If you do this, you will experience a peculiar thing about your memory. If you are like most people, you will be strongly tempted to include at least one word that was not in the original list. When read aloud in a classroom of 100 students, a substantial minority will misremember this word and other related words not on the list.

Figure 10.1 shows words within a path length of two from the target false memory in the above list. This is called a *two-hop network*. The edges are taken from the Small World of Words free association norms: An edge is formed between two words whenever presenting them with one word led them to produce the other. For example, when asked to produce the first word that comes to mind upon hearing "mattress," many people produce

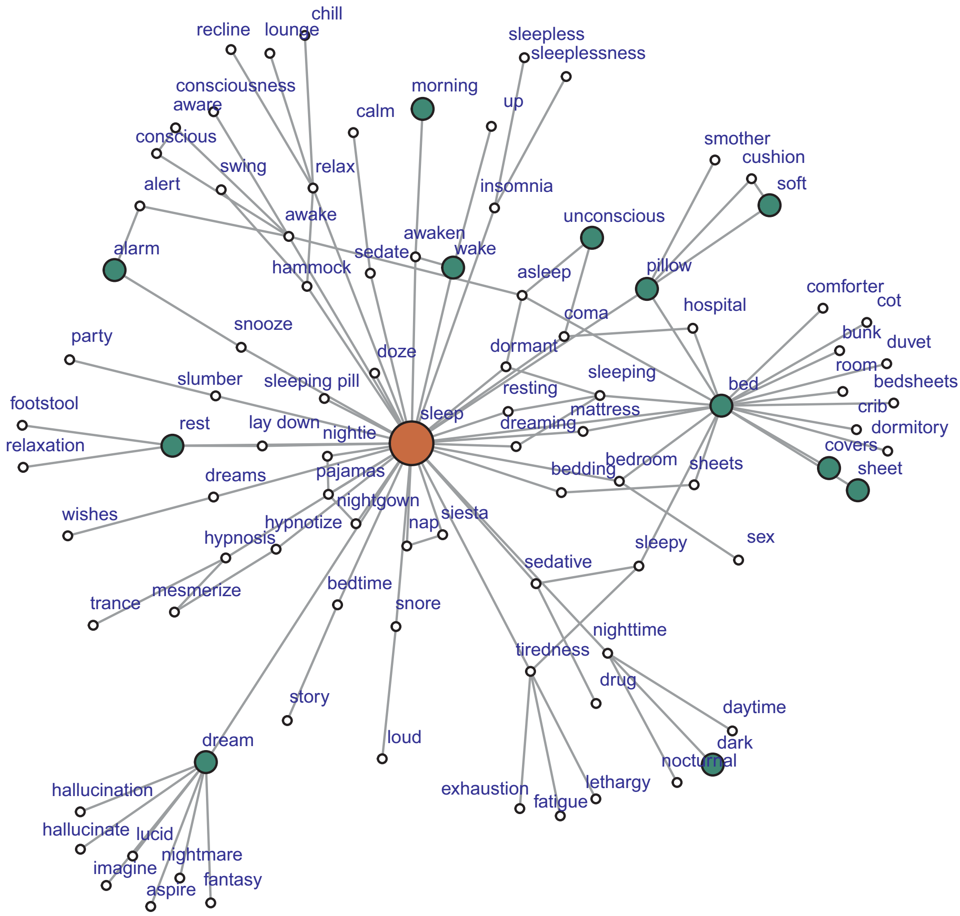

Figure 10.1 The semantic network around a target word for false memory induction. Words are shown from the Small World of Words free association norms that are within a path distance of two edges from the word "sleep." Edges are only shown for cue-response relationships produced more than 10% of the time. Lure words in the list provided in the main text are darkened and enlarged.

"bed." What the figure shows is that the words in the above list, chosen to create a memory of the target word, are also close associates of that word.

10.3.1 Phonological False Memories

Though the DRM false memory paradigm was initially developed for semantic information, Luce & Pisoni (1998) argue that much of the way we perceive words is influenced by the phonological communities in which they are embedded – that is, on similarities in sound. Phonological communities in language have fascinating network structure (Siew, 2013) and they also predict phonological false memories (Sommers & Lewis, 1999; Vitevitch et al., 2012).

Figure 10.2 gives an example of two-hop phonological networks around the words *fount* and *comma*. A phoneme is a basic unit of speech sound and the word *fount* can be classified as having four phonemes. These can be written in a computer readable

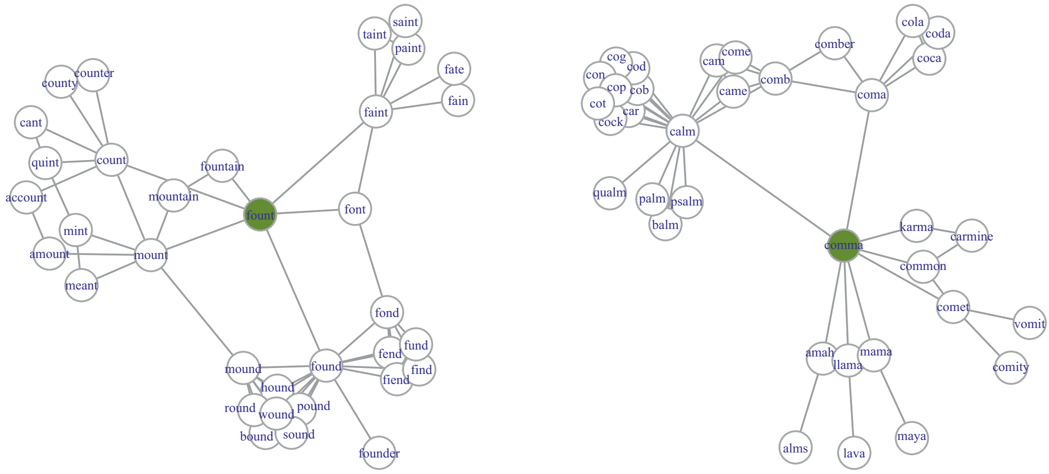

Figure 10.2 The phonological neighborhood network with 19,340 English words. Edges represent a Levenshtein distance of 1, meaning one phoneme addition, substraction, or substitution. Words and phonological translations are from the Hoosier Mental Lexicon (Nusbaum et al., 1984).

transcription as -f- -W- -n- and -t-. For the word *fount*, the -W- represents the sound *ow* as in *cow*. Using this approach, *about* becomes xbWt and *psychologist* becomes sYkalxJlst.

In the two-hop phonological networks, words share an edge if they are one phoneme away from their neighbor, either by substitution, addition, or deletion. This is called *Levenshtein distance*, or *string edit distance* (Levenshtein et al., 1966) and it is a common tool for determining distances between strings, which can then be used as edge weights. Here are two examples:

- *fount* and *mount* (mWnt) involve one substitution.
- *fount* and *fountain* are one addition or deletion away, each produced from the other by adding or deleting the -N- sound.

Once words are translated into their phonological equivalents, a network of neighbors like that shown here is easy to produce by placing edges between words that are a string-edit distance of one phoneme away. These neighbors represent a word's *phonological neighborhood* (Siew, 2013; Vitevitch et al., 2012).

If phonological memory is structured like this, then we can implant false memories by exposing people to a list of words that sound similar to the word we want them to recall. If the target word is a neighbor to all the words in the list, this effect will be even stronger. For example, hearing words like "chute," "got," "hot," "knot," "lot," "pot," "sheet," "shock," "shop," and "shut" will lead people to remember hearing the word "shot."

Vitevitch et al. (2012) showed this for the 30 target words shown in Table 10.3. Participants heard the cue words during a training phase. Later, they were shown a list of test words and had to decide whether they were "old" (i.e., studied in the training phase) or "new" (not studied). People tended to have false memories for words that sounded similar to those they heard in the training phase. Again, false memories were induced by experiences similar to those memories.

However, Vitevitch et al. (2012) also found something else. Target words with a lower clustering coefficient, C, were more likely to be falsely remembered. In other words, words in well-connected phonological communities were more resistant to false memories than those in more sparse communities. In a very simple sense, if something reminds you of many things, then it is less likely that any one of those things will be falsely remembered.

10.4 Modeling False Memories

Network science offers us more than just the ability to predict phenomena. We can also model their underlying processes and begin to understand why they work the way they do. This moves beyond simple prediction, where we say "I've seen this system at work before so I know what its going to do next," to saying "I know how systems like this work, so I know not only what it is going to do next, but why, and how we can change it in the future."

In the case of false memories, we can model this with two components: a network of relationships that explain how things are connected with one another, and a process for describing how activating one thing in the network activates other things nearby. Both of these are general tools for network modeling. The first – string-edit distance – allows us to build networks from any kind of strings (e.g., DNA, words, lists of ordered preferences, and so on). The second – spreading activation – is a hallmark of cognitive modeling.

10.4.1 The Phonological Network

By translating words into their phonological equivalents and placing edges between words that have a string-edit distance of 1, we construct a phonological network.

In the present case, we use the computer-readable phonological transcriptions of words described earlier: what insiders call klattese (Vitevitch & Luce, 2004). The Hoosier Mental Lexicon (Nusbaum et al., 1984) provides a dictionary of 19,340 words along with their phonological transcriptions. We can then compute the string-edit distance between all pairs of word transcriptions. Figure 10.3 shows the resulting phonological network with edges connecting all pairs of nodes for which the string-edit distance was 1. The 10,265 nodes shown on the periphery are isolates.

10.4.2 Spreading Activation

If we imagine a network of memories as being connected by springs, then we can visualize how the agitation of one memory would lead to the agitation of its closely connected neighbors. In cognitive science, this process of moving agitation along the network is called *spreading activation*. Activation spreads in proportion to the number of edges and their weights.

If activation facilitates retrieval from memory then we can see how spreading activation from a carefully chosen set of targets could lead to the retrieval of a false memory.

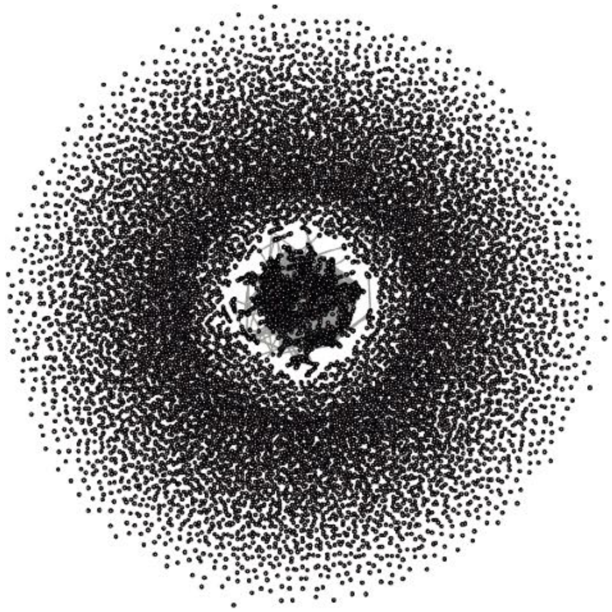

Figure 10.3 Spreading activation over a series of time steps. In the first time step, the activation is localized in the large central node. At each subsequent time step, the activation is split between the neighboring nodes. The retention of activation at each node, r, is set to 0, so each node shares all its activation with its neighbors.

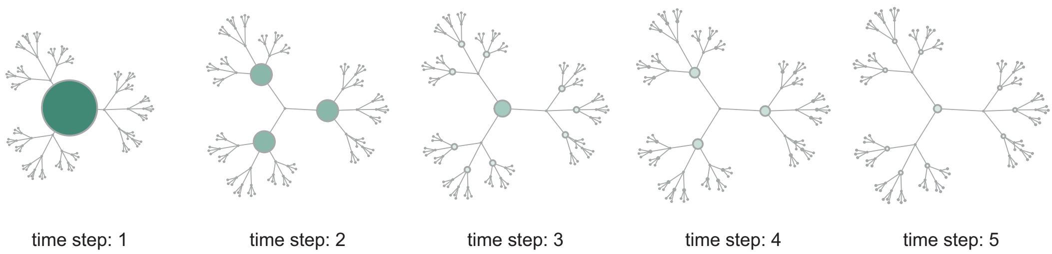

Figure 10.4 Two two-hop phonological networks around the words "fount" and "comma." Words share an edge if they are one string-edit distance away, meaning they are reachable either by adding, subtracting, or substituting one phoneme. Data to produce this table is from Siew (2019).

An R package developed by Cynthia Siew (2019a), called *spreadr*, simulates spreading activation over a network. The basic algorithm is as follows: At each time step, a node divides all of its activation, A, among its n neighboring nodes, each receiving activation equivalent to A/n. The network presented in Figure 10.4 gives a sense of how this works. Starting with activation at the large central node, with activation indicated by its size and color, spreading activation takes place at each subsequent time step. In this tree network, the central node has degree 3. Starting with activation $A = 100$ at the center node, each neighboring node will receive activation $A = 100/3 = 33.3$ at the second time step. In the third time step, the neighboring nodes that are now activated (each with degree 4) will split their activation, transmitting $A = 33.3/4 = 8.33$ to their neighbors. This will lead to the central node recovering 24.99 of its original activation. Nodes 2 steps away will have an activation of 8.33. At each time step the total activation over all nodes sums to 100.

This can all be modified, for example, by allowing nodes to retain some activation r or making the spreading activation proportional to edge weight.

10.5 Spreading Activation on Phonological Networks

Spreading activation simulated on phonological networks demonstrates how differences in the clustering coefficient, C, of target words are sufficient to explain differences in false memories (Chan & Vitevitch, 2009, 2010; Siew, 2019a; Vitevitch et al., 2012). As Vitevitch et al. (2012) explain:

> In the case of words with low C, activation spreads from the target word to the phonological neighbors. Because the neighbors are less interconnected, only a small amount of activation will circulate amongst the neighbors. The rest of the activation will spread from the neighbors back to the target word, and from the neighbors to other parts of the network. For words with high C activation again spreads from the target word to the neighbors. However, most of the activation will tend to circulate amongst the highly interconnected neighbors, with less activation spreading from the neighbors back to the target word and to the rest of the network. (p. 41)

This explanation implies that clustered nodes act as what I call *activation traps* (see the Open Questions). Activation traps are interesting in their own right, but to fully understand how they are working here we need to take a closer look. To do that, we will reconstruct the simulation from Vitevitch et al. (2012).

The 30 target words used in Vitevitch et al. (2012) were chosen so that 15 words had high clustering coefficients and 15 had low clustering coefficients, as shown in Table 10.3. Figure 10.5 visualizes the two-hop networks for each of the target words. These networks give us a concrete feel for the abstract structure of phonological memory and how clustering might influence spreading activation. But we will need the simulation to make any convincing arguments.

I also present the network metrics computed for these networks, shown in Figure 10.6. The degrees of the target words for the high- and low-clustering coefficient networks were chosen to be similar and the clustering coefficients were chosen to be different. This is confirmed in the figures. Somewhat visible in the network visualizations but shown more clearly in the metrics is that the number of nodes in the high C networks is higher than it is in the low C networks. The number of edges is higher as well. This provides additional pathways and locations to house activation away from the target node. Finally, the eigenvector centrality of the target words with high C is also higher. This would suggest that high C target words are likely to be connected to words of higher degree, consistent with the differences in edges and nodes.

These statistics support the explanation offered by Vitevitch et al. (2012). That is, when the clustering coefficient is higher, this increases the number of edges among neighboring nodes. These neighboring nodes also tend to be of higher degree, as indicated by the eigenvector centrality. With more nodes and more edges, the activation for high C targets is spread over more possible nodes, with less activation reaching the target.

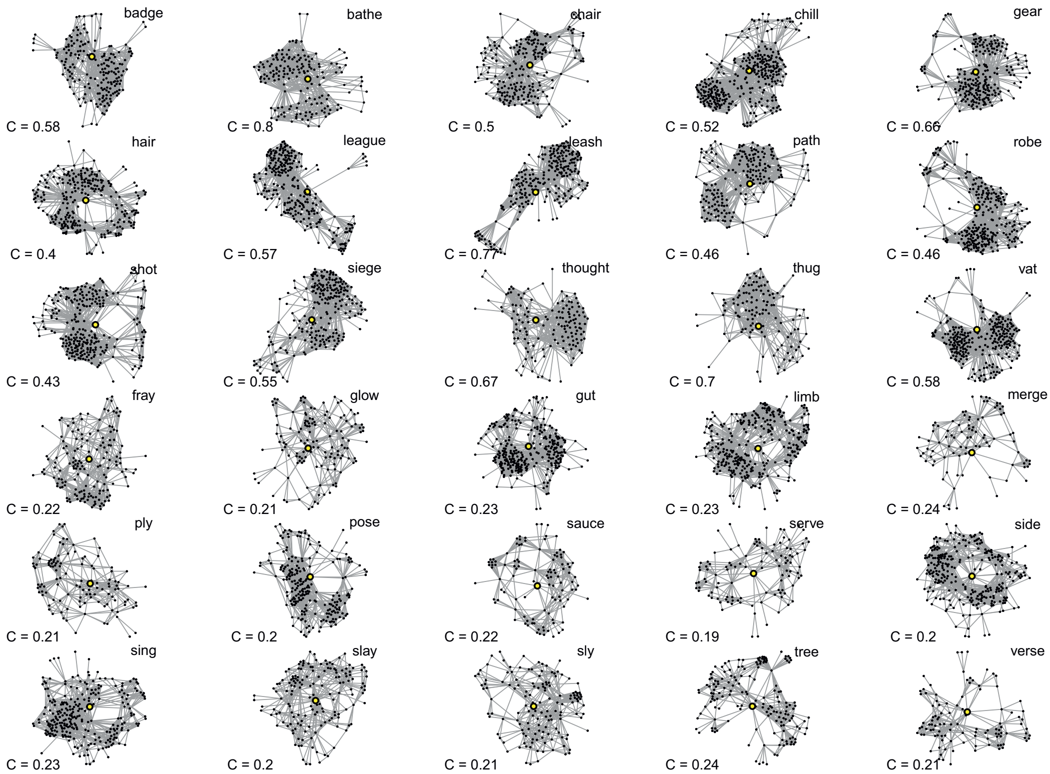

Figure 10.5 Two-hop networks for each cue in Vitevitch et al., 2012. The target node is named at the top right of each network and indicated by the larger central node. The local clustering coefficient for the target node is shown in the bottom left corner. The top 15 networks are the 'high' clustering coefficients targets.

This information is important. Clustering coefficient is a composite measure: it combines a number of underlying features like degree and the number of possible edges. By looking more closely at these features – adding them in, taking them out, modifying them, and so on – we can gain a better understanding of why clustering coefficient is important to false memory formation in this data. When we simulate it, as we will do, the picture becomes clearer still.

10.5.1 Simulating False Memories

Taking the networks provided earlier, we will simulate spreading activation from the cue nodes and track the activation in the target node over time. We are interested in how activation in the target node changes depending on its clustering coefficient, C. The target and cue words (represented by nodes in our network) are shown in Table 10.3. Note that the cue words have a string-edit distance of one from the target word, so they always share an edge.

In the simulation, I have chosen to use three-hop networks in order to allow activation to travel sufficiently far away from the target node. We will set the retention, $r = 0$, which

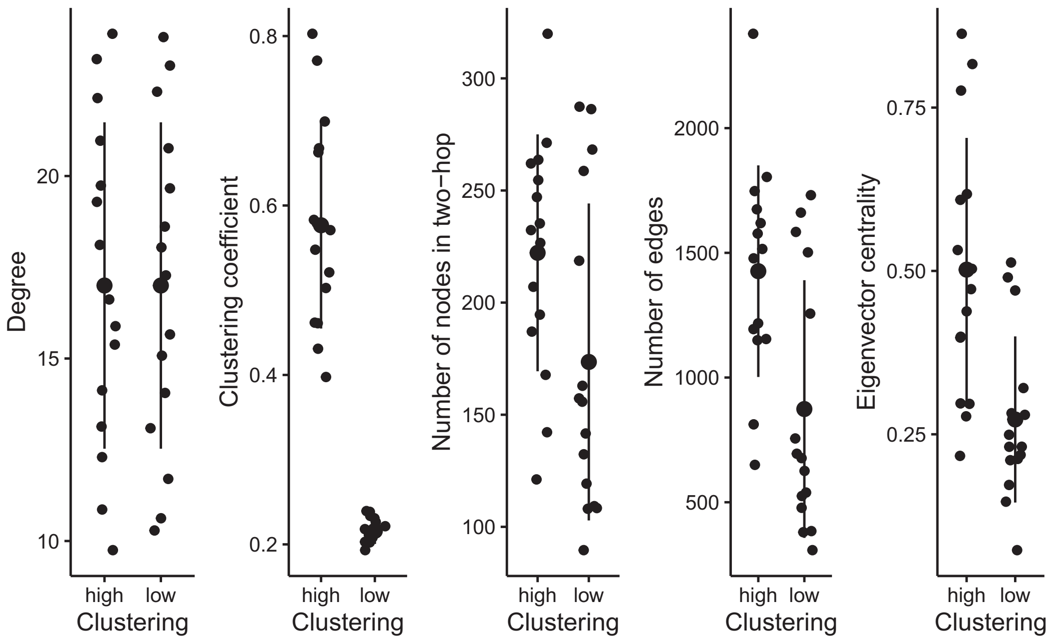

Figure 10.6 Network statistics for the two-hop networks around the 30 target words (represented by dots) in Vitevitch et al., 2012. Error bars are one standard deviation. Besides the deliberate difference in clustering coefficient, both frequentist and nonfrequentist (Bayesian) statistics support large differences in the number of edges and eigenvector centrality between the high- and low-clustering groups. For example, the Bayes Factors supporting the alternative hypothesis are both $BF > 10$ for the number of edges and eigenvector centrality (see the associated code).

means each node delivers all its activation to its neighbors. The number of time steps is set to 5 and we give each cue node a starting activation of 20.[1]

Figure 10.7 shows the cumulative activation across the different nodes over time. We can follow the process step by step. The thinnest line represents the target nodes, and these have their highest activation at the first time step and then rapidly fall to zero. The first-order cues transfer a large portion of their activation to the target node and

[1] Several cue words are missing from the 19,340 word phonological network. These are not symmetrically distributed across low and high words; *pear* is with *gear*, a high-clustering word, and *bray*, *flow*, *low*, *slow*, *pie*, *seed*, and *tea* occur across seven low-clustering words. Ideally, we want our simulations to reflect the symmetry in the original data as much as possible. The asymmetry of missing words (i.e., nodes) could be handled in at least four ways: (1) a random cue could be removed from all remaining target words so that the number of cues is equal across all targets, (2) an additional cue could be added to those target's missing words, (3) we can control for the number of cues in a regression, or (4) we can leave things as they are, noting that there is less total activation in the low-clustering networks on average and therefore we should expect less activation in the target node all other things being equal – then if we observe that high-clustering coefficient targets have lower activation we know it is unlikely to be an artifact of the fewer nodes. Ideally, one would do all of these and report and interpret the outcomes, whatever they are. Here, I've chosen option (2), replacing the missing words with phonologically identical words (e.g., replacing *pie* with *pi*), or (in the one case where this wasn't possible) with the nearest sounding neighbor to the target word (e.g., replacing *pear* with *peer*, *peace* with *pease*, and so on). *gear* also appears as a cue for itself in Vitevitch et al. (2012), and is replaced with the neighbor *leer*. (Thanks to M. Vitevitch, pers. comm.)

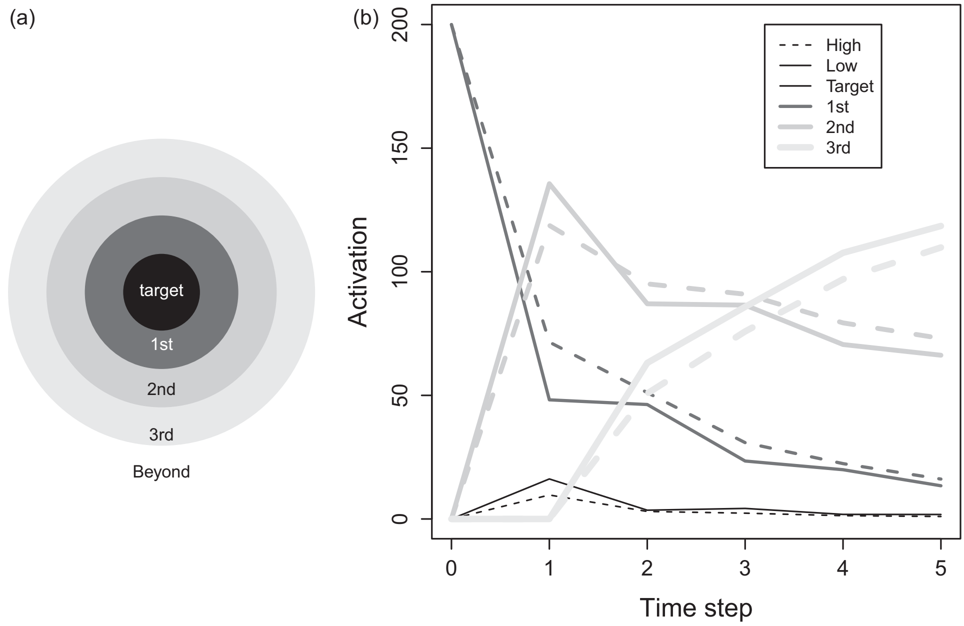

Figure 10.7 Simulating phonological false memories with varying clustering coefficients (high or low). The figure in panel (a) shows the relative locations and color indication of the target word, cue words (1st order neighbors with initial activation), neighbors-neighbors (2nd-order neighbors), and 3rd-order neighbors and beyond. The relative cumulative activation across these nodes is shown on the right. In panel (b), lines get thicker as they move away from the target. The high- and low-clustering coefficient are shown with dotted and solid lines, respectively.

2nd-order nodes. However, much is preserved among first-order nodes, and more is preserved among the high *C* networks (dotted line). This follows from there being more connections with other first-order nodes. Notably, this higher activation among first-order nodes is also preserved across the remaining time steps. The clustered nodes act as an activation trap, retaining activation for longer.

Second-order nodes gain activation at time step 1 but then slowly lose it over time to third-order nodes. Third-order nodes gain activation more rapidly for low *C* than high *C* networks. This again follows from the retention of activation among the highly clustered first-order nodes for high *C* networks.

Why is the activation initially higher in low *C* targets? All the cues are directly connected with the target, so at the first time step, the target's activation is the sum of the activation of its cues divided by their degree:

$$A_i = \sum_{j \in N_i} \frac{a_j}{k_j}, \tag{10.1}$$

where a_j is the activation of cue j, k_j is the degree of cue j, and N_i are the number of neighbor acting as cues for node i. The only way for low *C* target nodes to have higher activation is if their neighboring cues have lower degree, k_j.

Equation (10.1) tells us that the pattern we see in Figure 10.7 at time step 1 is the result of target nodes with higher clustering coefficient having cues with higher degree. Dividing their activation among more neighbors, the cues provide less activation to the target. Of course, many of these neighbors are other first-order neighbors, but that information is not relevant for Equation (10.1). This suggests that if a node has a higher clustering coefficient, then its neighbors have higher degree. Do they?

10.5.2 Do High C Cues Have Higher Degree?

Figure 10.8 shows that for the phonological network, nodes with higher clustering coefficients tend to have neighbors with higher degree. If cues of higher degree share their activation among their nodes, then the target will necessarily receive less activation. This

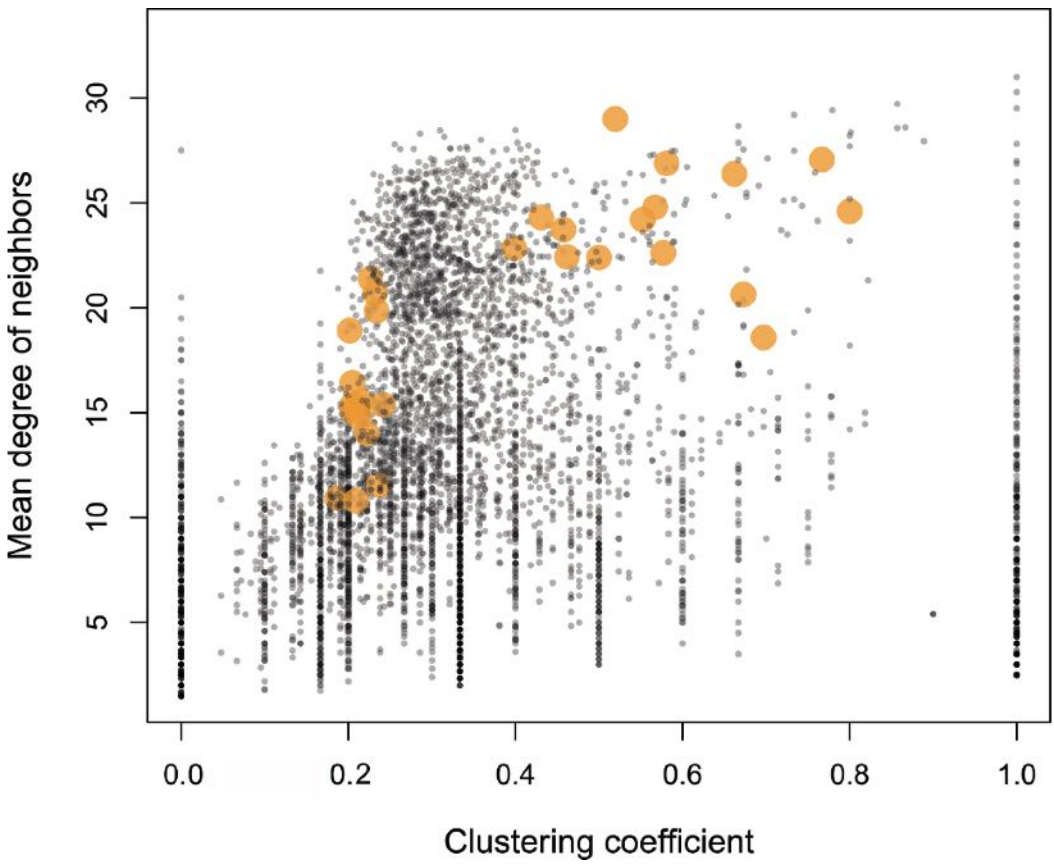

Figure 10.8 The relationship between a node's clustering coefficient and the mean degree of its neighbors. The general trend is positive across all nodes. The larger data points are the target words used by Vitevitch et al., 2012, showing those with higher and lower clustering coefficient. The target nodes with lower clustering coefficient also have lower average degree among their neighbors, meaning they divide their spreading activation among fewer neighbors, giving more activation to their low C targets.

should be true on average across all words in the Hoosier Mental Lexicon and possibly other networks as well.[2]

The figure also indicates which words were chosen by Vitevitch et al. (2012). The target words with higher C tend also to have cues with higher degree.

The relationship between neighbor's degree and clustering coefficient is true on average across all nodes, but the effect size is small ($R^2 = .02$). There is substantial variation here to choose target nodes with equal degree among their neighbors. This would mean that A_i would be equal for high and low C targets at the first time step. What happens after that?

10.5.3 Controlling for Neighbor's Degree

To see if controlling for neighbor degree would eliminate the high C lower activation finding, we can simulate the outcome of choosing target nodes with different clustering coefficients but cues with similar degree. To do this, we will take a slice of the distribution in Figure 10.8 with mean neighbor degree between 23 and 25 and simulate spreading activation over a series of time steps after choosing 10 random neighbors as cues. Among the $N = 242$ nodes in our slice, there is now no correlation between clustering coefficient and mean degree of neighbors ($p = 0.81$), making this a good test case.[3]

Figure 10.9 shows the resulting cumulative activation after five time steps. The difference in cumulative activation over the five time steps is not strong. The regression in Table 10.1 tells a more detailed story, and also allows us to control for the degree of the target's neighbors. At time step 1 there is no longer a correlation between clustering coefficient and activation. This is true even if we control for the mean degree of neighbors. Across all time steps, the effect of clustering coefficient on total activation is now statistically unremarkable.

These results offer several interpretations. Foremost, they suggest that clustering coefficient may not be playing a causal role here, but rather that it is an indicator of neighbors with high degree. Second, clustered nodes (k-cliques) should retain activation for longer (see Open Questions section at the end of the chapter). As a consequence, there may be additional impacts of neighborhood clustering that are relevant for memory activation that have not yet been explored.

10.6 Summary

Memory has structure. This structure helps explain and predict the influence of various memory effects, including elaboration effects and the formation of false memories. Phonological networks offer a useful framework for understanding these effects in fine detail. String-edit distance neatly solves the problem of edge construction and is also a general

[2] The relationship between clustering coefficient and neighbor's degree is likely to be a general property across many kinds of networks. Recall that a clustering coefficient of 1 means that all edges among neighbors are filled: this naturally requires a higher degree, all other things being equal. What Vitevitch et al., (2012) have therefore found is a property in phonological networks that potentially tells us about networks more generally.

[3] A more thorough investigation has much to do: one should look across all words, in different languages, across word lengths, and so on. Here we are taking only the first step.

Table 10.1 Activation of target node

	Dependent variable:			
	Activation		Cumulative activation	
	t = 1	t = 1	t = 1−5	t = 1−5
	(1)	(2)	(3)	(4)
C	1.672	3.059	−0.511	1.513
	(1.751)	(2.112)	(3.025)	(3.479)
Neighbor's <k>	−0.925***		−1.350***	
	(0.087)		(0.151)	
Intercept	31.940***	9.260***	50.569***	17.447***
	(2.226)	(0.741)	(3.846)	(1.220)
Observations	242	242	242	242
R^2	0.326	0.009	0.252	0.001
Adjusted R^2	0.320	0.005	0.246	−0.003

Note: *p<0.1; **p<0.05; ***p<0.01

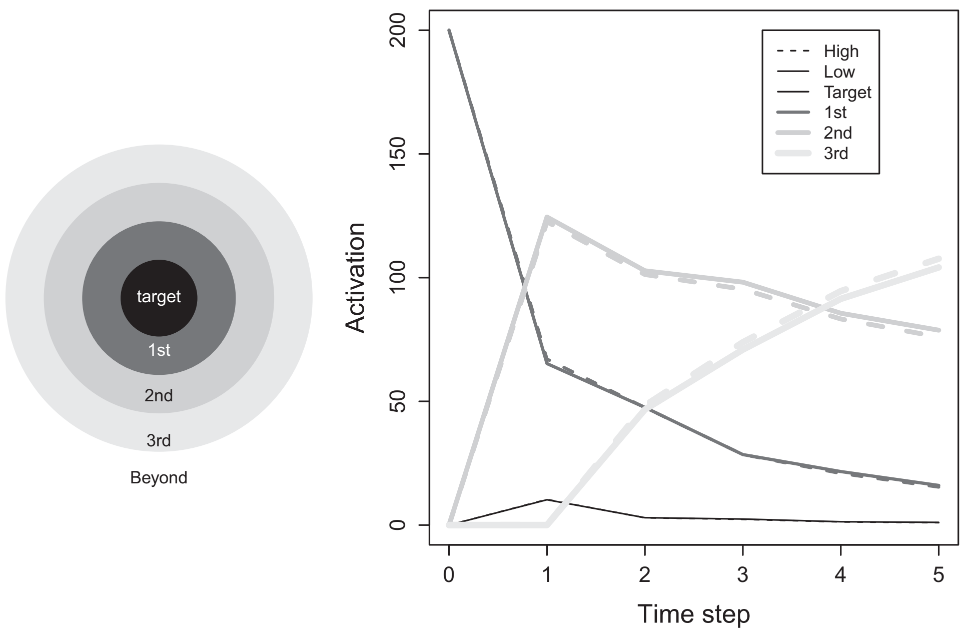

Figure 10.9 Simulating phonological false memories and controlling for the average degree of the target node's neighbors. This uses $N = 242$ nodes with neighbor degree between 23 and 25, but otherwise follows Figure 11.7.

method for building networks out of strings. By simulating spreading activation on these networks, we can see how false memories might form, and when and where we should expect to find them. What does this mean for real false memories? It may mean that false memories are most likely to be induced by cues that are primarily related to the target, but otherwise related to few other things.

10.7 Open Questions

10.7.1 Phonological Networks of Phonemes

This chapter explores one way to build networks out of strings, but there are many others. For example, given a list of words such as one might find in a piece of writing or in a multi-item recall task (e.g., "name all the animals you can think of"), words can be combined to form networks based on various rules regarding their colocation (Zemla et al., 2020; Zemla & Austerweil, 2018). Imagine how this might be applied to understanding phoneme structures in language. Phoneticians, phonologists, and typologists have exhaustively detailed the phonetic (sounds produced) and phonemic (sounds heard) diversity of the world's languages (e.g., Ladefoged & Maddieson, 1996) and the phonetic and phonemic inventories of specific languages (Dryer & Haspelmath, 2013). Study of this diversity led to the field of historical linguistics and the identification of "sound laws," such as *Grimm's law* (Grimm, 1822), which identified a number of structural explanations for regular patterns of sound change over time. One familiar example is the chain shift, such as the routine pattern of *p* sounds changing to *f* sounds: consider *paternal* and *fatherly* which come from the same root. This tradition has carried forward into William Labov's structural and sociolinguistic principles of sound change (e.g., Labov, 1994). Given that every word is a list of phonemes, how might one build a network from colocated phonemes to investigate how phonological colocations have evolved in language? For example, *bat*, whose phonology is described in the Hoosier Mental Lexicon as *b@t* might have links between *b* and @ and also @ and *t*. Done for many words, this would produce a phonological network indicating how phonemes organize themselves within a language. Given that words are the outcomes of language evolution, which responds to the communicative needs and limitations of its users, we should expect phonological relationships to be far from random (e.g., see Luef, 2022b). How could we explore this? What hypotheses could one form? Would these predictions be born out across languages?

10.7.2 Activation Traps

Communities of nodes that are well connected are potentially traps for activation. This should roughly follow a simple retention equation, R:

$$R = \frac{W}{W + O},$$

where W is the number of edges *within* the community and O is number of edges *outside* the community. This equation reflects the proportion of activation a community will keep within the community. It is also related to modularity, making it potentially a general property of networks that spread activation or other resources over the network. How might we compute this for high and low *C* nodes? Does it correlate with C across the network? Can we simulate the role of highly clustered nodes as activation traps and track their activation over time?

10.7.3 Negative Priming

Memory blocks occur when things are known but we are unable to retrieve them from memory. This is sometimes called *retrieval inhibition* and is commonly studied as a tip-of-the-tongue effect: when we cannot remember a name though we know we know it. Smith & Tindell (1997) studied retrieval inhibition using negative priming. Participants were given word fragments to complete and then either shown or not shown a negative prime. Table 10.2 shows a short list of examples. If you cover the left-hand column (Target), the negative prime inhibits your ability to complete the word fragments on the right. Without the negative prime, solutions are easier to find. How might we model this process of negative priming?

Table 10.2 Targets, negative primes, and word fragments from Smith and Tindell, 1997

Target	Negative prime	Fragment
ALLERGY	ANALOGY	A_L__GY
BAGGAGE	BRIGADE	B_G_A_E
CHARITY	CHARTER	CHAR_T_
VOLTAGE	VOYAGER	VO__AGE

Table 10.3 Target and cue words from Vitevitch et al., 2012. Each cue is a string-edit distance of one from the target, meaning one phoneme difference.

		Cues									
C	Target	1	2	3	4	5	6	7	8	9	10
high	badge	back	bad	bag	ban	bang	badger	bass	bat	batch	bath
high	bathe	babe	bail	bait	baize	bake	bane	base	bay	beige	lathe
high	chair	air	bare	care	check	cherry	fair	pair	rare	share	their
high	chill	bill	chin	chip	fill	hill	ill	kill	mil	pill	will
high	gear	beer	cheer	dear	ear	fear	leer	hear	mere	peer	rear
high	hair	air	bare	care	fair	head	hear	hell	pair	share	their
high	league	lea	leaf	leak	lean	leap	lease	leave	leg	legal	log
high	leash	lash	lea	leaf	leak	lean	leap	lease	leave	lied	lush
high	path	bath	math	pack	pad	pal	pan	pass	pat	patch	wrath
high	robe	aerobe	roar	rob	roe	role	rope	rose	rote	rub	road
high	shot	chute	got	hot	knot	lot	pot	sheet	shock	shop	shut
high	siege	cease	cede	sage	scene	seal	seam	seat	seek	seize	serge
high	thought	aught	bought	caught	fought	naught	sought	taught	thaw	thong	wrought
high	thug	bug	chug	dug	hug	jug	mug	rug	thud	thumb	tug
high	vat	at	cat	fat	hat	pat	sat	that	van	vast	vote
low	fray	fry	frail	frame	freight	gray	phrase	pray	ray	tray	brae
low	glow	blow	floe	glee	gloat	globe	glue	go	grow	lo	sloe
low	gut	but	cut	gait	get	got	gum	gun	hut	nut	shut
low	limb	dim	gym	him	lamb	lid	lime	limp	lip	live	slim
low	merge	dirge	emerge	midge	mirth	murk	myrrh	purge	serge	urge	verge
low	ply	fly	lie	pi	play	plea	plight	plough	ploy	pry	apply
low	pose	chose	hose	nose	pause	pease	pole	poor	pope	rose	those
low	sauce	boss	cease	loss	moss	saucer	saw	song	sought	souse	toss
low	serve	curve	nerve	salve	save	search	serf	serge	sieve	sir	verve
low	side	cede	cite	hide	ride	sad	said	sign	size	tide	wide
low	sing	king	ring	sick	sin	sink	sit	song	swing	thing	wing
low	slay	clay	lay	play	say	slate	slave	sloe	slain	stay	sway
low	sly	fly	lie	sigh	sky	slaw	sleight	slice	slide	sloe	spy
low	tree	free	t	three	tray	treat	trio	trow	troy	true	try
low	verse	curse	hearse	nurse	purse	vase	verb	verge	vice	voice	worse

11 Cognitive Foraging
Exploration versus Exploitation

11.1 The Problem

Is searching memory like searching space? William James (1890) once wrote that "We make search in our memory for a forgotten idea, just as we rummage our house for a lost object" (p. 654). Both space and memory have structure and we can use that structure to zero in on what we are looking for. In searching space, this is easy to see. A person hunting for their lost keys is not unlike a starling scouring the garden for wayward insects. But in searching memory, what is the map? And by what means does a person move from one memory to the next? In this chapter I will lay out the similarities between foraging in space and mind and then describe a research approach inspired by an ecological model of animal foraging. Using this approach, we will combine data from a memory production task with a cognitive map – a network representation – of memory derived from natural language. We will then use this to compare a suite of models aimed at teasing apart how memory search is similar to that of our garden starling.

11.2 Exploration versus Exploitation

At a fundamental level, every organism that can move has to make a decision about whether to stay where it is or go somewhere else. An animal that can move to known or unknown places has to decide whether it will stick to what it knows or explore the unknown. A foraging bumblebee has to decide whether to stay among the flowers it is currently in or fly to a new location. A foraging city dweller has to decide whether to revisit a familiar restaurant or to try something new. A researcher has to decide whether to continue with their current research project or abandon it for something else. Or, more narrowly, our attention has to stay focused on what is in front of it or look elsewhere. Managing these kinds of decisions are the bread-and-butter of healthy cognition.

The trade-off between exploration and exploitation touches every domain, from animal movement, to mating strategies, to the trajectory of human knowledge (Hills, Todd, Lazer et al., 2015; March, 1991; Mehlhorn et al., 2015; Mobbs et al., 2018; Todd et al., 2012). The prevalence of the exploration–exploitation trade-off has generated a diversity of terminology, such as *focused versus diffuse search* in visual attention, *high-temperature versus low-temperature search* in annealing problems, *depth-first versus breadth-first* search in artificial intelligence, *intensive versus extensive search* in animal foraging, and *local versus global search* in memory.

There are also many important nuances, such as distinctions between *directed exploration* and *random exploration*. For example, I can directly explore a particular option to learn more about it. Or I can randomly explore among many options because the set of potential alternatives is too large to search systematically (Gershman, 2018; Wilson et al., 2014). There are also heuristics for handling search in challenging domains, such as *satisficing* (find something good enough), *imitate the majority* (look where others are looking), and win-stay, lose-shift (if it works, keep doing it; when it stops working, try something else) (Worthy & Maddox, 2014).

This diversity of approaches and distinctions tells us that the exploration–exploitation trade-off is woven into our understanding of behavior. It also tells us that the basic strategy for solving this problem is surprisingly general: exploit when things are "good," explore when things are "bad." And, sometimes, explore when you don't yet know what "good" means.

This solution is also strikingly similar to a basic principle of psychology – Thorndike's *law of effect*: we repeat behaviors that have rewarded us in the past and avoid those that have not (Thorndike et al., 1927). Played out in space, the same rule describes a common animal foraging strategy called *area-restricted search*: focus on locations near those that have been successful in the past, explore outwards as success subsides (Dorfman et al., 2022). To a first approximation, every animal that moves – whether in space or mind – follows this logic (Hills, 2006).

Another observation this diversity highlights is that the difference between search in various domains is not really about who is doing it and where – rather, the differences amount to what it means for something to be "good." Most models of search behavior are completely agnostic with respect to whether search is happening in an open field, on the internet, or in one's mind.

Consider the *marginal value theorem* (Charnov, 1976), which has been successfully used to investigate everything from animal foraging to internet search (Hills et al., 2012; Pirolli & Card, 1999; Stephens & Krebs, 1987; Wolfe, 2013). The basic claim of the marginal value theorem is that individuals attempt to maximize the rate at which they get resources. What the resources are is irrelevant. There is only one decision: individuals either continue to exploit an existing solution, which is slowly getting worse (resources are *depleting*), or they can explore for something new.

The logic of the marginal value theorem is simple and general. When an animal is in a location – a patch – where it can collect resources, it collects them. The longer it stays in the patch, the more it collects. The rate at which it collects resources, however, worsens over time, like scraping the last scraps of popcorn from the bucket. The animal can always move, but it will take time to get the next patch.

The question the marginal value theorem answers is as follows: When should the animal leave a patch? Should it stay or should it go now?

Formally, the model looks like this:

$$R(t) = \frac{g(t)}{t + T}$$

What it says is that the overall rate, $R(t)$, is the amount of resources collected, $g(t)$, divided by the time it takes to collect them. The time is made up of the time in the patch, t, and the travel time it takes to move between patches, T. The goal is to find the optimal departure time t^* which maximizes the overall rate. The solution to this problem is to leave the patch when the rate in the current patch is equivalent to the long-term average across all patches.[1] In other words, leave a patch at the very moment when your instantaneous rate falls below what your long-run average would be if you went elsewhere, travel time included. Figure 11.1 provides a visual way to think about the problem and its solution.

The marginal value theorem makes numerous predictions. For example, it predicts that animals will leave patches even when they could do better in the short term by staying. That is, because animals should leave before patches are fully depleted, they could gain more in the short term by staying instead of traveling. However, most animals, including people, leave before things hit rock bottom (Stephens & Krebs, 1987; Wilke et al., 2009). The marginal value theorem also predicts that the quicker one can reach better pastures, the sooner one should leave. If the world is full of riches that are not far apart, an optimizing forager will abandon each of them quickly, skimming the cream off the top.

The marginal value theorem applies to any resource-providing environment that depletes over time and which requires some amount of time to move from one patch

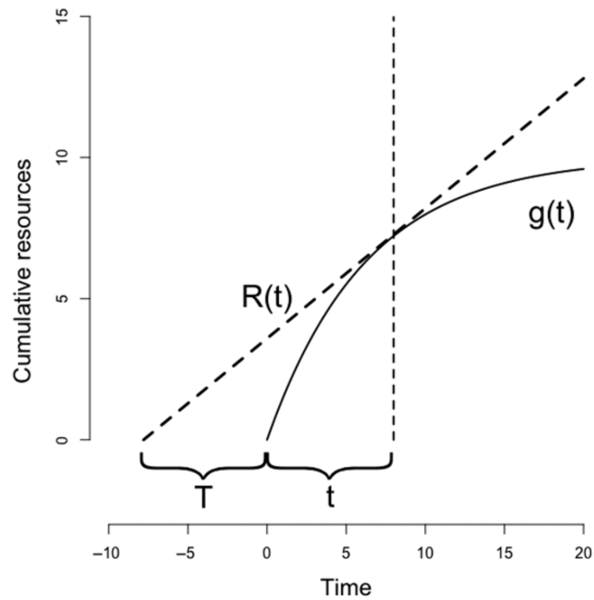

Figure 11.1 A geometric solution to the marginal value theorem. The travel time is added to the left of the origin. The time in the patch is added to the right. The optimal time in the patch is the time at which the marginal rate within the patch is the same as the overall rate across all patches including the travel time. That is the time that maximizes the slope of the dotted line, R(t).

[1] One finds the solution in the satisfyingly old-fashioned way by taking the derivative of the right-hand side, setting it equal to 0, and solving for t.

to the next. It is one of many such optimal foraging frameworks (Cohen et al., 2007; Daw et al., 2006; e.g., Houston & McNamara, 1999; Yechiam et al., 2005). However, the general point is the same: what matters is not what an environment is made of, but the capacity to navigate its structure effectively. Thus, the implications of these models apply to many things: vision, attention, information, holidays, doom-scrolling, and berry bushes (Pilgrim et al., 2021).

11.3 Searching in Mind

While studying how rats navigate mazes, Edward Tolman observed that they often took shortcuts along routes they had never taken before. Tolman suspected that the rats were putting memories together like puzzle pieces. This allowed them to build internal representations of their surroundings so they could envision pathways that they had never taken before. Tolman (1948) called these representations *cognitive maps* and argued that they provided a vast repertoire of behavior not conceivable under the standard theoretical frameworks at the time. In particular, Tolman speculated that this was a form of *vicarious trial and error learning*: the animals could learn from their own thoughts.

Today, we have a rich set of neurophysiological data revealing how animals search through their cognitive representations (Peer et al., 2021). Researchers now routinely record from single neurons in the brains of animals as they move through and "think about" their surroundings. The single neurons are chosen because they tend to fire when the animals are in specific locations. By observing the activity of these neurons when animals are deliberating, the researchers find that the animals string together mental locations into pathways through their surroundings. In other words, the animals are exploring possibilities in mental space without moving in physical space. This is called *episodic future thinking* (Pezzulo et al., 2014). More specifically, by activating hippocampal place cells that correlate with the animal's spatial experience, the animals can replay past experiences and explore routes they have never taken before (Pfeiffer & Foster, 2013). This confirms what Tolman could only hypothesize.

This view of mental search invites a map-and-vehicle framework like that shown in Figure 11.2. Memory structure implies there is a map. Searching that structure by some process assumes there is a vehicle. There is some *representation* and there is some

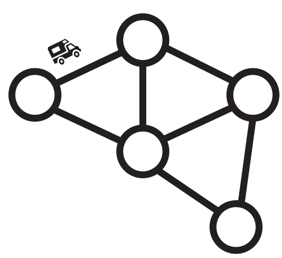

Figure 11.2 The map-and-vehicle framework is a general approach for considering how representations of information can be navigated by processes. This can generate memories, preferences, imagined futures, creative problem solving, recognition, and so on.

process that acts over that representation to produce behavior (Castro & Siew, 2020; Jones et al., 2015). This idea is prevalent, though often unspoken, in many cognitive models of the mind. One of cognitive science's ideals is to produce an account of both – of representation and process – and how the two interact (Estes, 1975; Johns et al., 2023). This distinction is important because the theoretical lens we take – whether we imagine something is a question of process or representation – can be strongly self-fulfilling (Hills & Kenett, 2022). Looking at both, we expand our hypotheses.

The prevalence of the map-and-vehicle framework is a testament to the importance of cognitive search. The shared structure of exploitation–exploration problems across domains, as well as the similarity of the neural architectures used to navigate them, suggests a shared evolutionary connection between search in mind and search in external space. In particular, internal search may be an *exaptation* – an evolutionary offshoot – of faculties initially adapted for external spatial foraging (Hills, 2006; Todd & Hills, 2020). We search our minds for ideas as we once searched our surroundings for food.

In Hills et al. (2012), we set out to track people as they foraged in memory. We did this by asking people to produce a list of all the animals they could think of. Then we constructed a variety of hypotheses about how they did this, including approaches that involved tracking people as they moved across internal maps of animal names. What we found was that people appeared to follow the logic of the marginal value theorem when searching their memory. In what follows I will spell out what we did and what we found, including throwing in a few new models and findings.

11.3.1 Category Fluency Tasks

If we apply the map-and-vehicle framework to memory search, then movement through memory is similar to moving via paths through a network. A network *path* is a set of edges that are connected through a series of shared nodes. To the extent that search in memory follows the structure of a network, we should be able to track the mental forager in action.

Identifying network paths requires asking people to retrieve a series of nodes in succession based on some relationship between them. One way to do this is to use a *category fluency task*. A category fluency task asks people to say all the things they can in a specific category. A fluency task is a kind of data dump: "tell me everything you know." As a consequence, it is a valuable tool for understanding knowledge structures and their access. They have been used to study things like the differences between experts and novices (Chi et al., 1981; Siew, 2019b, 2020; Siew & Guru, 2022), the semantic organization of children with cochlear implants (Kenett et al., 2013), mechanisms underlying mild cognitive impairment (Johns et al., 2018) and schizophrenia (Nour et al., 2023), and the structure of bilingual's mental representations (Borodkin et al., 2016; Taler et al., 2013).

One of the most frequently used fluency tasks in clinical practice is *the animal fluency task* (Thurstone, 1938). The animal fluency task asks people to "name all the animals" they can think of. Researchers often then use the number of animals produced in 60 seconds as an indicator of memory function. However, for a long time researchers have observed that people recall animals in clusters (Bousfield & Sedgewick, 1944): pets, African animals, sea life, and so on. This suggests that people are navigating a patchy internal environment. If we had a map of that environment, we could identify when people are moving from one patch to the next, much like a foraging animal.

11.4 Building Cognitive Maps

What is the best representation for studying memory? Many successful approaches have explored memory using free association norms (Abbott et al., 2015; De Deyne et al., 2013; Deese, 1959; Polyn et al., 2009; Steyvers et al., 2005). The assumption is that free association norms reflect a representation of the underlying semantic structure of cognition. However, on close examination, free association tasks are very similar to memory tasks. Indeed, the animal fluency task can be described as an iterated free association task. Its instructions could just as easily be the following: produce a series of animal names for which each animal name you say is always the first name that comes to mind after saying the previous animal name. In other words, free associate on your free associations. As a result, if we want to understand how cognitive processes access *cat* from memory after producing *dog*, noting that people tend to say *cat* after *dog* in a free association task is less an explanation than a redescription of what we already know. Free associations are incredibly valuable for theory construction, but not because they tell us how free associations are accessed (Jones et al., 2015).

Others have chosen to use models that derive representations from from natural language (Bhatia, 2017; Griffiths, Steyvers, & Tenenbaum, 2007). Natural language is simply unstructured language data like we would read in a textbook or hear spoken in a conversation. Arguably, it contains elements of cognitive processes, but these are less overtly similar to data generated in the animal fluency task. Moreover, it is also the data from which people develop their semantic memory representations which allow them to answer questions like “what comes to mind when I say *cat*?”

The most common method for deriving representations from natural language are *distributional semantic models*. These models build structured representations of meaning by exposing computational algorithms to vast amounts of natural language. Some early examples include BEAGLE (Jones & Mewhort, 2007) and Latent Semantic Analysis (Landauer et al., 1998). More recently, machine learning models involving trained neural networks have become popular, including examples such as word2vec (Mikolov et al., 2013), GloVe (Pennington et al., 2014), and BERT (Devlin et al., 2018). These methods vary in the way they extract meaning from natural language data, but they are all based on the common principle that patterns of word co-occurrence in language capture the semantic information necessary to infer what a word means (Kumar et al., 2022).

It is interesting to note that this is not done by simply connecting words in language with edges whenever they occur next to one another. The previous sentence would have connections between *not-done-by-simply* and so on. Doing this, one could also use a moving window that passed over the text, linking all words that co-occurred within the window. Done for a large corpus this would create a network of co-occurrences. Though this ultimately fails as a model of language learning (Pinker, 1979), it would capture the kinds of proximity relations found in natural language: *dogs* would wind up near *bones*, and *bees* would wind up near *honey*. This kind of relation is called *thematic* or *syntagmatic* (like *syntax*, meaning "put in order"). It is based on direct co-occurrence. Given a representation of direct co-occurrences one could infer a word's thematic context by looking nearby in the network.

But we are also interested in another kind of meaning. What is like a *bee*? What is like a *dog*? This is the kind of meaning needed to perform well on an animal fluency task, where the goal is to say things in the same category but that may never have been heard together in speech. This meaning is called *taxonomic*, *paradigmatic*, or sometimes *role similarity* (Estes et al., 2011). This kind of meaning is associated with substitutability: one kind of word can fill a similar role as another. *Wasps* are like *bees* and *cats* are like *dogs*. One can infer role relations from language but not necessarily because they occur together. One must instead look for words that share similar neighbors.

In network terminology, we say that words that have similar neighbors have *structural equivalence*. The sentences "a dog bites the man" and "a fly bites the man" reveal a structural equivalence between *dog* and *fly* even if the two words never appear together in language. People can learn both thematic and taxonomic relationships (Savic et al., 2022), and good distributional semantic models can deliver both.

If we have a vector representation of how a word co-occurs with other words – just as we have a vector of adjacencies between a node and its neighbors in a network adjacency matrix – then we can compute the similarity between word vectors to quantify their structural equivalence. This is effectively what all distributional semantic models do, in various ways. They each produce weighted edges between pairs of words indicating the degree to which they occur in the same language environments.

In Hills et al. (2012) we used the BEAGLE model developed by Jones & Mewhort (2007). In brief, BEAGLE is a *random vector accumulation model* that represents words with two vectors. One is an unchanging, randomly assigned high-dimensional vector that represents the word's features. The second is a memory vector that adds to itself the unchanging memory vectors of all the words it co-occurs with in a window of a fixed size. By accumulating information from the words it occurs with, a word's memory vector carries a signature of a word's history of use. With sufficient training, words with similar contexts take on similar memory vectors. Cosine similarity is then used to compute the relative similarity of memory vectors for different words. Cosine similarity measures the extent to which two vectors point to the same place. In the case of language models, cosine similarity is always a value between 0 and 1 because the count of word co-occurrences is always positive.

The BEAGLE representation for animals we use here was trained on a 400 million word Wikipedia corpus. After being trained on the corpus, cosine similarities were computed for all pairs of 765 animal names, which include those that our participants produced plus others. More recently, Zemla et al. (2020) produced a tool for studying animal fluency and networks with even more names. In our case, the resulting network was a weighted graph with 765 nodes. A subsection of the BEAGLE animal matrix is shown in Table 11.1. This nicely captures most people's intuitions: *cats* are most similar to *kittens*, and they are more like *dogs* than *squid*.

Table 11.1 A subset of the BEAGLE cosine similarity matrix.

	cat	dog	kitten	whale	squid	puppy	alpaca
cat	1.00	0.53	0.65	0.22	0.16	0.41	0.14
dog	0.53	1.00	0.49	0.26	0.21	0.63	0.19
kitten	0.65	0.49	1.00	0.25	0.18	0.46	0.17
whale	0.22	0.26	0.25	1.00	0.32	0.25	0.12
squid	0.16	0.21	0.18	0.32	1.00	0.13	0.15
puppy	0.41	0.63	0.46	0.25	0.13	1.00	0.13
alpaca	0.14	0.19	0.17	0.12	0.15	0.13	1.00

Table 11.2 Example participant data from the animal fluency task from Hills et al., 2012. Cosine similarity shows the similarity between the N and N+1 items.

Count	Name	Name+1	Cosine similarity
1	dog	cat	0.53
2	cat	lion	0.42
3	lion	tiger	0.57
4	tiger	bear	0.49
5	bear	monkey	0.42
6	monkey	panda	0.36
7	panda	cheetah	0.25
8	cheetah	jaguar	0.25
9	jaguar	horse	0.33
10	horse	coyote	0.44
11	coyote	bird	0.45
12	bird	koala	0.44
13	koala	flamingo	0.33
14	flamingo	giraffe	0.35
15	giraffe	elephant	0.53

11.4.1 Tracking Individuals in Mental Space

Table 11.2 shows the sequence of animals produced by one of the participants alongside the BEAGLE similarity rating for the word and the word produced after it (*n+1*). The average rating across the entire BEAGLE animal matrix is 0.14. Thus, most of the words produced by this participant are more similar than if the participant were sampling at random from memory space. This again points to structure in memory search.

One can also quickly see that Table 11.2 starts with familiar animals (like *dog* and *cat*) and then moves to less familiar animals. This suggests a potential role for frequency or typicality. *Frequency* can be measured by counting up the number of times the word is used in our Wikipedia corpus of natural language and it is a strong predictor of order of recall (e.g., Taler et al., 2020). *Typicality* is a measure of family resemblance: it captures how much an item is representative of members of its class. When asked to produce the first “bird” that comes to mind, few people say *ostrich* or *penguin*. They are much more likely to say a common bird, like a *robin*. Typicality is also a strong predictor in learning and recall (Rosch et al., 1976).[2] We will investigate both frequency and typicality in the models that follow.

Finally, we can also see that consecutive animals become progressively less similar over time. If we take similarity as a form of proximity, then items are becoming further apart the longer one searches memory. If this is happening locally, then patches of memory are depleting, and we can predict long-distance jumps to new patches.

11.4.2 Memory Is a Wave

If movement through memory follows a path through nearby items, then things that come to mind should be most similar to things that come to mind immediately before or after them. This wave can be seen in the similarities of animal names to their near neighbors in the sequence of retrieval. Figure 11.3 shows this wave by showing the average cosine similarities to the current item from items further backwards or forwards in the retrieval sequence. The peak in the center is analogous to a more focused search nearby in memory space.

This casual exploration of the data suggests two things. First, people appear to use the local structure of memory to search, producing clusters of items that are nearby in semantic space. And, second, global features like frequency – which in this case can

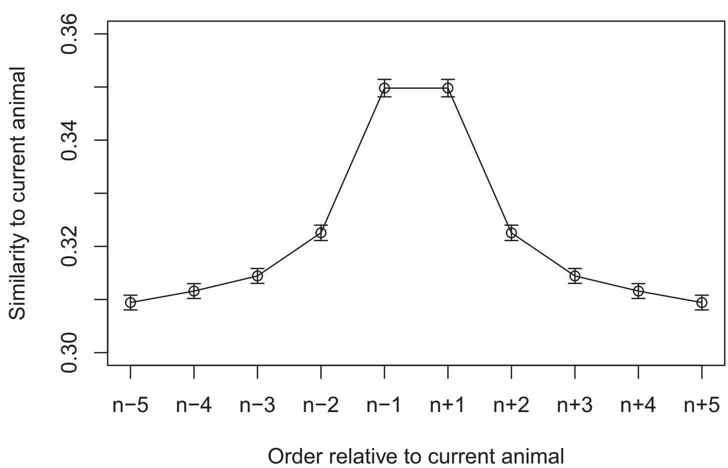

Figure 11.3 Memory is a wave. A word is most similar to the words produced nearby. A word two items back is more similar to the current word than a word three items back, and so on. Similarity is computed using the BEAGLE similarity matrix.

2 More typical examples of a class might also be considered more prototypical, in the sense that they represent ideal or averaged versions of their class. A long-standing controversy in psychology asks whether mental representations are *prototypes* or *exemplars*, with prototypes resembling something more like averages or ideals and exemplars representing specific instances. That is, when you think of a dog, what do you think of? An average dog? Or a specific instance of a dog? Most contemporary research suggests the answer is some of both and it may vary by person (Bowman et al., 2020; Nosofsky, 1986).

be classified as a node attribute – may also play a role. People tend to produce more familiar items earlier. We can formalize these in various models by mixing and matching processes to determine which best explain the data.

11.5 Modeling: The Value of Being Precisely Wrong

George Box once noted that "All models are wrong, but some are useful" (p. 2, Box, 1979). Or, as Paul Valéry put it, "The simple is always false. What is not [simple] is unusable" ("Ce qui est simple est toujours faux. Ce qui ne l'est pas est inutilisable"; Valéry, 2020, p. 107).

Above all, we want our models to be useful, and they achieve this in part by being simple. If they are simple and precise enough to follow Pauli's dictum of being "good enough to be wrong," then we stand to learn something. Even better is to have a collection of models, which tell us what matters and by how much. Indeed, the most informative models are often the ones we thought would work better than they do. These truly put guardrails on our ignorance.

In addition, good models should also be parsimonious – meaning they are as simple as they can be but no simpler. They are necessary *and* sufficient.

In general, useful parsimonious models will have the following features:

- They should be plausible. That is, they should reflect what we currently understand about the system that we are modeling. This is parsimonious with respect to existing theory.
- They should provide a good quantitative fit. To do that, they will need to be sufficiently well-described to measure how bad they are. Then one can choose models that minimize the error, given appropriate penalties for model complexity. Models that win this competition are parsimonious with respect to the data.
- They should explain the behavior from more or less first principles. This is parsimonious with respect to the process. The model should try to avoid sweeping complexity under the rug, but instead explain *how* the behavior comes about.

Any model that has these features is moving toward being useful. The modeling framework I describe next allows us to apply this framework to memory search.

11.6 Models of Mental Search

The Search of Associative Memory (SAM) model is a cognitive model developed by Raaijmakers and Shiffrin (1980) that aims to explain the process of retrieving multiple items from memory in series. In general, SAM assumes that memory search is achieved by accessing memory with cues which then activate memory representations that are similar to those cues. Though SAM has a specific form in its original description, here I will take it as a general framework for testing different assumptions about search through a network representation of memory. Starting from a simple framework, I will build toward more complex models. As it turns out, the original model proposed by Raaijmakers and

Shiffrin remains one of the best explanations of memory search. It also has the properties of area-restricted search in memory, making it a useful framework for comparison with animal foraging models.

To understand the model, it is useful to start with a concrete example. Let the cue be the category of "animals" and the representation be frequency. The cognitive process in SAM then selects a word in proportion to its frequency, $P(w_i) \propto f(F_i)$. Placing this into a Luce choice axiom (Luce, 1959) – a probabilistic choice rule similar to the *softmax function* introduced in Chapter 5 – we have the following:

$$P(w_i) = \frac{F_i}{\sum_{k \in W} F_k}.$$

In plain English, this means that the probability of word i, $P(w_i)$, is equal to the frequency of the word, F_i, divided by the sum of the frequencies for all words. This is indicated by summing across F_k for all $k \in W$ (which reads, k in W), where W is the list of all words. We can add a sensitivity parameter, β, which allows us to increase or decrease the importance of relative frequency in the retrieval process. This gives us the following:

$$P(w_i) = \frac{F_i^{\beta}}{\sum_{k \in W} F_k^{\beta}}.$$

If $\beta = 0$, then the model chooses randomly with respect to frequency. As β increases, the probabilities for the most frequent items grow larger compared to less frequent items. If β is negative, less frequent items become preferred relative to more frequent items. Using the maximum likelihood method, we can fit the β to the data for each participant, determining their relative reliance on frequency.

We can replace the frequency, F, with any metric, M, we hypothesize to be relevant to the search process. We can then write the model in a generic form as:

$$P(w_i) = \frac{M_i^{\beta}}{\sum_{k \in W} M_k^{\beta}}.$$

This approach levels the playing field for comparing different measures with respect to one another. Whatever pet hypotheses we may have, we can test them against one another by first optimizing the model parameters to minimize the error, and then comparing the error penalized for model complexity.

As described in Chapter 5, with each model's probabilities for the observed data, we can write the log-likelihood function as follows:

$$\hat{L} = \sum_{i}^{n} log(P(w_i)).$$

The goal is to maximize the log-likelihood, $\hat{L}$.[3] This will be zero if the model fits the data perfectly. Otherwise it will be negative, because we are adding the log values of numbers

[3] To get the probability of the observed data, we multiply the probabilities. Because probabilities are less than 1, multiplying probabilities will rapidly converge to numbers very close to 0. We can avoid this by taking the log, so that we can add the probabilities, which gives us the log-likelihood.

between 0 and 1. The goal is to find the negative value closest to zero (i.e., to maximize it). For the generic models noted earlier, this first means finding the β that maximizes $\hat{L}$. Models that produce different likelihood values can then be compared alongside one another. Note that if any model ever produces a probability of 0, this immediately makes the model infinitely unlikely: $log(0) = -\infty$. It is common practice to avoid this in maximum likelihood models by adding a small constant amount to values of *M* so there are no zero probability items. This has the same outcome as adding a small amount of noise to the process, such that the least likely items always have some nonzero probability of being chosen. If these values are never chosen, the β value will increase to compensate for the added noise. No harm done.

Using methods like the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) we can compensate for more complex models. This is important because more complicated models have more degrees of freedom and will, in almost all cases, fit the data better than less complicated models. For example, the most complicated models for our memory search data might have an individual parameter for each word produced, fitting the data exactly. This would overfit the data and tell us nothing about the underlying process. The penalties applied for AIC and BIC help compensate for models that are too ambitious by penalizing them for the number of parameters, *k*. Here we will use the BIC, which scales the penalty per parameter in proportion to the number of data points, *n*, the parameter is used to fit, as follows:

$$\text{BIC} = k \ln n - 2\hat{L}.$$

The BIC tends to be more conservative, favoring models with fewer parameters than AIC (see Chapter 5 for more details about AIC and BIC).

In summary, each model will produce a set of probabilities for the data that we observe. Using the maximum likelihood method and the log-likelihood function, we find the models' parameter values that maximize the probabilities. Then we can compare models by evaluating their log-likelihoods using the BIC to penalize models with more parameters.

Given this framework for model comparison, we will let the models proliferate. I produce a number of different models in the following sections. These provide some methodological richness to how different processes might search a network. Readers less interested in the model building can skip ahead to the "Model Results" (section 11.6.5), where all will be made clear.

11.6.1 Single-State Models: Global and Local Models

Single-state models meet each additional data point with the same assumptions. Thus, it could be that each retrieval from memory is like reaching into a bag and pulling out an item, with the probability of choosing an item based on some feature of the item (e.g., frequency or centrality). This data-generating model is unchanged throughout the course of production. This model could be based on *global* cues (like typicality), which do not consider "where" search is in the spatial representation, or it could be based on *local* cues (like the last item retrieved), which assume some form of structural nearness and farness.

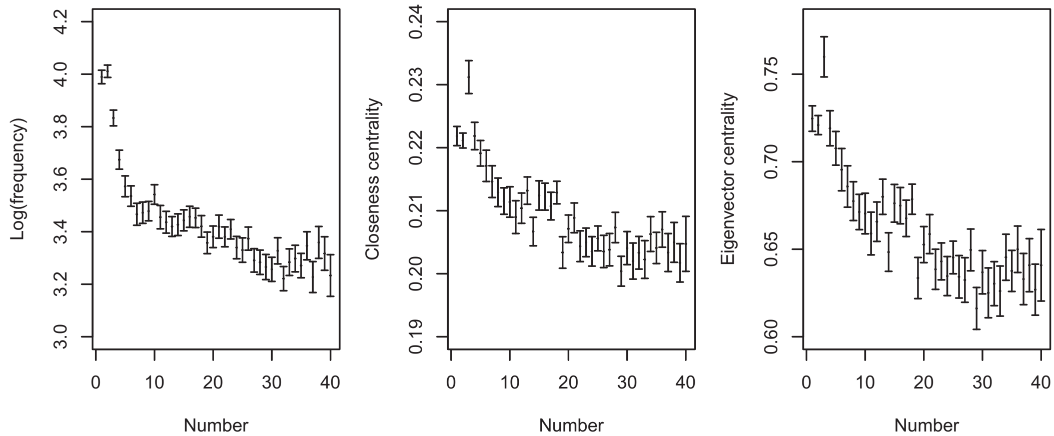

Figure 11.4 The frequency, closeness centrality, and eigenvector centrality by order of production for animals produced in the animal fluency task. This figure and subsequent data figures were constructed from the 141 participants in Hills et al., 2012. The mean number of animals produced per participant was M = 34.4 (SD = 6.2).

That memory is a wave gives us a motivation for testing local cues. Figure 11.4 shows that for each of the global cues listed (frequency, typicality, and eigenvector centrality) we have motivation as well. Items produced earlier tend to have higher values of frequency, typicality, and eigenvector centrality. As a first hurdle, we would like to establish whether or not local structure is important when considering memory search. After examining the individual cues, we will also combine the best of these in a combined cue model.

11.6.1.1 Frequency

Assume that participants sample items from memory in direct proportion to the frequency with which they have encountered those items in the past. This has been proposed in other models of memory (Anderson & Schooler, 1991; Murray & Forster, 2004). Frequency is a *global* search policy; it does not use local information to search memory. It treats all words independently without regard to how close or far they are from the last item produced: sampling one item does not increase the likelihood of sampling another nearby in semantic memory. The maximum likelihood formulation for this is as written earlier, but which I repeat here for convenience:

$$P(w_i) = \frac{F_i^{\beta}}{\sum_{k \in W} F_k^{\beta}}.$$

$F(w_i)$ is the frequency of word i and β is the sensitivity parameter. Frequency distributions are often long-tailed, with few items having large frequencies and most items having very small frequencies. It is common to take the *log* of long-tailed distributions as this produces a better fit to the data, which tells us something about scaling in cognition as well (Kello et al., 2010). The *sensitivity parameter*, β, is fit to each individual participant, as it is for all measures given in the remainder of the chapter.

11.6.1.2 Typicality

Typicality can be defined as the psychological tendency for some members of a category to be more representative of that category than others (Rosch et al., 1976). Typicality is a structural variable and it can be operationalized in a network as the average nearness of an item to other members in the network. Here we compute the normalized closeness centrality, c_i, which is 1 over the average distance to other nodes. For distance we use 1 over the similarity, $S_{i,j}$.

$$c_i = (N-1)/\sum_{j \in W} \frac{1}{S_{i,j}}.$$

Placing this into our modified choice rule, we have:

$$P(w_i) = \frac{c_i^\beta}{\sum_{k \in W} c_i^\beta}.$$

11.6.1.3 Eigenvector Centrality

The eigenvector values for a node provide a measure of how well-connected a node is to other well-connected nodes. Nodes that are well connected to other well-connected nodes have the highest eigenvector values. Griffiths, Steyvers, and Firl (2007) explored the directional eigenvalue (PageRank) for free recall by asking participants to produce the first word that came to mind that started with a specific letter of the alphabet. PageRank based on free association norms (Nelson et al., 2004) was an excellent predictor, outperforming both associative frequency (how often an item was produced as an associate – its indegree) and word frequency (how often a word is used in natural language). As our BEAGLE matrix is undirected, we compute the eigenvector centrality, x, for each node.[4] This provides another *global* variable for comparison and is inserted into the SAM model as follows:

$$P(w_i) = \frac{x_i^\beta}{\sum_{k \in W} x_i^\beta}.$$

11.6.1.4 Local Similarity

The local similarity model assumes that an individual searches for the next item in memory using the weighted edge similarities from the previously produced item. If a person produces word w_j and then w_i, the weighted edge between the two nodes, $S_{j,i}$ – taken from the similarities produced by BEAGLE – represents the semantic similarity between two words. Comparing this with all possible words that could be produced, we have the probability of producing word w_i after w_j:

$$P(w_i|w_j) = \frac{S_{j,i}^\beta}{\sum_{k \in W} S_{j,k}^\beta}.$$

[4] Some eigenvector values are 0 and produce 0 probability outcomes. To avoid the problem of producing infinitely unlikely models, the minimum eigenvector centrality is set to 0.01.

Table 11.3 Median Betas and improvement in BICs relative to a random model. Higher BIC improvements indicate the preferred model.

Model	Beta	(SD)	Δ BIC	(SD)
Random model	–	–	0	–
Single-cue models				
Frequency (global)	8.47	1.98	74.2	20.02
Eigenvector centrality (global)	4.82	1.23	49.21	15.59
Closeness (global)	7.43	1.54	45.3	14.32
Similarity (local)	4.34	0.89	74.2	23.65
Combined-cue model				
Frequency (global) +	6.15	2.14	97.59	27.76
Similarity (local)	3.23	1.08	–	–
Dynamic relaxation model				
Frequency (global) +	5.83	2.12	96.1	27.57
Similarity (local) +	3.25	1.08	–	–
λ	−1.67	0.91	–	–
Residual proximity model				
Frequency (global) +	5.83	2.12	95.07	27.55
Similarity (local) +	3.25	1.08	–	–
λ	2.37	6.74	–	–
Similarity drop				
Frequency (global) +	5.83	2.12	101.72	28.1
Similarity (local)	3.25	1.08	–	–

$P(w_i|w_j)$ indicates the conditional probability of producing w_i conditional on (indicated by $|$) having just produced w_j. Thus, the next item is always conditional on the previous item. Because the first item does not have a previous item, we will let the first item be chosen based on the best of the global models. This is important since model comparison requires that the models all be fit to the same number of data points. If we omitted the first item, the local models could outperform the global models simply because they fit less data and therefore accumulate less error.

We have not described all the models yet, but it is useful to see where things stand before moving to the combined models. The combined models will be a composite of the best single models. Table 11.3 shows the BIC improvements, ΔBIC, for each of the models, revealing the extent to which each model improves on a random model (with $\beta = 0$). Among the single-cue models, frequency and similarity are tied for being the best fitting model, and both are improvements over eigenvector centrality and typicality (closeness). Eigenvector centrality and typicality are nonetheless both better than a random model. The median β for participants is also positive across the board, which tells us that the higher the measure the more likely an item is to be produced. Together, this suggests that similarity and frequency might both be involved.

11.6.2 Combined-Cue Model

Taking the best two single-cue models, frequency and local similarity, a combined model can be constructed as follows:

$$P(w_i|w_j) = \frac{S_{i,j}^{\beta_s} F_i^{\beta_f}}{\sum_{k \in W} S_{i,k}^{\beta_s} F_k^{\beta_f}}.$$

This takes the product of the two measures, each modulated by their own sensitivity parameter. This could be expanded further. The main problem with adding more variables is that optimization over multiparameter models is limited by the amount of data. Here we have about 30 words per participant, each word representing an individual data point. There are different suggestions about the number of data points needed per model parameter, which range from 5 to 1 to 20 to 1, and so on (Kenny, 2015). In practice this is often a pragmatic question which depends on multiple constraints. Here we will limit our model to two parameters.

Table 11.3 shows that the combined cue model with frequency and similarity is an improvement over the single cue models. This is a comforting result as it suggests that frequency and similarity each add novel predictive power over either alone.

11.6.3 Multistate Models: Jumping from Patch to Patch

Multistate models change in response to the environment or recent success. Area-restricted search is a multistate process, which changes its search strategy based on recent successes. The marginal value theorem is similar in that it requires animals to transition between local and global foraging behaviors in response to the environment. For comparison, Lévy flights, which sample directions and path lengths always from the same distributions, are not state dependent. They treat animals as if they have no memory and are therefore effectively always lost (see Open Questions).

If memory search is like patch foraging, then it should be a multistate process. Taking our inspiration from the marginal value theorem, people should search locally within patches but travel between patches when local patches become depleted. In the following section I describe the state-dependent search model outlined by Hills et al. (2012). In that model, we showed that individuals tended to leave patches in memory when the rate of retrieval for the last item was approximately equal to what was predicted by the marginal value theorem.

11.6.4 What Is a Patch in Memory?

Prior work set out specific animal categories for the animal fluency task, such as pets and Arctic animals. That work focused on the clinical insights that the fluency task could provide by considering when people switch between categories. The categories were hand-coded based on visual inspection of participant data (Troyer et al., 1998). We can assume that these categories are patches and treat jumps between patches when people leave one category for another.

However, subsequent work found that patch strategy in memory is more nuanced. That is, it is unlikely to be determined by individuals traveling to predefined *categorical* patches such as those described by Troyer et al. (1998). For example, if a person produced DOG and then CAT, we might assume they are in a pet category. But if they then produce LION, did they switch?

An examination of different patch models supports a more *associative* definition (Hills, Todd, & Jones, 2015). Associative patches do not have a jump between CAT and LION, because CAT and LION have fairly high similarity. If we only use associative information to search memory, then we do not need to leave *pets* to move from CAT to LION. Fixed categories are irrelevant. What matters is how hard you have to look for the next closest thing.

Model comparisons between categorical and associative patches found that jumps between patches occurred when people found themselves in associative dead ends, not because a category in memory had been exhausted. This suggests that patches in memory are a very fluid concept. Memory search is like searching with a flashlight. After each item is retrieved one looks around for something nearby. If nothing is nearby, one jumps to another place on the map.

This is what we called a "skyhook" in Hills and Kenett (2022). It picks up the searching vehicle and moves it to a new and hopefully richer location on the map. These locations are detected using the similarity drop model described in Section 11.6.4.3. It is based on an algorithmic rule for identifying patch jumps after they occur. However, there is no reason to accept the skyhook as canon. Perhaps we can do better. Before presenting the similarity drop model, I will first develop two more fluid alternatives for comparison: the relaxing frequency model and the residual proximity model.

11.6.4.1 Relaxing Frequency Model

The *relaxing frequency model* has a tunable weight that can turn up or down the relative influence of frequency with respect to similarity. It follows the observation one can infer from Table 11.2 that higher-frequency items might follow low-similarity transitions. This model is state dependent because it changes its relative contribution of similarity and frequency depending on its last retrieval. Here I implement one of many ways to achieve this. The relaxing frequency model tunes the β on frequency as a function of the previous two items' similarity to one another, as follows:

$$\beta_f = \beta_b e^{-\lambda S_{-1}}.$$

The similarity of the previous jump is indicated by S_{-1}. It is larger when the words just produced were more similar to one another. The influence of this similarity is controlled by λ. When λ is positive, higher similarity items reduce the influence of frequency on the next sample from memory by lowering the sensitivity parameter, thus reducing the influence of frequency. When the items were less similar, as they would be after a long jump, frequency becomes more important. When λ is 0, this resolves to the combined-cue model with $\beta_f = \beta_b$. The output of this tuning, the sensitivity parameter β_f, is then incorporated into the combined-cue model presented earlier as the sensitivity parameter on frequency.

11.6.4.2 Residual Proximity Model

The residual proximity model is based on the observation that global transitions tend to take place at dead ends in memory. As Hills et al. (2012) observed, long-distance jumps in memory were more likely to take place when the last item produced was on average far from other items. This distance from all other items was called residual proximity because it is the distance to what remains to be retrieved. This is the same as the closeness, c, described earlier for typicality, but only includes items remaining in the set to be recovered. If we tune the sensitivity to frequency as a function of the residual proximity of the previous word, c_{-1} – in a similar fashion to what was done above for similarity – then we have:

$$\beta_f = \beta_b e^{-\lambda c_{-1}}.$$

This is then substituted into the combined-cue model, as the sensitivity parameter on frequency.

11.6.4.3 Similarity Drop

The similarity drop model is based on what the phenomenology of a patch switch should look like. Suppose an individual retrieves four items: A, B, C, and D. If a patch switch occurs between two middle items, B and C, then the similarity between B and C should be lower than the similarity between A and B. A and B were the last items in the old patch. B and C should be lower than the similarity between C and D, because C and D are the first items in the new patch.

These similarity drops are identified using the following algorithmic rule, which looks ahead in the data to identify jumps:

$$S(A,B) > S(B,C) < S(C,D).$$

Whenever this rule is satisfied, an individual uses the global frequency cue to jump to a new location in memory, briefly leaving the local similarity surface.

Formally, this is written as follows:

$$P(w_i) = \begin{cases} \dfrac{F_i^{\beta}}{\sum_{k\in W} F_k^{\beta}}, & \text{if S(A,B) > S(B,C) < S(C,D).} \\ \dfrac{S_{i,j}^{\beta_s} F_i^{\beta_f}}{\sum_{k\in W} S_{i,k}^{\beta_s} F_k^{\beta_f}}, & \text{otherwise.} \end{cases}$$

This equation tells us that memory search uses the local proximity cue unless a similarity drop occurs, and then the searcher uses only the global frequency cue to move to a new location, where it starts the local search once again. Similarity drop is flawed in that it always assumes patches contain at least two items.

11.6.5 Model Results: Fortune Favors the Flipper

Table 11.3 shows the BIC for the dynamic relaxation, residual proximity, and similarity drop models. Both dynamic relaxation and residual proximity are worse than the combined cue model, which has fewer parameters. The BIC penalty here is approximately

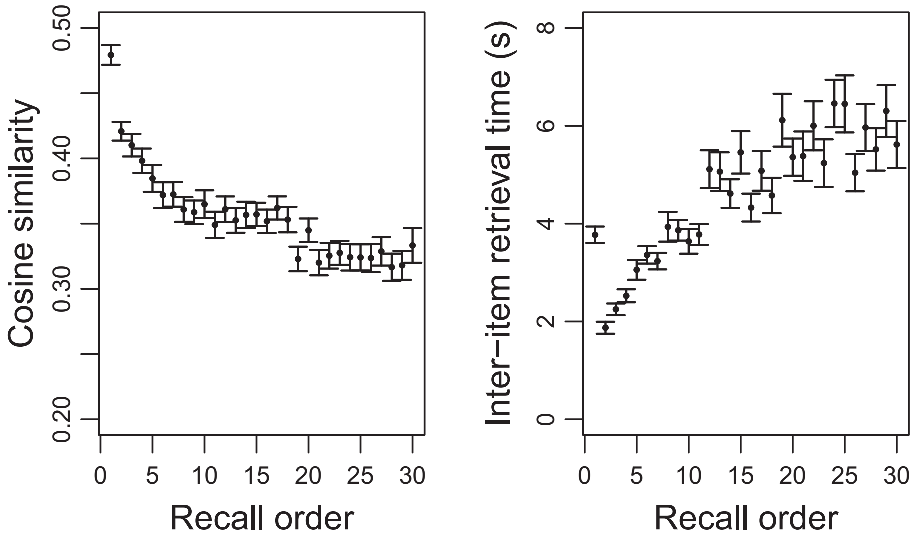

Figure 11.5 Things grow further apart as memory search continues and the time taken to retrieve them increases. The pairwise cosine similarity and inter-item retrieval time is averaged for all items in the same order of recall. This suggests that memory, though it has local depleting patches, is also depleting globally over time.

$ln(30) \approx 3$, which makes both models better than the combined cue before the penalty. This is as it should be unless our model is particularly bad. However, after the penalty neither improves things beyond the combined cue model. Notably, both are *useful* in the George Box sense. First, we learn that the relaxation parameter for the dynamic relaxation model is negative. There is already a negative sign in front of the λ parameter in the model. So this means that, despite our expectations, when the previous similarity is higher, S_{-1}, so is the sensitivity to frequency. This may reflect the observation that frequency is higher overall for earlier items, as is the similarity. These items also take the least time to produce (see Figure 11.5). Thus, the overall picture is that the two dominantly work in tandem.

The residual proximity model is slightly worse than the dynamic relation model, but here the residual proximity parameter is in the expected direction. When residual proximity is lower, the β weighting on frequency is higher. This tells us that retrieving animals that are further away in memory from other animals leads to search focusing more on frequency than similarity. Still, this is not better than the combined cue model, which tells us that modulating frequency in this way by residual proximity is not warranted by the data.

The best of the models is the similarity drop model. It suggests that memory search has two components: a local component, when search follows the network, and a global component, when it jumps between patches in memory.

As far as models go, similarity drop is not beautiful. It is not beautiful in the sense that it does not describe the conditions that would give rise to a switch before it occurs. It is based on what the phenomenology of a patch switch should look like, but it requires knowing what people have produced in order to detect the patch switch. Ideally we would like to explain what features of the cognitive landscape cause a switch to occur before it happens. I have always imagined that a model could be developed that would achieve that goal, but I have yet to see it.

Beauty aside, the model has given rise to various explorations of control processes in cognitive exploration. The model suggests that memory search involves a strategic multistate process, and the presumption is that this is managed by cognitive control. The research exploring this, which increasingly involves neuroimaging, is strongly suggestive that this is the case (Hills, Mata, et al., 2013; Lundin et al., 2022; Marko & Riečanskỳ, 2023; Ovando-Tellez, Benedek, et al., 2022; Pachur & Schulze, 2022; Rosen & Engle, 1997).

11.7 Foraging in Memory: The Marginal Value Theorem Revisited

Now that we have identified likely switches in memory search, we can ask if these switches occur at locations predicted by the marginal value theorem. We can do this because Hills et al. (2012) also collected the inter-item retrieval times, allowing us to ask if the rate of memory retrieval changes over time and if individuals transition when the rate within a patch approaches what it would be on average for all patches including travel time.

We need to know three things to make this evaluation:

1. The time it takes to move from one patch to another. This is the travel time T.
2. The cumulative resource intake within a patch over time, $g(t)$.
3. The departure time, $\hat{t}$.

Given this, we can reproduce Figure 11.1 for our participants. We will use a wisdom-of-the-crowds approach here and aggregate the data across participants, to ask if, collectively, they leave patches near the optimal departure time.

We need to make a number of approximations to arrive at these values. We do not know exactly when people enter a patch or when they leave the previous patch. These numbers are confounded in the data. Here, we will assume that travel time between patches starts with the last item before a patch jump and ends with the first item in a new patch. Thus, travel is associated with similarity drop. This means that people enter memory patches when they find the first item. Thus, the y-axis starts at 1. The travel time is $T = 5.65$ seconds. The average time within a patch is $\hat{t} = 14.75$ seconds. We can get the cumulative number of items retrieved from memory over time by averaging the number of items people have recovered from memory at various times (in seconds). Finally, we plot a line from the travel time T through the cumulative curve at $g(\hat{t})$. The outcome is shown in Figure 11.6. This is a nice visual confirmation of the marginal value theorem in memory.

11.8 Summary

When people and other animals search for things, they adapt their behavior to the structure of the environment. In clustered resource environments, this often means focusing search around locations where resources have been found in the past. This behaviour is

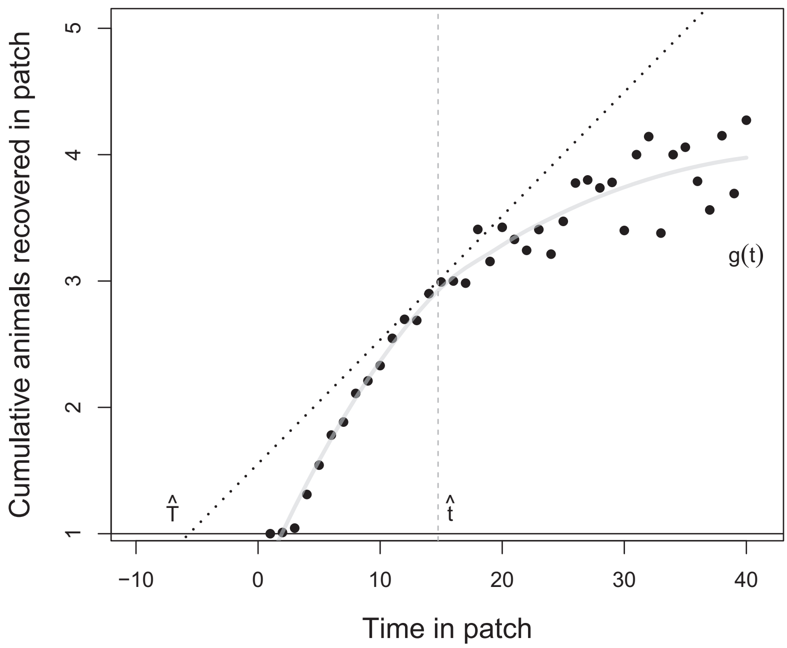

Figure 11.6 The marginal value theorem behavior of the participants in Hills et al., 2012. The dotted line indicates the maximizing rate, with a rise equivalent to the cumulative patch value and a run equivalent to the between patch travel time and the within-patch foraging time. The dots are the average cumulative number of memory productions for participants across the time spent within a patch. The line is a local polynomial regression fit to the observed averages. The dotted line intersects the cumulative number at a point where the average rate across all patches is approximately equal to the marginal rate within a patch.

often called area-restricted search and it is found, to a first approximation, among all animals that move (Dorfman et al., 2022). It is also well adapted to clumpy resource distributions. If memory has clumpy structure, then we should expect that internal foraging will follow similar adaptive search strategies to those found for organisms searching in space.

Numerous methods exist for building structured network representations from natural language or people's responses to free association tasks (e.g., Zemla et al., 2020). In this chapter, we used a model trained on a Wikipedia corpus to generate a network representation of memory. This provided us with a weighted network on which to track people as they foraged through their memory. Search in memory has many features similar to those assumed by the optimal foraging model, the marginal value theorem. This includes patch-like structure, search that follows the local structure of memory within a patch, and the addition of long-distance jumps that are analogous to the kind of between-patch transitions seen in patch-foraging animals. The quantitative representation of memory allowed us to compare different models of memory search against one another in order to look for the one that best hypothesized processes that would generate the data. This in turn identified patch boundaries, allowing us to confirm that movement through memory followed the marginal value theorem.

11.9 Open Questions

11.9.1 How Are Random Walkers Different from a Dual Process Model?

Random walkers are often proposed as cognitive henchmen who wander around through cognitive representations to generate activation when they accumulate in sufficient numbers. They are mathematically elegant, hence their popularity, and are closely matched to our intuitive notions of spreading activation. Indeed, spreading activation and random walker models converge as the number of random walkers increases (Siew, 2019a). Their mathematical grounding has also allowed them to be applied to a host of domains from physics to biology (Berg, 2018). An open question is how are random walker models and models like the state-dependent SAM model similar or different?

Abbott et al. (2015) developed a random walker model on a free-association network and found that it was capable of replicating the marginal value theorem pattern found in Hills et al. (2012). This represents a difference in both the process and the representation: free associations instead of a distributional semantic model and random walkers instead of a two-state SAM. When Abbott et al. (2015) used a random walker model on the BEAGLE representation used by Hills et al. (2012), they found that the random walker model was a poor fit to the data and did not show marginal value theorem behavior. This invites a number of valuable questions. First, what are the differences between the BEAGLE semantic space and a free association network? Perhaps the clumpiness created by the local and global processes in the similarity drop model are embedded in the free association data. In that case, we should be able to generate one representation from the other. Second, how does the best fitting SAM model perform on a free association network, and how does this differ from that found using the BEAGLE representation? Can we compete these against one another, for example, by comparing them with the inter-item retrieval times (Avery & Jones, 2018)? Does this tell us anything about the processes hidden in the free association data? Third, how might one incorporate executive control processes found in other memory search research into a random walker model?

11.9.2 Lévy Walks in Mental Space

Lévy flights and *Lévy walks* are search paths that have long-tailed distributions of path lengths. The probability of a path length is proportional to its size:

$$P(l) \sim l^{-\mu},$$

where l is the size of the path and μ is the power-law exponent. The distribution is considered a Lévy walk when $1 < \mu <= 3$.

The difference between Lévy flights and Lévy walks depends on whether the path is instantaneous (Lévy flights) or takes time proportional to its length (Lévy walks). Thus, one can measure event times to infer Lévy walks, or path lengths to infer Lévy flights. In practice, the choice is often down to the data type. For example, in memory retrieval we can measure time between retrievals but must infer distance traveled.

Many animals, including humans, have long-tailed distributions of path lengths or travel times (Garg & Kello, 2021; Humphries et al., 2010; Reynolds & Rhodes, 2009; Viswanathan et al., 2002). The long tails have led some to propose that animals use a Lévy walk or similar Lévy process to forage.

This is a fairly controversial claim, for a variety of reasons (Benhamou, 2007; Edwards et al., 2012; Plank & James, 2008). A Lévy process is, strictly speaking, a rule that involves choosing each path length independently from a power-law distribution (as described in the preceding equation). Path lengths are therefore uncorrelated in time. They are also chosen independently of whether one has just encountered resources or not. Lévy processes are thus memoryless. If an animal finds a resource and it is using a Lévy process, it is as likely to jump away from it – choosing a long path length – as it would be if it had not found a resource. This makes Lévy flights poorly adapted to search for clustered resources (Dorfman et al., 2022; Hills, 2006), and research shows they are not optimal when resources are clustered or depleting (Ferreira et al., 2012; Plank & James, 2008). Finally, like many other power-law processes, one cannot infer a Lévy process from the distribution alone: Long-tailed power-law path distributions are also created during area-restricted search (Hills, Kalff, et al., 2013).

Nonetheless, many behaviors exhibit power-law distributions and they can do so even in the absence of environmental feedback (Kello et al., 2010; Kölzsch et al., 2015; Zhu et al., 2018). This may reflect a special role of Lévy walks as an intrinsic state underlying behavioral preparation (Reynolds, 2015). For example, Lévy walks could be the natural default state of systems capable of wide-ranging and finely tuned behavior (Abe, 2020).

Might people exhibit Lévy walks in memory search, as they have been proposed to do in other domains (Garg & Kello, 2021; Raichlen et al., 2014)? Rhodes and Turvey (2007) investigated this idea by having people recall animal names for 20 minutes. Their results showed that people largely had inter-item retrieval time distributions that fit power-laws with μ values in the interval proposed for Lévy walks. Might these results be generated by the SAM model described here? Would different results be generated on free association networks versus distributional semantic networks? Finally, recent work has asked how Lévy walks might evolve under natural selection (Campeau et al., 2022). Do these conditions apply to memory? Could similar simulations be adapted to foraging on memory representations to ask what kinds of search processes might evolve there?

12 Age-Related Cognitive Decline
A Network Enrichment Account

12.1 The Problem

There are two contrasting views of cognitive aging. One view sees aging as a process of cognitive decline, a natural consequence of biological aging. The other sees aging as a process of lifelong learning: older adults show conspicuous improvements in vocabulary across the lifespan as well as in many other knowledge-related domains. Of these two views, one is based on an underlying process of decay; the other is based on enrichment. Here we will investigate how understanding the nature of structural changes across the lifespan can help align these views, demonstrating how age-related cognitive decline can be explained as a process of network enrichment caused by lifelong learning.

12.2 Cognitive Aging

From a behavioral perspective, aging is a curious and often contradictory process. Across the adult lifespan, many measures of working memory, processing speed, and long-term memory decline in performance from approximately the age of 20. At the same time, measures of vocabulary and other kinds of general knowledge increase (see Figure 12.1; Park & Reuter-Lorenz, 2009; Salthouse, 2004). This distinction between the ability to solve novel problems in a fast and accurate way, called *fluid intelligence*, and the quantity of one's prior knowledge, called *crystallized intelligence*, is a classic division of intelligence (Cattell, 1987; Park & Reuter-Lorenz, 2009; Salthouse, 2004) and one that also divides the old from the young.

The transition from fluid intelligence to crystalized intelligence has led to arguments that aging is a process of increasing reliance on prior knowledge. This explanation is supported by corresponding changes in the brain. With age, there is increased coupling between areas of the brain associated with access to prior knowledge – the default mode network – and areas associated with goal-directed behavior – the executive control network (Spreng & Turner, 2019). The default mode network is associated with the kind of internal thought that occurs when we are mindwandering (Andrews-Hanna et al., 2014). The executive-control network is associated with getting things done. A simple interpretation of this pattern is that as we age goal-directed cognition becomes more informed by prior knowledge.

This transition matters. As Spreng and Turner (2021) put it: “if access to semantic knowledge is relevant to the current task, greater default-executive coupling [informing

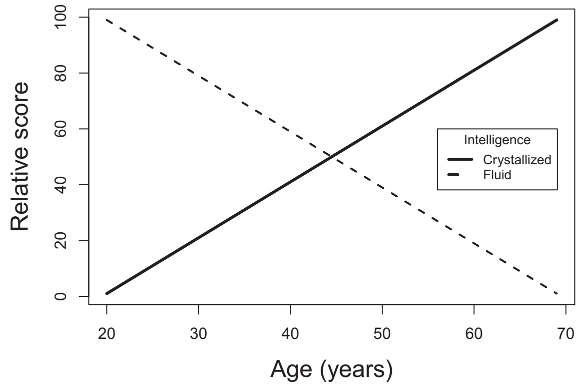

Figure 12.1 A stylized view of the decrease in fluid intelligence and increase in crystallized intelligence commonly associated with the aging mind. Measures of executive function and fluid intelligence consistently show reductions from early adulthood across the lifespan. Measures of crystallized intelligence generally show increases over the same time period. There is controversy about the upper limits of crystallized intelligence later in adulthood caused by the difficulty of producing general measures for specialized knowledge.

goals with prior knowledge] should be adaptive [...]. In contrast, if prior knowledge is irrelevant or distracting, this pattern should lead to poorer performance (p. 525). With many tasks, the trick is tuning the amount of reliance on prior knowledge to the task at hand. Older adults, having more experience, default toward using more of it (Spreng & Turner, 2021).

One might interpret the union of goal-directed cognition and prior knowledge as increasing specialization: cognition focuses on matching the external context to past experience. But one could just as easily argue that the union is an unwelcome consequence of local desegregation: A breakdown of community structure within brain regions causes a loss of functional distinctiveness, with past thoughts leaking into present experience. The first story is one of adaptation. The second is one of age-related decline.

Two measures of network connectivity show that the facts of neural connectivity fit both interpretations. Network measures of the neural connectome – the wiring diagram of the brain – show age-related reductions in modularity and local efficiency. *Local efficiency* is a measure of how close a node's neighbors are to one another: 1 over the ASPL between a node's neighbors: $e_i = \frac{1}{k_i(k_i-1)} \sum_{l \neq m \in k_i} \frac{1}{d_{l,m}}$. Modularity is a measure of community loyalty. That both of these decrease with age is a reflection of the impoverishment of local communities occurring alongside greater between region connectivity: A reduction in functional segregation is associated with wider integration (Damoiseaux, 2017).

This pattern is also seen in the age-related disbanding of what is called the *rich-club network* (Cao et al., 2014). The rich-club network is the observation that high-degree "brain hubs" tend to be more well connected with one another than one would expect by chance (Van Den Heuvel & Sporns, 2011). Formally, the rich-club coefficient, $\Phi(k) = \frac{2E_{>k}}{N_{>k}(N_{>k}-1)}$, measures the extent to which high-degree nodes are connected with one another, with $> k$ indicating the number of nodes, N, and edges, E, that remain among nodes with degree higher than k. This is analogous to computing the density among nodes of degree higher than k. A rich-club network is present when as k increases so too increases the connectivity among the remaining nodes. As individuals age, the number of nodes in the rich-club network shrinks in size (Riedel et al., 2022).

Is this all a break down in the mind's republic or the rise of a more democratic neurosocialism? Before we investigate the evidence for and the application of networks to this problem, we can first take a look at the evidence for changes in the structure of the mental representations, or what is sometimes called the aging mental lexicon.

12.2.1 The Aging Mental Lexicon

One of the largest studies to investigate the structure of the mental lexicon across the lifespan was conducted by Dubossarsky et al. (2017). This work asked more than 8000 people, ranging in age from roughly 10 to 70, to provide 3 free associates for each of 420 words. With approximately 1000 people in each 10-year age group, data was aggregated within age-groups to produce networks among the 420 words, with edges representing a weighted function of common associates. That is, the bipartite cue-associate network was projected onto the 420 cue words. The properties of these resulting networks was evaluated as they changed across the lifespan.

Figure 12.2 visualizes the 420 word networks across the adult age groups. The statistics provided in Dubossarsky et al. (2017) confirm that, from around age 30, older networks have lower degree, higher ASPL, higher node entropy for associations (less predictable associations), and lower clustering coefficient. It is as if the semantic lexicon relaxes with age.

Zortea et al. (2014) undertook a study with a smaller group of participants and cues, and found a similar pattern for all metrics but clustering coefficient. With a still smaller group of participants (n = 8) but far more cues (n = 3000), Wulff, De Deyne, et al. (2022) found this pattern yet again, this time corroborating the falling clustering coefficient.

Wulff et al. (2019) collected a large set of semantic fluency data from old and young adults and built networks using the method described in Goñi et al. (2011). This method forms edges between nodes that appear near one another in the semantic fluency task.[1] By comparing the resulting networks, Wulff et al. (2019) found that, as before, older networks had lower degree and higher ASPLs. Modeling supports these results. Hills, Mata, et al. (2013) found that when searching memory in the animal fluency task older adults use similarity less than younger adults.

Cosgrove et al. (2021) used *percolation analysis* to evaluate the resiliency to attack of the fluency networks of old and young individuals (see Chapter 14 for a more detailed description of this method). This involved subjecting networks to specific patterns of attack and then evaluating the number of k-cliques remaining using the clique percolation method (see Chapter 5). The networks of older adults tended to break apart faster.

[1] For more information about ways to build networks from fluency data see Zemla and Austerweil (2018) and Christensen and Kenett (2021).

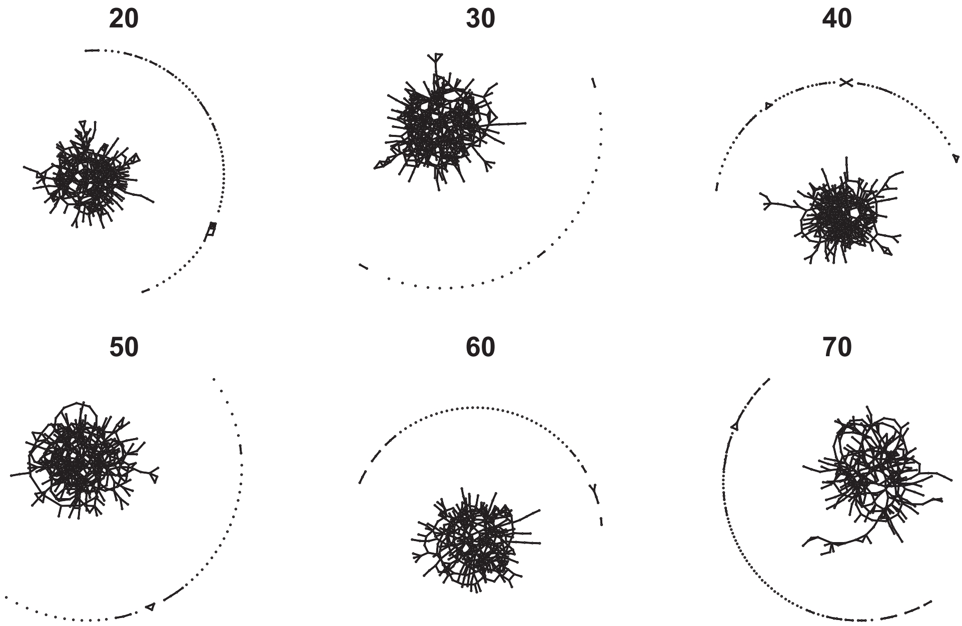

Figure 12.2 Adult networks of 420 cue words collected across individuals ranging in age from approximately 20 to 70. Edges are formed when cues produced similar patterns of free associations. Giant components are shown in the center of each representation, with isolates and smaller components shown in the hemispheres along the edge. From age 30 onwards degree gradually falls as ASPL increase. The figure is reconstructed from data in Dubossarsky et al., 2017.

This apparent degradation of the lexical network appears to influence the way people perceive objects in the world around them. Wulff et al. (2019) also asked younger and older adults to judge the similarity of different animals. Individuals were presented with two animals at a time and rated the animals for similarity on a scale from 1 to 20. Figure 12.3 shows several views of this data, both computed within individuals and aggregated within age groups. The results are clear: older adults judged animals as less similar.

In what follows we will explore various network explanations for these patterns before asking what learning alone can achieve.

12.2.2 Decay

Age-related decline is often interpreted as damage to the cognitive or neural architecture. For example, the *common cause theory of cognitive decline* argues that physiological markers of aging in the brain and cognitive decline (e.g., processing speed deficits) both arise as the result of common age-related biological causes, such as oxidative stress and telomere shortening (Deary et al., 2009). The premise is that cognitive aging occurs because the brain is breaking down.

Some models of aging, and especially those involving pathology, find evidence for degradation and model it accordingly. For example, removing edges by thresholding networks at various values shows that a variety of neural and age-related pathologies

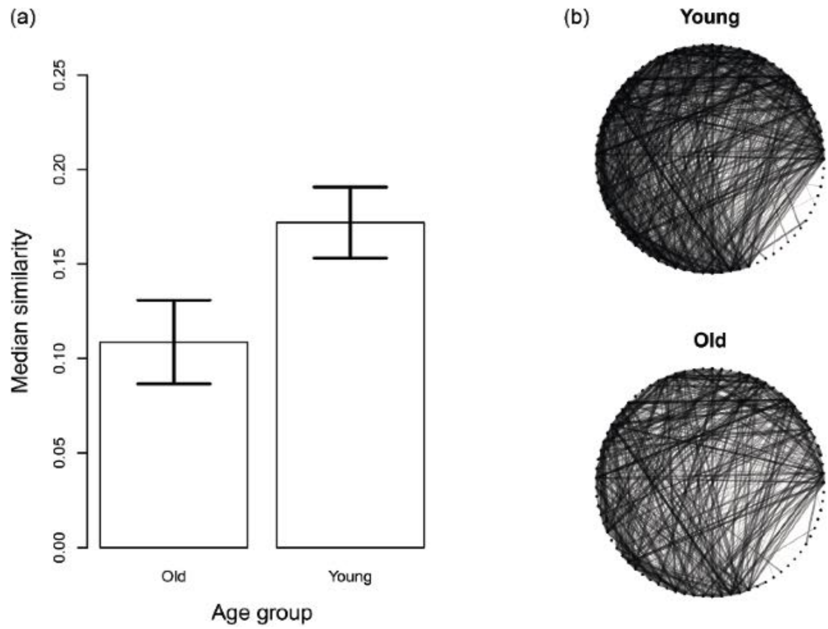

Figure 12.3 Median similarity among animals for old and young groups in Wulff et al. 2022b. The median similarity rating is computed for each individual. The median is more appropriate here because similarities are long-tailed – most are at 0. However, the mean similarity shows the same pattern. Older adults rate animals as less similar than young adults, $t(66.4) = -2.3$, $p = 0.02$. Panel (b) shows the similarity networks for the 63 animals aggregated for the old and young individuals separately ($n = 36$ participants in each group). The darkness and width of the edge value indicates the similarity. Each network is the aggregated data for the young or old individuals with edge width representing the median similarity value for the two animals. This makes the relative density of the similarity ratings clear to the eye, which is higher for the young than the old. The figure is reconstructed from data in Wulff et al., 2022b.

reduce brain connectivity (Kaiser et al., 2007; Liu et al., 2008; Stam et al., 2007). Borge-Holthoefer et al. (2011) used a degraded network architecture to model Alzheimer's Disease and its associated phenomenon of *hyperpriming* – above normal semantic priming for closely related words. The disappearance by degradation of some edges, by removing competition, makes strong edges stronger by comparison, thus driving enhanced local activation that mirrors hyperpriming. Recall that this follows what one would predict from activation spreading outward from lower degree nodes. When the degree of the source node is lower, more activation reaches the nodes that remain connected (see Chapter 10).

Damage-related explanations are also proposed for why processing speed slows with age. Salthouse (1992) explains possible causes of slowing as including "a slower speed of transmission along single (e.g., loss of myelination) or multiple (e.g., loss of functional cells dictating circuitous linkages) pathways, or . . . delayed propagation at the connections between neural units (e.g., impairment in functioning of neurotransmitters, reduced synchronization of activation patterns)" (p. 116). Such biological explanations based on loss and impairment readily motivate degradation and other damage-based accounts.

A common approach to evaluating network damage is to measure the impact of error, attack, or degradation. In network science, *error* (or failure) is manifested as random node or edge removals – removing nodes or edges without preference. *Attacks*, on the other hand, focus on targeting nodes or edges with specific features (e.g., high degree nodes that facilitate network connectivity, such as nodes in the rich-club network). Degradation subjects edges to increasing amounts of damage, with edge weights slowly decaying over time. A classic work in this domain is Albert et al. (2000), showing that scale-free networks have a high degree of error-tolerance to random node failure. They are, however, more susceptible to targeted node attacks than random graphs. Specifically, the diameter of scale-free networks remains fairly impervious to random attacks, but exceeds that of a random graph when nodes are targeted based on degree. This might be taken to suggest that semantic networks (being more scale-free) should be particularly robust to random error – though, they may be more susceptible to targeted attacks on hubs, like those in the rich club.

12.2.3 Enrichment

Rarely proposed is the alternative account that age-related cognitive decline is the natural result of knowledge enrichment. This assumes that learning continues unabated across the lifespan, which creates computational limits on the ability to sort and process cognitive information.

Enrichment follows naturally from combining a rise in crystallized intelligence with the cognitive science tradition of seeing the mind as a computer: an unboxed computer is fast and handles computational challenges with ease; the same computer a few years later is easily winded and responds slowly to the same challenges. Is the slower computer a function of component degradation? Rarely. Rather, its slowing is associated with increased processing demands.

Ramscar et al. (2014) in *The myth of cognitive decline* argue that "older adults changing performance reflects memory search demands, which escalate as experience grows" (p. 5). There are many potential ways in which this could happen, but one source of evidence they provide is based on a careful analysis of *paired-associate learning*.

In paired-associate learning individuals are shown a list of word pairs and then must recall one when shown the other. Trained on word pairs like *up–down*, *come–go*, *necktie–cracker*, and *jury–eagle*, participants are later shown the cue word (*up*, *come*, *necktie*, or *jury*) and must recall the associated target word (*down*, *go*, *cracker*, or *eagle*). Desrosiers and Ivison (1986) observed that older adults perform more poorly on paired-associate learning than younger adults. But what you may notice about these word pairs is that they are not all the same. Some are fairly common associates, like *up–down* and *come–go*. Others are unrelated, like *necktie–cracker* and *jury–eagle*. Ramscar et al. (2014) point out that older adults largely show impaired learning for unrelated terms. This is a straightforward prediction of learning: "the discriminative processes that produce 'associative' learning teaches English speakers not only which words go together, but also which words do not go together. This process both increasingly differentiates meaningful and meaningless word pairs and makes meaningless pairs harder to learn" (Ramscar et al., 2017, p. 3).

The key claim here is that many tasks that are designed to differentiate old from young individuals are not independent of prior experience. Any task that requires lexical processing may therefore show differences between old and young, not based on decline but on additional experience. To demonstrate this further, Ramscar et al. (2017) tested bilinguals who had less experience with word-pairs and showed that they performed better on paired-associate learning than more experienced individuals of the same age. Ramscar et al. (2017) went on to show that second-language learners perform better in the paired-associate task than first-language learners for specifically those word pairs with which they had less experience.

12.3 Explaining Aging as a Process of Enrichment

Can we use network analysis to explain cognitive aging as a process of enrichment? Specifically, can we explain the age-related changes in similarity and entropy found in Wulff et al. (2019) and Dubossarsky et al. (2017). In what follows, I will first introduce a well-accepted learning model, the Rescorla–Wagner model. We will then examine how this model explains age-related decline in the paired-associate learning task, as described by Ramscar et al. (2017). Finally, we will apply this framework to learning a network of associations across the lifespan and see how the resulting representations reproduce the similarity and entropy results described earlier.

12.3.1 Rescorla–Wagner

The Rescorla–Wagner model was developed to model associative learning, commonly called *classical conditioning* or *Pavlovian conditioning* (Rescorla & Wagner, 1972).[2] The general logic of classical conditioning follows the familiar Pavlovian narrative: a hungry dog that drools when presented with food learns to drool in response to a bell that signals the arrival of the food. The presence of food is the *unconditioned stimulus* (US). The bell is the *conditioned stimulus* (CS). The two become associated through repeated experience. The salivating is the *conditioned response* (CR). Thus, pairing a CS with a US eventually leads the CS to bring about a preparatory CR. In other words, pairing a bell with the arrival of food eventually leads the dog to salivate in response to the bell.

Classical conditioning is conducive to a network representation. Cues and outcomes are nodes and associations between them are edges. A simple model creates a link between the conditioned and the unconditioned stimulus (CS–US). That link could be proportional to the number of times the CS and US occur together.

[2] Classical conditioning is part of the behaviorism paradigm associated with such notable characters as B.F. Skinner and Edward Thorndike. Classical conditioning should not be confused with operant conditioning. Classical conditioning associates an unconditioned response with a conditioned stimulus. Operant conditioning uses reward and punishment to shape behavior. Classical conditioning is passive: the consequences do not depend on the animal taking action to produce an outcome, the animal simply learns an association. Operant conditioning is active: the consequences the animal experiences depend on the actions the animal takes.

However, this simple model is incomplete because it fails to predict phenomenon like *blocking* and *inhibition*. Blocking occurs when prior learning about a cue-outcome relationship inhibits future learning about a new one. Once the dog learns that a bell signals the impending arrival of food, learning about a second cue alongside the bell, such as the addition of a light, is reduced. Inhibition occurs when a cue signals the absence of an outcome. Thus, classical conditioning is implicitly a structural phenomenon.

The Rescorla–Wagner model captures these structural effects using *prediction error*, which is now commonplace in models of reinforcement learning (Dayan & Abbott, 2005; Hoppe et al., 2022; McClelland & Rumelhart, 1981; Sutton & Barto, 2018). Given a set of cues, if the animal expects what comes next, then no learning occurs. If the animal fails to expect what comes next, the animal updates its expectations. This difference can be considered a measure of surprise. Indeed, Rescorla and Wagner (1972) claim it was Kamin's (1969) notion that learning was driven by the *surprisingness* of an outcome that inspired the Rescorla–Wagner model.

Minimizing surprise corresponds nicely to a host of other things we now understand about learning, such as attention being focused on information that reduces uncertainty (Gottlieb, 2023), the Bayesian brain hypothesis (Knill & Pouget, 2004), Friston's entropy minimizing free energy principle (Friston, 2010), and neural signals that indicate surprise, such as dopamine neurons that are activated both with underpredicted and overpredicted events (Waelti et al., 2001).

To minimize surprise, the Rescorla–Wagner model tries to assign association values between cues and outcomes so that cues predict outcomes perfectly. For every cue, i, and outcome, j, there is a cue-outcome association strength, V_{ij}. The difference between the correct expectation of the outcome, λ_j, and the association strength is the prediction error, $(\lambda_j - V_{ij})$. The goal is to minimize the prediction error. Learning minimizes prediction error by changing the association strength.[3]

The change in the associative strength is indicated by ΔV_{ij}, which is equivalent to the prediction error times the learning rates of the cue and the outcome:

$$\Delta V_{ij} = \alpha\beta(\lambda_j - V_{ij})$$

The α corresponds to cue *salience*. This follows the evidence that some cues are easier to learn about than others. The β indicates the rate of learning associated with the outcome, based on a similar logic. The α and β values are confined to values between 0 and 1. If they are both 0, no learning occurs. If they are both 1, learning is instantaneous. Intermediate values of learning allow integration of past information with present information, which better tracks a noisy environment and is closer to how we actually learn.

Because this is an *iterative model*, learning is the change from time t to time $t+1$. After learning, the updated associative strength is simply what it was before plus the change:

$$V_{ij,t+1} = V_{ij,t} + \Delta V_{ij,t}$$

[3] To quantify this relationship, the association strength represents the value of the expected outcome when encountering the cue measured in some arbitrary unit like "intensity." In reality, when we hear a siren, we expect an emergency. When we smell smoke, we expect a fire. The Rescorla–Wagner model simplifies these outcomes into a single value.

When the outcome is present, V always moves toward λ. Prediction error is negative when $V > \lambda$, which adjusts V downward. It is positive when $V < \lambda$, which adjusts V upward. The result is associative learning.

When the outcome fails to occur, the prediction is not λ_j but 0.

$$\Delta V_{ij} = \alpha\beta(0 - V_{ij})$$

As a result, when cues appear without outcomes, the model learns to disassociate cues with outcomes. This accounts for extinction – the loss of a previously learned response.

When multiple cues appear at the same time, we have a *compound cue*, C. The predicted value of the outcome is then the sum of the association strengths, V_{ij}, for each cue, i, for all $i \in C$:

$$V_{Cj} = \sum_{i \in C} V_{ij}$$

The compound cue prediction for λ_j is now V_{Cj}. We can incorporate this into the prediction error equation above to get ΔV_{Cj}, which is then used to update all the cues in the compound cue.

This model produces blocking. If a cue already perfectly predicts an outcome, then when a cue with an associative strength of 0 is added to create a compound cue, no new learning will take place for either cue. That is, $\Delta V_{Cj} = 0$.

The model also produces inhibition. When a second nonpredictive cue is added to a predictive cue to form a compound cue and the outcome then fails to appear, both cues will lose association strength and the second nonpredictive cue will become negative. A negative association strength predicts the absence of an outcome.

Here are the four prominent phenomena associated with classical conditioning. The right arrow indicates "signals" and *ik* is the compound cue of i and k.

- Associative learning: $i \to j$
- Blocking: $i \to j$ followed by $ik \to j$
- Extinction: $i \to j$ followed by $i \to 0$
- Inhibition: $i \to j$ followed by $ik \to 0$

Figure 12.4 shows the results of simulating the Rescorla–Wagner model for each phenomena.

12.3.2 Applying Rescorla–Wagner to Paired-Associate Learning

Paired-associate learning is generally considered to be independent of prior learning. If this were true, then if older adults do more poorly than younger adults, it is because older adult learning is impaired relative to younger adult learning. However, if a lifetime of learning teaches older adults that pairs like *jury* and *eagle* are unrelated, then this association should be harder to learn. This is what Ramscar et al. showed for aging in the simulation we will reproduce in the following sections.

This negative relationship is also a core lesson of Pavlovian conditioning. That is, if one learns that two things are unrelated, it is more difficult to learn later that they are related.

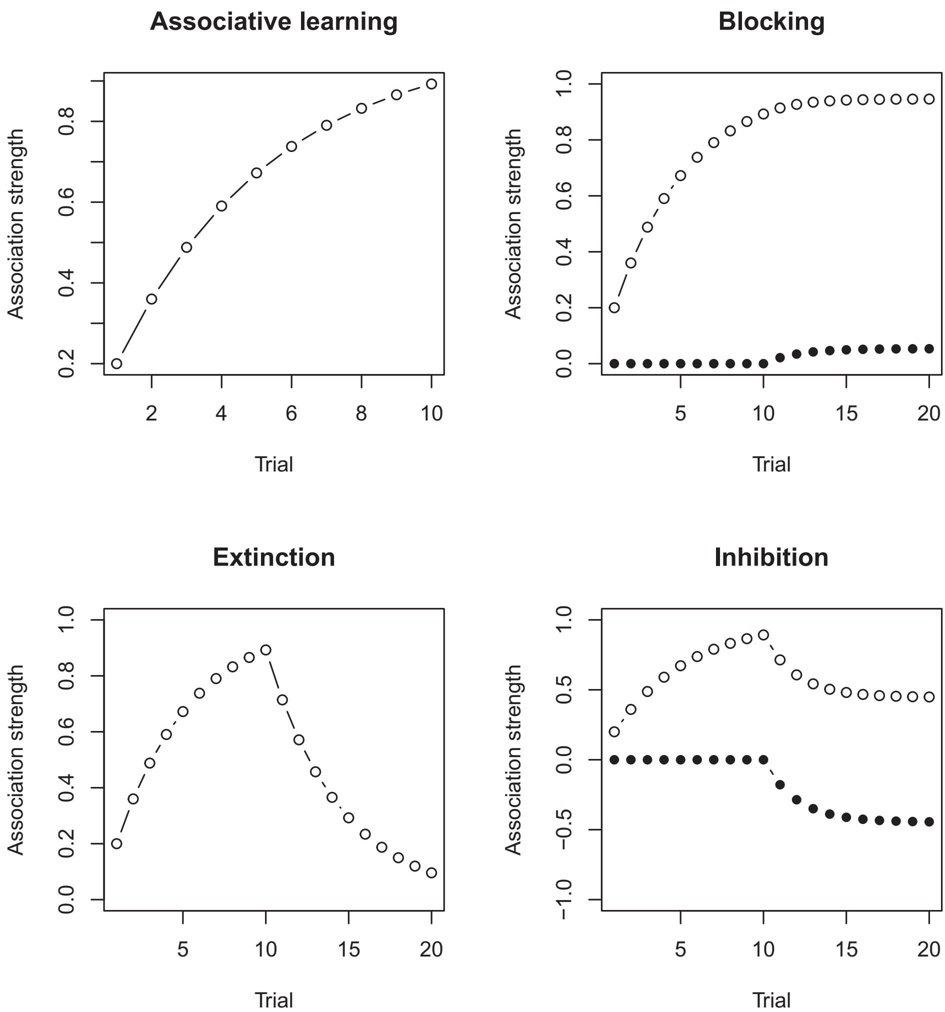

Figure 12.4 The Rescorla–Wagner model simulations of four classical conditioning phenomena: associative learning, blocking, extinction, and inhibition. White dots indicate cue *i* and black dots cue *j*. For each model, $\alpha = 1$ and $\beta = 0.2$. For associative learning, the *i* and *j* are paired 10 times. For blocking, *i* is paired with *j* 10 times and then the compound cue *ik* is paired with *j* 10 times. For extinction, *i* is paired with *j* 10 times and then *i* is paired with 0, no outcome, 10 times. For inhibition, *i* is paired with *j* 10 times, then *ik* is paired with 0, no outcome, 10 times.

This is an important but often overlooked addition to the typical behaviorist mantra that learning happens via pairing with rewards and punishments. Importantly, *learning also happens when rewards and punishments are absent* (Rescorla, 1988). If a cue appears, the absence of a subsequent outcome is just as informative as its presence.

Ramscar et al. (2017) used the Rescorla–Wagner equations to demonstrate how this impacts paired-associate learning. To reproduce their example, we train the Rescorla–Wagner model with a randomized exposure to *jury–duty* (40 times), *jury–summons* (80 times), *baby–cries* (60 times), and *baby–eagle* (60 times). In addition to these pairings, we will also keep track of the values between *jury–eagle* and *baby–summons*, which never receive a training trial. The cues each appear alongside a *context* cue which represents the

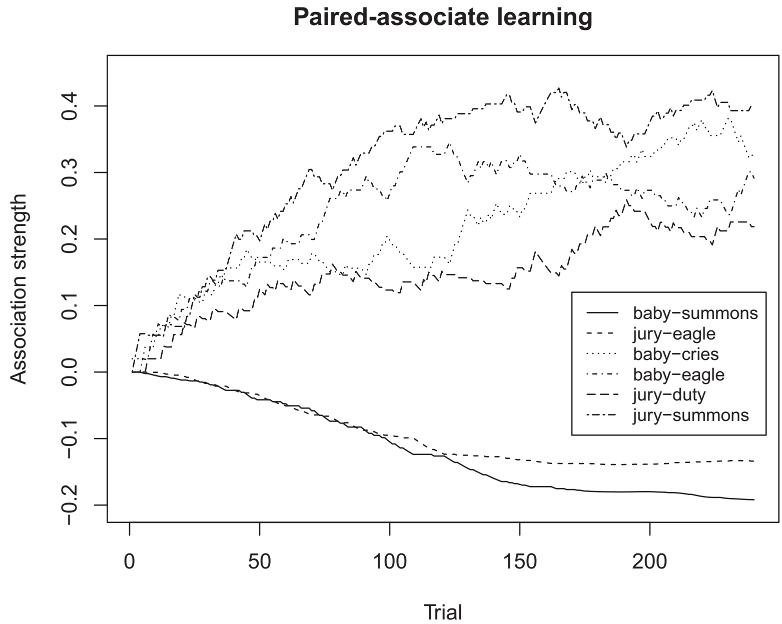

Figure 12.5 The Rescorla–Wagner model trained on paired associates. The model is trained on (*jury*, *duty*) 40 times, (*jury*, *summons*) 80 times, (*baby*, *cries*) 60 times, and (*baby*, *eagle*) 60 times. The pairs the model is trained on develop an increasing associative strength. The model also learns to disassociate the pairs that do not appear together, making them harder to learn in the future. This provides a learning-based account of the age-related performance declines found in Desrosiers and Ivison, 1986. This simulation follows Ramscar et al., 2015.

general environment in which learning takes place (see Danks, 2003).[4] All association strengths start at 0, $\lambda = 1$ when the outcome is present for all outcomes, and $\alpha = 1$ and $\beta = 0.02$.

Figure 12.5 shows that the cues and outcomes that appear together rise in their association strength. This fits the standard notion of Pavlovian conditioning: learning occurs via the pairing of cue and outcome. But we can also see that occurrence of a cue in the absence of an outcome is informative. When *jury* occurs without *eagle*, we learn that these are unrelated. If people with more experience are later asked to remember pairings,

[4] The context cue is a standard addition to the environment and allows cues to become associated with outcomes even if they never appear together. This is a reverse inhibition effect. It occurs when we learn that a cue inhibits an outcome predicted by another cue, and then the predictive cue is replaced by a nonpredictive cue. This happens in relation to the context cue as follows. First, the context cue becomes negatively associated with an outcome: If a cue appears that predicts an outcome, but the outcome is absent, the association between the context and the outcome will become negative. This negative context–outcome association strength can then make a different cue positive for the outcome: The context predicts a negative value for the outcome when a 0 value should be predicted instead. If a new cue appears it will pick up the positive prediction error. As an example, if Louise talks about politics a lot except when Thelma is around, then we can predict that Thelma inhibits political talk. What should we predict when we encounter Thelma without Louise? Even less political talk. But if Darryl shows up with Thelma, and there is still little political talk, then it must be that Darryl by himself would talk about politics – or so predicts the model. This property of Rescorla–Wagner is not unique: it is found in distributional semantic models as well.

they should do more poorly on pairings that they learned were negatively associated. From this we can infer that learning explains learning difficulties.

This is all perfectly intuitive. When someone cries wolf in the absence of a wolf, we eventually learn to ignore them, and it will be much harder to learn later that they have changed their tune.

12.4 An Enrichment Model of Cognitive Aging

We are now in a position to use network science to ask how learning could generate the rising entropy of associations and reduced similarity judgments described earlier for older individuals. I will envision this process as happening in three stages:

1. *Environment*: The environment is a structured set of possible experiences (i.e., network associations). This set grows over time.
2. *Representation*: Learning from the environment generates a cognitive representation that continues to develop across the lifespan.
3. *Behavior*: Behavior is the outcome of recovering information from the cognitive representation. This generates free associations, memory search, similarity judgments, and so on.

Figure 12.6 makes explicit what is required to translate experience with the environment into behavior, with arrows representing processes. Without such an arrangement, one is left imagining that behavior *is* the representation (e.g., using free associations to model recall), or that behavior *is* a response to the environment (behaviorism). The notion of a mind in the middle, complete with representations and information processing for handling those representations, is the primary contribution of the cognitive revolution and its ongoing development (Griffiths, 2015; Miller, 2003). We handle each of these three stages in the following sections.

12.4.1 Environment

The environment is represented as a *training network*. This contains all the possible relationships an individual could experience.

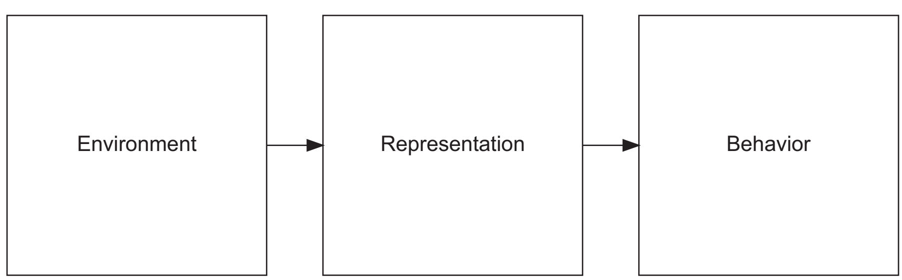

Figure 12.6 The processes of translating experiences with the environment into a representation and then behavior. Arrows represent processes that translate one domain into another.

To construct the training network, I use a fitness model I call *rank-based construction*. This assumes a power-law relation in the frequency of possible experiences. To achieve this, a lexicon of 1,000 words is created which are assigned a rank, r, from 1 to 1,000. Then two words are chosen from the lexicon each with probability $p \propto r^{-a}$ and an edge is created between them. Here a is set to 1 and we create 2,000 edges.

This representation is revealed to the learner over time. Brysbaert et al. (2016) estimate that the average 20-year-old knows about 42,000 lemmas (equivalent to dictionary entries). As we age, we learn about 180 new lemmas per year up until age 60. Studies of human communication based on recordings throughout the day estimate that college students speak about 16,000 words a day (Mehl et al., 2007). Brysbaert et al. (2016) estimate that 20-year-olds have experienced about 84,000 word types while 60-year-olds have experienced about double this number (157,000). One explanation for this vocabulary expansion is that lexical experience expands with age, following Herdan–Heaps' law (Brysbaert et al., 2016).

Herdan–Heaps' law (also called Herdan's law or Heaps' law) is the finding that the vocabulary contained in language grows according to a power relation with the total number of words (Petersen, Tenenbaum, Havlin, Stanley, & Perc, 2012; Serrano et al., 2009). That is, the vocabulary size, $V(n)$, required to recognize all the words in a document of size n words is:

$$V(n) = Kn^{\beta}.$$

When fit to real language, the value of β is typically around 0.5. The value of K is usually between 10 and 100. What this means is that if a document grows by two orders of magnitude, its vocabulary grows by approximately one order of magnitude: $V(100) = 10 \times 100^{.5} = 100$ and $V(10000) = 10 \times 10000^{.5} = 1000$.

To achieve the growth in experience proposed by Brysbaert et al. (2016), we will allow the size of the experience network to grow according to Herdan–Heaps' law as shown in Figure 12.7.

12.4.2 Representation

The Rescorla–Wagner model is trained on the experience network by sampling an edge and presenting one of its nodes as a cue and the other as the outcome. This is repeated 1000 times in each training epoch, for four epochs. Each epoch is treated as a development stage, which I use here as a proxy for age. The experience network grows according to Herdan–Heaps' law following each epoch of training.

After each epoch of training, the representation network is produced from the cue-outcome association matrix created by the Rescorla–Wagner training. This is a weighted network representation of encoded experiences. It is the representation from which information will be retrieved and processed to generate behavioral responses. For the purposes of computing behavior, this network is transformed into an undirected network by summing the activation in reciprocal edges.

Figure 12.8 shows the outcome of simulating lexical growth following Herdan–Heaps' law. What the figure illustrates is an increasing lexicon over the learning trials, with the giant component growing over time. This fits our understanding of the growing lexicon

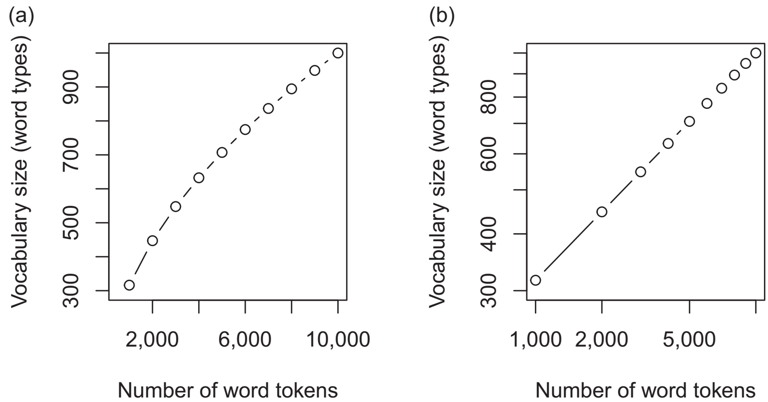

Figure 12.7 Herdan–Heaps' law relating the number of word types (unique words) and the number of word tokens. Types can be considered the vocabulary size required to understand all words. The power exponent is approximately 0.5, or the square root of the volume of tokens in the language. This number varies slightly across languages. Panels (a) and (b) differ only in that the x and y axis are on log scales on the right, producing the linear relationship associated with Herdan–Heaps' law.

and suggests increasing interconnectivity among words. Note that the learned lexicons are more dense than the environment networks: there are more edges in the learned lexicon than the experienced lexicon. This is driven by small positive edge weights forming between nodes that are not in the training set, but which arise through standard Rescorla–Wagner dynamics on growing networks.

12.4.3 Behavior

We will measure behavior based on the representation in several ways. One aspect we are interested in capturing is the predictability of neighbors. In the aging free association networks presented in Dubossarsky et al. (2017), the entropy of associations rises with age; the predictability of free associations decreases.

12.4.3.1 Entropy

We will measure entropy using *Shannon's information entropy*. This measures the extent to which we can predict which of a node's neighbors will be chosen as an associate in a free association task. Because the output of Rescorla–Wagner learning is a weighted edge, we can apply this to information entropy to each node as follows:

$$H = -\sum_{i=1}^{k} p_i log(p_i).$$

For each of a node's edges, we divide its weight by the node's strength to get the proportion of a node's strength dedicated to that edge, p_i. That is, $p_i = \frac{w_i}{\sum_k w_k}$. The contribution of an edge to the node's entropy is the product of $p_i log(p_i)$. This is summed over all the node's k edges. If a node has only 1 edge, its entropy is 0. If it has one edge with a much higher weight than other edges, then its entropy is closer to 0 than if it has edges of equal

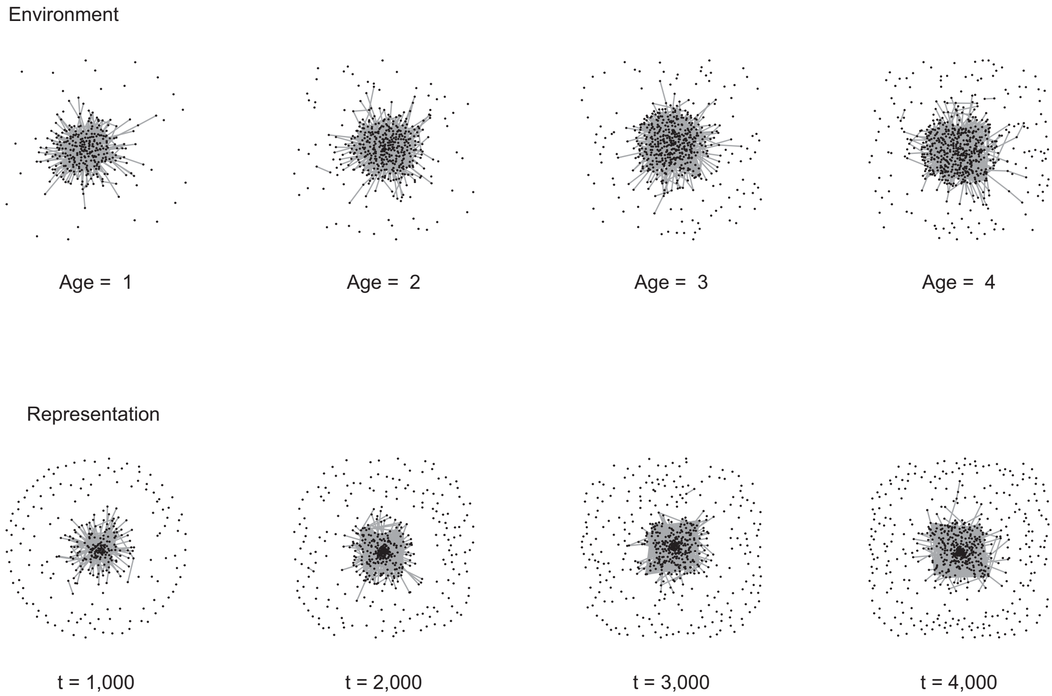

Figure 12.8 The environment and the learned representation. As individuals age they experience more words according to Herdan–Heaps' law. The environment networks are created by sampling 2,000 pairs of words from a 1,000 word pool, each word sampled in proportion to its rank ($P(r) \propto r^{-1}$). Edges are created between pairs. At each age, the environment network is made up of the subgraph of the first n ranked words following the pattern shown in the previous figure for Herdan–Heaps' law. The first 316 words are included at age one, the first 447 words at age two, and so on. The representation is created by forming a 1,000 event training set of cue-target relations from the environment network, each time constraining the training data to the age-appropriate environment network. Each training event is learned according to the Rescorla–Wagner model described earlier, creating a weighted network that represents learned associations. The top networks show the environment. The bottom networks show the learned lexical representation.

weight. For a fixed degree, entropy is maximized when all the edges are of equal weight. When all weights are equal, entropy rises with the number of edges. That is, entropy rises as it becomes harder to predict which edge will be chosen for the production of an associate.

Behavioral patterns of word associations were used to form edges in both Dubossarsky et al. (2017) and Wulff et al. (2019). Dubossarsky et al. (2017) directly measured rising entropy with age, but we can also infer it. Rising entropy means the strongest edges will have weaker weights as those associates are produced less predictably. This reduces the likelihood of an edge after thresholding. Word pairs that are produced less often will therefore have a lower likelihood of forming an edge in the network. As a consequence, higher entropy behavioral networks will be more sparse.

12.4.3.2 Similarity

Wulff, Hills, et al. (2022) found that pairs of animals were rated as less similar by older people than younger people. To measure this from the representation, we need a mechanism for evaluating similarity between nodes. There are many potential mechanisms for doing this. If we take the edge weight alone, we miss out on the competition among other nodes. We could compute some number of shared neighbors, Manhattan distance, or cosine similarity, but these have similar problems. Competition is clearly observed in the processes explored in Chapters 10 and 11 and is a predictable outcome of many approaches, including softmax choice rules, spreading activation, and random walks (Abbott et al., 2015; Hills, 2012; Siew et al., 2019).

The approach used here incorporates competition by capturing similarity as a measure of coactivation. Allowing spreading activation to leave one node and measuring its activation at the other node, we can measure the extent to which one word coactivates the other. Doing this for both nodes, we take similarity, S, as the summed coactivation:

$$S = M_{j\to k} + M_{j\to k},$$

where $M_{j\to k}$ is the maximum activation at the target node k after spreading activation is released from node j.

We will measure this similarity for 20 node pairs randomly sampled from the training data set experienced during the first epoch of learning. This means that the pairs will be chosen from the top most frequent words across the entire lexicon. This is also an experimental requirement if we are choosing words that are familiar enough to both young and old to evaluate on similarity.

12.4.4 Enrichment Produces Cognitive Aging

Figure 12.9 shows the results for entropy and similarity ratings across ages. With age, the average entropy increases. This follows the late-life pattern of rising entropy in free associations found in Dubossarsky et al. (2017) and the rising sparsity in memory search networks in Wulff et al. (2019).

With age, similarity decreases. Two randomly chosen nodes learned early in lexical development grow increasingly less likely to activate one another as the network continues to learn. This follows our intuition that an increasing number of associations diffuses spreading activation away from initially related targets and corresponds to the results found in Wulff et al. (2019).

These results offer an enrichment-based account of some phenomenon often associated with cognitive decline. Alongside the results presented by Ramscar et al. (2017) for paired-associate learning, the argument for an enrichment-based account of cognitive decline becomes increasingly strong. Experience trains the cognitive representation to favor certain outcomes over others. No one finds this objectionable. The curiosity is that the proliferation of associations created by learning diffuses spreading activation, increasing entropy and reducing the similarity of coactivation between pairs of nodes. Both have clear implications for cognitive slowing. Both rising entropy and falling similarity indicate that increased amounts of activation are needed to process relationships across the network.

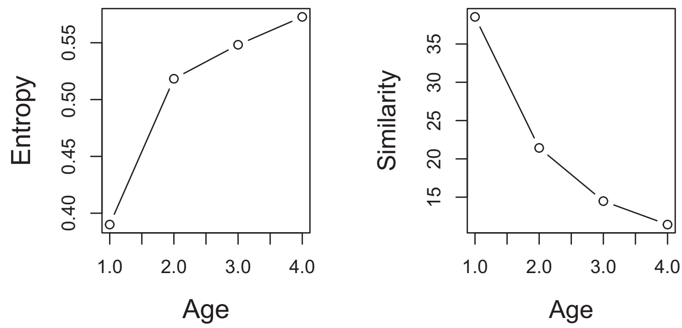

Figure 12.9 The rising entropy and falling similarity of the aging lexicon as a function of learning.

12.5 Summary

This chapter focused on how we might model aging as a process of increasing knowledge and recover many of the behavioral phenotypes commonly associated with cognitive decline. Representation networks built by training a standard prediction error model on a set of expanding experiences were then used to generate behaviors. These in turn produced the rising entropy and falling similarity judgments empirically observed among older individuals.

The enrichment model presented here explains declines in fluid intelligence as a direct consequence of increases in crystallized intelligence. Learning over the lifetime leads to a changing cognitive representation that places constraints on what can be learned later. In particular, as the representation becomes more enriched, the degree of nodes increases as does their entropy, leading to greater diffusion of spreading activation (see Figure 12.10), as suggested by others (Buchler & Reder, 2007). Experiments demonstrate similar reductions in processing speed caused by learning in the *fan effect* (Anderson & Reder, 1999) – learning many relationships with a target concept reduces the speed of accessing any one of those relationships.

Of course, it could be that declines in fluid intelligence are independent of representational changes. This is the standard account, which often sees crystallized and fluid intelligence as separate cognitive faculties. This way of thinking has led to models that explain aging via process-based accounts and that take no notice of representational changes across the lifespan (e.g., Hills, Mata, et al., 2013).

A butcher might use Occam's razor to argue that the enrichment account is better because it requires fewer processes. Common cause accounts require that whatever unspecified mechanisms account for biological aging in the brain somehow account for fluid degradation but miss crystallized intelligence. Such a model must predict that biological aging only impacts some areas of the brain, missing others. In addition, common cause accounts must posit two separate mechanisms for declining fluid intelligence and rising crystallized intelligence. In contrast, enrichment-based accounts explain changes in fluid intelligence as a consequence of rising crystallized intelligence. They require nothing more than the assumption that learning continues across the lifespan, something we have ample evidence for. Still, there is a gap between demonstrating the influence

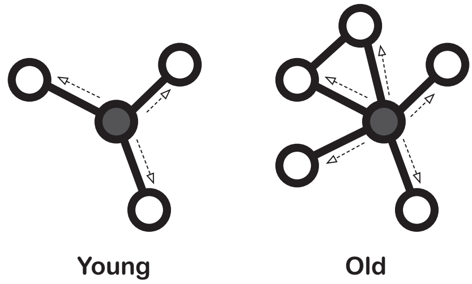

Figure 12.10 A visual representation of the enrichment account of cognitive aging. As individuals age, the enrichment among relations in their lexical network increases diffusion of spreading activation from nodes. This increases entropy and reduces similarity.

of representational enrichment on behavior and showing that this would also explain the wide range of tasks on which older adults' performance declines. To do this, the butcher will need to become the surgeon.

12.6 Open Questions

12.6.1 Aging as Degredation

The common cause theory of cognitive decline argues that cognitive aging and its associated declines in performance are the result of biological aging (Deary et al., 2009). Some accounts of pathological aging have also asked to what extent these effects might be explained by degradation of the semantic network (Borge-Holthoefer et al., 2011). Might such decay-based accounts explain healthy aging of the semantic lexicon? What assumptions are required to formally describe the degradation or decay-based models of healthy aging that reproduce the patterns of age-related changes described earlier? Can one create models that allow crystalized intelligence to increase while impairing fluid intelligence?

12.6.2 Inverted U-Shaped Patterns in Aging

Dubossarsky et al. (2017) presented data for free associations from individuals aged between 10 and 70 years of age. Various network measures indicate a U-shaped pattern of changes across the lifespan, peaking at around 20 to 30 years of age. For example, entropy falls and then rises again. Degree rises and then falls.

On the face of it, such an inverted U-shaped pattern suggests at least two processes. The usual story is that enrichment occurs up until midlife followed by a pattern of degradation. But if the model of enrichment is correct, then what explains the early fall in entropy? There are many potential explanations. Could the change be due to a change in the rate of language exposure? Might a change in the rate of novel language associated with Herdan–Heaps' law produce the U-shaped change?

12.6.3 Graph Learning and Graph Search

Could we learn with the same process by which we search? In Chapter 5, I briefly described graph learning. This describes how we learn graphical representations by experiencing them one node at a time, in the same way that we experience language. The Rescorla–Wagner model described here offers a form of graph learning based on prediction error. But it is interesting to note that a prediction approach always involves the first step of a search process. Potential outcomes are activated before they occur in an effort to minimize surprise. To what extent does the capacity to search a cognitive representation underlie our capacity to learn a cognitive representation? Do we see individual differences in search correlated with encoding? Further, we may ask how search itself influences graph learning. This latter question, as well as many others, are addressed in Lynn and Bassett (2020).

13 Creativity

How Noisy Processes Create Novel Structure

13.1 The Problem

Compared to people who are rated as less creative, more creative people tend to produce ideas more quickly, with more novelty, and more actively engage regions of the brain associated with cognitive control. Both inside and outside the laboratory, the evidence is clear: The creative mind is a productive mind. Structural analysis of what more creative people produce has led to two different proposals for how this is achieved. One is based on differences in the underlying knowledge representation – the structure of semantic memory – called the associative theory of creativity. The other is based on more effortful cognitive control – how semantic memory is accessed – called the executive theory of creativity. Evidence supports both, but there is a need for models that integrate these two ideas. Network analysis offers some inroads into how to tackle this problem and invites some creativity of its own.

13.2 Be Fertile in Hypotheses

> The genius of discovery depends altogether on the number of these random notions and guesses which visit the investigator's mind. To be fertile in hypotheses is the first requisite, and to be willing to throw them away the moment experience contradicts them is the next. (James, 1880, p. 456)

In his lecture on the parallels between social evolution and Darwinian theory, William James portrays the great mind as the creative mind. Able to escape "treadmill routine," it is the curator of accidents. In James' view, such a mind is the producer of conceptual mutations that sustain our social evolution. As biological evolution is driven by the frequency of random mutation, so creative thought is fueled by the fertility of the unexpected.

James identifies the creative mind as a combination of systematic productivity married to unsystematic bumbling. Nietzsche held a similar view: "The imagination of the good artist or thinker constantly produces good, mediocre, and bad, but [their] judgment, most clear and practiced, rejects and chooses and joins together" (Nietzsche, 1878/1996, p. 159). Many great artists have embodied this view across a lifetime of work. Pablo Picasso is estimated to have produced more than 20,000 paintings. Van Gogh produced approximately a painting a day in the most productive years of his life. Paul Erdös (of the Erdös–Rényi random graph) wrote more than 1,500 mathematical articles. If failure and success in the uncertain domain of creative exploration are both unpredictable outcomes of trying, then more failures predict more successes. A similar logic motivates the rule of nine in comedy: If you want one good joke, you have to write nine bad ones.

James, Nietzsche, and many others, have imagined that creativity is such a *variance-selection process* (Campbell, 1960; Simonton, 2022). The mind generates alternatives, roughly at random, and then – presumably some other part of the mind – selects the best options. These two components correspond to what Dietrich (2004) might refer to as spontaneous and deliberate creativity. It is the same framework that is proposed to underlie the more general ability to search for and identify alternatives during deliberation (Hills, 2019a): a stochastic process, invoked by effortful aspects of cognition, throws up alternatives that are then selected. Empirical studies support this relationship and have even suggested neural correlates (Benedek & Neubauer, 2013; Finke, 1996; Jung et al., 2015).

The idea that creativity follows productivity has been called the *equal-odds rule* (Simonton, 1997, p. 73): "the relationship between the number of hits (i.e., creative successes) and the total number of works produced in a given time period is positive, linear, stochastic, and stable." If you want to be more creative – so the thinking goes – be more productive.

But the variance-selection and equal-odds approaches potentially oversimplify matters. For one, they pose a useful null hypothesis: creativity is not a matter of thinking differently, but of thinking more. This is a claim about which various contemporary theories of creativity have much to say. It also something that network science allows us to test.

13.3 Theories of Creativity

What causes some people to be more creative than others? This is a challenging question because creativity is not easily defined. Nor is it easy to find measures that capture what everyone would agree is creativity. Guilford (1972) illustrates the point: "Since creative talent … is composed of numerous special abilities, and since criteria of creative performance in everyday life are also complex, no one test of a creative ability can be expected to correlate highly with those criteria" (p. 86).

To further the point, researchers have developed a variety of ways to measure creative aptitude. These include various tasks that can be performed in the laboratory. Convergent thinking tasks, like the remote associates task ("what word is common to 'rubber,' 'game,' and 'soft'"), and divergent thinking tasks, like the alternative uses task ("how many uses can you think of for a paperclip"), are both commonly used and combined into batteries that attempt to make summative evaluations of creative ability. In addition, there are survey measures that ask individuals to self-report on creative activities and achievements, as well as studies that focus on individuals for which creativity is a component of their profession, such as entrepreneurs and artists.

One key finding using these various measures is that higher performance on measures of creativity is associated with faster productivity. That is, as the equal-odds rule would predict, greater creativity goes hand in hand with greater productivity.

In divergent thinking tasks this productivity is often called *fluency*. In the alternative uses tasks ("how many uses can you think of for a paperclip"), greater fluency is correlated with greater novelty of uses. Novelty also increases with each new production: something called the *serial order effect*. With rising novelty of uses comes novel strategies, such as disassembling the item (e.g., melting the paperclip down and making a necklace) which is speculated to involve more effortful cognitive control (Beaty & Silvia, 2012; Gilhooly et al., 2007). The increased speed underlying this creativity is not limited to producing more creative solutions: greater fluency in the alternative uses tasks is also associated with faster judgments about the uniqueness of concepts (Vartanian et al., 2009). This suggests that more fluent production is also associated with faster evaluation – a useful combination. Already in this brief overview we see evidence of three features of greater creativity: speed, novelty, and effortful cognitive control.

Two prominent theories of creativity have been offered to account for these differences. These two positions epitomize the map-and-vehicle framework introduced in Chapter 11. There is a map and there is some process that operates on the map. For creativity, one theory, *the associative theory of creativity*, claims that the difference between high and low creatives is in the map – the structure of the knowledge we represent in our minds. Creative people think differently because the relationships among the things they know are different. The other theory, *the executive theory of creativity*, claims that creativity is not a consequence of the map, but is an outcome of the processes that people use to navigate that map: creative people think differently because they use what they know in different ways. One may imagine colorful mental vehicles such as the school bus of mediocrity and the clown car of creativity.

The general consensus is that both theories – both map and vehicle – underpin creativity. That is, both bottom-up associative and top-down executive processes play a role (Beaty & Kenett, 2023; Benedek et al., 2023). But we do not yet understand how they might work together. As Lloyd-Cox et al. (2023) note, we need better models to integrate these. To achieve that, we must first understand these theories in more detail.

13.3.1 The Associative Theory of Creativity

The associative theory of creativity argues that a person's creativity is a consequence of mental structure: "The organization of an individual's associations will influence the probability and speed of attainment of a creative solution" (Mednick, 1962, p. 222). A common metaphor is that of a rugged landscape. If we envision ideas in one's mind as sitting atop peaks on the landscape, then differences in the difficulty of traveling from one peak to the next is the difference between high and low creative individuals. Creative individuals, Mednick (1962) proposed, have more accessible "flat" associative landscapes, while less creative individuals have less accessible "steep" associative landscapes. Figure 13.1 visualizes this difference. Plainly put, the flatness of the creative mind puts otherwise distant concepts close at hand. Meanwhile, the uncreative mind is stuck on James' treadmill of routine, less able to cross the steep canyons to new ideas.

This simple metaphor for the distance between ideas has been supported by numerous studies, which show that individuals who score higher on various measures of creativity

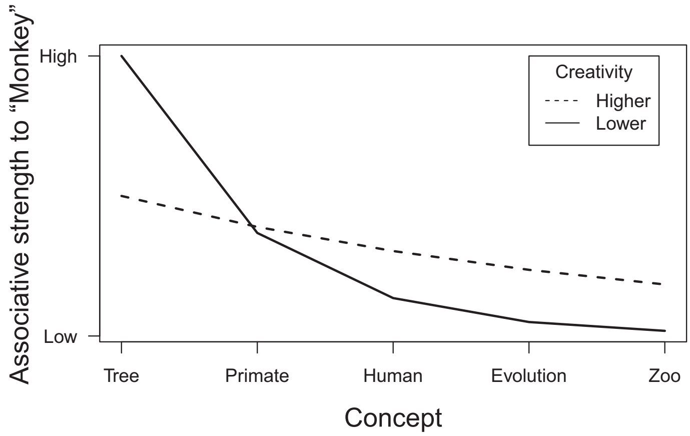

Figure 13.1 Associative landscapes for high- and low-creative individuals, as proposed by Mednick. The associative strength indicates the strength of the relationship between a concept and the cue – in this case the cue is "Monkey." The lower the associative strength, the less likely the cue and target concept are to be co-activated. The figure is adapted from Mednick, 1962, p. 223.

also tend to produce differently structured knowledge representations. For example, when asked to produce associations, they produce more associations, more quickly, and with more variability than less creative individuals (e.g., Benedek & Neubauer, 2013; Kenett et al., 2014). Even when cued to provide a single creative word – a verb in response to a noun – the creativity of the response, as measured by latent semantic analysis, correlates with a variety of individual creativity measures (Prabhakaran et al., 2014).

In a clever application of network science to this problem, Kenett et al. (2014) evaluated the semantic structure of free associations produced by high and low creatives. People also completed several creativity tasks, including the remote associates task (RAT) (Mednick, 1962), the Tel-Aviv University Creativity Test (TACT) (Milgram, 1963), and the Comprehension of Metaphors task (CoM) (Faust, 2012). These tasks were then used to classify individuals into high and low creatives.

To study their semantic networks, Kenett et al. (2014) had participants produce a list of free associates in response to different cue words, with 60 seconds for each word. Each participant did this for 96 cue words. The free associations produced by participants for the 96 cue words were then used to produce networks for the high and low creative individuals in the following way. First, a matrix of cues by associates was produced for each group. This matrix has 96 rows for the cues and as many columns as there were associate words. Each cell in the matrix represented the number of individuals within the high or low creative group who produced that associate in response to that cue. If 10 people in the high creative group produced the word "food" in response to "banana," then the value in the "banana, food" cell of their matrix would be 10.

In a different approach to projecting feature networks, the cue by associate matrix is used to produce a correlation matrix for the cue words, producing a 96×96 cue-by-cue matrix. Each cell was the output of the Pearson correlation for the corresponding rows in the cue by associate matrix (see Figure 13.2).

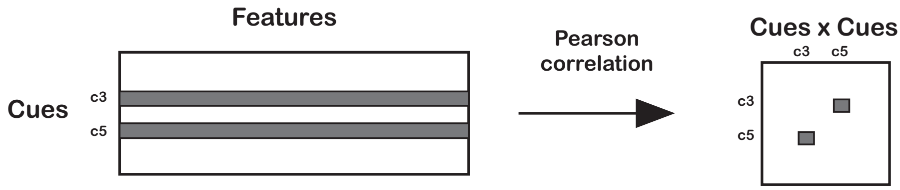

Figure 13.2 Kenett et al.'s method for turning cue-by-feature (target) networks into undirected and weighted cue by cue networks. (Left) The cue by feature matrix gives an indication of which features are associated with each cue. Cue 3 and cue 5 are represented by c3 and c5. The Pearson correlation between the vectors for c3 and c5 is used to fill out the corresponding values for the cue by cue matrix shown on the right.

This cue-by-cue correlation matrix is a weighted graph with a value between 0 and 1 in each cell. This was converted into a simple graph using the Planar Maximally Filtered Graph method (PMFG) (Tumminello et al., 2005). Briefly, this method orders edges by strength and then keeps only the strongest edges that can be projected onto a plane without overlap. I describe this and other methods for filtering networks in more detail in the Open Questions section.

Kenett et al. (2014) compared the resulting PMFG networks by computing the average shortest path length, L, the average degree $\langle k \rangle$, the clustering coefficient C, the network diameter D, the modularity of each network Q (following Newell & Simon, 1972), and the small-world index S (following Humphries & Gurney, 2008).[1] This comparison showed that high creatives had lower L, lower C, lower D, lower Q, and higher S. Because the PMFG method tends to control for $\langle k \rangle$, no differences were found there.[2] That is, following the prediction of the associative theory of creativity, nodes in the semantic network of higher creatives are nearer to one another, among other things.

Since there are only two networks here – the high and low creative networks – this makes comparing the networks statistically challenging. In effect, as is often the case when comparing the macroscopic structure of networks, each network is but one data point. So if one has two networks produced by two different populations, then the statistical analysis is based on an n of two. However, Kenett et al. (2014) tackled this problem using three additional approaches:

- *Impact analysis*: each node in the low and high creativity networks was removed, recomputing L, after each removal, and then taking the difference between L before and after the removal. This provided a distribution of differences that indicates how the nodes in the network contributed to the overall distribution of path lengths. See Vitevitch and Goldstein (2014) for a similar approach.
- *Bootstrapping*: This involved producing 1,000 randomly selected subgraphs. For each subgraph, half of the target nodes were randomly chosen and then the graphs were recomputed to produce new cue-by-cue networks. This allows for two networks to be compared statistically based on the distribution of their subgraphs.

[1] Described in Chapters 2 and 4.

[2] There are excellent off-the-shelf tools for investigating semantic network structure (e.g., Christensen & Kenett, 2021).

- *Unique productions*: The number of unique target words for each cue word were computed for each group.

These three tests each supported distinct differences between the high- and low-creative semantic networks. High-creative individuals produced nodes which played a stronger role in the connectivity of the overall graph, produced networks with statistically different populations of subgraphs (reconfirming the original differences), and produced more unique responses per cue word.

13.3.1.1 Percolation Theory

Percolation theory offers another approach to comparing networks and it attempts to answer the question "Is there a path between two points?" (Saberi, 2015). A classic percolation problem is the Swiss Cheese Problem: imagine you have a block of cheese, full of tiny holes, and you pour a liquid on one side of the cheese. What is the probability that the liquid will pass through the empty spaces to get to the other side? To pass through to the other side, there must be a continuous series of internal spaces connecting one side to the other. Percolation theory focuses on understanding this probability and how it is arrived at in different systems.

One of the central findings in percolation theory is that there is a critical value for the probability of holes, p_c, called the *percolation theshold*. Near the percolation threshold, long-range paths rapidly appear or disappear as one increases or decreases the probability. This general finding applies to a wide variety of structured systems and therefore reflects a feature of complex systems. It also has many behavioral applications: Can a rumor reach a distant person through a set of acquaintances? Is there a supply line that can reach a population in need? If a neuroinvasive disease attacks one area of the brain, what is the likelihood that it will threaten areas of the brain elsewhere? The nonlinear approach to the percolation threshold will be true in all of these systems and reflects a sudden change in system structure associated with what the complex systems literature calls *criticality*.

Percolation problems are applied to networks either as site-percolation (removing nodes), or bond-percolation (removing edges). Here, we focus on bond-percolation and ask what happens to a network's structure as the probability of removing edges increases or decreases. Figure 13.3 shows a series of 2D lattices each subjected to different probabilities of random edge removal. As the probability of edge removal increases, the size of the largest component (shown in black) decreases. The lower panel shows the average size of the largest component for 100 simulations at each probability. The inflection point, at $p = 0.5$, is the critical value for percolation of this network.

Kenett et al. (2018) applied a percolation analysis to the association network data from Kenett et al. (2014). They did this by asking how the networks for those two groups would break down as edges below a moving threshold were removed. What Kenett et al. (2018) found was that higher-creative individuals had networks that were more robust to edge removal – their percolation threshold was higher. For each value of the edge threshold, they recompute the size of the largest component. As the threshold increases, the network of the low creative individuals breaks apart first and proceeds to decay into increasingly smaller subcomponents.

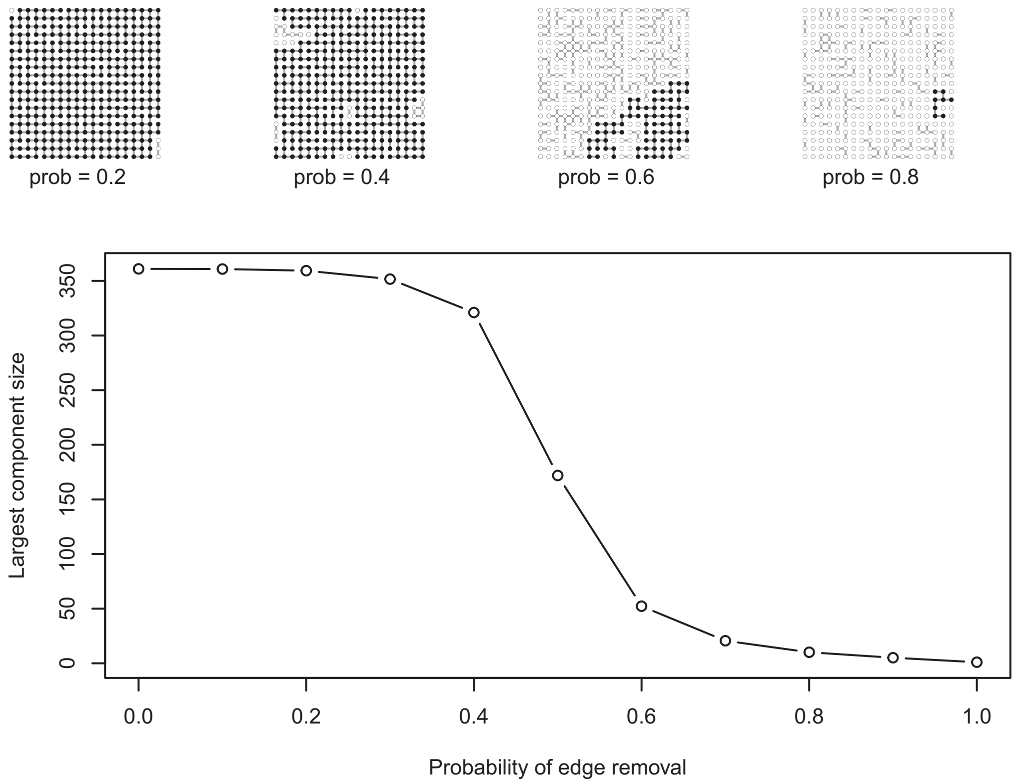

Figure 13.3 Percolation of a 19 × 19 lattice. The top panels show the largest component in black for a series of networks with increasing probabilities of edge removal. The bottom panel shows the size of the largest component across a series of edge removal probabilities, with 100 simulated networks at each probability.

Kenett et al. (2018) established that this result is not merely a consequence of higher creatives having higher edge weights. They demonstrated this by randomly shuffling the edge weights within each network (see the configuration model in Chapter 4). This removes the structure in the network. The result of this shuffling is that lower-creative networks become *more* robust than higher creatives.

The associative theory can also be evaluated using divergence within a participant's sequences of responses. Gray et al. (2019) showed that various measures of divergence – including one called *forward flow*, based on path length – were correlated across tasks and with real-world creative activity (see also Beaty et al., 2021). A similar and more direct measure of deliberate creativity, the *divergent association task*, asks participants to produce 10 nouns that are "as different from one another as possible" and then computes the average distance between the first seven valid words. Olson et al. (2021) found that this average distance correlated with measures of flexibility, originality, and fluency in the alternative uses task. These various path measures are consistent with more creative individuals traveling further across the mental landscape to reach more divergent associations.

Still other methods have demonstrated that divergence is related to effortful cognitive control. The *associative chain test* asks participants to produce separate lists of similar and nonsimilar associations. Because producing nonsimilar associates is more effortful, one can compute the retrieval time differences between the two lists to get a measure of cognitive control. This time difference was found to correlate with semantic relatedness judgments and is influenced by the addition of a secondary task (i.e., a cognitive load) – a clear indication of the influence of cognitive control (Marko et al., 2019; Marko & Riečanskỳ, 2021, 2023).

Broadly, the evidence reported here demonstrates that high and low creatives retrieve different kinds of associations. This is consistent with the associative theory of creativity, but it may also reflect processing differences, which we turn to next.

13.3.2 The Executive Theory of Creativity

The executive theory of creativity focuses on the relationship between creativity and measures of cognitive control. The central idea is that creativity is mediated by a top-down control process (Benedek et al., 2017). This process is proposed to involve effortful cognitive control, which involves mental effort and can be interfered with by having the participant do something else at the same time. Effortful cognitive control is proposed to influence how knowledge is navigated and combined. In principle, control could also influence the structure itself, for example, by altering edge weights, thresholding, or some other means (see the Open Questions). Whatever the mechanism, the executive theory proposes that two individuals who have similar knowledge representations could use different strategies to access that knowledge, leading one to produce more creative responses than the other.

Contributions of effortful control to creativity are supported by a wide variety of evidence. For example, intelligence and other measures of executive function correlate with creativity (Benedek et al., 2014; Nusbaum & Silvia, 2011). In addition, a great deal of neuroimaging work finds relationships between creativity measures and brain regions associated with cognitive control (reviewed in Beaty et al., 2016). For example, resting activation levels in the brain of higher creative individuals show increased connectivity between key regions of control and representation, which Beaty et al. (2014) interpret as "pointing to a greater cooperation between brain regions associated with cognitive control and low-level imaginative processes" (p. 92). Indeed, one of the most replicable neural patterns associated with creativity is that higher creatives show greater interconnectivity between the executive control network and the default mode network (Beaty et al., 2014; Evans et al., 2020; Ovando-Tellez, Benedek, et al., 2022): The former is associated with goal-directed processing and the latter associated with imagination (among other things).[3] This pattern even extends to musicians engaged in musical improvisation (reviewed in Beaty, 2015). Thus, control appears to be a key contributor to creativity and one that touches on a wide variety of laboratory and real-world tasks.

One piece of evidence that speaks more directly for the executive theory and against the associative theory is from Benedek and Neubauer (2013). This work looked at patterns of continuous free associations to a set of six cue words, with participants providing

[3] We also saw this pattern associated with older individuals in Chapter 12.

associates for 60 seconds in response to each cue. Higher creatives produced the familiar pattern of more rapid and more uncommon associates. However, the uncommoness of their associates appeared to be driven more by production rate than semantic structure. Both high and low creatives produced their strongest associates with equal association strength (measured here as frequency of associates summed across participants). Moreover, by limiting the total number of productions made by the creatives, the uncommonness of their associates largely disappeared. A simple interpretation of this result is that both groups have similar semantic structures, but higher creatives navigate their structures more quickly, leading them to exhaust the more common associates and move on to the less common associates before the task is complete.

Other studies have been less straightforward. Benedek et al. (2017) investigated the relationship between creativity, semantic structure, and measures of executive cognitive control. The creativity measure used was the alternative uses task and the executive cognitive control measures were fluid intelligence (ability to solve novel problems) and broad retrieval ability. Benedek et al. (2017) also had individuals produce individual semantic networks by asking them to rate the similarity between 28 concepts. Benedek et al. (2017) found that some network transformations of the similarity network showed structural correlations with divergent thinking in the alternative uses task. Higher creatives also produced more similarity judgments and had shorter average shortest path lengths, consistent with prior work (Kenett et al., 2014; Rossmann & Fink, 2010). Finally, Benedek et al. (2017) also found a correlation between creativity measures and broad retrieval ability, but not fluid intelligence.

In summary, we see evidence of semantic network differences combined with various forms of evidence for contributions from executive control, but it is not yet clear exactly how they might work together.

13.4 The Riddle of Creativity

The above results are consistent with both associative and control theories. There are also numerous reliable patterns in the data: compared with less creative individuals, higher creatives (1) more rapidly produce associates and produce more of them per cue word, (2) produce less common associates, on average, (3) report greater similarity between pairs of items, 4) have more interconnected semantic networks – either measured by associate count (degree) or edges above threshold – with shorter average path lengths between nodes, (5) have more neural interconnectivity between areas of the brain associated with control and the default mode network (among other regions), and (6) show behavioral contributions from cognitive control.

Embedded within these results is a riddle. If we ask ourselves how processing might work to spread activation from a cue node to a target node, then it should strike us as odd that associates are being produced faster *and* they are more interconnected across the network as a whole.

Being more interconnected and producing greater activation in individual nodes violates our understanding of how activation moves through a network. If a fixed amount of activation leaves a node, less of it will reach any specific node if the activated node is of higher degree. This is easy to understand: if a target node were connected to only one other node, the other node would receive all the activation that leaves the target node. If the target node were connected to 10 other nodes with equal edge weights, they would each receive 1/10th the activation.

The observation that higher-degree nodes spread less activation to each individual neighboring node was demonstrated in Chapters 11 and 12. It is also a result that is supported in a wide variety of experiments that lie beyond the domain of creativity research. For example, in the *fan effect*, people are taught many or few relationships with various target concepts. The more relationships one learns with a target concept, the longer it takes to retrieve any one of those relationships (Anderson & Reder, 1999; Beaty et al., 2023). Whether we consider random walkers, spreading activation, or some other mechanism, greater connectivity should lead to lower activation of potential targets.[4]

This is true unless something additional is at play. Cognitive control is the obvious candidate and offers several possibilities for how it might remedy this problem. We can benefit from describing them in detail.

One way to increase the production rate in free association tasks is for higher creatives to release more activation from the cue node. This would lead targets to rise above threshold and be produced more quickly. Where would this extra activation come from? If cognitive control is involved, it could come from cognitive control better maintaining attention to the task at hand.

If this is all that is going on, then this would be reflected in a *raised* version of the low creatives' association strength for all items (see Figure 13.1). In other words, we create *faster-but-proportional* retrieval by increasing the area under the curve, leaving the relative proportions of different words the same. This would drive up the production rate for all items equally and follow the pattern suggested earlier by Benedek and Neubauer (2013). According to this faster-but-proportional hypothesis, we need only additional activation to explain creativity.

However, the faster-but-proportional hypothesis would not explain the flatter curve proposed by the associative theory, as shown in Figure 13.1 for higher creatives. The flatter curve implies easier access to more distant words. However, as noted, a simple change in structure involving more interconnectivity would *decrease* speed. We would therefore still require the added activation to overcome the increase in degree. According to this model, activation must be both *faster-and-further*.

We can frame these two hypotheses relative to the original Mednick proposal. These are presented alongside one another in Figure 13.4. To further formalize these views, I let *faster-but-proportional* be a proportional increase in association strength – equivalent to raising the initial intercept and increasing the area under the curve. Otherwise the curve follows the decay of lower creatives. For *faster-and-further*, the curve reflects the flatter association curve proposed by Mednick, but raises the activation across concepts indicating the need for an additional process to explain the faster production rate with greater interconnectivity.

[4] These methods are described in detail in Chapters 10, 11, and 12.

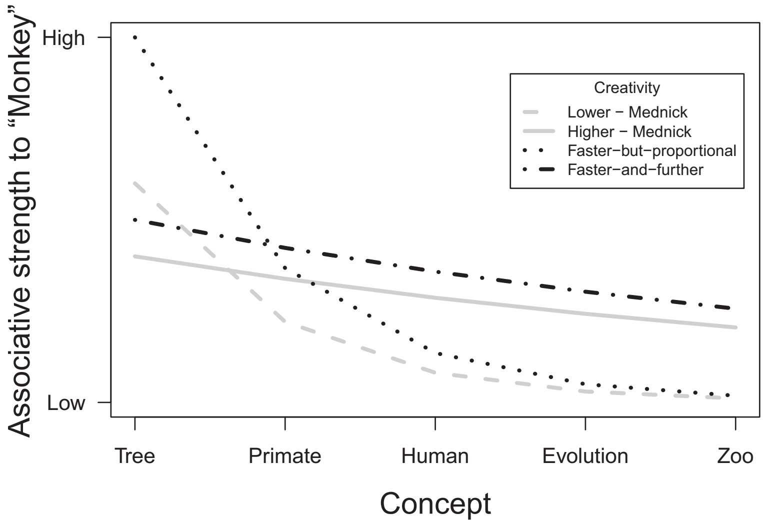

Figure 13.4 Two hypotheses to explain greater creativity based on the evidence for faster productions and less common productions among higher creatives. The original Mednick curves are shown in gray. The faster-but-proportional hypothesis proposes that higher creatives have additional activation strength, but are otherwise similar to lower creatives. This leads to a proportion increase in activation across all concepts. The faster-and-further hypothesis proposes that activation is faster but also reaches further through greater interconnectivity, flattening the distribution. I represent this by raising the activation strength relative to Mednick's original curve, indicating that some additional source of activation strength is required to overcome the greater interconnectivity. The associative strength indicates the strength of the relationship between a concept and the cue ("Monkey"). The lower the associative strength, the less likely the concept is to be activated by the cue.

The results from Benedek and Neubauer (2013) described earlier are suggestive of the faster-but-proportional view. But network differences are strongly suggestive of the faster-but-further view. In what follows, we will first visit more recent work by Ovando-Tellez, Kenett, et al. (2022), which offers the opportunity to revisit these hypotheses with a new data set before evaluating more formally the underlying mechanisms that might generate them.

13.4.1 More Creative People Are Not Only Faster, They Are Also Less Predictable

Ovando-Tellez, Kenett, et al. (2022) examined differences in similarity judgments between pairs of words among individuals who self-reported being more or less creative in activities and achievements. Each individual provided pairwise similarity ratings between 35 different items. By neuroimaging individuals as they made this comparison, Ovando-Tellez, Kenett, et al. (2022) found that more interconnectivity between brain regions (e.g., control and default mode network) predicted structural differences in their similarity judgments, which in turn predicted more creative behaviors.

Using the online data provided by Ovando-Tellez, Kenett, et al. (2022), we can compare people's semantic relatedness judgments with their creativity. Using the similarity judgments, we create networks for each individual, using the judgments as edge weights. It is then straightforward to recompute the network measures and confirm the findings in the original article. For example, using the nonparametric Kendall's rank correlation, we find that individuals who self-reported more engagement in creative activities also had shorter average shortest path length in their weighted similarity judgment network, $\tau = -0.20$, $p < 0.01$.[5]

Figure 13.5 also shows that the mean similarity judgments provided by each individual correlate positively with their creative activities and achievements. The right-hand side of the figure shows the distributions of mean-similarity judgments for the top and bottom quartiles of creatives, providing a visual representation of how the groups differ in their distributions of judgments. Both groups show a preference for low similarity with a small tendency to give high ratings to a select group of pairs. However, higher creatives are shifted rightwards in their similarity judgments. This further confirms past work showing that higher creatives judge items to be more similar (Benedek et al., 2017; Rossmann & Fink, 2010).[6] However, it also appears that higher creatives' similarity ratings are more equally distributed, consistent with a flatter curve.

We can measure inequality more directly using the Gini coefficient. The Gini coefficient quantifies the relative absolute differences among all pairs in the population, with higher numbers indicating more unequal distributions. Using the Ovando-Tellez, Kenett, et al. (2022) data, we find that lower creatives have a higher Gini coefficient ($G = 0.31$) than higher creatives ($G = 0.27$). This suggests that higher creatives have flatter association curves than lower creatives.

These results are most consistent with the faster-and-further hypothesis presented in Figure 13.4. Both faster-but-proportional and faster-and-further would produce higher similarity judgments. But only faster-and-further would produce a flatter association curve for higher creatives.

We can go further. Taking our inspiration from Benedek and Neubauer (2013), we can ask to what extent higher and lower creatives agree with other members of their group in terms of what they rank as the most similar pairs of items. We will evaluate this by running a moving threshold across similarity-based edge weights, limiting them to only the top *n*th percentile of edges, and then take correlations between individuals and other members of their group. As the threshold increases, we expect the correlations for both groups to rise – stronger similarities should be easier to agree on than lower similarities. But if higher creatives produce more diverse similarity judgments – even among the strongest associates – they should also be less well correlated with one another. Figure 13.5 shows that this is the case. Comparing how well the participants in the lower and upper creativity quartile correlate with one another, the figure shows that higher creatives are less well correlated with one another than lower creatives. It also shows that this increased disagreement among higher creatives does not go away as we limit the analyses to only the strongest similarity judgments – they are more likely to disagree even among their most similar items. We can conclude from this that faster-and-further is the more likely explanation.

[5] Ovando-Tellez, Kenett, et al. (2022) used permutation tests to evaluate the statistical significance. This involves comparing the observed data with a distribution of shuffled versions of the same data to establish how unlikely the observed data is. Here I use Kendall's rank correlation.

[6] People tend to look for different things when asked to identify reasons for or against something. This might mean that high creatives could identify more differences, if asked to do so. The data provided here suggests they were looking for similarities. But if they are more fluent whatever the task, they should also be able to produce more differences.

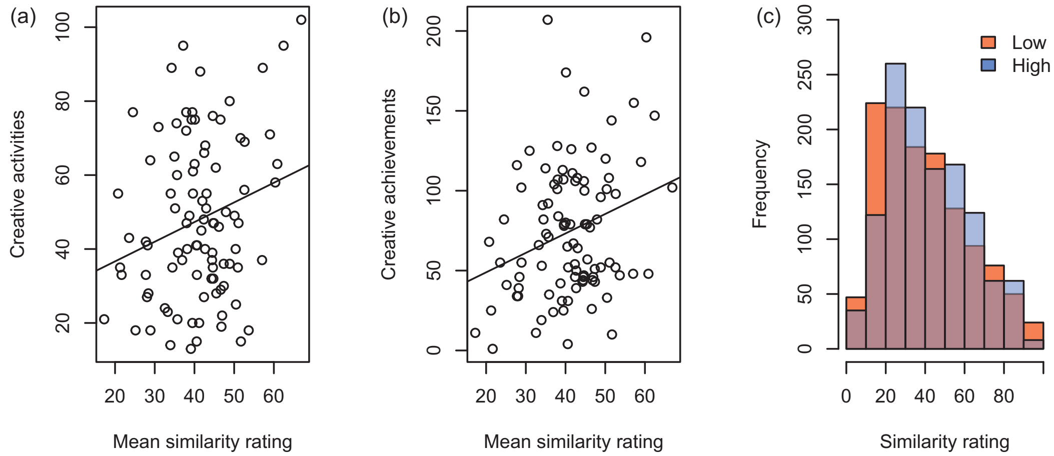

Figure 13.5 Relationship between mean similarity rating and creativity. (a and b) Mean similarity judgments correlate positively with creativity measures. Each dot represents an individual. (c) The disribution of mean-similarity ratings for the top and bottom quartile of high and low creativity indivdiuals. Higher creatives tend to produce higher similarity ratings.

For comparison – and to point out a pitfall – I provide the correlations between the similarity ratings for the aggregated networks of the high and low creative quartiles. The aggregate networks are created by averaging similarity ratings within each group. Aggregation effectively filters out the within-group variance. For example, if high-creative individuals tend to be less consistent with one another in within-group comparisons, but similar to low creatives *on average*, then aggregating the network will hide the within-group variance and make the two groups look similar. As panel (b) of Figure 13.5 shows, as we raise the threshold, the aggregate networks look more similar. What is more challenging to conclude from this aggregate comparison is whether high creatives are less predictable than low creatives, because there is no point of comparison. Panel (a) provides that comparison, and shows that high creatives are less predictable (one from the other) than low creatives even when the threshold is high.

This all supports the faster-and-further view. If we take similarity judgments as reflecting what the underlying semantic networks represent – which would be consistent with much of the work described here – then this rules out certain kinds of models. For example, it cannot simply be that more creative individuals have greater activation at the starting point of spreading activation. They must also have differently structured networks or spread activation in a way that leads to different patterns of retrieval. How could this work?

13.5 A Noisy Model of Creativity: Getting Structure from Control

There are many potential ways to model faster-and-further. I will outline two here, which provide different origins for the interconnectedness among higher creatives.

First, if we stick closely to the interpretation of the associative theory of creativity, we know the ingredients of creativity require two things and we know roughly what

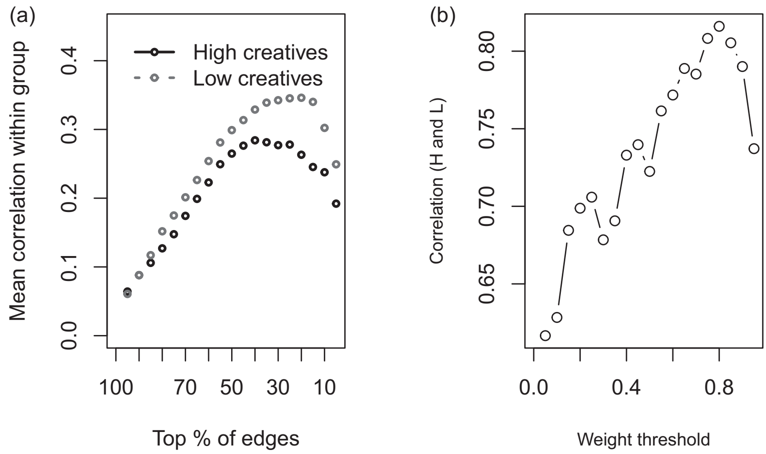

Figure 13.6 Two comparisons of high- and low-creatives networks, limited to the top and bottom quartiles. (a) Within group correlations for high and low creatives limiting the edges to the top Nth percentile. This shows us that creatives are still more divergent even among the strongest edge. (b) the correlation between aggregate high and low creatives rises as the edge threshold increases. This shows us that judged pairs with the highest ratings are more likely to be shared by the high and low creatives than pairs with lower ratings.

they are: developmental differences and increasing activation. Development or learning differences are required to create a more interconnected network. And additional activation is required so that neighboring words reach activation sooner. Together these achieve a faster-and-further pattern of production: *faster* is created through additional activation and *further* is created through more interconnected semantic networks. This *development-plus-added-activation model* requires thinking about network development.

One developmental pathway is to expose learners to different experiences. Different patterns of experience produce different patterns of connectivity. This follows the Hebbian learning rule that neurons that fire together wire together. An alternative pathway is to assume that individuals have the same experiences but a different cognitive mechanism for processing those experiences into different representations.

Both of these developmental explanations still must confront the problem of greater connectivity *and* faster activation. But one of these two must be true to support the associative theory as we currently understand it: the different representation must come from somewhere, and it either comes from what people experience or how they encode what they experience.

Can we get around the requirement of having different developmental pathways for higher creativity? That is, can we produce different free association production patterns for high and low creatives from the same underlying representation? To set an even higher challenge for ourselves, can we do it without assuming greater activation? The answer is yes, we can.

One way to achieve this is to have higher creative individuals increase the noise in their retrieval process. Let higher creatives be better curators of noise.

A noise curation model is consistent with many different kinds of evidence. For example, it is inspired by the evidence that humans can increase random exploration in response to environmental uncertainty (Wilson et al., 2014). Similarly, a number of studies show that cognitive control supports random sequence generation (Baddeley, 1998; Oomens et al., 2021; Towse & Cheshire, 2007; Vandierendonck et al., 1998). We also know that random exploration (and unpredictable behavior more generally) in a variety of animals – including birds, humans, monkeys, and rodents – is an adaptive response to environmental change supported by variability in neural activation (Hills, 2019a; Wilson et al., 2021).

There are also many theories of creativity that focus on the stochastic nature of creative productivity and the influence of leaky or deliberate focusing of attention to external noise (Malthouse et al., 2022; Simonton, 2003; Zabelina et al., 2016). Finally, creativity can also be increased through conscious (deliberate) control (Green et al., 2015; Weinberger et al., 2016). The curation of noise therefore also offers us a mechanism that is consistent with cognitive control.

Incorporating a noise curation process into creativity leads to what I will call the *the noise curation model*, which I offer here for further exploration. The central idea is that higher creatives have noisy activation of potential associates, but maintain the same average level of activation.

The noise curation model is designed to reproduce the semantic associate results described earlier as follows: Starting from a given cue node, other nodes are sampled for retrieval, each with a probability inversely proportional to their path length from the cue node – nodes that are further away are sampled less often. When sampled, the node receives an activation increment, δ. The sampling process is repeated until the sum of one node's activation exceeds a threshold. At that time, the word exceeding the threshold is reported.

This is a kind of *race model*, in which different potential targets compete to reach some threshold level of activation, each accumulating activation over time (Heathcote & Matzke, 2022). In sum, retrieving a word is equivalent to adding incremental levels of activation to words chosen randomly as a function of their distance and retrieving the first word that passes a threshold level of activation.

Creativity is modeled by adding noise to the retrieval process. We achieve this by adding Gaussian noise, ξ, to the activation increment: $\delta + \xi$. The Gaussian noise has mean 0 and standard deviation σ. The difference between high and low creatives lies in the standard deviation of the noise, with higher creatives having more noisy activation, $\sigma_{high} > \sigma_{low}$.

I simulate this process 10000 times for high and low creatives. We will assume all individuals sample nodes from the same underlying representation: an Erdös–Rényi random graph with $N = 500$ and edge probability $p = 0.005$. The network is shown in panel (b) of Figure 13.7. To compute the distance table, distances are computed from a randomly chosen cue node. All nodes start with 0 activation. Nodes are then sampled repeatedly with probability inversely proportional to their distance from the random cue. Each node sampled receives an activation increment that is added to its current activation. This is repeated until a node's activation exceeds a threshold, $\tau = 5$, at which point the node is reported. Here we let $\sigma_{high} = 2$ and $\sigma_{low} = 0.5$.

The mean reaction time is shown in the bottom left panel of Figure 13.7. It is computed as the number of words sampled before one reaches the threshold. The reaction time is

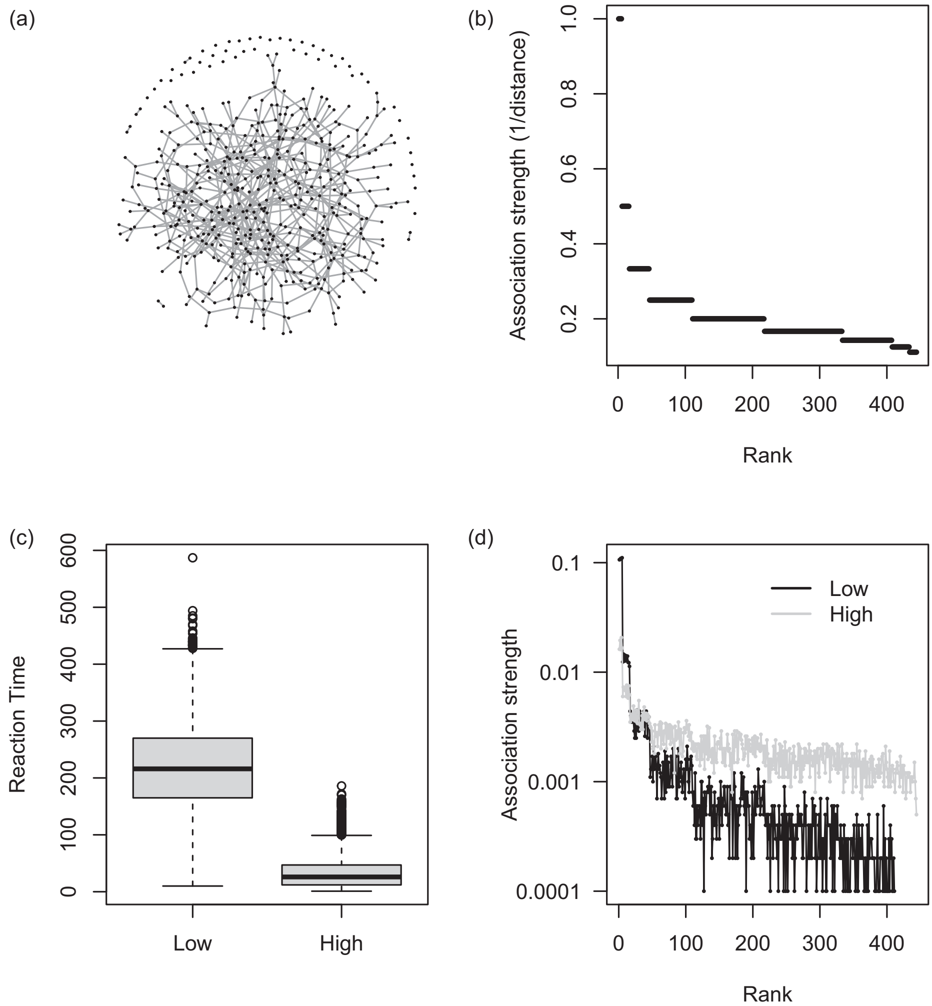

Figure 13.7 The noise curation model of creativity. (a) The network representing the common underlying cognitive representation shared by high- and low-creative individuals, (b) The distribution of target node distances from a cue node in the large component in A. The probability of sampling a node is inversely proportional to its distance from the cue node. (c) The average retrieval times for the first 1000 retrievals for the high and low creatives. Retrieval times reflect the number of samples before a word reaches threshold. (d) The association strengths produced by the noise curation model for high and low creatives. The association strength is the frequency of a target word retrieval divided by the total number of retrievals. This normalizes the curves for high and low creatives, who each retrieve different numbers of words. The only difference between high and low creatives is that high creatives add more Gaussian noise to their activation increment for each sampled word.

faster for the noisier retrieval process of the higher creatives, and thus they are more productive.

Panel (c) in Figure 13.7 shows the retrieval proportion for the various target nodes for the high and low creatives. Total productions are normalized to account for the additional number of retrievals of high creatives. The pattern of activation produced closely follows that of the faster-and-further hypothesis, with high creatives producing items further away from the cue node.

These results suggest that some form of noise upregulation could be responsible for the faster-and-further phenomenology. This model is parsimonious in that it does not require two separate explanations for higher creativity: a different developmental process for generating a different representation and different activation strengths for overcoming the diffusing effect of higher connectedness on spreading activation. It demonstrates that creativity can be the consequence of turning up the noise in the retrieval process – something we suspect cognitive control can achieve. For a more thorough treatment of this model see Hills & Kenett 2024.

13.6 Summary

Many theories of creativity have been based on theories of productivity: creative people tend to be more productive. But network science applications to creativity have helped quantify structural differences in creative productions. They have also suggested potential contributions of cognitive control. Structural differences include different patterns of free association production and different patterns of similarity judgments. These both correlate in consistent ways with a wide variety of creativity measures and patterns of brain activation. These support the notion that more creative individuals genuinely "think differently" – they do not just think faster. And it is possible that both are driven by the same underlying process.

We explored several explanations for how this thinking differently might come about. These included developmental explanations and alternative processing accounts. Perhaps the most parsimonious of these is the noise curation model: a model based on the control of noise. This revisits views from James and Nietszche, and creativity as a variance-selection process, but also suggests that more creative individuals are better at amplifying the variance. This also increases the speed of production. Thus, we learn that structural differences in production do not require structural differences in the underlying representation. Processing differences can achieve the same results. Developing predictions from these various models will help to refine our questions to better tease them apart in the future.

The noise curation model does not require us to overlook the obvious. We know that creativity can be taught and used strategically (Gibbert et al., 2012; Goldenberg et al., 1999). We also know that many of those hailed as icons of creativity immersed themselves in the creative products of their culture. Artists like Vigée Le Brun, Van Gogh, Picasso, Tom Morello, and Herbie Hancock spent a surprising amount of time copying the works of the masters who came before them. This undoubtedly changed the way they represented artistic products, but also helped them to see them in a broader cultural context. To be culturally creative is almost assuredly to understand the landscape well enough to know where to look for new ideas.

13.7 Open Questions

13.7.1 Network Filtering

Many network science researchers create networks via data transformation, routinely using various transformations of the same raw data to produce a variety of networks (e.g., Benedek et al., 2017; Kenett et al., 2014). This speaks to the richness of network methods to elucidate structural differences. Indeed, one should think of different approaches to transforming networks as different kinds of filters that let various kinds of information pass through while holding other kinds back. Each method provides a different view on what is structurally important in the information it represents. As is often the case when theory is limited, a more exploratory approach can help reveal the patterns that are consistent across studies, which can help improve theory going forward. This has been precisely the case in creativity research.

Weighted networks are especially amenable to transformation, as one can deal with the weights in different ways. Moreover, computing structural statistics on weighted networks is underdeveloped compared with unweighted networks, making results from weighted networks often more challenging to interpret. If all edge weights are nonzero, or have values ranging over orders of magnitude, this can make some measures meaningless (e.g., degree) or subject to subtle assumptions (e.g., clustering coefficient). If edge weights are highly skewed, with many edges of small weight, then it is useful to consider ways that focus on extracting the information most relevant to the problem at hand.

There are various ways to go about this. Some of the methods used by Kenett et al. (2014) and Benedek et al. (2017) are described below. There are many other approaches (Christensen et al., 2018; Epskamp et al., 2018), but these provide a basis for comparison.

- *Fixed edge number*: This method keeps the N strongest edge weights. Only networks that have at least N nonzero edges are kept in the analysis. Benedek et al. (2017) used the top $N = 100$ relatedness judgments. This method has the benefit that all networks have the same density. Its drawback is that different individuals will have rated the remaining edges with different degrees of similarity.
- *Fixed minimum relatedness*: This method keeps all edges above a set threshold, with the same threshold applied to all individuals. A similar approach is to use a moving threshold and statistically evaluate the networks at each threshold (Wulff, Hills, et al., 2022). The benefit here is that the same threshold is applied to everyone. The drawback is that different individuals will have different numbers of edges in their network for different thresholds.
- *Planar maximally filtered graph*: This information filtering method produces a network that can lay flat on a surface without any crossing edges (Tumminello et al., 2005). It achieves this by ranking edges in the correlation matrix from strongest to weakest. Then it produces a new network by adding an edge corresponding to the highest

ranking correlation, connecting the two most strongly correlated cue words. As it goes down the ranked edges, it adds each edge if and only if the addition of that edge would still produce a planar graph, which can fit on a sphere without edge overlap, skipping all those edges that would violate the planar requirement.

What kinds of structural problems are each of these methods most appropriate for? For example, planar maximally filtered graphs tend to minimize differences in degree (because cliques are limited to 4-tuples to avoid edge overlap). This can hide differences in degree, effectively transforming them into other structural properties. Yet, at the same time, to compare networks we need to constrain them in ways that limit their variation along some parameters so we can evaluate them along others. When is that appropriate? What structural features do the different methods amplify and what do they diminish? This is a problem about which we know very little (but see Christensen et al., 2018).

13.7.2 Network Flexibility

Another explanation for creativity is that the representation is changed at the time of search. Consider the way one looks for a rarely used and now missing key. A good start is to look in the location where it usually is found, perhaps in the top drawer in the kitchen. If this fails, one will start to look in other places. One may look on counter tops, in other drawers, the pockets of jackets, backpacks and so on, searching increasingly less likely places. How one manages this exploration-exploitation dilemma is critical to an efficient search and it is a route along which people are likely to vary. Some individuals may choose to explore more quickly than others, and this would lead to greater exploration and thus potentially more creative outputs. Others may choose to exploit uncovered locations to thoroughly examine their potential contributions. As a result, their creativity may focus more on the details. This strikes me as a creative application of "breadth first" search – which first explores more diverse avenues – and "depth first" search – which first explores the more derivative possibilities of a single avenue, possibly exploiting variations on a prominent pre-existing solution.

How might we apply the exploration-exploitation dilemma to creativity using network science? Two possibilities come to mind. One is that the search process changes in response to the difficulty of the problem. Specifically, the sensitivity parameter, β, in the search model could change: $P(w_i) = \frac{S_i^\beta}{\sum S_j^\beta}$.[7] This could change as a function of time, $\beta(t)$, or production rate, $\beta(r)$. This could in turn decrease more rapidly – searching more uniformly – for some people than it does for others. This would nicely correspond to the change from common to more creative solutions over time described earlier (e.g., Gilhooly et al., 2007).

A corresponding way to change the output is to change the edge weights over time with a moving threshold, $\tau(t)$. This can be achieved in a number of ways. One simple way is to use a moving threshold for edges. Edges below a threshold value are removed: they

[7] See Chapter 11 for more details on this model.

are given a value of zero. Edges above this value are kept: they are given a value of one. As the threshold increases, the density of the network decreases and weaker connections are removed. As the threshold falls, the network becomes more interconnected and more distant nodes are more likely to be retrieved. How does changing the β and changing the τ make different predictions with respect to spreading activation?

13.7.3 Aesthetic Foraging and Generative Creativity

Another way to study creativity is to look at how people generate creative products. When this is targeted toward some external criterion, like making a plane that flies further, it is a form of creative problem solving associated with cumulative cultural evolution (Caldwell & Millen, 2008). However, when it is directed at some internal aesthetic criterion, it is similar to that seen in the evolution of art forms.

Hart et al. (2017) developed a paradigm for studying aesthetic foraging by having individuals produce tiny works of art constructed from 10 pixels. Participants moved one of the 10 pixels at a time to produce various images and they saved the images they liked. There are 36,446 possible shapes on the pixelated canvas and by studying the way people explore and save images from this shape space, Hart et al. (2017) were able to gain insight

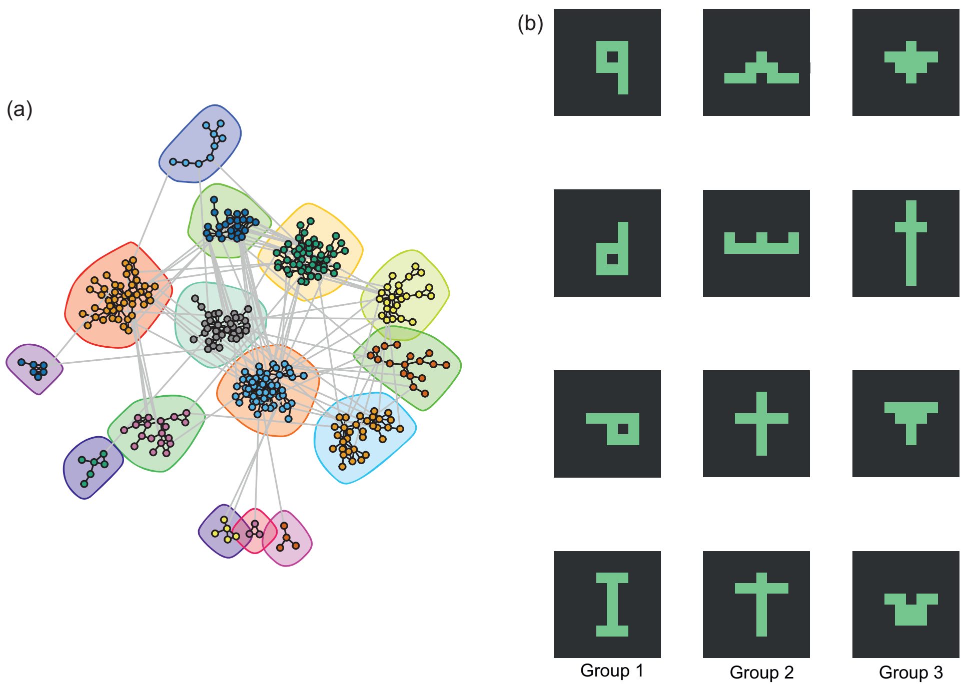

Figure 13.8 In the creative foraging task, participants produce 10-pixel works of art they can save during their exploration. Participants will often save a series of similar images (e.g., numbers or planes) before transitioning to a new category. Nodes in the network in panel (a) represent clusters of saved shapes, with nodes sharing an edge when they share a common shape. Encircled nodes represent categories identified using the fast-greedy clustering algorithm from Clauset et al., 2004. Image samples from three different groups are shown in panel (b). The image is reproduced from data provided by Hart et al., 2017.

into aesthetic exploration. For example, people tend to produce clusters of valued images in aesthetic space – much like Picasso had a Blue Period, a Rose Period, and so on.

Figure 13.8 shows a network visualization of the clusters that people produce. Each node represents a cluster of images that participants saved. These were identified using an algorithm that inferred clusters from the times at which images were saved. Edges between nodes indicate the clusters share at least two shapes.

The superordinate groupings, indicated by color grouping, were identified using Clauset et al. (2004)'s fast-greedy clustering algorithm. Starting by defining each node as its own community, it joins communities into larger communities in a way that maximizes the modularity at each join. The pixel images on the right show columns of images that are in the same cluster as defined by the algorithm.

What drives people's search in such domains? If aesthetic foraging is like other kinds of foraging – with its apparent patch foraging dynamic – what resource are people trying to optimize? Hart et al. (2017) note that people tend to leave clusters long before they are depleted. Yet, people still produce patterns of foraging that briefly exploit local regions of the space before transitioning to explore for new regions (Hart et al., 2018). The capacity to so clearly articulate the structure of the aesthetic landscape makes it amenable to studying individual differences in this aesthetic foraging, including their neural correlates and developmental pathways (e.g., Hart et al., 2022).

Part IV

Society

14 Network Illusions

How Structure Misleads Us

14.1 The Problem

The Brexit referendum and the US presidential election of Donald Trump were a surprise to many, on both sides of the fence. In this information age, it is useful to ask how so many people could be so wrong about issues of so much importance. If we all knew what everyone else was thinking there could be no surprise. We are, in fact, wrong about many things when it comes to estimating the beliefs of the majority. Many of our errors arise not because our brains are tricking us, but because our appreciation of structure is underdeveloped. For example, the majority of people experience heavier than typical traffic and sit through classes with more than the average number of students. This is because, by definition, the most crowded places are attended by the most people. Similar illusions lead us to overestimate how many friends the average person has, confuse us into making backward inferences about class and gender divisions, and allow politicians to misrepresent their populations. All of these are guaranteed outcomes of certain kinds of structure. An understanding of networks helps demystify these illusions.

14.2 Cognitive Illusions

Many of the things we think we know are understood in such a lukewarm way we can hardly call it knowledge. People's confidence in their understanding of questions like "How does a helicopter move from hovering to forward flight?" falls dramatically after they try to explain it (Rozenblit & Keil, 2002). Perhaps this is why we tell people if you want to really understand something, you have to teach it; a false sense of comprehension is easy to nurture when no one is asking questions. Worse still, a number of studies suggest that people with more limited knowledge are often more overconfident about what they know. This has been shown for things like antivaccination and genetically modified foods and is known as the *Dunning–Kruger effect* (Fernbach et al., 2019; Kruger & Dunning, 1999; Lackner et al., 2023; Motta et al., 2018).

Overlooking what is left out of our understanding is pervasive. Consider the *survivorship bias*, which is the tendency to overlook failures when focusing on successes. To illustrate, consider planes returning from combat in World War II, which mostly had bullet holes in the wing tips and certain areas of the fuselage – clear evidence that these planes were suffering damage in those locations. Based on the data, where might we suggest reinforcing the plane's armor? The damaged locations, of course. But that would be misguided. What is missing from the data is all the planes that do not return, which presumably were suffering unsustainable damage elsewhere.

Positive examples of survivorship bias exist as well, such as in studies of mutual funds or successful industries (e.g., Elton et al., 1996). If we throw a million darts out of an airplane, a few of them might still hit the bullseye of a dartboard that a nearby dartsman misses. Inferring that darts thrown out of planes are more accurate than a dartsman on the ground is woolly logic, to be sure. Yet this is exactly the logic many investment companies hope their investors apply when they see how well the investment company performed on some of its highlighted investments.[1]

Not paying attention to how data does or does not make it into a data set is a basic problem of *data quality*. But many illusions of attention occur when the data is present and our attention is elsewhere. This *inattentional blindness* was made famous in an experiment by Simons and Chabris (1999). People asked to count the number of passes between a group of basketball players fail to see a gorilla walking in their midst. There are many other examples of this effect, such as pilots failing to see planes on runways or people failing to hear their own name (Conway & Cowan, 2001; Mack, 2003). Directing attention to one thing makes unattended things less salient. And this happens even when they are perfectly visible to anyone not so misdirected (Mack, 2003; Neisser, 1979).

Our attention can be equally misguided by the structure of questions. The order in which we read choices on a menu can influence what we choose (e.g., Schmidtke et al., 2019). Such order effects can also influence what we claim to believe. People's responses to whether or not they think Al Gore is honest and trustworthy depend on whether or not they have just been asked the same question about Bill Clinton (Wang et al., 2014). This is why surveys produced without input from experts on survey-design are often useless or misleading (Schwarz, 1999; Tourangeau & Rasinski, 1988). How healthy will people find a new low-fat yogurt? That depends on whether you tell them it is 75%-fat-free or 25%-fat (Sanford et al., 2002). Different rating scales can also lead to different answers: When asked "How successful would you say you have been in life?" people provided with a scale from −5 to 5 selected answers in the bottom half (−5–0) 30% of the time. But when provided with scale from 0 to 10, only 13% chose a value in the bottom half (0–5) (Schwarz, 1999). Different words that objectively mean the same thing can also mean different things to different people: Republicans in the US are fine with paying a "carbon offset" but are extremely averse to a "carbon tax" of the same value (Hardisty et al., 2010).

Information source also matters – and the most important source is usually ourselves. Free recall-based scales (allowing people to say whatever comes to mind) or recognition-based scales (requiring individuals to rate a particular value) can produce substantially different results. For example, asking people to list and then rate the frequency and positivity (i.e., valence) of emotions they have felt in the last month produces different measures of emotional well-being than being asked to rate the intensity of experienced

[1] This is a variation of a pervasive illusion called *base-rate neglect*, in which people fail to consider the relative proportions of different categories of data (Kahneman & Tversky, 1973).

emotions provided in a pre-defined list (Li, Masitah, et al., 2020). Similarly, when asked if the economy matters with respect to the next election, many people agree that it does. But if people are simply asked "what matters," the number choosing the economy drops by half.[2] As requoted in Kahneman et al. (2006), "nothing in life is quite as important as you think it is while you're thinking about it."

Asking questions in different ways can also lead to *preference reversals*. Suppose you are asked to choose between two potential sets of parents in a child custody decision: Parents A and Parents B. Parents A have a close relationship with the child and an active social life, but they also have a lot of work-related travel and health problems. Parents B have a reasonable relationship with the child, stable social life, and average income, working hours, and health. Which parents should be awarded the child? When asked this question, most people select Parents A. However, if asked to choose which parents should be denied custody, most people *also* choose Parents A. The study's author, Shafir (1993), explained this as a consequence of how people look for evidence. When asked who should be awarded the child, decision makers seek reasons *for* a particular set of parents. When asked who should be denied the child, they seek reasons *against* a set of parents. Parents A have both.

In a classic demonstration of these kinds of *framing effects*, people are asked to choose between two treatments for a disease to be applied to a village of 600 people (Kahneman & Tversky, 2013). They are told if they choose treatment A, 200 people will survive. If they choose treatment B, there is a 1/3rd chance that everyone will survive. Given this choice, what do you choose?

Now consider a different choice for a neighboring village of the same size. If you choose treatment A, 400 people will die. If you choose treatment B, there is a 2/3rd chance that everyone will die. Now what do you choose?

When posed with the first question, most people choose to save 200 people for sure. They choose treatment A. When posed with the second problem many more choose treatment B, giving themselves a 2/3rd chance that everyone dies. Mathematically, the two questions are identical. Both are choices between 1/3 living and 2/3rds dying versus 200 living and 400 dying. The difference is in the framing: when asked in terms of saving lives, people prefer the sure thing. When asked in terms of losing lives, people would rather avoid certain losses. Kahneman and Tversky (2013) explain the results using *prospect theory*, which states that people compare decision outcomes relative to a *reference point*. Relative to the reference point, people tend to prefer certain gains and uncertain losses.

This is not meant to represent a complete list of the many ways that cognition can be misled. There is a fascinating menagerie of such effects (Kahneman, 2011; Oeberst & Imhoff, 2023; e.g., Pohl, 2016). Moreover, many of these apparent biases are perfectly sensible heuristics for dealing with our boundedly rational minds in environments with incomplete information (Gigerenzer et al., 2011). Still, we can lay these distortions out on the examination table and see them for what they are – illusions created by the way

[2] Pew Research Center has advice for creating questionnaires for polling: https://www.pewresearch.org/methods/u-s-survey-research/questionnaire-design/.

information is presented and processed. The sufferers of these illusions are often blind to their existence. Against this backdrop, we can turn our attention to network illusions, which lead to equally confounding effects.

14.3 Network Illusions

Network illusions arise because we fail to take structure into account. This can happen in two ways: Either we are packaged in the structures we are examining – as is a person who examines their local surroundings to take the temperature of the world – or we are looking at the whole without realizing that it is the sum of varied parts. This leads to two broad classes of network illusions: degree-attribute paradoxes and aggregation illusions.

Degree-attribute paradoxes occur whenever node attributes are correlated with their degree. When an attribute is correlated with degree, then nodes will on average have neighbors that have more of that attribute than there are on average. Therein resides the paradox. Most people will experience a social environment more extreme than it actually is. To give a concrete example, if height were correlated with social network degree, then the average person will have friends who are taller on average than the average person.

The friendship paradox, the class size paradox, and the majority illusion are all degree-attribute paradoxes. These each occur on social networks and all are driven by the same cause. In each case, something that is correlated with a node's degree becomes over-represented in the experiences of individuals in that network.

14.4 The Friendship Paradox

The *friendship paradox* is the observation that, on average, most people's friends have more friends than they do. This is a simple consequence of network structure. People with more friends have more friends, by definition. But this means they are connected to more people, which means that more people will have them as friends. Unless everyone has the same number of friends, this distributional anomaly is unavoidable.

Consider a *star network* with one central hub and all other individuals connected only to the hub. The hub individual experiences friends who all have only one friend, him. But every other individual experiences a friend who is friends with everyone. Thus, in this dystopian case, the average person has friends who have more friends than they do. The rub is that almost every network is dystopian in the sense that it creates a similar anomaly.

Figure 14.1 provides an example of a random network, where each node is a person and each edge is a friendship. Table 14.1 shows the number of friendships for each node alongside the average number of friendships of its neighbors. Six of 8 nodes have friends who have more friends than they do. As shown in the bottom row, the average number of friends an individual has is fewer, on average, than the average number of friends their friends have.[3]

[3] Feld (1991), who first characterized the friendship paradox, documented it for children using James Coleman's data published in *The adolescent society* (Coleman, 1961). Coleman asked children in 12 different

Table 14.1 The friendship paradox.

node	myFriends	theirFriends
1	2.0	2.00
2	3.0	3.33
3	2.0	4.00
4	5.0	2.40
5	2.0	3.50
6	3.0	3.33
7	1.0	2.00
8	2.0	3.50
Average	2.5	3.01

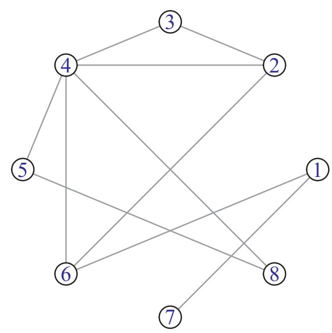

Figure 14.1 The friendship paradox is the observation that most people will have fewer friends than their friends do on average. Each node's label, its degree, and the average degree of its neighbors are shown in Table 14.1.

14.5 The Class Size Paradox

The *class size paradox* is the observation that students experience average class sizes to be larger than the average class size (Feld & Grofman, 1977). Larger classes have more students, and thus more students experience them.

The class size paradox is the bipartite extension of the friendship paradox. Figure 14.2 provides a simple demonstration with a random network, with classes on the left and students on the right. Each student takes each class with an equal probability ($p = 0.4$). But because bigger classes are taken by more students, the majority of the students experience their class sizes to be larger than the average class. Indeed, 90% of them do as shown in Table 14.2.

One can generalize from this phenomenon to many real world contexts. Most people experience more crowded beaches than the average beach. Most people experience more traffic on the highway than occurs there on average. Most people experience more

high schools to name their friends. Feld then took the reciprocated friendships, in which two individuals named one another, as his data.

Table 14.2 The class size paradox.

Students		Classes	
Node	Avg. Class Sizes	Class	Size
1	5.5	a	3
2	3	b	8
3	4.67	c	3
4	8	d	3
5	3		
6	5.5		
7	4.25		
8	8		
9	8		
10	8		
Average	5.79		4.25

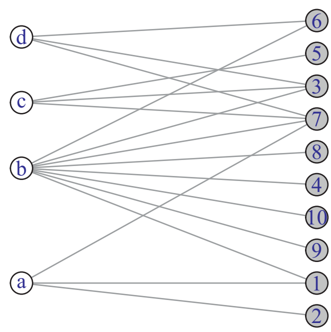

Figure 14.2 A random network depicting the class size paradox. Classes are indicated by letters on the left, and students are indicated by digits on the right. Each person is a member of each class with probability $p = 0.4$. Individuals will tend to overestimate the average class size if their inferences are based on their own class sizes.

large parties, crowded nightclubs, overbooked airplanes, and long queues than there are on average.

14.6 The Majority Illusion

The quintessential degree-attribute paradox is the majority illusion. Suppose someone asks you whether you are more or less attractive than average. Suppose you do what most people do when confronted with social questions of this kind: you consider yourself in relation to all the people you know. For the sake of simplicity, assume you know three people. Two of them are more attractive than you are, and a third is less attractive. Easy enough: you conclude that most people are more attractive than you are.

There is no apparent illusion here yet. This is not the friendship paradox. You are not evaluating the friends your friends have. You are evaluating their attractiveness. But in doing this, you are covertly evaluating degree because attractiveness is correlated with degree. Attractive people enjoy many beauty premiums and the fact that they tend to be more popular is one of them (Langlois et al., 2000). Because attractiveness is correlated with higher degree in the social network, then the degree-attribute paradox kicks in: our friends will be more attractive on average than the average person.

This goes on. If people who read books tend to be introverts, then the average person will interact with people who read less often than they do. If riskier drivers tend to drive more often, then most of us will drive among riskier than average drivers. And if people with more extreme views are more likely to find platforms for expressing those views, then most of the views we hear will be more extreme than the average person's views.

This is called *the majority illusion*. It arises whenever nodes' attributes are correlated with their degree. Because nodes with higher degree are experienced more often by others, their attributes will be overestimated in the population (Hodas et al., 2013; Lerman et al., 2016).

Figure 14.3 provides a visual demonstration of the majority illusion based on attractive and unattractive people. The two networks are identical except for which nodes are attractive. They both have the same objective minority of attractive (lighter) nodes, but the subjective experience of nodes is different in the two networks. In the top network, among the unattractive nodes, nine of their neighbors are attractive and only two are unattractive. The bottom network reverses this trend and, in fact, underrepresents the distribution of attractive nodes.

The majority illusion is surprising in part because it does not take much to create it. Given a real-world social network such as that found in the Enron email network, even a small number of nodes can overrepresent a minority trait if those nodes have high degree (Lerman et al., 2016). This is further amplified by two things:

- More disassortative networks will be more subject to the majority illusion.
- Networks with higher degree–attribute correlations will be more subject to the majority illusion.

The friendship paradox, the class size paradox, and the majority illusion all stem from the same *degree–attribute paradox*. Things associated with high-degree nodes are overweighted in the collective experience of the network's nodes. In the majority illusion, some feature or attribute is correlated with degree. In the friendship and class size paradox, degree is itself the attribute being experienced.

The majority illusion is a network version of a broader class of illusions called *selection bias*. In the survivorship bias, only subjects who survive the selection process are included in the analysis. Another example is scientific publication bias: if publication is more likely for results with statistically significant findings or that support more surprising hypotheses – as opposed to studies that are inconclusive or attempt to replicate past findings – then this will increase the absolute number of false positives (Type I errors) (Ioannidis, 2005). This bias will be further amplified in proportion to the absolute number of people attempting to publish research (Hills, 2019b). The majority of research findings may then look surprising and "statistically significant," even though the average research finding is neither.

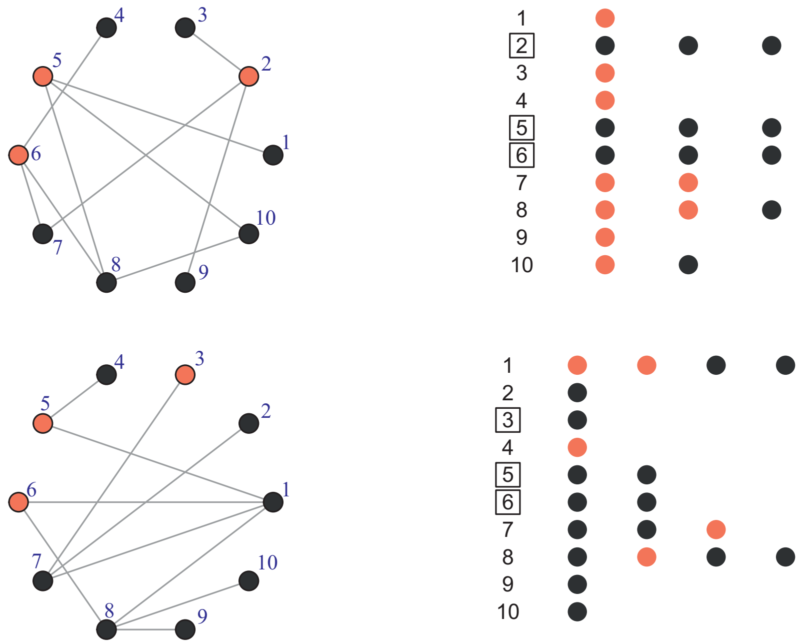

Figure 14.3 The majority illusion is created when node features correlate with node degree. The top and bottom networks are identical except for the location of the "attractive" (lighter) nodes. The tables on the right-hand-side show the status for each node (indicated by number) of each of its neighbors. In the top network, though most nodes are unattractive, the majority of unattractive nodes' neighbors are attractive. Only node 10 has an equal distribution of unattractive and attractive neighbors. In the bottom network, the majority of nodes' neighbors are unattractive.

It is straightforward to predict implications of degree–attribute paradoxes. Well-supported cognitive models of human decision making argue that when we go to make judgments about the frequency of various things (such as disease or causes of mortality), we use a *social circle heuristic*, which involves sampling in succession our social networks, starting with those closest to us (Galesic et al., 2012; Pachur & Schulze, 2022). If people use their social environment to estimate the prevalence of things that are correlated with social network degree, this will lead to systematic overestimation. Many people judge their own alcohol consumption, diet, and criminal activity (e.g., illegal downloads) not based on medical advice or moral inclinations, but rather based on how their behavior ranks relative to other people they know. These *rank-based comparisons* explain a wide variety of other effects, including well-being and mental health (Brown et al., 2022). Such rank-based comparisons are already problematic without considering the global structure of the social network (Aldrovandi et al., 2013), but if these behaviors are correlated with degree, they will be overestimated by the average person and therefore subject to a rising tide.

The friendship paradox, the class size paradox, and the majority illusion happen because people make inferences from within the data. Not seeing all the data, they draw sensible but incorrect conclusions from what they observe. The network distorts these observations because lower degree nodes are less visible. The reverse of this paradox happens when we lump all the data together, treating it as if it were all of the same kind. In this scenario, we have *aggregation illusions*: a consequence of assuming the data we see is equally mixed all the way through, when in fact it has structure within. For degree–attribute paradoxes, we are on the inside looking out. For aggregation illusions, we are on the outside looking in.

14.7 Simpson's Paradox

Sometimes when we look at a group, we fail to see the groups within it. It is like assuming a tree looks like a forest. This can be excused if the group looks like its constituents. But in many cases, it does not. What we get when we average over observations is a Frankenstein, made up of different groups doing different things, who, on average, look like no one.

A now classic and frequently used example of this phenomenon, called Simpson's paradox (Simpson, 1951), stems from the graduate admissions figures of UC Berkeley in 1973. Notably, across all applicants, men were more likely to be admitted than women (men: 44%; women: 35%). This can be easily interpreted as an inequality in the way men and women were being treated. It is, however, a difference in the way they behave.

Men and women did not apply with equal probability to the same departments.[4] Women were more likely to be accepted than men in most of the departments to which they applied. However, they also applied to more competitive departments, which accepted fewer students overall. This drives up the number of rejections for women compared to men. Bickel et al. (1975) described the data as follows:

> If the data are properly pooled, taking into account the autonomy of departmental decision making, thus correcting for the tendency of women to apply to graduate departments that are more difficult for applicants of either sex to enter, there is a small but statistically significant bias in favor of women. The graduate departments that are easier to enter tend to be those that require more mathematics in the undergraduate preparatory curriculum. The bias in the aggregated data stems not from any pattern of discrimination on the part of admissions committees, which seem quite fair on the whole, but apparently from prior screening at earlier levels of the educational system. (p. 403)

A more careful investigation of the structure in the data gives us a better understanding of what is happening. In this case, individuals distribute themselves differently across departments which have different acceptance rates.

14.7.1 Simpson's Paradox and the Hunter-Gatherer Australian Martu

Hunters and gatherers often show a division of labor by gender, which neatly demonstrates Simpson's paradox in an intuitive way. The traditional view is that women tend to gather more reliable resources and men hunt for riskier resources. This division happens for a variety of reasons, which include providing reliable resources, social bonding, cultural norms, and evolutionary pressures favoring more competition among males (Bird & Bird, 2008; Fischer & Hills, 2012).

[4] This is a bipartite network, with directed edges that may or may not be reciprocated. According to Bickel et al. (1975), the probability of a reciprocated edge is slightly higher for women than men, but different departments have different numbers of outgoing edges (admissions) at their disposal.

Table 14.3 Simpson's paradox among hunters and gatherers.

	Total	Successes	Failures	P(Success)	P(Failure)
Total foraging attempts	100	47	53	0.47	0.53
Women	50	30	20	0.60	0.40
Men	50	17	33	0.34	0.66
Lizards–Women	40	28	12	0.70	0.30
Lizards–Men	10	8	2	0.80	0.20
Kangaroo–Women	10	2	8	0.20	0.80
Kangaroo–Men	40	9	31	0.22	0.78
Lizards	50	36	14	0.72	0.28
Kangaroo	50	11	39	0.22	0.78

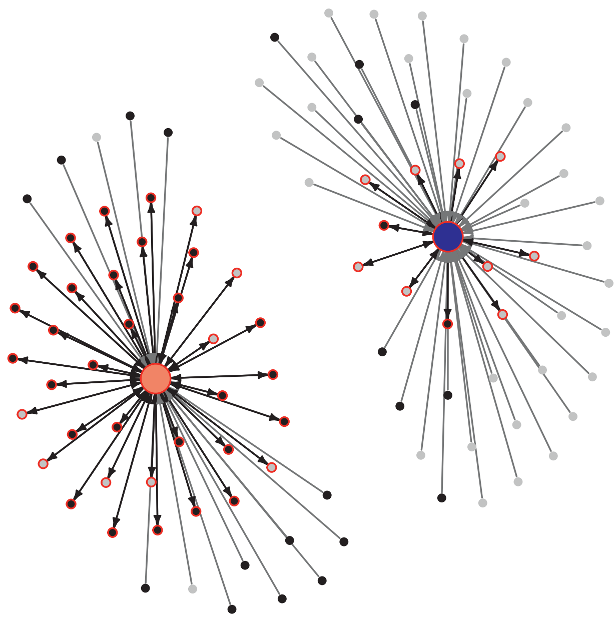

Figure 14.4 A simulation of Simpson's paradox among the Australian Martu. The simulation produces 50 women and 50 men, who each choose one of two foraging strategies: predictable small lizards or risky large kangaroo. Men choose kangaroo 80% of the time and women choose lizards 80% of the time. Lizard strategies lead to successful outcomes 80% of the time; kangaroo strategies are successful 20% of the time. Men are indicated by small gray nodes; women are indicated by small black nodes. The large nodes indicate foraging strategies: small lizards (lighter node) and large kangaroo (darker node). Direction arrows toward strategies indicate attempts and reciprocated arrows from strategies to individuals represent successes (also indicated with a red circle). The corresponding table indicates the relative success of women and men and shows a clear Simpson's paradox.

To show how this matters, we can take advantage of the fact that Simpson's paradox has a straightforward bipartite network representation. Figure 14.4 provides a simulated demonstration inspired by men and women hunting and gathering among the Australian Martu (Bird & Bird, 2008). Among this community, women tend to hunt for more reliable

Table 14.4 Simpson's paradox among hunters and gatherers with multiple strategies per individual.

	Total	Successes	Failures	P(Success)	P(Failure)
Total foraging attempts	491	307	184	0.63	0.37
Women	252	192	60	0.76	0.24
Men	239	115	124	0.48	0.52
Low risk–Women	209	178	31	0.85	0.15
Low risk–Men	53	45	8	0.85	0.15
High risk–Women	43	14	29	0.33	0.67
High risk–Men	186	70	116	0.38	0.62
Low risk	262	223	39	0.85	0.15
High risk	229	84	145	0.37	0.63

game (goanna lizards), whereas men tend to hunt for larger and riskier game (kangaroo). Goanna lizard hunting is successful about 90% of the time, whereas kangaroo hunting is successful around 25% of the time.

As the simulation shows, men hunt in an environment that leads to far fewer successes, mirroring the women who applied to more competitive departments at UC Berkeley. As a result, they are largely unsuccessful. On the other hand, women, who hunt for more reliable game, are more successful. But comparisons only make sense within a strategy. And even then structure matters because women and men show seasonal differences in their hunting strategies and strategy difficulty changes by season (Bird & Bird, 2008).

Table 14.3 summarizes the paradox. Women are more successful overall. However, men are slightly more successful at both hunting kangaroo and hunting small lizards. We could wrongly chalk this up to men's competitive nature, but the simulation does not distinguish between men and women after they choose a strategy. All that matters is the prey. Another run of the simulation will produce a different outcome.

Figure 14.5 shows the outcome of a more realistic situation described by Bird and Bird (2008), with men and women choosing multiple strategies out of a set of 10 that vary between high risk (darker big nodes) and low risk (lighter big nodes). Here, again, women favor low-risk strategies and are more successful, shown by the dark black reciprocating edges representing successful foraging. Men favor riskier strategies and are less successful on the whole. But as revealed in Table 14.4, the genders are only different in success if we fail to distinguish between the challenges they face.

14.8 Gerrymandering

The seriousness of our misrepresentations depends on the circumstances. But when people make decisions based on them, the outcome can be far from benign (Wagner, 1982). An example that affects the political policies to which we are subjected is *gerrymandering*. Gerrymandering is the intentional altering of regional boundaries to distort the representation of people within those boundaries. It is typically used by political parties

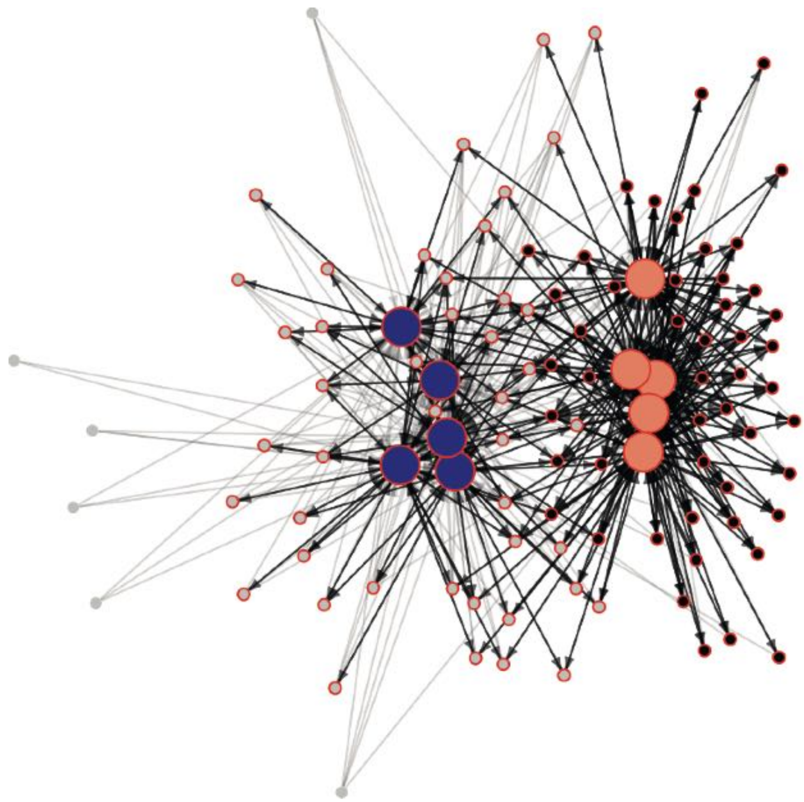

Figure 14.5 The outcome network for 100 simulated Australian Martu (small nodes) who can each choose multiple strategies (large nodes) from a set of 10 possible strategies, which are high (dark) and low (light) risk. Individuals are divided equally between men (gray) and women (black). Men prefer high risk strategies and women prefer low risk strategies. Each chooses each preferred strategy with probability $p = 0.8$, choosing the alternative strategy with $p = 0.2$. Unsuccessful strategies are shown with light gray edges, successful strategies are shown with dark edges. Individuals that are successful at least once are circled in red. Low- and high-risk strategies are assigned a risk level from a beta distribution.

to dilute the vote of the opposition party by redistributing it in such a way that it is in the minority in most subgroups.

Consider a state with 1,000 people split across 10 districts, with 100 people in each district. Suppose half the people support Party A and half support Party B. Ideally, if we believe that representation should be proportional – one-person-one-vote – then a government made up of representatives from each district should represent each half: fifty–fifty.

But district boundaries can easily be redrawn to distort this ideal. For example, one district can be drawn so that it contains mostly Party A members, electing Party A 95 to 5. The rest of the districts can be drawn so that they elect Party B 55 to 45. The result is that Party A gets one representative and Party B gets nine.

The process of redefining district boundaries in this way is called *packing and cracking*: Pack as many members of the opposition party as possible into one district, and then

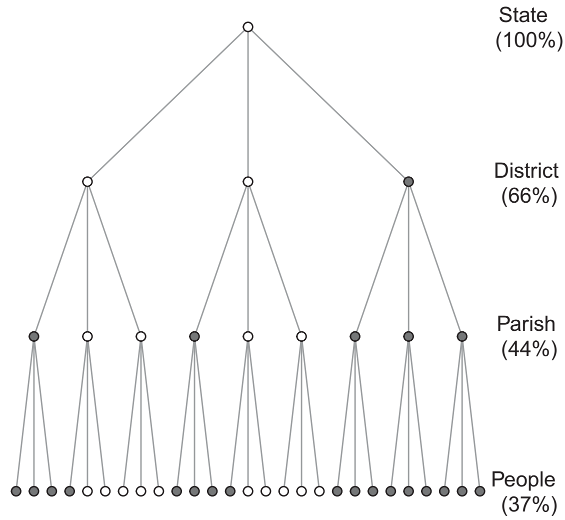

Figure 14.6 A visual illustration of gerrymandering. Gerrymandering takes place by reassigning people and regions to higher-level representatives in such a way that the representatives misrepresent the majority. It is clear from the image that there are more gray nodes at the bottom than white nodes, but the state partitions itself in such a way that it represents the white node minority. This is a multipartite assignment of people within regional boundaries. In this example, the majority of districts elect white node representatives, even though white nodes are outnumbered by gray nodes 17 to 10.

crack the remainder across districts so they never represent a majority anywhere else (Herschlag et al., 2020).

Packing and cracking is a network problem: the challenge of packing and cracking is one of identifying partitions of a network in such a way that the outcome of regional elections will overrepresent the desired party. Because of legal requirements that partitions receive equal representation – and thus contain roughly the same number of people per representative – and only include regions that are contiguous in space, the problem is one of mapping subregions of the network (nodes) into districts in such a way that they pack and crack the undesired voters (McCartan et al., 2022).

In this sense, gerrymandering is a *multipartite network* problem. In any representational government, people can be assigned to local governments, local governments can then be assigned to regional governments, regional governments to provinces and so on. At each level, the lines defining regions can be redrawn. Figure 14.6 gives an example of this. This can happen for any representational system, such as for departments within universities or divisions within a business.

When populations are gerrymandered, trying to infer what the underlying population prefers based on their representatives is misguided. Just as in Simpson's paradox, the overall average need not be reflected in local averages.

The situation can be made worse. At least in the above situation, districts are of equal size in terms of proportion of the population represented. Proportional representation limits the bias, though it does not eliminate it. But this proportional representation is often not preserved at the national level. The US Electoral College, for example, which represents state's presidential votes by voting on their behalf, compounds this structural problem in two additional ways. First, some states assign votes on a winner-take-all basis. In the 2016 election, Trump had 52% of the votes in Texas, while Clinton had 44%. Trump received all 38 of the state's electoral votes – misrepresenting almost half the state's population. The second way the Electoral College misrepresents American voters is that electoral votes are not equally distributed across states in proportion to their population. Votes in Wyoming and Alaska count roughly three times more toward the presidential election than the votes of Georgians or Floridians. Lest one think there is something special about the USA, many other nations have seen similar nonmajority representation, such as the 1951 general election in the UK in which Labour won the popular vote, but the Conservatives won more seats (Siaroff, 2003).

The friendship paradox, the class size paradox, Simpson's paradox, gerrymandering, and the misrepresentation of the US Electoral College are all cases where structural differences in the way individuals are allocated play a large role in how individuals are represented and what they experience as a result. If people draw conclusions from these experiences, they can easily be wrong.

14.9 Information Gerrymandering

We when deliberately combine gerrymandering with the majority illusion we are gerrymandering information for individuals. Suppose there are three Republicans (red nodes) and three Democrats (blue nodes) as shown in panel (a) of Figure 14.7. If the Republicans could control the structure of the network how might they allocate edges to maximize their influence?

One way is for Republicans to connect to a weakly tied Democrat. If the rule is that people are persuaded by the majority view in their local networks, then looking around themselves the Republicans will see a majority of Republicans. However, the Democrat to which they are all connected will see a majority of Republicans as well.

This process of creating an asymmetric influence network is called *information gerrymandering*. Analogous to political gerrymandering, it manipulates people's experience by influencing who they are grouped with.

Figure 14.7 Information gerrymandering. If the red nodes in panel (a) connect themselves to one of the weakly tied blue nodes, they gain an influence gap over the blue nodes. In this case, the information assortment prior to the connection is 1 for each node, as each node only shares an edge with members of the same party. But after the red nodes connect to the blue node, their information assortment is $a_i = 0.66$ for each red node. The single blue node connected to them is $a_i = -0.75$

Stewart et al. (2019) proposed a way to evaluate information asymmetry using a measure of *information assortment*:

$$a_i = \begin{cases} \Delta_i & \text{if } \Delta_i \geq 1/2 \\ -(1-\Delta_i) & \text{otherwise} \end{cases}$$

Δ_i reflects the proportion of node i's neighbors that are in the same party as i. For example, if 60% of node i's neighbors are in the same party as i, then $a_i = 0.6$. If only 40% of its neighbors are the same party, then $a_i = -0.6$. Once this measure of information assortment is computed for each node, it can be averaged over individuals in both groups to compute a measure of information gerrymandering called the *influence gap*:

$$\mathcal{G}_D = \frac{1}{N_D}\sum_{i\in D} a_i - \frac{1}{N_R}\sum_{j\in R} a_j$$

where D and R in this case represent the two parties. The influence gap is simply the difference in the average information assortment for the two parties. When it is positive, that party benefits from information gerrymandering.

Information gerrymandering has the interesting property that once one party starts to use it, the other party can only defend itself by increasing its information assortment. To do that, it must reduce its ties with the alternative party, forming an echo chamber. These echo chambers form as individuals try to maintain the dominance of their own views. This has much in common with Thomas Schelling's model of segregation (Schelling, 1971), where in individuals can swap neighbors to maintain a majority of their own racial group, which in turn leads to large-scale segregation. This influence of network structure on opinion dynamics is discussed in detail in Chapter 15.

Another defense mechanism against gerrymandering is the creation of *zealots*: individuals whose opinion will not change when exposed to alternative views. The strategic introduction of zealots can shift the tide of public opinion. But both parties can do this. Indeed, it is cheap and easy to create zealot bots to do the job and many people and governments do just that to information gerrymander online environments (Menczer & Hills, 2020; Pomerantsev, 2019).

14.10 Summary

Information illusions, whether of a cognitive or network variety, are both around and inside of us. They arise because we overlook structural features of the data. This includes overlooking its representativeness and subcategorizations within the data that lead different categories of individuals to experience different environments. Broadly, these fit into degree-attribute paradoxes and aggregation illusions. Like cognitive illusions, the oversight becomes obvious once pointed out. They are more than just parlor tricks, though. They can effect everything from our well-being to our political representation. They can also be put to work to influence or misrepresent the beliefs and attitudes of others. Look out.

14.11 Open Questions

14.11.1 Curation of Information

One of the most pervasive problems in modern information environments is the challenge of evaluating information quality. This has become especially problematic as the established gatekeepers of information production have radically changed. The International Telecommunications Union (International Telecommunications Union, 2017) estimates that more than 4 billion people now have access to mobile-broadband, collectively giving us the power to receive, create, and share information on scales previously unheard of – turning us all into gatekeepers. This leads to important questions about how to protect ourselves from fake news, misinformation, paltering, and other forms of alternative facts or information dilution. But it also creates questions about how best to create productive dialogue useful for effective democracies, which break through ideological boundaries. Lorenz-Spreen et al. (2020) discuss a number of ways to achieve this, including methods that inform people about their position within a network, "dislike" buttons, and better informing people about the behavior of others. Lee et al. (2019), by demonstrating structural illusions associated with over-estimating and under-estimating the size of minority groups, also showed that local information can be substantially improved by taking into account the perceptions of one's direct neighbors. In addition, crowdsourcing at various levels may also be a reliable mechanism for efficient fact-checking (Pennycook & Rand, 2019). All of these suggestions have structural implications. They will be affected by how information is aggregated, either locally or globally, potentially producing degree–attribute paradoxes or aggregation illusions. Lee et al. (2019) present an excellent exploration of this impact, but there is much to be done here.

14.11.2 Illusions of Diffusion

Herbert Simon once noted that a "wealth of information creates a poverty of attention" (Simon, 1971, p. 41). There is good evidence that this is true. For example, in a simulation of meme sharing in which networked agents preferred high-quality information but also had limited attention to evaluate it, more information led to lower-quality-information sharing (Qiu et al., 2017). This makes sense. As the amount of information we receive increases, at some point we become limited in our ability to evaluate it. It is also the case that when we give people more options to explore before making a decision, they explore each individual option less (Hills, Noguchi, et al., 2013). This suggests that nodes with higher indegree – who receive more information from others – might be more susceptible to attention limitations and information overload. This extends to network density, with higher-density networks predicted to produce lower-quality information overall if there is a limit on attention. This is a different kind of illusion than size or aggregate illusions, as it is based on the capacity of an individual node (or collection of nodes) to evaluate high quantities of available information. If this illusion is a consequence of low quality processing (e.g, as a result of information overload) then it may lead us to overestimate information quality even as we become poorer at evaluating it. There are many potential network implications of this yet to be explored. However, it is also possible that greater quantities of information lead us to select information that more strongly appeals to our biases (a form of cognitive selection; Hills, 2019b). Book publishers know this and choose book covers and titles that stand out, to cut through the noise. "If it bleeds it leads" is a similar trope for the news. How these various forms of cognitive selection might impact high-degree nodes or high density networks is an open question.

15 Group Problem Solving
Harnessing the Wisdom of the Crowds

15.1 The Problem

How can groups best coordinate to solve problems? The answer touches on cultural innovation, including the trajectory of science, technology, and art. If everyone acts independently, different people will explore different solutions, but there is no way to leverage good solutions across the community. If everyone acts in consort, early successes can lead the group down dead ends and stifle exploration. The challenge is one of maintaining innovation but also communicating effective solutions once they are found. When solution spaces are smooth – that is, easy – communication is good. But when solution spaces are rugged – that is, hard – the balance should tilt toward exploration. How can we best achieve this? One answer is to place people in social structures that reduce communication, but maintain connectivity. But there are other solutions that might work better. Algorithms, like simulated annealing, are designed to deal with such problems by adjusting collective focus over time, allowing systems to "cool off" slowly as they home in on solutions. Network science allows us to explore the performance of such solutions on smooth and rugged landscapes, and provides numerous avenues for innovation of its own.

15.2 The Wisdom of the Crowds

The *wisdom of the crowds* is the observation that when we combine estimates from a number of independent observers, the average estimate is generally quite good and often better than most, or sometimes all, of the individual estimates. Francis Galton (1907) first described this for a group of festival goers who entered a competition to guess the weight of a "fat ox." Each person paid a fee for entry and wrote their guess on a paper card. About 800 people entered the competition, and the median guess was 1207 lbs. The true weight of the ox was 1198 lb – a difference of 9 lbs, or less than 1% error.

Clearly, crowds can be wise when they are not hysterical. And there has been ample discussion about how best to aggregate their wisdom (e.g., Lorenz et al., 2011). Should we take the mean, the median, the geometric mean, or something else? All this fine tuning is just further evidence that we are on to something. That something is even something we can internalize: people can improve their reasoning by asking themselves to "consider the opposite" – taking different views on hard problems – leveraging the wisdom of the crowds inside their own head (Herzog & Hertwig, 2009).

Not all problems are amenable to crowd wisdom. If the wisdom of the crowds applied to all problems equally well, there would be no point in experts. We could all simply vote on everything and solve our problems democratically. If our collective wisdom were infallible, Charles Mackay would have never written a book entitled *Memoirs of extraordinary popular delusions and the madness of crowds* (Mackay, 1869). Some kinds of problems do require specialist knowledge and in those cases the majority wisdom is often wrong. For example, is Philadelphia the capital of Pennsylvania? Most people say yes, but a few people know this is incorrect (Pennsylvania's capital is Harrisburg). Those who know that Philadelphia is not the capital of Pennsylvania also know that many people nonetheless believe that it is. This insight led Prelec et al. (2017) to develop a novel method for aggregating the wisdom of the crowd which works for both hard and easy problems. Ask people for their guess, but also ask them for a second-order guess – what they think other people believe. The correct answer to the question turns out to be the *surprisingly popular* answer – it is the answer that people guessed more often than other people thought they would. In the case of Harrisburg, people who don't know Harrisburg won't predict its popularity, and the few who do know Harrisburg will know that Philadelphia is more popular. This will make Harrisburg surprisingly popular, given that its predicted popularity will be fairly low.

15.3 Group Polarization

The wisdom of the crowds is most wise when people's beliefs are arrived at independently. But this rarely happens. We pay close attention to one another and to one another's beliefs – especially from those of our ingroup. Indeed, people's appetite for ingroup information seems inexhaustible. Given all the other things people could pay attention to, the reality is that as our capacity for sharing social information has grown so has our appetite for consuming it, with a variety of unwanted consequences – ranging from extremism to distorted views of risks (Hills, 2019b).

Our ability to learn from one another has been characterized as an evolutionary turning point. It has been extremely difficult to find evidence of comparable social learning among other closely related species. Belyaev's wild foxes, selected over a course of 50 generations for their friendliness, turned into social beings capable of learning from humans (like domestic dogs). Hare and Woods (2021) claim that a similar self-domestication may have created our own exceptional abilities in communication and social learning. Of course, the flip side is that we now also find it difficult to ignore one another. Some have even speculated that the advent of smartphones has tapped into our evolutionary predisposition for social information, giving rise to *hypernatural social monitoring* (Veissière & Stendel, 2018).[1]

[1] Our ability to follow others might also be seen as a requirement to follow the ingroup, to the point where it is an inability to follow the outgroup: Henrich (2015) tells of how, in 1845, the crews of the HMS Erebus and HMS Terror became icebound as they searched for a Northwest Passage between the Arctic and Pacific

Social information is also extremely competitive in relation to other forms of evidence. In an experiment focused on asking what kinds of information is most memorable, social media status updates easily outcompete things like newspaper headlines and sentences from books (Mickes et al., 2013). When other people make claims, even if they are inconsistent with our own experience, we are also more likely to make similar claims ourselves. Solomon Asch's now classic experiments on group polarization showed people lines of various lengths and asked them which was longer. If as many as three other people (confederates with the experimenter) claimed that an obviously shorter line was the longest, the likelihood of the experimental subject making the same claim increased to 30% (Asch, 1955).

This social influence creates problems for our collective wisdom because when people's beliefs are not independent, the wisdom of the crowds rapidly breaks down. Lorenz et al. (2011) found that when people could see what other people guessed in a group estimation task, social influence reduced the overall range of guesses, increased the error, and led people to be overconfident. This social herding has been documented numerous times, demonstrating contagion of beliefs, emotions, and behaviors (Raafat et al., 2009).

One particularly compelling study was done by Salganik et al. (2006), who allowed people to listen to fragments of songs and then download a song of their choice. In addition to listening to the tunes, some people were exposed to social information – they could see what other people had chosen before them. Groups of independent decision makers were similar to other independent groups, sharing a similar diversity of views. However, groups exposed to social influence rapidly diverged from one another – they became less similar. Social groups also produced more inequality among song preferences. By following one another, social groups demonstrated the characteristic phenomenology of "going viral," with rich-get-richer choice patterns. Another way to describe these two results is that social information both added noise and amplified it. Early differences in individual preferences were amplified by the group choices that came later.

The same socially fueled preferential attachment is potentially behind long-tailed distributions of citation counts among scientific papers (Clauset et al., 2017) and observed in online auctions in which bidders follow bidders instead of other quality indicators (Huang & Chen, 2006; Simonsohn & Ariely, 2008).

This propensity to follow one another plays out in interesting ways when we focus on how groups solve problems.

15.4 Characterizing Problem Difficulty: Smooth versus Rugged Landscapes

How can we define problem difficulty? One way is to think of solutions to problems as residing on a landscape. If higher points on the landscape are better, then individuals can

Oceans. The crews began to starve. The local Netsilik Inuit population, who were fairly permanent residents of that part of the world, attempted but failed to get the crews to eat seal meat. In the end, the Netsilik could only look on in horror as the crews ate one another.

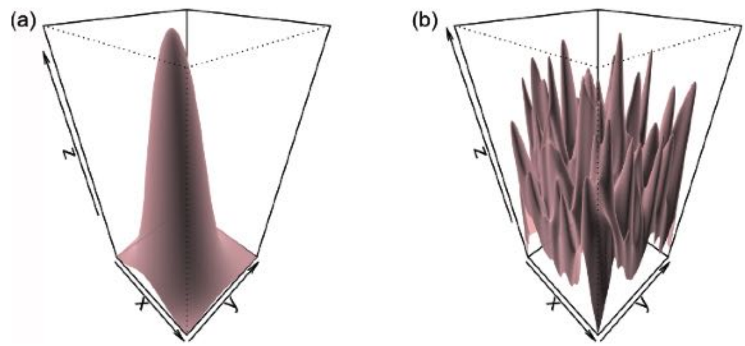

Figure 15.1 Visual metaphors of smooth and rugged landscapes. The smooth landsdcape in panel (a) always indicates the path to the global maximum by its gradient. Moving up the gradient leads to the global maximum. The rugged landscape in panel (b) requires exploration. Most peaks are local maxima: following the gradient leads to a less than optimal solution.

often search for the best solutions by moving uphill. By making small or large changes to existing solutions, individuals can navigate the landscape by making small or large jumps in the solution space.

We can define two broad classes of landscape types (see Figure 15.1). *Smooth landscapes* are associated with simple problems: one can find good solutions by moving uphill. *Gradient ascent* algorithms solve these problems easily by finding the gradient (slope) at the current location and then moving in the uphill direction. If the landscape is smooth enough – with only one hill – gradient ascent will eventually arrive at the tallest peak, the *global maximum*.

In contrast to smooth landscapes, *rugged landscapes* represent complex problems that have many local maxima (little hills). Rugged landscapes have lower *spatial autocorrelation*: nearby positions on the landscape are less similar to one another than on smooth landscapes. As a result, small changes to a solution lead to larger changes in the payoff (or fitness). This makes the space itself less predictable and therefore less easy to navigate. Gradient ascent algorithms get trapped in *local maxima*.

15.5 Exploration, Exploitation, and Group Problem Solving

There is a rich body of experimental and theoretical work on collective search (Centola, 2022; Lazer & Friedman, 2007; March, 1991; Shore et al., 2015). The intuition behind much of this work is that the structure of social information should influence collective problem solving. Good solutions can be communicated rapidly in efficient networks, but poor communication can preserve solution diversity.

In some of the first experimental work on this, Mason et al. (2008) invited people into the laboratory and allowed them to earn money by choosing values along a one-dimensional landscape: Individuals were given a payoff by guessing a number between 1 and 100, which mapped onto payoffs determined by a smooth or rugged payoff landscape, as shown in Figure 15.2. People guessed each turn, allowing them to improve their guesses over time. People were also connected in different network

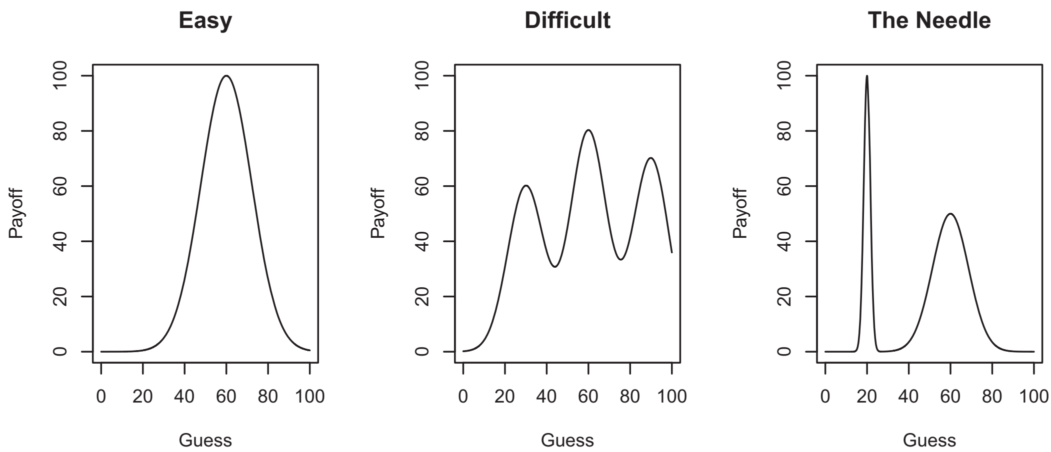

Figure 15.2 Payoff functions for three collective search problems in Mason et al. 2008. Participants would each guess a number and receive the payoff associated with that number.

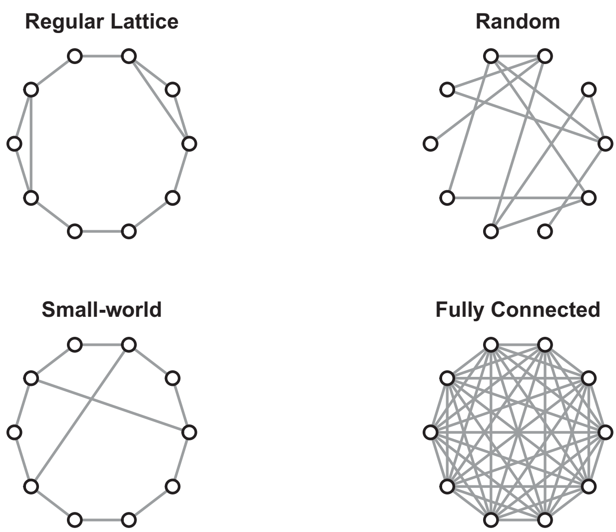

Figure 15.3 The four different treatment groups used in Mason et al. 2008. Each network represents a different configuration for information sharing. Nodes represent individuals in the experiment and edges indicate the ability to see another individual's guesses and performance. Networks were recreated based on the description provided in Mason et al. 2008.

configurations, including a near regular lattice, fully connected, random, and small-world networks, as shown in Figure 15.3. If two individuals were connected by an edge, they could see one another's guesses and their performance, using this information to change their guesses if they wished.

What Mason et al. (2008) found was that for smoother payoff functions, fully connected groups fared better and did so more quickly than lattice networks. However, as problem difficulty increased, less well-connected groups performed increasingly well. For the most difficult problem, what the authors called "the needle," the least well-connected lattice network performed best.

The observation that less well-connected (less efficient) networks perform better for complex problems has been found repeatedly. For example, Derex and Boyd (2016) explored the way groups search for a solution to a problem that offers multiple pathways to innovation. In their experiment, groups of individuals were allowed to combine ingredients to create a remedy to stop a virus. The solution space was designed such that there were two possible pathways for early innovation, the A path and the B path. Progress along both paths would lead to new innovations. However, with sufficient innovation along both paths, players could combine the products of both paths to create still more effective "composite" solutions.

Derex and Boyd (2016) found that fully connected groups were faster off the mark at finding innovations, but partially connected groups were more effective in the long-run at finding composite solutions. This is because the partially connected groups were more likely to find innovations along both pathways which they could later combine to create the composite solution. Individuals in a fully connected group were more likely to engage in local exploration around the best solution so far. By abandoning one of the pathways, this impaired their ability to discover composite solutions later.

In a fascinating variation, Hahn et al. (2020) juxtaposed social information against empirical truth. They used a Bayesian agent-based model to evaluate the ability of different network topologies to arrive at the "truth." Simulated agents updated beliefs based on connectivity, but they also updated their trust in others at the same time. In addition, agents also had access to outside sources that had a more accurate relationship with the truth. Using regression models, Hahn et al. (2020) found that the best predictors of convergence toward the truth are low degree networks with low-clustering coefficients. Networks with similar average degree can benefit in their pursuit of the truth from lower clustering (i.e., with larger effective network sizes, see Chapter 18). This may remove the tendency to form self-reinforcing "echo chambers." As Hahn et al. (2020) put it, "The higher the connectivity in the network, the more the misled majority can drag down the whole, or parts of, the network" (p. 1521).

The intuition from these studies is that less efficient networks (with higher average path lengths) are better at maintaining exploration, whereas more efficient networks are better at exploiting known solutions. That isn't the whole story, however. Mason and Watts (2012) examined the performance of several hundred networked groups searching for a target in a rugged landscape. Networks had $N = 16$ individuals, each with degree $k = 3$, but the networks varied in efficiency (i.e., roughly the inverse of the average path length) and clustering.[2] In contradiction to what was found in the previously mentioned studies, the results of this study found that more efficient networks were better at finding solutions: more well-connected networks found difficult-to-find targets more often.

Though on the face of it, the results from Mason and Watts (2012) challenge the "less efficient networks are more exploratory" intuition, a closer look reveals a key consistency. Mason and Watts (2012) found that (1) when networks had higher clustering, an

[2] The network structures were designed using the adaptive network search algorithm described in Chapter 18: starting from a regular lattice, a network is randomly rewired, only keeping rewirings that increased or decreased the network along a chosen metric. The metrics that were used were betweenness, clustering, closeness, and constraint, which were further modified in relation to whether the algorithm increased or decreased the minimum, average, or maximum of the value being changed by the rewiring.

individual's neighbors tended to have similar values to one another, and (2) when an individual's neighbors were similar, individuals copied their neighbors more often – suggesting a form of what is called *complex contagion* (Centola, 2010). Because of the network generation method used by Mason and Watts (2012), less efficient networks had more clustering. This may mean they engaged in less exploration and were less likely to find the global maximum. In effect, local communities of similar individuals reduced exploration. Interpreted in this way, the results from all the studies support one another in showing the influence of social proof on performance. Social proof, in the form of collectives of like-minded neighbors, reduces the capacity for exploration derived from diversity. When neighbors are connected in a local community (either in groups of three like Mason & Watts (2012) or in fully connected groups like Mason et al. (2008)), people tend to follow one another more and explore less. Generalizing from the Asch experiment described earlier, groups of individuals can be divergence-reducing machines.

15.5.1 Learning Strategies: Follow the Best versus Follow the Crowd

Different learning strategies can influence the way information flows through a network. Integrating the information from one's neighbors in different ways should lead to different patterns of information propagation. Barkoczi and Galesic (2016) explore two different social learning strategies: follow the crowd (conformity) versus follow the best neighbor. They found that the relative performance of inefficient versus efficient network structures depends on the learning strategy. Complex problem environments benefit in the long run from fully connected networks *if* the learning strategy is "follow the crowd." However, if the learning strategy is "follow the best," then less efficient networks do better.

Barkoczi and Galesic (2016) simulated local conformity by having nodes evaluate a random sample of three or nine neighbors and then following the strategy that was most prevalent. If there is no crowd to follow, individuals continue local exploration. They also did this if they were following the best and there was no one better. Note that random sampling of neighbors effectively weakens edges by reducing the frequency with which they are used, limiting information flow. This is also likely to limit early formation of communities of similar strategy users. These local communities have to form through repeated random sampling, which also limits their occurrence, and therefore increases exploration. On the other hand, the follow-the-best strategy diffuses good (but local) strategies quickly through the network. Follow-the-best diffuses these often suboptimal strategies most rapidly in fully connected networks. Thus, when follow the best is the strategy, less efficient networks are better at finding solutions on rugged landscapes because they maintain exploration for longer.

As Barkoczi and Galesic (2016) put it, "the network structure and social learning strategies jointly affect the levels of exploration and exploitation in the population. [When problems are hard] … if both strategy and network promote high levels of the same activity (either exploration or exploitation), performance is likely to drop; however, if network and strategy promote opposite behaviours, performance is likely to rise" (p. 4). I added the "when problems are hard" because when problems are easy, fully connected networks with follow the best neighbor do best.

15.5.2 Partial versus Full Copying

Another kind of information-limiting learning strategy is to modulate the amount of copying among multielement solutions. Campbell et al. (2022) studied this by looking at partial copying on networks. Partial copying mimics genetic recombination. If one individual chooses to follow another individual, instead of copying their entire solution, they copy elements of that solution each with some probability. If a solution is a sequence of elements – like a genome or a recipe – partial copying replaces its own element with one from its target with probability p. As p approaches 1, partial copying becomes full copying. As p approaches 0, partial copying becomes individual search.

What Campbell et al. (2022) find across many different network configurations is that partial copying performs better than full copying in the long run for hard problems. Once again, this maintains diversity in the population for longer. Partial copying also reduces the difference between efficient and inefficient networks – network structure is less important when individuals only partially copy. What Campbell et al. (2022) also find is that there can be too much diversity. This leads to the familiar inverted U-shaped function, with optimal performance somewhere between too much and too little. As landscapes become more rugged, the peak of the inverted "U" leans toward more diversity.

15.6 Adaptive Network Annealing

The networks approaches I've described tend to be fixed – neighbors are neighbors and they remain that way throughout the search. However, some approaches to solving hard problems involve modulating divergence over time. One such approach is called *simulated annealing* (Kirkpatrick et al., 1983). Annealing is a process used in metallurgy to increase the stability of metal by cooling it slowly. The slow cooling allows the molecular structure to arrange itself appropriately so that it can find a global molecular equilibrium that maximizes strength. Cooling too quickly prevents the structural components of the metal from arranging themselves in coordination with one another, leading to a weaker metal. Similar cooling-controlled annealing happens in polymerase chain reaction (PCR), a method for replicating DNA in the laboratory by heating and cooling strands of DNA in ways that allow them to bind with new genetic material.

Simulated annealing is the computational equivalent to literal annealing and uses a metaphorical temperature. When the temperature is high, the algorithm engages in more exploratory search, with less dependence between subsequent solutions. High temperature exploration is therefore less *auto-correlated* – less correlated with itself. Previous attempts at a solution are less likely to predict future attempts. By starting the algorithm out at a high temperature, the algorithm explores a greater breadth of the landscape. As the temperature is lowered, the algorithm searches near the best solutions found so far. Slower cooling rates are more likely to find better solutions, but they take longer to do so.

Many intelligent systems routinely alter their interconnectivity to explore and coordinate action. This reorganization can be envisioned as a network architecture in which edge weights change dynamically. Examples of this include brain networks altering their connectivity (e.g., Gutnisky & Dragoi, 2008) and schools of fish that alter the sensitivity of their behavioral response to close neighbors (Sosna et al., 2019). If this connectivity improves the behavior of the system, we can consider this a kind of adaptive collective intelligence. For networks, this means the edge weights change in a way that enhances the ability of the network to solve problems. This can either increase exploitation of good solutions or increase exploration for novel solutions.

A study by Almaatouq et al. (2020) demonstrated the utility of this dynamic connectivity. After being taught the basics for evaluating scatterplot correlations, participants saw scatterplots and then were asked to guess the correlation. They either did this alone or with shared information from other individuals they were connected to in a network. The other individuals saw different distributions of data, but representing the same correlation. People who saw information from other individuals in their network prior to making a final guess made better estimates than people who solved the problem in isolation. Indeed, even the best individuals improved their guesses after seeing other individuals in their network, further reducing error by more than 20%.

However, a third group were able to dynamically alter their connectivity with other individuals over successive rounds. This was achieved by allowing people to see how others had performed, and allowing people to follow or unfollow up to three others. This dynamic group did better than either of the other two groups. These dynamic networks showed improvements in their best individuals as well, but they also performed better as a group than even the best individuals, reducing both their error and their variance.

Analogous to simulated annealing, we can similarly describe processes of network annealing. Network annealing varies the density of the network, or the edge weights, such that high temperature networks have lower density than high temperature networks. If individuals, represented by nodes, are influenced both by feedback from the environment as well as information from neighboring nodes, then network annealing will allow networks to learn first from the environment and then coordinate solutions across the network as a whole.

15.7 Simulating Network Annealing on Rugged Landscapes

To create an environment for exploring some of the approaches described above, we will simulate agents as they search for solutions on a rugged landscape.

The networks will each have $N = 30$ agents. The structures we will investigate are a fully connected graph, a ring lattice, an empty graph with all agents acting independently, and what I will call *network annealing*, which starts off empty and slowly adds edges over time according to a cooling parameter, γ. Specifically, at each timestep after the first, the annealing network will, with probability γ, add one edge between a randomly chosen pair of nodes. This probabilistic edge addition continues throughout the group search process.

For all networks, each agent starts off with a random guess. At each time step thereafter, agents see the performance of their own guess and that of their neighbors. If one of their neighbors performs better, they copy the better performing solution. If there are no neighbors with better solutions, the agent explores. It explores by flipping one of the elements (bits) in its solution (to be described in more detail) and determining if the outcome is better than its current solution. If the outcome is better, it takes up the new solution. If the outcome is worse, it keeps its old solution. In this way, agents only improve solutions in a gradient ascent like process, but the ascent may also be influenced by copying better performing neighbors.

We will implement this collective search on rugged landscapes using the NK model, following Barkoczi and Galesic (2016).

15.7.1 Building Rugged Landscapes with the NK Model

NK environments were developed by Kauffman and Levin (1987) to mimic the ruggedness of evolutionary environments. In such environments genes are altered at scales ranging from base pairs to entire chromosomes. However, even a small change at the level of a base pair can have a large effect if there are many (epistatic) interactions between it and other genes. If small changes had small effects, the environment would be more smooth. When small changes can have large effects, the environment becomes more rugged.

To generate environments where the impact of small changes can be scaled from small to large, the *NK Model* allocates fitness to different gene-like solutions based on their sequence. A solution is N elements long, each taking a value of 0 or 1. Thus, a solution might look like $\{0,1,0\}$. K indicates the number of interactions between elements. When $K = 0$, each of the N elements acts independently, adding a random fitness contribution to the overall fitness based on whether it has a value of 0 or 1. Interactions between elements occur when $K > 0$. When $K > 0$, the fitness of an element depends on K other elements. Here I implement this as the Nth element in the sequence also being influenced by $N + K$ elements, which wraps around to the beginning at the end of the sequence. Thus, in the $\{0,1,0\}$ sequence given earlier, if $K = 1$, then the fitness of the first element is determined by the sequence $\{0,1\}$, the second $\{1,0\}$, and so on. Random fitness values are assigned to each of these subsequences for each new starting position, and the total sequences fitness is the average of the subsequences, as shown in Figure 15.4.

NK landscapes can be visualized as networks, as shown Figure 15.5. Nodes are genomes (or bit sequences) and edges between nodes are possible one-step transitions between neighbors, reachable by flipping one *bit* in the sequence. Edges are directed because paths through the sequence space are only valid when they improve fitness. Such a configuration has the feature that nodes with zero outdegree are local maxima.

15.7.2 Simulation Results

In the simulation, $N = 15$ and $K = 10$. Each network is run for 150 timesteps and the average fitness and total diversity of solutions are tracked over time. Each simulation for each network is also repeated 200 times, with new fitness landscapes generated every 10 repetitions. We build two annealing networks, using two values for the cooling parameter: $\gamma = 1.0$ and $\gamma = 0.75$ – the first adds an edge after every time step and the second every 3 out of 4 timesteps.

(a)

S1	S2	G1	G2	G3
0	0	0.2	0.9	0.5
0	1	0.7	0.9	0.5
1	0	0.6	0.1	0.6
1	1	0.2	0.8	0.2

(b)

P1	P2	P3	Fitness Score
0	0	0	0.53
0	0	1	0.57
0	1	0	0.43
0	1	1	0.7
1	0	0	0.67
1	0	1	0.57
1	1	0	0.27
1	1	1	0.4

Figure 15.4 NK example subsequence payoff matrix (a) and fitness matrix (b) for $N = 3$ and $K = 1$. $S1$ and $S2$ refer to the elements in sequence starting from a certain position in the genome. $G1$, $G2$, and $G3$ correspond to the fitness of a subsequence that starts at position 1, 2, and 3. If the genome were $\{010\}$ then the corresponding sequence for $G2$ would $S1 = 1$ and $S2 = 0$. The sequence, $\{10\}$, starting at the second position ($G2$) has a payoff of 0.1. The total fitness is the average across the N positions. The table on the right hand side shows the full genome fitness for each possible genome. $P1$, $P2$, and $P3$ refer to the three positions in the genome, corresponding to the fitness contributions $G1$, $G2$, and $G3$ in Table 1. The fitness score is the average across the three positions from Table 15.4a.

The results of the simulation are shown in Figure 15.6 and confirm the general intuition that more fully connected networks achieve better outcomes early but poorer outcomes later. The fully connected network makes a swift early transition to its current best solution, and then cruises up to a local maximum. The less efficient ring network overtakes the full network, but is then itself overtaken by the annealing networks. The annealing network that cools faster reaches higher faster than the one that cools more slowly, though the slower annealing may catch up. This tells us that slower cooling is not always better if we have both speed and performance to consider.

The explanation for the performance differences is provided by networks' abilities to preserve unique solutions. The fully connected network immediately jettisons all but the first best guess – at timestep 2 the population operates in unison. Each individual can explore independently, but all others will then follow the best guess. The less efficient ring network preserves sufficient diversity to help it linger on the hillsides of higher maxima. The success of those who find the maxima eventually trickles to other members in the ring. The annealing networks take this to the limit by allowing isolated individuals to search local regions of the rugged landscape before they start communicating, eventually converging on a collective solution that exceeds that of the other network configurations.

The rapid transition to a local peak but slow transition to global peaks is indicative of tortoise-and-hare like processes often observed on rugged fitness landscapes (Nahum et al., 2015). This trade-off is analogous to other forms of trade-offs such as overfitting versus underfitting and specialists versus generalists. A collective move to a local peak is a kind of overfitting to the observed data. It is also a kind of specialization in a local environment that may make transitioning to a new or changing environment more challenging.

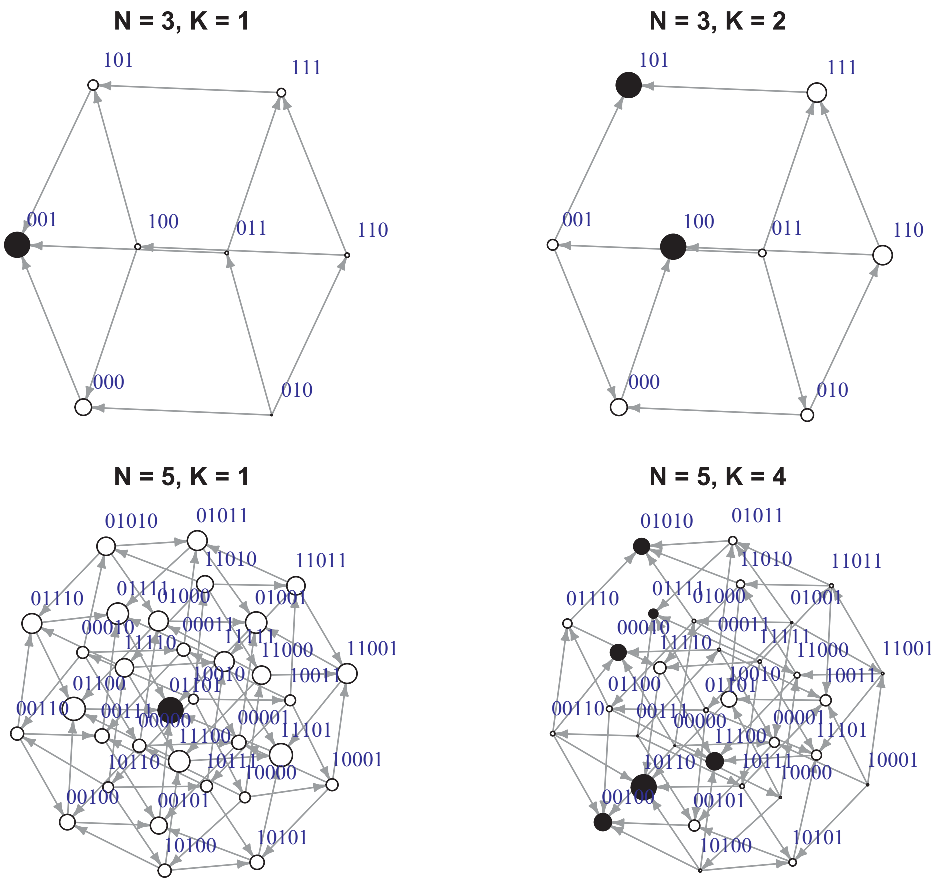

Figure 15.5 Plotting the NK landscape as a network. Arrows show possible one-step transitions in the space from genomes of lower fitness to genomes of higher fitness. The solid black nodes are local minima with zero outdegree, and inescapable using gradient ascent. The size of the node corresponds to its relative fitness. N = 3, K = 1 and N = 5, K = 1 both have one maximum. N = 3, K = 2 has one local maximum and one global maximum. N = 5, K = 4 has six maxima, five of which are local.

15.8 Summary

The ability of groups to find solutions to problems often depends on their communication structure. When groups demand top-down compliance, shut down dissenting voices, or otherwise facilitate social herding, they inhibit the kind of innovation and diversity necessary to identify good solutions. When solutions are easily found, this matters little. The harder the problem the more independent exploration is needed.

The implication for networks is that when information flows quickly through a network, this rapidly reduces diversity and leaves populations with suboptimal solutions. Decentralized networks can maintain diversity, as can more dynamically tuned networks that anneal over time. Such annealing networks mirror systems that use temperature to identify good solutions in rugged landscapes. By allowing communication to slowly

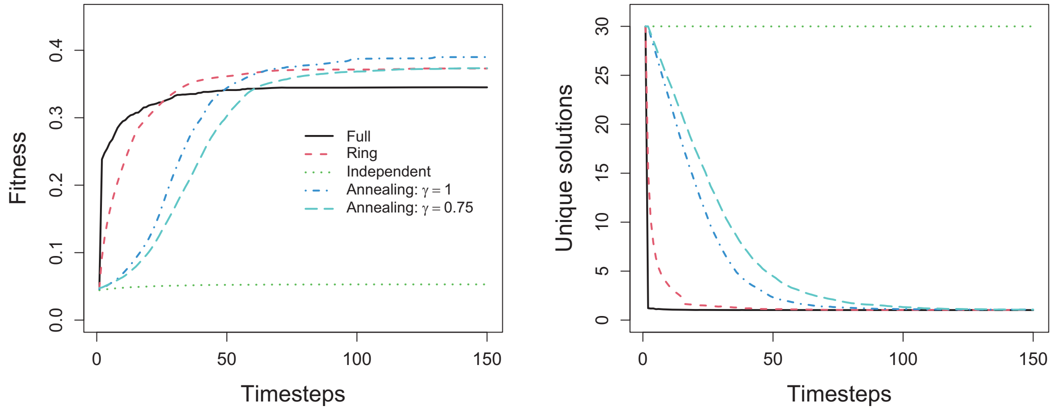

Figure 15.6 Performance trajectories for each of the network configurations. "Fitness" represents the average performance of all the individuals in the group. "Unique solutions" gives an indication of the exploratory potential of the group.

increase, these networks preserve the diversity needed to identify the most well-hidden solutions before they are inundated by suboptimal settling. Once these independent attempts have a chance to get off the ground, then combining them allows a comparison that improves the welfare of the group. The collective forum is likely to improve solutions still further as it recombines independently developed ideas and puts good – but not yet great ideas – to the challenge of collective scrutiny (Derex & Boyd, 2016; Mercier & Sperber, 2011).

15.9 Open Questions

15.9.1 Where Does Diversity Come From?

Conformity reduces diversity. If all social information systems were focused only on copying solutions, diversity would be eliminated. So what generates diversity?

One observation is that systems under stress tend to diversify. For example, if resources are depleting, then foraging individuals will tend to transition from exploitation to exploration (as we saw in Chapter 11). Similar cognitive systems may drive adaptive access to more exploratory solutions in the domain of creativity (as we saw in Chapter 13). Species can do this genetically as well. Various species, such as yeast and bacteria, engage in diversity-creating DNA sharing or mutation when environments become stressful (Galhardo et al., 2007; Moxon et al., 1994; Wagner, 2019). These are random attempts to diversify when the current solution isn't good enough – a kind of "Hail Mary" pass for organisms that might otherwise reach the end of the line. The same kind of risk-sensitivity is found in individual decision making. Many organisms (including humans) transition to riskier strategies as the payoffs for safe and stable solutions fall below sustainability (Weber et al., 2004).

This all makes sense. Increasing solution diversity is useful for exploring the solution space. This is necessary when the environment changes, which can happen via resource depletion, increasing competition, or gradual changes in the fitness landscape. Changes in predators, temperature, oxygen levels, and other resources offer benefits to those who can maintain the capacity for exploration, either via learning or genetic diversity (Fisher, 1930; Hinton & Nowlan, 1987).

How networks of individuals might adapt to changing environments is less well understood. Experiments on individuals in changing environments show they can adapt, and groups in fixed environments adapt to one another. But might these two forces oppose one another as environments take a turn for the worse? Might social proof lead groups to overexploit bad solutions, impairing their ability to track environmental change? Or does good communication better inform the group when change is needed? How might we model this?

15.9.2 Particle Swarm Optimization and Optimized Annealing

The group problem solving described here is a variation of what is called *particle swarm optimization*, which allows individuals to search independently but also to share information with a select set of neighbors (Kennedy & Eberhart, 1995). Both particle swarm optimization and simulated annealing (Kirkpatrick et al., 1983) are tunable in various ways and both have inspired a tremendous amount of research aimed at optimizing them for various problem spaces. In practice, the performance of these various approaches are highly sensitive to the structure of the landscape – small differences in N and K can have large differences in relative performance. Thus it makes sense to ask, across a range of NK problem spaces, varying the values of N and K, what structures best manage the tradeoff between speed and performance. For the annealing networks, what value of cooling parameters best achieve this. One can also ask how treating networks as collections of communities might change things. For example, classrooms and businesses often break into subgroups that work on the problem independently before coming together. This suggests another variation of the annealing network based on a dimension of hierarchical structure, with individuals forming lower level groups and then higher order groups over time. This might better preserve diversity while also maximizing group performance. Finally, how does group size interact with structure? Thinking in the limit, a network size of 1 will look the same for all network structures. To see structural effects, networks must be sufficiently large that agents land near better maxima. But networks that are too large will put the probability near 1.0 that at least one agent identifies the global maximum on the first guess. This again makes all structures similar. Somewhere in the middle lies the difference. We can ask how problem structure and network structure influence where exactly that middle is.

16 The Segregation of Belief

How Structure Facilitates False Consensus

16.1 The Problem

The false consensus effect is the observation that people tend to overestimate the number of people who share their views. In modern environments we also see growing evidence of greater polarization. For example, according to the Pew Research Center (DeSilver, 2022) over the past five decades, congressional US Democrat and Republican ideologies have increasingly diverged, with an ever shrinking middle ground. This also appears to be reflected among US citizens, with a disappearing center hastened by growing anarchist and antiestablishment ideologies. Many have speculated that this polarization is a global phenomenon (e.g., Mansfield et al., 2021). The question we pose here is how beliefs and network structure interact to facilitate both false consensus effects and rising polarization.

16.2 Social Influence

Many people get their information from other people. This can include direct communication as well as inferences we make about what is appropriate or normal, called *social norms*. For example, models of *rank-based comparisons* find that people judge their own alcohol consumption, diet, and criminal activity (e.g., illegal downloads) not based on their absolute engagement in these activities, but rather by where they rank relative to others that they know (Brown et al., 2022). If an individual drinks 20 units of alcohol a week, what is relevant to their self-assessment is not that that the average alcohol consumption is 14 units per week, but rather how many of the people they know drink more or less than 20 units per week. If their behavior is ranked near the middle of their social community's behaviors, then they are likely to see themselves as well adjusted. If, however, all the members of that community drink less (or more) than them, they are likely to evaluate their own drinking as extreme.

Social cues are especially effective because they tend to have primacy in cognitive processing (Fiske, 1993; Hills, 2019b). Some researchers suggest that this makes us vulnerable in the modern world: our social-media rich environments induce hypernatural social monitoring on par with other forms of addiction (Veissière & Stendel, 2018). In addition, people often alter information in ways that make it more engaging, by embellishment or hyperbole, such that information that passes through people – as one person communicates to the next – becomes increasingly engaging and difficult to correct (Jagiello & Hills, 2018; Moussaid et al., 2015).

The stickiness of social information makes it a centerpiece of influence and behavior change. The iconic *Influence: The psychology of persuasion*, by Robert Cialdini et al. (2009), introduces six principles for influencing other people's behavior: social proof, scarcity, reciprocity, commitment, authority, and liking. All of these take their influential strength from social information. Scarcity signals the preferences of others ("Only one room left"). Reciprocity creates an obligation to another, which people feel motivated to repay. Social proof signals a social norm ("This is what other people do"). Commitment is based on the idea that we judge our own behavior partly through what we believe other people expect us to do ("In the past you chose X"). Authority and liking are both implicitly social cues, which are based on our valuing of other people and, as a consequence, their behavior.

Cialdini is not the only one with such views. The behavior change mnemonic developed by the UK's Behavioral Insights Team is EAST – which stands for Easy, Attractive, Social, and Timely – summarizes all of Cialdini's effects in one word: "social" (Halpern, 2015). The *theory of planned behavior*, on which much behavior change theory is based, devotes one of its three components to social norms (Armitage & Christian, 2003). Its other two components are efficacy (the belief that we can achieve a result through our behavior) and attitude.

Social comparison is pervasive. It influences such things as charitable giving (Frey & Meier, 2004), whether or not we vote (Gerber & Rogers, 2009), and employee productivity (Bandiera et al., 2010). What other people believe can lead us to claim that things are false even when, in the absence of other people, we would know they are true (Asch, 1955). That is, other people can lead us to hide our true beliefs. In the political sphere this observation led Elizabeth Noelle-Neumann (1974) to develop her theory of the *spiral of silence*: that is, political beliefs often go unexpressed when people fear their own ideas are not shared by the majority. This, in turn, can lead these views to go further unexpressed. *Pluralistic ignorance* is that sorry state in which we all keep our views to ourselves under the false assumption that no one else shares them (Prentice & Miller, 1996).

What we think about other people can be as valuable a predictor of our behavior as our own stated preferences. One example of this is a story of the German federal elections in 1965, described by Noelle-Neumann and Petersen (2004). For the entire 10 months prior to the election, polls querying voting intentions consistently showed that the Christian Democrats and the Social Democrats were head-to-head. Neither party had a clear advantage. However, in the month prior to the election, polls started asking people not about their own preferences, but who they thought would win: "Of course, nobody can know for sure, but what do you think: Who is going to win the election?" These polls showed a clear margin favoring the Christian Democrats over the Social Democrats: 50% to 40%. The election results confirmed this prediction.

The power of this *second-order inference* is revealed in the success of *Bayesian truth serum* – in which the wisdom of the crowds is enhanced by asking people "what do other people think" (Prelec, 2004). This has in turn led to prediction markets in which people buy and sell contracts on future outcomes as if they were buying and selling stocks

(Wolfers & Zitzewitz, 2004). This attention to our social community is further reflected in the *social circle heuristic*, which is our tendency to treat our local social networks as if they are representative of the larger social community in which we are embedded (Galesic et al., 2012; Pachur & Schulze, 2022).

How might these many aspects of social dynamics alter social structure in ways that facilitate polarization? Social structure is rarely distributed randomly. People tend to find themselves in the company of like-minded others. This goes by the term *homophily* and is summarized in the proverb: "birds of a feather flock together." We birds flock together in part because numerous biological, social, and cultural sorting mechanisms are built into society; ethnicity, religion, work, education, age and behavioral choices all act to filter us toward people of similar dispositions (McPherson et al., 2001; Shrum et al., 1988). Homophily in beliefs and behavior can also arise because people who spend more time together become more alike, through various mechanism of social influence (Brown et al., 2022; Hills, 2019b). In early translations from the Rabbinic literature, the birds of a feather proverb once ended "so honesty comes home to those who practice it" (Ben Sira 27:9) – meaning that our behaviors are returned to us. This has the recursive logic of social influence: We are active participants in the creation of our flock.

16.3 The False Consensus Effect

The *false consensus effect* is the observation that most people tend to overestimate the number of people who share their views. In one of the first studies to introduce the false consensus effect, Ross, Greene, et al. (1977) had individuals estimate how other people would decide on various issues, such as allowing themselves to appear on TV, support funding an unmanned space program, prefer white or brown bread, or pay an improperly reported speeding fine. Following people's estimate of others, the researchers later asked the people what they would themselves choose. What Ross and colleagues found is that people's beliefs about others were strongly biased in favor of their own views. People who themselves favored public spending on an unmanned space program believed that that view was more popular than it actually was, and vice versa. People who preferred brown bread believed that 52% of other people preferred brown bread, whereas people who preferred white bread felt that brown bread preferences were closer to 37%. This effect tends to be larger for those holding minority opinions, for example, among climate change deniers (Krueger & Clement, 1997; Maier et al., 2022).

What causes the false consensus effect? A statement about the beliefs of others could be caused by many things. For example, it could reflect an intentional misrepresentation aimed at convincing others that one's own view is normal (Mullen et al., 1985). It could also be a form of *motivated reasoning* or *self-serving bias* (Kunda, 1990). For example, people might think something like this: "I am normal. I do this behavior. Therefore, this behavior is normal." It reminds one of the George Carlin joke: "Anybody driving slower than you is an idiot, and anyone going faster than you is a maniac."

False consensus might also be driven by *the availability heuristic*: people tend to overestimate the likelihood of things that come easily to mind (Tversky & Kahneman, 1973). For example, when people are asked to imagine a future event for someone and are then asked for the likelihood of that event, they rate the event as more likely than if they had not imagined it (Ross, Lepper, et al., 1977).[1] One's own behavior is certainly a good source of things that are likely to come to mind. One prominent psychological theory is based on a similar idea: Bem's (1972) *self-perception theory* argues that people come to understand themselves by evaluating their past behavior. One of the best predictors of what someone will do in the future is what they have done in the past – that is, unless they have thought exceptionally clearly about an alternative (Gollwitzer & Sheeran, 2006).

One of the most well accepted explanations for the false consensus effect is that humans artificially create communities around themselves that share their views. Pursuing friendships with like-minded others and unfriending the rest will eventually result in self-serving communities, or echo chambers (Sasahara et al., 2021). This is a form of *cognitive segregation*, similar to that proposed by Thomas Schelling (1971) for racial segregation and described later in this chapter. When we're talking about beliefs and attitudes, this is also a form of self-serving information gerrymandering, as described in Chapter 14: we curate our cognitive environments by controlling our external environments. This includes the artifacts we place around ourselves (media, motivational sayings, and coffee pots) as well as the people we engage with. The theory of *extended cognition* takes this idea to its logical conclusion: The distinction between what is inside our heads and what is outside our heads is fairly artificial, as we tend to use external resources to facilitate and guide internal cognition (Clark & Chalmers, 1998).

Indeed, even if one were in a community of only one other individual, just hearing the same thing repeated over and over again increases the likelihood we will believe it is true.[2] *The illusory truth effect* describes how our minds are changed by hearing falsehoods repeated (Hasher et al., 1977). It works even when people know that the statements are factually wrong (Fazio et al., 2015). For example, when people are asked what is the name of the process by which plants make their food, most will provide the correct answer ("photosynthesis"). However, if they are first asked to rate how interesting is the statement "chemosynthesis is the process by which plants make their food," they will later claim that statement is more true than if they hadn't previously rated it.

The growing research area of *deliberate ignorance* adds additional insularity to our internal majority. There are many reasons why people might simply not want to know about views that differ from their own (Hertwig & Engel, 2021; Hills, 2022). For example, emotional regulation, regret avoidance, and eschewing responsibility can all lead people to avoid social contacts that might inform them of something they would rather not know. As Caplan (2000) notes, irrationality and ignorance are goods like any other, and people are willing to trade them for wealth.

Indeed, there are a number of ways in which our social environments can help support views that are satisfying, comfortable, and wrong. We can ask exactly how far such social effects might go. Can they explain false consensus effects and polarization? Taking it one step at a time, we can demonstrate that they do.

[1] For example, ask yourself what is the likelihood of a female leader of Russia? Now ask yourself, what is the likelihood that the current leader of Russia is incapacitated by his health and his wife takes over.

[2] This suggests that the majority illusion (described in Chapter 14) might be better represented as a weighted network, with edges representing speaker intensity.

16.4 Schelling's Segregation Model

Thomas Schelling is renowned for offering simple but surprising insights on topics ranging from nuclear diplomacy to group coordination. One contribution that speaks directly to polarization is his segregation model (Schelling, 1971).

Schelling's segregation model is an *agent-based model*, meaning that it focuses on how collections of individual agents, who each make independent decisions based on their state and environment, can produce collective outcomes that simulate real-world phenomena. The collective outcome is a consequence of relationships between agents and not the decision of any single agent. There is no eye-in-the-sky or central control system guiding individual behavior. In agent-based models, individuals make independent and often reasonable decisions based on evaluations of their local environment. Yet the outcome often appears planned and global. Birds flocking, fish schooling, bee hives, termite nests, traffic, opinion dynamics, stock markets, and neural development are all standard examples of the emergent behavior that can arise from agent-based models. Many interesting research questions develop around asking what kinds of complex phenomena can be explained as the outcome of simple decisions among agents with only local information.

In Schelling's model, agents (i.e., individuals) are represented as colored nodes (black or white) and they share edges with their neighbors. Agents pay attention to only two things in this world: their own color and the relative distribution of the color of their neighbors. If the neighborhood is not to their liking, they have only one option: move. More specifically, if agents are in a color minority among their neighbors, then they would prefer to move. Otherwise, they are happy where they are. If they would like to move, they can swap positions with another unhappy agent.

Many different variations of Schelling's model have been proposed. These include varying the structure of the environment, how many groups there are, how agents make decisions, and how moving is achieved. It is easy to think of variations. Here I will use a simple variation of a standard method (Brown et al., 2022; Clark & Fossett, 2008): Two unhappy nodes are randomly chosen that differ in color and they swap position. This continues until there are no more improvements possible through swapping.

Under what conditions should individuals swap? Schelling examined a variety of cases, but one of the most interesting is that people swap when they are in the minority. That is, agents are perfectly happy to find themselves in mixed neighborhoods with equal numbers of white and black nodes. But they will move if they are in the minority. With roughly equal size populations of white and black nodes, this leads to heavily segregated communities.

Figure 16.1 shows the basic result. Starting from a random distribution of white and black nodes (panel a), migration rapidly leads to segregation (panel b). This result continues to be surprising, even among those who know it well: agents happy to live in mixed neighborhoods rapidly wind up in segregated communities if they have a preference to avoid being in the minority.

However, it is easy to show that Schelling's model as I have described it here does not fully explain segregation. If one population is much smaller than the other – as is

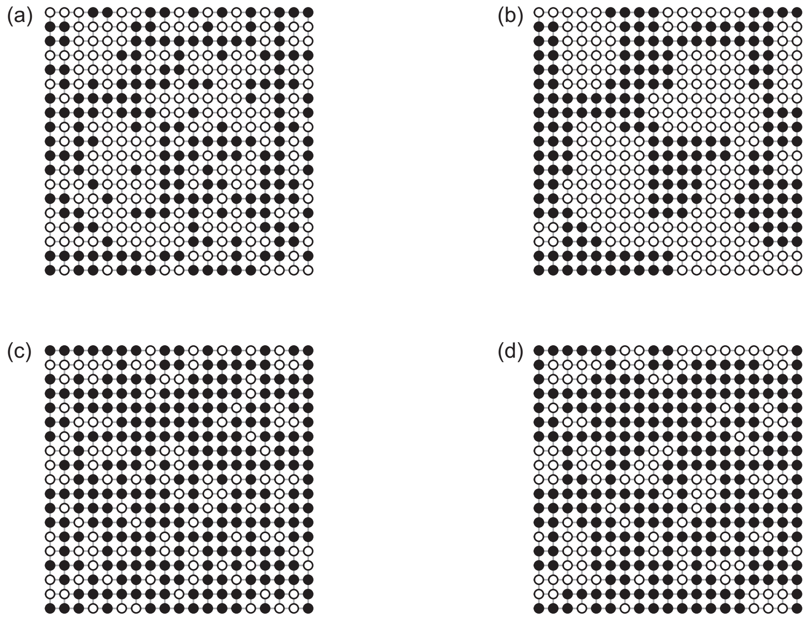

Figure 16.1 Thomas Schelling's segregation model. All agents prefer to move if they are in the minority color among their neighbors. At each timestep, one white and one black agent who are willing to move are chosen at random and swapped. Panels a and c show the random starting arrangements. Panels b and d show the arrangements after all available swaps are made to the corresponding panel on the left. Panels a and b show a 50:50 distribution of white to black nodes. Panels c and d are 30:70, white to black.

true by definition for minority populations – the populations remain more mixed after all available opportunities for swapping are exhausted. Figure 16.1 (panels c and d) shows the outcome when the starting population is split between black and white nodes in a ratio of 70:30. The lack of mixing in panel d, when the population sizes are asymmetric, is down to greater satisfaction among the majority nodes. There are still 43 white nodes who would swap if they could, but there are no black nodes willing to move. In panel b (with equal populations of black and white nodes), there is only one remaining dissatisfied white node who would move.

Schelling's model can be adapted in a number of ways to recapture segregation. For example, we might consider the effects of allowing individuals to move to empty spaces or to squeeze between others (as Schelling did). It is this process of exploring how specific models do or do not explain phenomena, and what changes would be needed to cross that divide, that is the power of computational models, demonstrating their ability to help us think and ask better questions about the subjects of our study.

16.5 Social Sampling Theory: Giving Agents Beliefs

If we replace race with beliefs in Schelling's model, then segregation by belief already creates a tendency for individuals to maintain a false consensus. Very few individuals would have a neighborhood representative of the population as a whole – most would have a neighborhood favoring their own opinion. In such a world, if agents infer majority opinions based on inferences from their local community, the majority of individuals will believe their opinions are more prevalent than they actually are, even if the population is split 50:50.

We can pursue this idea further by giving agents beliefs that respond to social influence. Brown et al. (2022) present a model of belief representation that is ideal for our purposes.[3] Their model, *social sampling theory*, explains numerous findings in the psychological literature, including social influence, polarization, consensus, and backfire effects, which I describe later in the chapter.

Social sampling theory assumes that for anything a person could believe, they believe it more or less intensely, ranging in value between 0 and 1. Individuals also hold this belief with more or less confidence, which could be based on biological predispositions or personal experience. For example, if an individual holds a confident but well-balanced view between left and right-wing politics, we might assume they have good but well-balanced reasons for and against both sides of the political divide. If they hold the same view with low confidence, we can assume they have balanced but limited evidence and are therefore more sensitive to outside influence. An ideal way to represent these beliefs is using a beta distribution.

16.5.1 Beta Distributions as Beliefs

This characterization of beliefs between 0 and 1 is well described by a β distribution. β distributions are powerful tools for representing Bayesian minds. That is, they represent the Bayesian optimal belief given a set of evidence in favor of one or another outcome, they characterize degrees of belief over all possible outcomes, they update when new evidence becomes available, they can characterize a wide variety of beliefs, and they have many elegant mathematical properties.

In simple terms, a β distribution represents the intensity of one's beliefs about a two-sided phenomenon: heads or tails, left or right, up or down, chocolate or vanilla. The β distribution characterizes this belief with two *shape* parameters, α and β. The shape parameters have an intuitive interpretation: they are the number of items of evidence supporting each outcome. For example, suppose we are attempting to estimate the probability that a coin will come up heads when flipped. If we flip it 10 times and it comes up heads 4 times and tails 6 times, then our belief can be represented with a β distribution with $\alpha = 4$ and $\beta = 6$. If the coin is flipped again and comes up heads, the α parameter changes to $\alpha = 5$.

Figure 16.2 shows how we can visually represent and interpret this distribution. Looking at the solid line, the density plot shows the relative belief we should have for various probabilities. The belief for which we have the most evidence is that the coin comes up heads 40% of the time ($p = 0.4$). But the distribution also tells us that we still consider other probabilities as being reasonable. If we flip the coin 40 more times and come up

[3] Their model focuses on attitudes, but I take these to be synonymous with beliefs in regards to the model implementation described here.

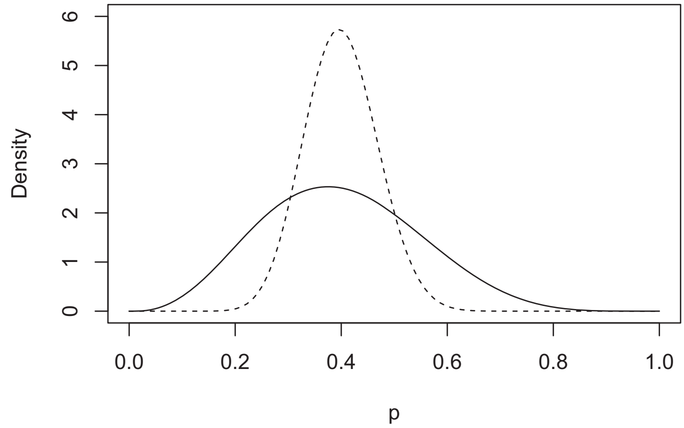

Figure 16.2 Beliefs as beta distributions. The two distributions shown in the figure represent different degrees of belief. The horizontal axis has a range between 0 and 1, and a position along that axis indicates a belief (e.g., the coin has a 0.5 probability of coming up heads). The height of the distribution indicates the relative strength of the belief. The area under the curve indicates the probability of a particular set of outcomes. Therefore, the sum under the curve across the entire belief distribution sums to 1. The solid line has shape parameters $\alpha = 4$ and $\beta = 6$. The dotted line distribution has shape parameters $\alpha = 20$ and $\beta = 30$. Both distributions peak at 0.4, but the 20:30 distribution is more confident in this outcome. It assigns less weight to the probabilities of 0.6 and 0.2, for example.

with an additional 16 tails and 24 heads, we can simply add these values to our previous evidence, resulting in $\alpha = 20$ and $\beta = 30$. This is shown by the dotted line. Our confidence around $p = 0.4$ has increased and our belief that the coin is fair, $p = 0.5$, has reduced. More evidence increases confidence, narrowing the distribution around our strongest belief.

16.5.2 The Trade-Off between Authenticity and Social Influence

Social sampling theory represents an individual's authentic beliefs using a beta distribution. Different individuals have different distributions reflecting different beliefs. Individuals also express behavior, but that behavior is a trade-off between their own authentic beliefs and the expressed behavior of their neighbors in their social community. This creates a recursive process in which people act based in part on the actions of others, who then change their own actions in response, with no one knowing exactly what anyone else thinks, a perfect representation of pluralistic ignorance.

More specifically, an individual's expressed behavior is a compromise between their *authenticity preference* – their desire to act in accordance with their own beliefs – and their *social extremeness aversion* – their desire to not deviate too boldly from the beliefs of their community. The individual pays a penalty for deviating from their authentic preference. They also pay a penalty for deviating from the perceived social norms of those around them. Their expressed behavior is the behavior that minimizes the combined cost of those two penalties.

Formally, the cost of the authenticity preference is zero when individuals express the median value of their authentic beliefs. Deviations from authentic beliefs are penalized in relation to an individual's confidence in those beliefs. That is, they are penalized in proportion to the area under the curve between the behavior they express, A_i and the median of their authentic beliefs. A small deviation from a confidently held belief is more costly than the same deviation from a less confidently held belief. This deviation from the authenticity preference, d_A, can be written as

$$d_A = |I_{A_i}(\alpha_i, \beta_i) - 0.5|,$$

where $I_{A_i}(\alpha_i, \beta_i)$ is the cumulative distribution function for the expressed behavior A_i: the area under the curve from the median value of their authentic beliefs to the expressed behavior. If the expressed behavior represented 0.6 of the distribution under the curve in the individual's authentic beliefs, they would deviate by 0.1 from the median (0.5) behavior.

The deviation from the social environment can be written in a similar way. The deviation from the beta distribution representing the social environment, d_S, is

$$d_S = |I_{A_i}(\alpha_s, \beta_s) - 0.5|$$

where $I(\alpha_s, \beta_s)$ reflects the beta distribution that best fits the expressed behaviors of one's social, s, neighbors. The least socially deviant behavior is the one that reflect the median value of the social environment.

Each deviation ranges in value from 0 (no deviation) to 0.5 (maximum deviation). The maximum of 0.5 reflects the fact that the deviation is measured from the median of the distribution, and therefore cannot be larger than half of the distribution.

The deviation from authenticity and the social environment both have a cost. The costs are written as C_A and C_S, respectively, and more generally as

$$C = e^{-\gamma(0.5-d)}.$$

with γ indicating the *insensitivity* of the increase in the deviation. Larger values of γ require larger deviations to achieve the same cost. When the deviation is at its maximum, 0.5, then the cost is 1.

The final value, or *utility*, of an agent's expressed behavior can then be written as a weighted function of the costs of deviating from authentic beliefs and the behavior of others, C_A and C_S:

$$U_{A_i} = 1 - w \times C_A - (1 - w) \times C_S$$

The w term is the degree to which individuals prefer to minimize the costs to their authenticity preference (C_A) over the costs of their social extremeness aversion (C_S). When $w = 1$ individuals are indifferent to social extremeness aversion. When $w = 0$ individuals are indifferent to authenticity preference. When $w = 0.5$ costs are weighted equally.

The goal of any agent is to choose a behavior, A_i, which maximizes its utility in the preceding equation. Figure 16.3 shows the utility of different expressed behaviors given a set of authentic beliefs and a social environment, with $w = 0.5$ and $\gamma = 20$.

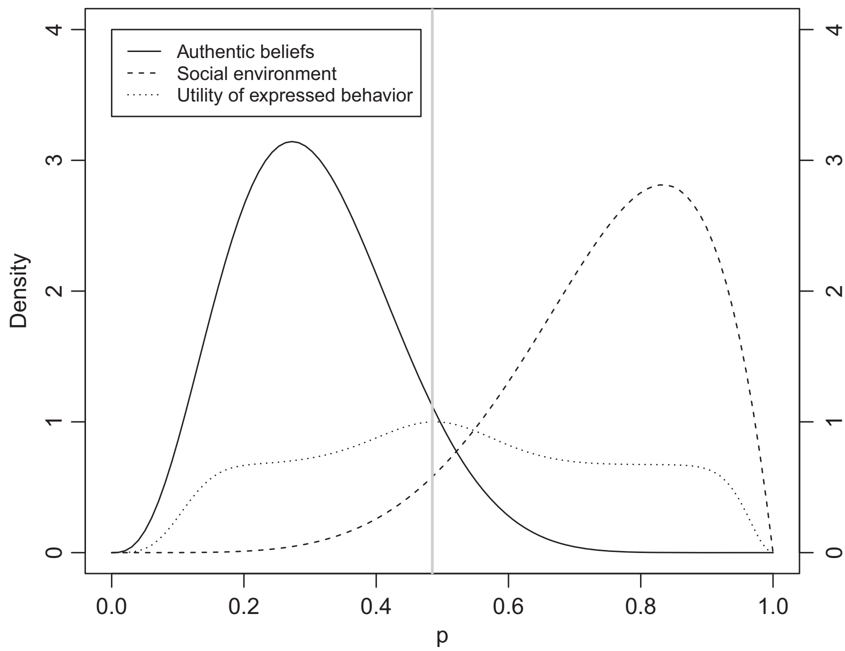

Figure 16.3 Social sampling theory. An agent with authentic beliefs represented by a beta distribution encounters a social environment represented by a beta distribution. By minimizing the combined influence of its authenticity preference and social extremism aversion, the agent is able to identify the expressed behavior with the maximum utility, indicated by the vertical gray line.

The beauty of this approach arises in the many complex ways in which expressed behavior responds to differences in authentic attitude and social environment. Social sampling theory also tells us how satisfied agents are with their environment. The utility of the expressed behavior is a measure of an agent's satisfaction with that behavior. An agent trapped in an environment that poorly reflects their authentic attitude will have lower utility. The lower the utility, the less satisfied they are with their surroundings. This can be used to measure the overall average satisfaction of a network structure, as well as to determine when agents would prefer to leave one place for another.

16.5.3 Social Influence, Conformity, and Backfire Effects

Social sampling theory reproduces various well-known psychological effects. These also apply directly to social networks, as social extremeness aversion is derived from the influence of social neighbors. Figure 16.4 presents the transition from authentic behavior to the four effects described here:

- *Authentic behavior* reflects what an individual would do in the absence of social influence.
- *Social influence* occurs when one or a few other individuals drive one's behavior in the direction of the social norm. The model presented by Brown et al. (2022) derives the distribution of social effects by fitting a beta distribution to the expressed behavior

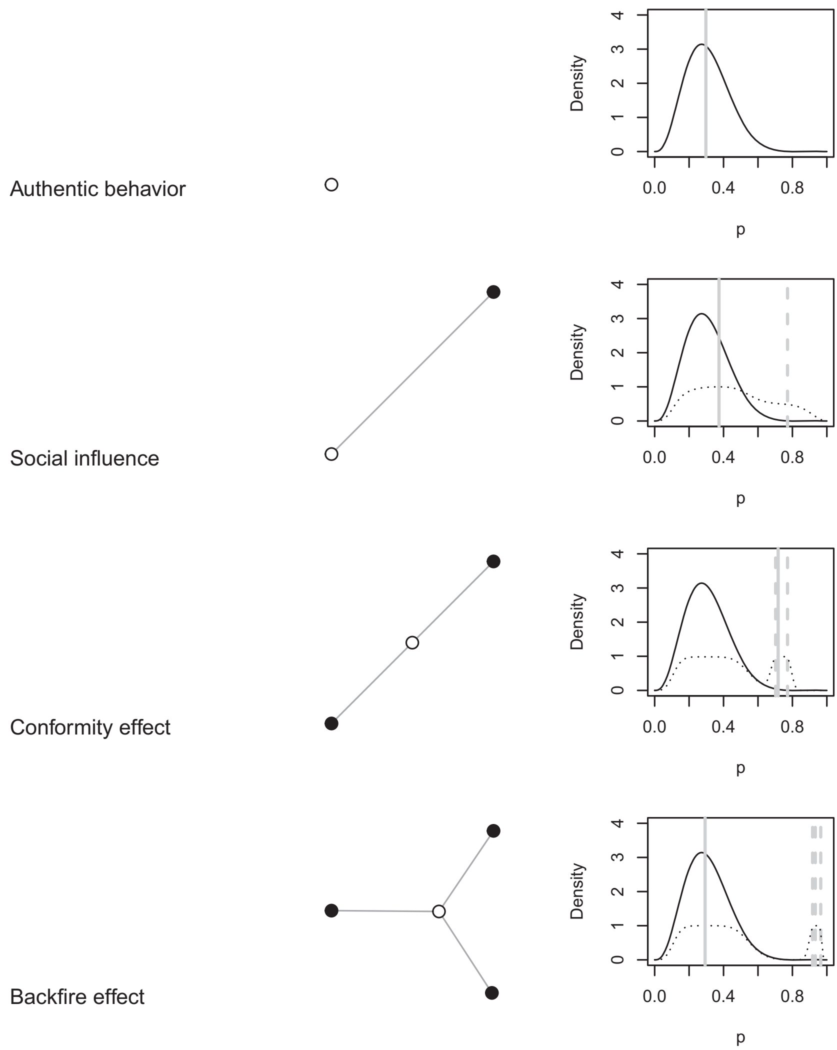

Figure 16.4 Giving agents psychologically plausible beliefs with social network influence. Social sampling theory is presented for four cases: no social influence leading to authentic behavior, one neighbor producing a social norm effect, two neighbors producing a social conformity effect, and three neighbors producing a backfire effect. Social norm, conformity, and backfire effects are all possible with different numbers of neighbors – what is presented here should be taken as a stylized representation. The individual's belief distribution is presented along with the expressed behaviors of their neighbors (gray vertical dotted lines), the value of the individual's potential expressed behaviors (black dotted line), and the value maximizing expressed behavior (vertical solid gray line).

of an individual's neighbors. In the case of only one neighbor, I combine the single neighbor's expressed behavior with the target individual's median authentic behavior to derive the social distribution.

- *Conformity effects* arise when an individual's neighbors share a consensus. This was famously studied by Asch (1955) in his series of conformity experiments in which an individual was placed in a setting where they had to make a judgment with a clear and apparent objective truth. However, the individual was first exposed to the judgments of others who all made the same incorrect choice. The results indicated that when faced with a majority, many people publicly expressed an opinion different from their own objective judgment.

- *Backfire effects* or *boomerang effects* are observed when individuals become more committed to their beliefs after receiving information that is inconsistent with those beliefs. This is commonly observed around political topics (e.g., Bail et al., 2018; Hart & Nisbet, 2012). Predicting when conformity effects will transition into backfire effects is an area of ongoing study.

Each of these effects falls out of the model naturally with no changes in the parameters ($w = 0.5$ and $\gamma = 20$). All that is required is a change in the social environment. Brown et al. (2022) describe these and other many other effects, including the role of zealots (individuals with $w = 0$ who are uninfluenced by social extremeness aversion).

16.5.4 Social Sampling Theory on a Network

Figure 16.5 shows social sampling theory on a 10×10 lattice. The bottom left individual is initially lightly colored in generation $G = 0$, revealing its expressed behavior if it were true to its authentic beliefs. In generation 1, it changes its expressed behavior to move closer to its neighbors. However, its neighbors have changed in turn to accommodate their social surroundings. In generation 2, node 1 changes back toward its authentic behavior because its neighbors seemed less extreme, having adapted toward node 1. All of the nodes are adapting and readapting as this out-of-phase accommodation continues across the three generations shown. Notably, however, the system also settles down toward a steady state. After 100 generations, little of the initial diversity remains, with all individuals showing a muddy mediocrity that barely registers differences in individual beliefs.

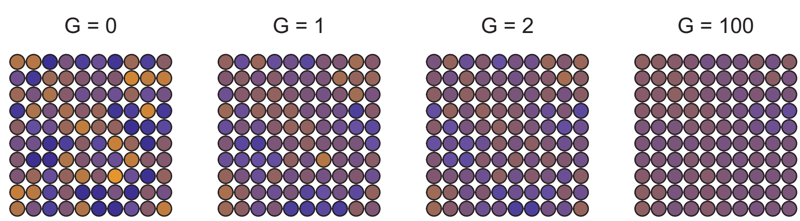

Figure 16.5 Social sampling theory over repeated generations. Individuals express their median authentic behavior in generation G = 0. All agents then update their expressed behavior based on their environment for G equal to 1, 2, or 100 generations ($w = 0.4$ and $\gamma = 5$). Careful inspection reveals how the iterative nature of the adaptation to one's neighbors, with them adapting at the same time, creates a subtle oscillation. An individual perceives its neighbors to be more extreme, so it expresses more extreme behavior. But they perceive their neighbors to be less extreme, so they express less extreme behavior. Now the first individual experiences less extreme neighbors, and so on. This oscillation settles down over generations, with individuals eventually expressing a behavior that is both a reflection of their authentic beliefs and their social environment.

16.5.5 Social Sampling Theory and Migration

What if individuals could move to places with more accommodating attitudes, in a kind of ideological Schelling migration? We can achieve this in our modeling framework by allowing some individuals to switch places each generation. Here we randomly sample 5% of nodes and assign them to pairs. Pairs evaluate one another's neighborhood. If both members find that they would produce an expressed behavior with a higher utility in their neighbor's location, they swap positions. Otherwise they stay where they are.

Figure 16.6 demonstrates how adding this Schelling migration preserves more extreme beliefs. After 100 generations we can readily see how the distribution of expressed behaviors varies across a wider range than it did in Figure 16.5. Without migration, after 100 generations, the minimum and maximum expressed behaviors are 0.40 and 0.59, respectively. With migration, after 100 generations the minimum and maximum expressed behaviors are 0.31 and 0.64, respectively. After 1,000 generations of migration, the minimum and maximum are 0.21 and 0.68, respectively.

The ability to migrate to neighborhoods of like-minded individuals preserves and potentially amplifies behavioral polarization (Bramson et al., 2017; Flache et al., 2017). That is, if polarization can be equated to the revealing of more divergent, but authentic, beliefs – as opposed to pushing people toward these beliefs – then migration is sufficient for polarization. If human behavior works at all like that described in social sampling theory, we never have access to people's authentic beliefs. Indeed, people may not have access to their own authentic beliefs. Placed in the right environment, however, those beliefs are more likely to be expressed.[4]

Another useful product of social sampling theory is the measurement of collective utility. This too increases with migration. After 100 generations without migration, the mean utility is 0.78. After 100 generations with migration the utility is 0.80. After 1000 generations with migration, the utility is 0.84. Thus, the capacity to move among neighbors who share one's authentic beliefs (as inferred through expressed behavior) both preserves behavioral variance and increases experienced utility. We might hope that utility is synonymous in this case with things like well-being and mental health.

16.6 The Impact of Social Search on Polarization and Perceived Normality

Schelling migration involves moving into neighborhoods. But networks also allow us to focus on the formation and removal of edges. In social media, this would be analogous to following and unfriending. Sasahara et al. (2021) developed a model designed to emulate social media behavior, in which agents can share original social media posts with their

[4] This speaks to the power of social structures to moderate extremism. Thomas (Hobbes, 1651/2016) argued for the power of government, the *Leviathan*, to prevent "a war of all against all." The Hobbesian position is often interpreted as people having authentic attitudes or beliefs that are socially undesirable. Jean-Jacques Rousseau, however, felt that people's authentic attitudes were inherently good. If Rousseau is right, the model is missing the ingredient that moves people outside the range of their collective authentic preferences. It's an open question as to what that would be.

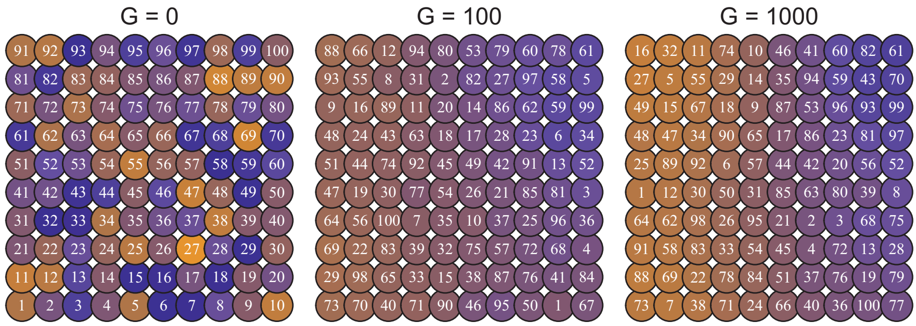

Figure 16.6 Social sampling theory with Schelling migration. Individuals are sampled in pairs, approximately 5% each generation, and allowed to evaluate the expressed behavior among their partner's neighbors. If both parties would have higher utility if they swap position, then they replace one another on the network. In the figure, $G = 0$ shows the behavior individuals would express if they expressed their authentic beliefs. Each generation thereafter includes two phases. The first phase involves mediating the trade-off between authenticity preferences and social extremeness aversion. The parameters are $w = 0.4$ and $\gamma = 5$, slightly favoring social extremeness aversion. The second phase involves migration. Individuals that swap position move to their new location with the expressed behavior they would have in that location and with their corresponding utility. Numbers are preserved across panels to show how individuals can move to locations at $G = 1{,}000$ that better allow them to reveal their authentic behavior shown at $G = 0$. However, even at $G = 1{,}000$ there are still many individuals who would benefit from migration (e.g., number 16).

neighbors and also repost information they receive from others. Opinions are formed via an initial random assignment and via social influence, based on the information agents receive from their neighbors in their social media feed. This is modulated by a tolerance parameter, ϵ, and an influence strength parameter, μ. Finally, if an agent's opinions are too divergent from a post they receive – they can remove the edge to that neighbor, unfollowing them, with probability q. When they unfollow one node, they form a new edge to nodes whose posts they receive as reposts from their neighbors, via recommendations to others with similar opinions, or at random.

What Sasahara et al. (2021) found is that echo chambers quickly form in this environment. The speed at which they form is increased by factors such as influence strength and probability of unfollowing. The total number of opinion clusters and diversity of opinions is influence by tolerance. Lower tolerance means a greater diversity of opinions and more echo chambers.

The Sasahara et al. (2021) model is based on a host of evidence about real-world echo chambers, online polarization, and the observation that many people show limited diversity in online media consumption (Bakshy et al., 2015; Conover et al., 2011; Del Vicario et al., 2016). This all follows naturally from the combination of social influence and belief-directed segregation.

What happens if we combine ideas around online echo chamber formation with social sampling theory? There are many possibilities for achieving this. We will explore one of these variations: agents can sample from a subset of potential partners. In Schelling migration, partners are chosen at random. Here, agents will be allowed to search *through* the social network (i.e., through their neighbor's neighbors). On social media this is similar to surfing friendships by seeing who our friend's friends are. People that many of our friends know are more likely to become contacts than people who our friends do not know. The people we know are also likely to mention other people who share our values. Or more simply, our ability to discover new people and things online is partly influenced by who we know. People don't wind up on the dark web by accident.

This has the flavor of Granovetter's (1973) *strength of weak ties*. As Granovetter notes, "the stronger the tie connecting two individuals, the more similar they are, in various ways" (p. 1362; discussed further in Chapter 18). Therefore, the further we can navigate away from the people we know, the more likely we are to find what we least expect. In the language of social sampling theory, this might mean we find people who better share our authentic preferences.

Formally, we will let search be limited by the path length, S. S indicates the depth of search. If agents sample among their neighbor's neighbors, then $S = 2$. If they sample among their neighbor's neighbor's neighbors, $S = 3$, and so on. For some value of S one searches the entire network. Recall the *six degrees of separation* often claimed to connect all of humanity. True or not, we'll explore a simple case here that starts on a 10×10 lattice (diameter = 18, for the curious).[5]

At each time step, agents update their authentic behavior as described earlier based on their local community ($w = 0.4$ and $\gamma = 5$). Following this update, one agent will be chosen at random to search among their neighbors to depth S. Among those neighbors they will identify the neighbor with the expressed behavior that most closely matches their authentic attitude. If that behavior is a better fit to their authentic attitude than the worst expressed behavior among their current neighbors, then they form an edge to the new individual and remove the edge from their worst current neighbor. Otherwise, they do nothing. If they do swap edges, they do not update their expressed behavior or utility until the next round. Finally, any nodes that become isolates are joined to a new node at random. Isolates are possible because some nodes may suffer repeated edge losses.

Finally, note that if individuals search over short distances but do so over repeated generations, they may eventually search through the entire network. This is true as long as the network is still connected. In the limit, we might then expect that search on networks may all wind up in the same place given enough time, regardless of search depth. Therefore, what we will focus on here is how the networks change over a shorter interval of time. As we will see, over short intervals we already observe dramatic differences.[6]

[5] Brown et al. (2022) study social sampling theory on a torus, which is described further in the Open Questions section.

[6] When studying things that take time, we have to consider how long things take. If we give the processes too long, they may all look the same or settle down to different stable states (sometimes called steady states). For example, if we're asking how drinking coffee influences the likelihood of death, it is probably obvious that what we're really asking is how drinking coffee increases or decreases that likelihood over a given interval of time. In the limit of infinite time, we all die with probability one. However, if coffee makes some people immortal, we won't notice if we don't give the mortals enough time to die. More generally, whenever a parameters can take different values (e.g., of time, space, or some other value), it's insightful to consider what happens when those values are as large or as small as we can make them. This is called "thinking in the limit," with the limit being the boundaries of the possible (very large or very small amounts). At the same time, it's also important to consider what happens in between, which is what we do here.

We can ask numerous questions about the outcome of this exploration:

- Does increasing search depth increase the likelihood of insular communities (e.g., will the number of components grow with search depth?)
- Will increased search depth enhance authenticity? Will greater search depth lead individuals to have higher utility (i.e., happiness) as a result of expressed behaviors more true to their authentic attitudes? If it does, then will we also see an increase in polarization visible as an increase in the variance of expressed behavior across the population?
- Finally, we can ask if search leads individuals to find themselves in communities that place their authentic attitude at the center of their communities. Following rank-based approaches to attitude formation, we might infer that people who rank nearer the center of their social distribution see their own beliefs as more representative of the broader community, facilitating a false consensus.

Figure 16.7 shows representative outcomes after 100 and 400 simulations for search at depths of $S = 2, 4$, and 6. The basic trends made visible in the figure are further supported by Table 16.1, which shows the outcome of averaging the results for 50 different networks running 200 iterations each for social sampling sheory applied to different movement strategies: 1) no movement, 2) Schelling migration with pairs chosen at random from across the network, and 3) Search at depth 2, 4, and 6.[7]

The results show us that utility increases when moving from no migration to Schelling migration. It rises further with increased search depth. Variance in behavior also rises, revealing what might appear to some as rising extremism. This makes sense if we imagine that search increases the ability of individuals to find other individuals who better support their authentic attitudes.

The network also becomes increasingly connected with more search, indicated by the reduced number of components at larger search depth. This is likely to be counterintuitive to some. It was to me. One might imagine that increased search would create more insular communities, as individuals can find others who are more like themselves, to the exclusion of everyone else. However, greater search depth enhances connectivity relative to more shallow search. The explanation one must infer from Table 16.1 is that deeper search allows individuals to more rapidly find their place among and thereby connect to and maintain the diversity of beliefs. This needs to happen before beliefs become too well adapted to their local communities through consensus effects, which is what happens when search is more shallow.

In sum, as we increase agents abilities to define their own social community, this enhances utility and behavioral variance. As agents become better able to investigate potential future neighbors, this increases utility further still, but also increases connectivity across the entire network. Moreover, adding search to the Schelling model greatly enhances behavioral variance and thus polarization.

[7] A more thorough investigation of these trends should look at hundreds of simulations instead of the 50 we look at here. However, the standard deviations give us some confidence that we're on the right track.

Table 16.1 Impact of search on social dynamics. The table shows the impact of search depth on various aspects of belief (utility, variance, and normality) and network structure (components). Results show the average of 50 network simulations after 200 iterations each. Schelling migration picks two individuals at random as in Figure 16.1.

Depth	Utility	SD	Variance	SD.1	Components	SD.2	Normality	SD.3
No movement	0.778	0.000	0.034	0.000	1.00	0.000	0.448	0.000
Schelling migration	0.783	0.014	0.065	0.022	1.00	0.000	0.607	0.095
2	0.829	0.013	0.153	0.021	4.90	1.403	0.619	0.030
4	0.843	0.010	0.158	0.016	1.68	0.713	0.650	0.033
6	0.846	0.010	0.156	0.018	1.36	0.525	0.650	0.032

16.6.1 False Consensus: Moving to the Middle of Perceived Normality

What can be said about false consensus? Individuals in worlds with greater search depth have higher utility, which reflects their observation that their neighbors more closely share their authentic attitude. But are they more likely to view their beliefs as more representative? One way to quantify this is to ask where individuals rank in their own communities. If they are near the center of the ranking, then we may expect that they see their behavior as being more reflective of the average person's beliefs.

To compute this we can ask for each node in the network where it lies relative to its neighbors in the ranking distribution of expressed behaviors. We can create a normality statistic to achieve this. Ideally, this statistic should do the following: if an individual holds the lowest or the highest expressed behavior among its neighbors, it should have a value of 0. If it holds a value that is in the middle of its neighbors, it should have a value of 1. A value of 1 reflects perfect perceived normality. After some trial and error, we can arrive at a function for this perceived normality as follows:

$$\mathcal{N}_i = 1 - \left| 1 - \frac{2(r_i - 1)}{k_i} \right|$$

where r_i is the rank of the agent among its k_i local neighbors including itself. If this ranking is 1 (the highest value) or $k + 1$ (the lowest value), then $\mathcal{N} = 0$. If the ranking is in the middle, $\frac{k_i+1}{2}$, then $\mathcal{N} = 1$.

Table 16.1 shows that the average normality increases when individuals can migrate. This is computed first for the average node within a network and then averaged across networks of the same type. Normality increases further when agents can search. If we take perceived normality as reflecting a false consensus, then movement and search increase the false consensus effect. People able to search their near environment for like-minded compatriots are more likely find themselves at the center of their local distribution and may thus feel that their authentic attitude is more representative of their community at large.

16.7 Summary

By looking at how social influence and social structure interact, we can see three ways in which individuals might be mislead by their social environment into a false consensus that is directly related to increasing polarization.

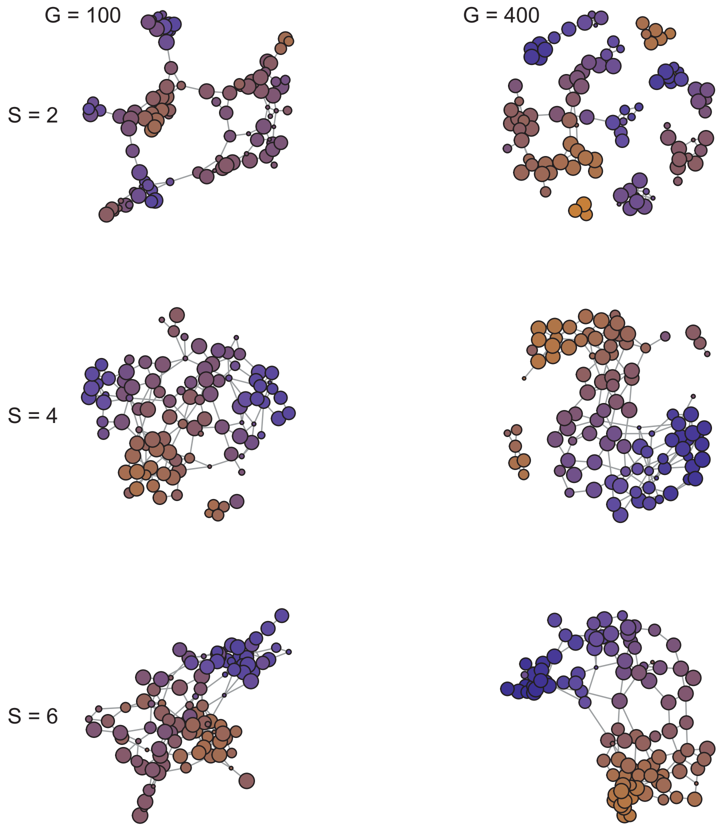

Figure 16.7 Search and social sampling theory. Starting with a 10 × 10 lattice, agents adjust expressed behavior to their local neighbors and authentic attitude. Then agents search through local neighbors to identify the individual with the expressed behavior that is the smallest distance from their authentic attitude. If this behavior is closer to their authentic attitude than the most distant behavior among their existing neighbors, they rewire the edge to ther existing neighbor, swapping it to the neighbor who is more similar. Agents search to a path length, *S*, of 2, 4, and 6 edges. Colors indicate the expressed behavior and size indicates the relative utility.

The first is via movement. When individuals can move to social environments that better reflect their own beliefs, they may start believing that most people share those beliefs.

The second is via search. The false consensus that corresponds to an increased perception of normality is amplified by the ability to search for ideal locations in the social environment. As the ability to more accurately search the local network increases, the average utility of expressed behaviors increases. Thus, people are more satisfied with their expressed behaviors and perceive them to be more central to their local distribution, even when the behavior is more likely to be extreme relative to the global distribution.

The third misleading element is the mirage of social consensus. When individuals adapt to their social contacts, this shrinks the authentic diversity of opinions. The consensus is not an expression of authentic attitudes. It reflects individuals only after processing them into the whole. And in some cases, such as pluralistic ignorance, a group's attitudes may reflect none of its members.

Search may not be strictly negative in terms of separating individuals. When individuals can rewire their local network, increased search depth does not increase their disconnectedness. Limited search depth, on the other hand, does. Greater depth maintains higher levels of connectivity, at least in the framework set out here. As we learned in Chapter 15 on group problem solving, the maintenance of diversity is critical for identifying solutions to hard problems. Segregation helps maintain that diversity. But connectivity between groups is critical for the sharing of good solutions. Search helps maintain that connectivity as well. Empirical evidence also shows that online social media networks are often more connected across the political divide than we might expect (Bakshy et al., 2015), and this might be a result of people's ability to search across the divide for more nuanced appeal.

One cannot help but wax philosophical. Identity loses its contrast in the soup of collective opinion. As the philosopher David Parfit might put it, our identity is nowhere if it is not tied up in the ever-changing context of our existence (Parfit, 1971). Or, to rephrase Douglas Hofstadter (2007), we are mirrors mirroring mirrors. A true authentic preference might be a kind of unicorn, especially if it is ultimately wrapped up in the identities we perceive in others. And yet, under the right conditions, it may come to life.

16.8 Open Questions

16.8.1 Updating Authentic Beliefs

How might allowing authentic beliefs to update influence the nature of online communities? That is, how can we model change in authentic attitudes, and what influence does this have on the social dynamics? Because authentic attitudes are beta distributions, there are various ways these might be updated either through the individual's choice of behavior or through encounters with the expressed behavior of others. These can be taken as additional evidence for the authentic attitude, effectively making individuals' attitudes less flexible over time by altering the α and β parameters.

16.8.2 Multi-Dimensional Preferences

SST can also reflect multi-dimensional preferences. Multiple beta distributions could characterize the complex interests of an individual: their music preferences, book preferences, food preferences, and so on. How might this influence social dynamics? Kullback–Leibler distance or a similar method could be used to measure the proximity of individuals' preferences and social polarization more broadly (Adams et al., 2022).

16.8.3 Behavioral Immune Systems and Polarization

Social dynamics as presented here are created by individuals seeking like-minded others. There is also value associated with making contact with individuals outside of one's community, as I will discuss in Chapter 18. However, humans are also exposed to threats, such as disease, that might make them more likely to avoid outsiders. The behavioral immune system is proposed as an adaptive mechanism for limiting exposure to such threats, and most specifically to disease. Brown et al. (2016) explore a model of how disease threat influences openness to others. This model has two opposing forces: one

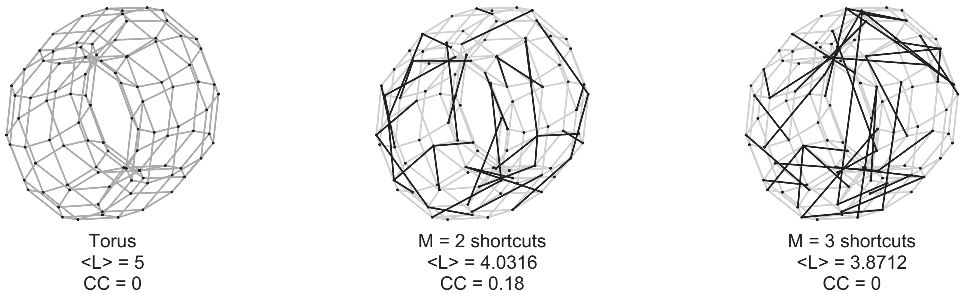

Figure 16.8 A 10×10 torus: A two dimensional lattice with periodic boundary conditions. This is the starting point for the Newman–Watts small-world model. Shortcuts are then added with some probability between all pairs of nodes. The torus on the left is the original lattice. On the right are two networks with different length shortcuts (M = 2 or 3) added with probability 0.2, shown in black.

associated with the benefits of encountering individuals outside one's local network, and another associated with the increased risk of infection. A simple prediction borne out by the evidence is that greater intensity of disease leads to more insular communities. This has numerous potential implications, including impacts on religion and ideology (Thornhill & Fincher, 2014). One additional threat associated with social sampling theory is that parasites reduce polarization and feelings of normality by limiting individuals to their local communities. How would this work and what evidence is there to support it, if any?

16.8.4 Ring World: A World on a Torus

Many models that use lattices prefer to use lattices that have no boundaries, or what are called *periodic boundary conditions*. One structure that achieves this is a *torus*. If we imagine a two-dimensional lattice as a sheet of paper, then we can fold the lattice over on itself to form a tube and then connect the two ends of the tube, as shown in Figure 16.8. We can then use a variation on Newman and Watts (1999)'s model developed for studying the transition between a lattice and small-world. It is itself a variation of the small-world model developed by Watts and Strogatz (1998). In the Newman–Watts model, we start with a lattice of N vertices with periodic boundary conditions: a one-dimensional lattice forms a ring, and a two-dimensional lattice forms a torus. *Shortcuts* are then added between random pairs of nodes, chosen uniformly over the graph, with probability ϕ for every existing edge. Unlike the Watts–Strogatz model, no edges are removed, which prevents the formation of isolates. However, in the original Watts–Strogatz model, the average degree remains the same after *rewiring*, because an edge is removed for every edge that is added. In the Newman-Watts model, the new degree is $k(1 + \phi)$, as $k\phi N$ new edges are added, in direct proportion to the number of existing edges, but no edges are removed. We can allow an additional variant proposed by Liu et al. (2009), which involves adding random edges only at specific path lengths. To do this, paths of length M are identified on the original lattice and then a random sample of these equivalent in number to $k\phi N$ are replaced with shortcuts. Figure 16.8 visualizes three of these networks with $\phi = 0.2$ and $M = 2$ and 3. How does these different structure influence the dynamics of social influence?

17 The Conspiracy Frame

Coherence through Self-Supporting Beliefs

17.1 The Problem

Conspiracy theories explain anomalous events as the outcome of secret plots by small groups of people with malevolent aims. Is every conspiracy unique, or do they all share a common thread? That is, might individual conspiracy explanations stem from a common higher-order belief? That belief would bind all explanations together under a common umbrella. We can call this belief the conspiracy frame. Network science allows us to examine the existence of this frame at two different levels – by examining the structural coherence of individual conspiracies and by examining the higher-level interconnectivity of conspiracy beliefs as a whole.

17.2 The Conspiracy Frame

Conspiracy theories explain events by invoking secret and malevolent agents that control outcomes from behind the scenes. The kinds of events in question cover a wide range of topics, including the deaths of notable individuals, the outcomes of elections, global pandemics, and climate change. The individuals controlling these events range from ordinary to superhuman agents. In some cases they are prominent figures (such as Bill Gates or George Soros), in others they are collectives (such as the New World Order or Illuminati), in still others they are aliens (reptilians), or variations of all three.

A great deal of research has gone into understanding the environmental and dispositional factors underlying conspiracy belief (Douglas & Sutton, 2023; Ecker et al., 2022). For example, conspiracy theories are more prevalent in times of crisis (Van Prooijen & Douglas, 2017). People who believe in conspiracies also tend to be vulnerable in other ways. They harbor feelings of paranoia, anxiety, and powerlessness, combined with a sense of hostility (Abalakina-Paap et al., 1999; Darwin et al., 2011).

What is perhaps most surprising is that conspiracy theorists are not a category but a continuum. And it is a continuum on which we all reside. The majority of people endorse at least one conspiracy and some theories are supported by the majority of people (Oliver & Wood, 2014). For example, Goertzel (1994) found that 69% of randomly selected individuals felt that John F. Kennedy’s assassination was “likely” to be a conspiracy. Conspiracy thinking also knows no borders: people in Europe and the USA are surprisingly similar in conspiracy thinking (Walter & Drochon, 2022). Some theories are frequently held – like JFK's assassination – while others are held by very few – such as flat earth theory (Dieguez & Wagner-Egger, 2021). The more conspiracies one believes in, the more likely one is to believe in the existence of secret elites working toward odious ends (Brotherton et al., 2013; Bruder et al., 2013). Some conspiracy beliefs may therefore act as gateway beliefs, making people more susceptible to believing in other conspiracies (Granados Samayoa et al., 2022).

This all supports conspiracy belief as lying along a multi-dimensional continuum. However, many conspiracy beliefs also seem to resolve down to a single shared idea: the mainstream story is a deception. This is not that out of the ordinary. The incompleteness of mainstream accounts is something most of us can get behind. Even scientists often see themselves as the enemies of certain standard dogmas, with an explicit aim to reveal inconvenient truths. Many believers of conspiracy theories see themselves in a similar light, preferring to be called "truth seekers." Indeed, some argue that the term "conspiracy" is itself a conspiracy promoted by government elites aiming to shut down arguments opposing the mainstream narrative (Franks et al., 2017). What differentiates conspiracy believers from others is the claim that the missing explanatory ingredient is the existence of secret plots by individuals with nefarious aims.

Believers in conspiracies are therefore not marked by a strong belief in the details of any particular claim, but "a more general belief in the deceptive nature of official explanations" (Franks et al., 2017, p. 2). We can call this belief the *conspiracy frame*. The conspiracy frame trumps mainstream explanations because it can always claim that behind the mainstream explanation lurk evil actors who, by secretly manipulating events, benefit from their consequences. The conspiracy frame offers a boilerplate explanation that can be adapted to a wide variety of anomalies. It may even support contradictory explanations for the same event (Lukić et al., 2019; Wood et al., 2012). The conspiracy frame should also motivate the pursuit of anomalies, since the mainstream story is unreliable. And if anomalies fail to appear, believers can claim that the absence of evidence is evidence of a cover-up.

The conspiracy frame is impossible to disprove. The underlying premise is that the covert bad actors are always one step ahead of us. Whatever evidence one might find to disprove the anomaly can be explained away with "That's what they want you to think." Indeed, for any n-level causal explanation – X causes Y causes Z – a conspiracy theorist can always posit n+1 levels – W causes X – with the initiating event housing the bad actors. This way of thinking has much in common with religious worldviews that argue for supernatural or otherwise exceptional agents capable of bringing about any given set of events for a higher or lower purpose (Franks et al., 2013). If the devil (the ultimate conspirator) can plant evidence in the Earth's crust to confuse scientists about the Earth's age, then all evidence is inadmissible in the court of sufficiently enthusiastic opinion.

17.3 Why Does Evolution Allow Us to Believe Things That Are False?

The cost of conspiracy beliefs can be high. They impact vaccination uptake, condom use, carbon footprints, political violence, crime, and much more. In some cases, conspiracy theories are used as political tools to motivate doubt around key issues – such as smoking and climate change. These have substantial costs to the public (Leiser & Wagner-Egger, 2022; Oreskes & Conway, 2011). These costs are measurable in money and lives and are routinely cited in conspiracy research. Yet, if conspiracy beliefs are so costly, why is it that so many people believe in them?

One might argue that it should be adaptive, above all else, for humans to seek the truth. Therefore, evolution should select against costly errors in reasoning. Perhaps. But many beliefs are the outcomes of trade-offs over multiple issues. If a belief benefits the believer, it can be maintained even if it is false (Abbott & Sherratt, 2011; Foster & Kokko, 2009). In other words, people have no trouble believing in costly things if, in the cost–benefit analysis, the benefits outweigh the costs (Hills, 2022). And there are many potential benefits to false beliefs, ranging from avoiding harm to belonging to a community of like-minded fantasists.

The classic example of this trade-off is our hard-wired fear of snakes. Why do we fear snakes so much, given that many of them are harmless, and in some cases what we fear isn't even a snake? The answer is simple. If you mistake a harmless piece of rope for a poisonous snake, you may be laughed at by your friends, but you will also live to laugh about it with your children. If, however, you mistake a poisonous snake for a harmless piece of rope, you may never laugh again. Evolution tends to err on the side of caution by favoring beliefs that enhance survival, even if they are wrong. In the case of a poisonous snakes, a *Type I error* (a false positive) that leads to an overreaction to rope is a lower cost error than a *Type II error* (a false negative), mistaking a snake for harmless rope. The benefit of occasionally being right far outweighs the cost of often being wrong. The same logic is behind Pascal's wager: Belief in the existence of God is a gamble worth making if the cost-benefit analysis is measured in eternity.

The truth-value of a belief can also be largely irrelevant compared to other benefits, such as social-value. People show favoritism toward others they see as similar to themselves (Tajfel, 1970). Moreover, if sharing the customs and beliefs of people around you is a way to enlist their help in times of need, whether their customs and beliefs are "correct" in any objective sense may be meaningless. This logic extends to conspiracy beliefs as well. Wagner-Egger et al. (2022) provide ample evidence that one of the primary values of conspiracy belief is social. Following their disengagement from mainstream beliefs, many conspiracy theorists find a social identity among a community of like-minded believers. Many are directly helped on what some call their "spiritual quest" by more informed enthusiasts. It is not surprising then that people who believe in conspiracies do not think they are alone (Chayinska & Minescu, 2018).

17.4 Conspiracies, Worldviews, and Overfitting

Worldviews reflect a general set of beliefs that motivate how we understand reality (Koltko-Rivera, 2004). When trying to understand a new set of "facts" about the world, our worldview determines which of the facts are admissible and which can be dismissed. As a worldview, the conspiracy frame is aimed at identifying and exposing evidence for deceptive agents (Franks et al., 2017; Miani et al., 2022a). Moreover, because beliefs gain strength through their ability to explain a variety of things, the more anomalies that the conspiracy frame can identify and explain, the greater its *explanatory breadth*. All other things being equal, we should prefer worldviews that have greater explanatory breadth (Thagard, 1989).

One way to achieve explanatory breadth is through interconnectivity. One set of events helps explain another set of events (Wojtowicz & DeDeo, 2020). That is, interconnectivity is the glue that helps bind a worldview together. Using the conspiracy frame to explain anomalies associated with Princess Diana's death helps support that frame to explain anomalies associated with the deaths of Elvis and Michael Jackson. Explanations based on population control in one domain support explanations based on population control in other domains.

If any given explanation is improbable with respect to what it aims to explain, that is not a problem. The interconnectivity provides an explanatory coherence that overshadows the incoherence associated with specific events. Lewandowsky et al. (2018) put it the following way: "The incoherence does not matter to the person rejecting the official account because it is resolved at a higher level of abstraction" (p. 179).

What might be an alternative to the conspiracy frame as an interconnected worldview? One potential candidate is the *overfitting hypothesis* proposed by Hattersley et al. (2022). According to this hypothesis, believers in implausible conspiracies are cognitively predisposed to overfit data from small samples. Hattersley et al. (2022) demonstrated this using several judgment tasks that measure people's propensity to integrate evidence before making a decision. Two of the measures used were the *jumping-to-conclusions task* and *the cognitive reflection test*. The jumping-to-conclusions task measures how much information people collect before coming to a conclusion. The cognitive reflection test presents people with questions that have intuitive but incorrect answers. A classic example is the following: *A bat and a ball cost $1.10 in total. The bat costs $1.00 more than the ball. How much does the ball cost?* People's intuitive first answers are usually wrong (most people initially and incorrectly think the ball costs 10 cents). By comparing people's performance on these tasks with people's belief in various conspiracy theories, Hattersley et al. (2022) found that people who more strongly believed in highly implausible conspiracies (e.g., 9/11 was orchestrated by Western governments and the earth is flat) also sampled less data in the jumping-to-conclusions task and were more likely to provide incorrect answers in the cognitive reflection test.

These results suggest that belief in conspiracies may stem from a more general psychological predisposition. For example, some people have high *uncertainty intolerance* (Moulding et al., 2016) and *need for cognitive closure* (Marchlewska et al., 2018). People with high need for cognitive closure prefer order, structure, low ambiguity, decisiveness, and predictability, as well as preferring to remain ignorant about contrary views (Webster & Kruglanski, 1994). The general predisposition may also reflect a kind of impulsiveness, as more impulsive individuals show greater propensity for conspiracy belief (Bowes et al., 2021).

The conspiracy frame hypothesis and the overfitting hypothesis offer different predictions. Hattersley et al. (2022) argue that, because each conspiracy theory is an over-explanation of all the details of an event, implausible conspiracy beliefs should not generalize to new conspiracies. Each conspiracy could in principle be unique and unconnected to other conspiracies. Connectivity is unnecessary. This view best fits the data from the jumping-to-conclusions task and the cognitive reflection test, which are based on making judgments with limited evidence.

The conspiracy frame hypothesis argues that a conspiracy worldview provides a ready-made alternative to mainstream accounts which are emboldened by their interconnectivity. As such, the conspiracy frame is sufficiently flexible that it can fit any data without adjustment. Moreover, one does not need to collect more data if one already has an answer that can explain everything.

To put this in context, consider an often noted anomaly in the JFK assassination. The noisy acoustic evidence recorded at the time of the assassination suggests that there were four gunshots. But Lee Harvey Oswald, the suspected shooter, had apparently fired only three. A nonconspiracist might argue "Well, the acoustic evidence is noisy and there is no other evidence of a second gunman, so we can conclude that Oswald was the only shooter." But an overfitting explanation demands that we explain the anomalous data, which is perfectly achieved by invoking a second shooter (overfitting) and a cover-up (the conspiracy frame). This example also demonstrates that the conspiracy frame and overfitting hypothesis are not mutually exclusive. A one-size-fits-all conspiracy worldview can motivate the overfitting of errant data.

17.5 Costly Signalling and the Evolution of Self-Deception

Conspiracies may be frames for understanding reality, but they are also things people share with one another. They are therefore social signals. Social signals lend themselves to the development of communities. They achieve this because shared beliefs provide a mechanism by which people can identify one another (Wagner-Egger et al., 2022). When like-minded individuals can identify one another, the value of the coalition can easily overshadow the truth value of the associated epistemology. One could convincingly argue that a great many human beliefs are adaptively meaningless beyond social signalling (e.g., which movie director/painter/hair-style/politician/musician/religion is the best?).

Because of their content, conspiracy beliefs fall into the class of *costly signals*. Public acknowledgment of belief in conspiracies can carry negative consequences, more so than mainstream beliefs. Other kinds of ritualized behavior and public displays of commitment are similar in this regard (Wagner-Egger et al., 2022). The gang member with the full body tattoos, the hazed frat boy with the underwear on his head, and the businessman with his suit and tie all pay a cost to be accepted into their ingroups. But that cost is compensated by the value of the ingroup membership.

The concept of a costly signal follows from Zahavi's *handicap principle* (Zahavi, 1975). The handicap principle points out that for signals to carry information about the quality of their sender, they must also be costly. The classic example of the handicap principle is the peacock's tail. If all peacocks could grow iridescent tails of equal size and beauty, the tail would be meaningless. To be of value to the peahen, the tails of peacocks must discriminate the good from the ugly (and therefore, bad). If all tails are equally easy to produce, there is no selective advantage to having a better tail – and therefore it would not have evolved. In order for the peacock's tail to evolve, it must be costly to be a valid signal of peacock quality.

For beliefs to follow the handicap principle, they must be costly to maintain. The costs of beliefs can be amplified by invoking components that will be unpopular with outgroups. This is what Williams (2022) calls *strategic absurdities*. Strategic absurdities, by appearing odd to outgroup members, also increase defection costs. They are a form of commitment device. Like the tattoos of criminal gangs, commitment to absurd positions binds individuals who share those commitments more tightly. This creates an ideological Alamo for members of the ingroup, who must fight for their position because they have left themselves nowhere to run.

This has a further consequence. If you are going to fight side-by-side for your beliefs with members of your ingroup, you had better be sure they are as committed as you are. Funkhouser (2017) describes how this social dependence can eventually drive individuals to believe the absurdities they espouse. The logic is as follows: If important beliefs can be faked, then humans should evolve mechanisms for detecting fakes. But if you can tell who is faking, then signals must be truly believed by the sender in order to overcome detection.

This makes strategic absurdities a powerful tool for detecting outgroup members because they must be believed. It is no surprise then that costly signals are often called *honest signals*. Here the honest and costly signal is belief in the absurd. Such self-deception is not rare. It is a member of a broad class of adaptive self-deceptions that seep into our everyday routines. Things like overconfidence, self-serving beliefs about fairness, and confirmation biases are but a few examples (Von Hippel & Trivers, 2011). In sum, the evolution of costly signals of belief requires that devotees genuinely drink the Kool-Aid.

17.6 Beliefs and Structure

How is costly signalling related to structure? There are at least two hypothetical pathways. First, structured beliefs should be more difficult to fake. The more difficult they are to fake, the more they can enlist trust. Sterelny (2012) describes how committed individuals can easily produce complex signals that are extremely costly for others to fake. Interconnectedness among beliefs adds to the complexity required for mastery of the lore. Recall that the number of possible relationships (edges) between nodes in a network grows roughly as the square of the number of topics.

Returning to our JFK example, if all it took to be a genuine conspiracy theorist was a belief that JFK's assassination was a conspiracy, then most people would be conspiracy theorists. Belief that JFK's assassination involved many people is an entry-level conspiracy belief that the majority of people share (Dieguez & Wagner-Egger, 2021). Indeed, a JFK conspiracy was proposed by the United States House of Representatives Select Committee on Assassinations (HSCA), which investigated Kennedy's death. To present a more convincing claim to being a card-carrying conspiracist, one might want to be able to add that the HSCA also investigated Martin Luther King's (MLK) death and came to a similar conclusion. Add this to your knowledge of FBI and CIA surveillance of both victims and you become more convincing. Like the Trekkie's ability to recall all episodes containing Captain Pike (Kirk's predecessor), the conspiracy theorist's mastery of the complex interconnections between conspiracies provides an additional cost that is hard to fake.

Second, interdependence also provides more persuasive arguments. Mercier and Sperber (2011) have argued that reasoning is primarily argumentative, not epistemic. In the domain of conspiracy belief what this means is that language around conspiracy should primarily be focused on convincing the listener of the coherence of the position, not the truth of the position. Therefore, the nature of human argumentation when supporting a given point of view should primarily be one of motivated reasoning. The point of argumentation is to demonstrate that the communicator has good reason to believe in their own position. The more evidence the better.

17.7 Conspiracy Networks

If conspiracy beliefs are a collection of interconnected and self-supporting narratives, then different conspiracy-related topics should be mentioned together more often in conspiracy writing than in mainstream writing. We can test this using the Language of Conspiracy Corpus (LOCO) developed by Miani et al. (2022b). This corpus was created by collecting a set of common conspiracies (see Table 17.1), which were then used as seeds to search for websites that contained those seeds. Websites were constrained to be among those identified by an independent source as either associated with conspiracy or mainstream reporting. This generated a *matched corpus*, with conspiracy webpages and mainstream webpages matched for topics. In natural language processing, researchers call each individual text sample a document, which is what I will do here. The corpus has approximately 88-million words and 100,000 documents (i.e., webpages), with 23,937 of those documents collected from conspiracy websites.

Using this corpus, Miani et al. (2022a) compared the language features of mainstream and conspiracy webpages. They found that conspiracy documents were less coherent by a variety of measures, including semantic similarity within a document. They also explored the interconnectedness of ideas between conspiracy topics using the method we will follow here.

Conspiracy and mainstream networks were created from the matched seed topics. Edges represented how often seeds co-occurred with one another in webpages. The LOCO corpus is first subdivided into a mainstream subcorpus and a conspiracy subcorpus. Here we will use the part of the mainstream corpus that does not use the term "conspiracy" to avoid mainstream articles that might report on conspiracy beliefs (Miani et al., 2022b).

The next step involves creating a document-by-term matrix where each cell in the matrix indicates that a particular seed returned that document during the corpus generation process. Thus, if a particular webpage is returned from a particular website when both "michael jackson" is entered as a search term, and the same webpage is returned when "5G" is entered as a search term, then the document will be tagged with

Table 17.1 List of conspiracy topics.

seed
michael jackson
5g
barack obama
saddam hussein
cancer
global warming
coronavirus
moon landing
reptilian
osama bin laden
september 11
cia cocaine
gmo
fluoride water
drug companies
mind control
vaccine
aids
population control
zika virus
new world order
planned parenthood
illuminati
sandy hook
chemtrails
george soros
alien
princess diana
big foot
ebola
flat earth
mh370
bill gates
george bush
jfk assassination
paul mccartney
elvis presley
pizzagate
jonestown suicide

both “michael jackson” and “5G”. We use this information to create a conspiracy network with weighted edges indicating how often seeds co-occur with one another across the various webpages.[1]

Figure 17.1 visualizes the networks associated with the matched conspiracy and mainstream corpora. The conspiracy corpus is visually more dense than the mainstream corpus. The average degree of nodes in the conspiracy network ($M = 429.0$, $SD = 357.1$) is also markedly higher than in the mainstream network ($M = 99.7$, $SD = 138.3$). This fits the argument that specific conspiracy worldviews are supported through their interconnections with other conspiracies.

Figure 17.2 shows the degree of seeds in the conspiracy and mainstream corpora. Conspiracy seeds are always more connected than mainstream seeds. The most well-connected seeds are associated with “vaccines,” “coronavirus,” and “new_world_order.” As the corpus was collected during the Covid-19 pandemic, this suggests that the conspiracy and mainstream websites were likely to represent more current events. An individual arriving at a webpage containing information about one of these topics is likely to find mention of a variety of other conspiracy related topics as well.[2]

The basic pattern supports an interconnected conspiracy worldview. This is all the more surprising given that we are only looking at the count of co-occurrences and there are many more mainstream documents than conspiracy documents.[3]

One may ask if these patterns of co-occurrences for conspiracy seeds reveal valid information about those seeds, or if they merely co-occur at random. There are at least two ways we can look at this. One, pointwise mutual information, addresses the information value of individual edges. Another, degree incoherence entropy, asks how predictable are the set of co-occurrences for individual seeds.

17.8 Separating the Signal from the Noise: Pointwise Mutual Information

Pointwise mutual information (PMI) is a way to determine if items co-occur with one another more often than one would expect by chance.[4] It is computed by comparing what is observed in relation to what is expected: taking the ratio of the number of observed co-occurrences over the number of expected co-occurrences if the two items occurred

[1] Miani et al. (2022a) also did the same analysis with topics identified by latent Dirichlet allocation (or topics analysis) and found the same pattern recreated here.

[2] If one were interested in statistic confirmation, one could use a *sign test* on the degree of the matched pairs. That is, all other things being equal, what is the likelihood that one corpus would always have higher degree than the other corpus? The probability of all 39 seeds being higher in the conspiracy subcorpus is 0.5^{39} if they each had equal probability (0.5). This is like flipping heads with a coin 39 times in a row. This statistical method is not entirely appropriate here, as the nodes are not independent. Higher degree on one node implies higher degree on other nodes. This is a common challenge when comparing networks. Nonetheless, a sign test is often a useful benchmark to consider in cases of binary outcomes.

[3] One can normalize the degree. The supplemental code for this chapter provides an example.

[4] PMI is one of many potential distance metrics. See Gomaa et al. (2013) and Pradhan et al. (2015) for some reviews of methods in the information retrieval literature.

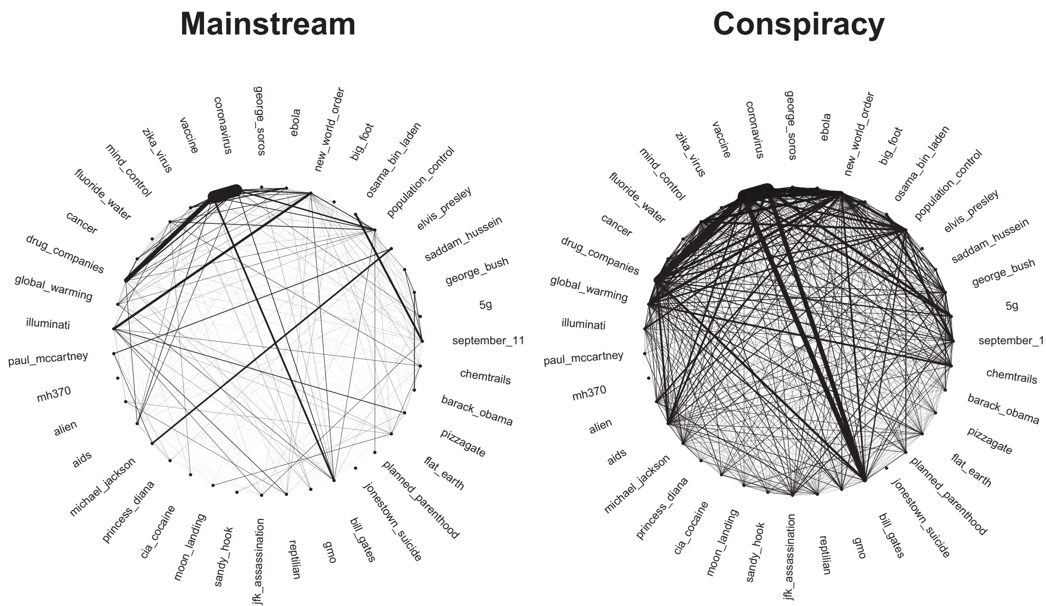

Figure 17.1 Networks from conspiracy and mainstream documents. Nodes are the seeds used to find documents and edges connect seeds that identified the same document. Edge weights are indicated by the thickness of the edges. Conspiracy networks are visibly more dense than mainstream networks identified using the same seeds.

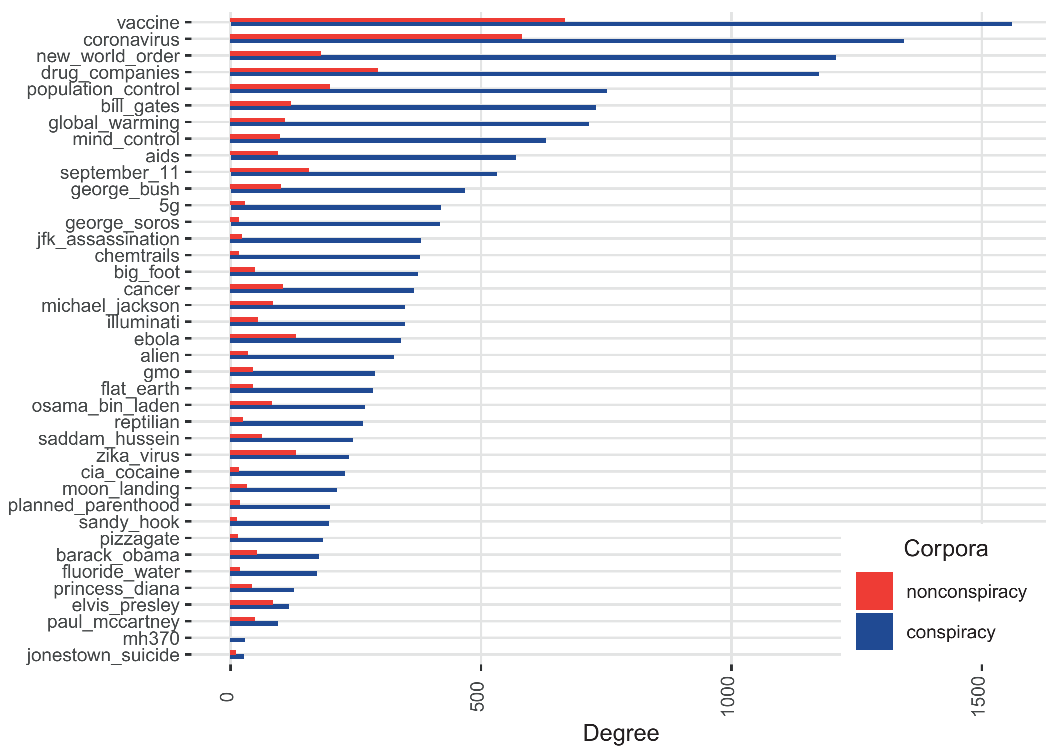

Figure 17.2 The degree of each seed in the conspiracy and mainstream networks.

independently of one another. The number of expected co-occurrences for two items is the product of their individual probabilities, just as the probability of flipping heads on a coin and rolling a 1 on a six-sided die at the same time would be $P = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$.

Formally, PMI is the log of the probability of x and y occurring together divided by the probability of x and y occurring independently:

$$PMI = log \frac{P(x, y)}{P(x)P(y)}.$$

Values lower than 0 indicate that x and y occur together less than one would expect by chance. Values greater than 0 indicate that the two co-occur more often than chance would predict.

PMI has been shown to be an excellent predictor of human similarity judgments. With sufficient data, it outperforms more sophisticated semantic distribution models (Recchia & Jones, 2009).[5]

To provide but one example, Turney (2001) used the hits from an online search engine to compute the PMI for various words. The PMI rating for various words was then used to compute answers for the Test of English as a Foreign Language (TOEFL). A typical question on the TOEFL looks like this:

- Which word below is closest in meaning to the word *levied*?
- A. believed
- B. imposed
- C. requested
- D. correlated

Let *levied* be x and the i^{th} answer is y_i. Since Turney (2001) was only interested in the relative values of the PMI, the probabilities can be replaced with page hits. Typing any word or pair of words into a search engine will produce a number of page hits. This gives:[6]

$$Score(y_i) = \frac{hits(x, y_i)}{hits(y_i)hits(x)}$$

Using this value, Turney (2001) found that the synonym with the highest PMI score was the correct answer about 75% of the time. This and other results suggest that PMI is remarkably versatile and predictive of human inferences (see Recchia & Jones, 2009).

We can use PMI to evaluate the edge weights in our conspiracy and mainstream networks to determine which of the many relationships co-occur more often than we would expect by chance. This will help remove the background frequencies from the relationships and amplify the signal of more meaningful relationships.

[5] One problem with PMI is that low-frequency events can produce extreme PMI values. Consider that an item that only occurs once either does or does not occur with another item: $P(x, y) = 0$ or 1. Various solutions have been developed to deal with these issues (Bouma, 2009). Here we limit our analysis to seeds that occur more than 100 times.

[6] $hits(x)$ can be removed since it is the same for all answers.

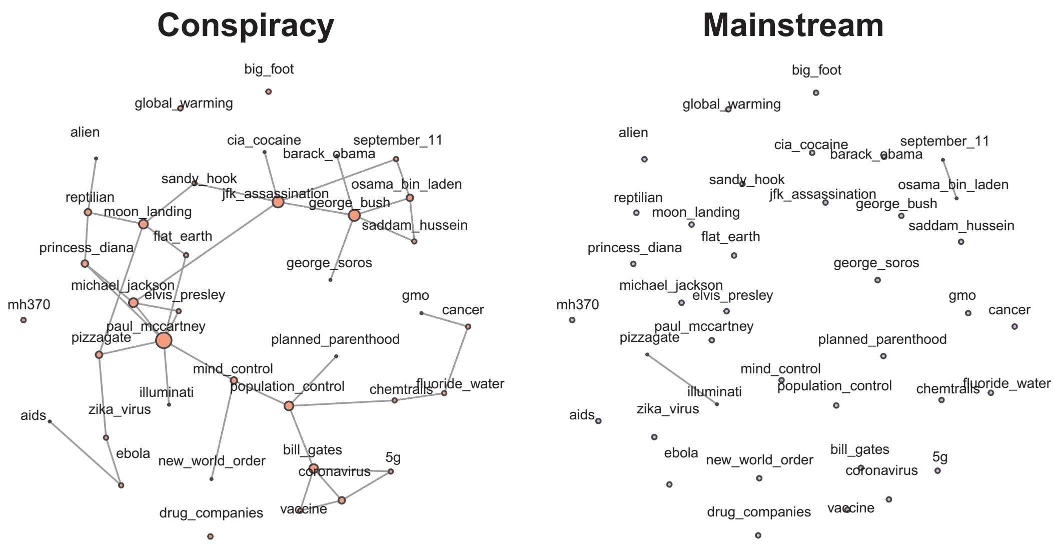

Figure 17.3 Network of positive PMI values for the conspiracy and mainstream subcorpora. The size of the node indicates the weighted degree of the node in the undirected network.

To compute the PMI, we need to compute the probability of each event by dividing by the number of possible webpages that a word could occur in. This is not the number of webpages Google currently indexes (25 billion), but something nearer the number of webpages within the conspiracy or mainstream websites that were searched. Since we do not have this number, we can approximate it as the number of total webpages retrieved for each subcorpus (conspiracy: $n = 23,937$, mainstream: $n = 72,806$).

After computing the PMI for all pairs of seeds, we threshold the lower end of the PMI values at 0 to remove negative values, which all co-occur less than chance. The remaining matrix is used to create the networks shown in Figure 17.3. These represent pairs of conspiracy beliefs in Figure 17.1 that would be unlikely if the beliefs were simply occurring independently.

What is immediately clear is that mainstream connectivity among these seeds is basically non-existent. The only two nonzero edges are "September 11"–"Osama bin Laden" and "illuminati"–"pizzagate." The first represents a relationship in the mainstream story. The second suggests that some conspiracy beliefs get reported in the mainstream media. This is to be expected: the pizzagate conspiracy eventually led an individual to attack the pizza restaurant with a rifle. Beyond these two edges however, nodes in the mainstream network are completely isolated.

The mean PMI for conspiracy theories is $M = 0.025$ compared with the mean for mainstream media $M = 0.0005$. This is roughly 46 times stronger for conspiracies than mainstream beliefs. A paired t-test drives the point home, $t(1443) = -7.16, p < .001$, with a Bayesian hypothesis test producing a Bayes factor of about 250 million in favor of conspiracy seeds being more coinformative than mainstream seeds.

The evidence from the PMI network further supports the conspiracy frame hypothesis, as conspiracies are more interconnected than mainstream content on the same topics. But the results also show that many of the co-occurrences in the conspiracy and mainstream networks occur randomly, no more frequently than we would expect by chance given their base rates for occurring independently. Such an investigation refines the conspiracy frame hypothesis as not all conspiracies provide equal evidence for other conspiracies. We can compare the nature of this support by looking more closely at the properties of individual nodes.

Using these PMI networks, we can compute various metrics to compare the different conspiracies in relation to where they are in the conspiracy network. Table 17.2 provides the following metrics:

- Normalized Betweenness: The proportion of shortest paths between all pairs of nodes that pass through a node.
- Strength: The sum of a node's weighted edges.
- Degree: The number of edges a node has with PMI values greater than 0.
- Incoherence entropy: The relative unpredictability of patterns of co-occurrence with other seeds across documents (discussed later in the chapter).

Paul McCartney is a standout among these metrics. Paul has the highest betweenness, strength, and degree. Not far behind is JFK's assassination. This lends some structural support for the empirical findings in Dieguez and Wagner-Egger (2021) indicating that JFK may be a gateway belief. In comparison, flat earth stands near the tail of the distribution. We can take these various metrics as hypotheses for further study.

17.9 A Network Measure of Incoherence in Conspiracy Beliefs

The overall connectedness of conspiracies either by straight co-occurrences or PMI shows that conspiracy beliefs are more interconnected than mainstream beliefs. Can we also use networks to tackle incoherence within individual conspiracies, mirroring the within document incoherence found by Miani et al. (2022a)?

Many different entropy measures have been developed for networks (Omar & Plapper, 2020; Safar et al., 2011). For example, a straightforward measure is to compute the entropy for a node by treating the normalized degree weights as probabilities (see Chapter 12 and Dubossarsky et al., 2017). Here, I develop a different measure, which I call *incoherence entropy*. This takes inspiration from information coherence measures that ask how information "hangs together" (Bovens & Hartmann, 2004; Harris & Hahn, 2009). In the present context, the question incoherence entropy answers is how predictably does a seed appear with sets of other seeds within documents.

Suppose that the evidence for one conspiracy is always a specific set of other conspiracies. For example, one might always mention MLK in relation to JFK, but not other conspiracies. This would have low incoherence entropy. However, suppose mentions of JFK sometimes involve MLK, sometimes Princess Diana, and sometimes the moon landing, but always in different combinations and rarely the same twice. This would suggest that the links between these topics are not supported by tight logical connections, in which one always leads to the other. Rather, they all have something vaguely in common, but nothing that directly binds any subset of them.

Incoherence entropy follows this logic by taking into account the observation that multiple seeds occur together at the same time on a single webpage. This allows us to

Table 17.2 Values of conspiracy seeds along various centrality metrics. Entropy is incoherence entropy.

seed	Betweenness	Strength	Degree	Entropy
paul mccartney	0.44	4.44	7	0.57
mind control	0.38	0.95	3	0.45
population control	0.36	0.89	4	0.50
jfk assassination	0.31	0.87	5	0.45
princess diana	0.25	1.59	3	0.34
michael jackson	0.22	3.20	4	0.43
george bush	0.18	1.50	5	0.43
pizzagate	0.18	0.67	3	0.41
bill gates	0.14	1.87	4	0.57
moon landing	0.14	1.86	4	0.46
chemtrails	0.14	0.38	2	0.37
sandy hook	0.11	0.63	2	0.34
fluoride water	0.09	0.41	2	0.34
zika virus	0.09	0.72	2	0.41
cancer	0.05	0.26	2	0.36
reptilian	0.05	1.81	3	0.41
ebola	0.05	0.52	2	0.40
5g	0.05	0.31	2	0.39
barack obama	0.00	0.40	1	0.37
saddam hussein	0.00	0.92	2	0.32
global warming	0.00	0.00	0	0.33
coronavirus	0.00	1.73	3	0.43
osama bin laden	0.00	1.94	3	0.38
september 11	0.00	1.05	2	0.40
cia cocaine	0.00	0.41	1	0.38
gmo	0.00	0.18	1	0.35
drug companies	0.00	0.00	0	0.36
vaccine	0.00	1.53	2	0.46
aids	0.00	0.04	1	0.41
new world order	0.00	0.04	1	0.47
planned parenthood	0.00	0.10	1	0.31
illuminati	0.00	0.47	1	0.37
george soros	0.00	0.15	1	0.42
alien	0.00	0.44	1	0.35
big foot	0.00	0.00	0	0.41
flat earth	0.00	0.94	2	0.36
mh370	0.00	0.00	0	0.23
elvis presley	0.00	2.49	2	0.41

compute how often seeds reliably co-occur together. For example, consider a pattern of co-occurrence with seed *O* as follows:

- O: AB
- O: AC

- O: B
- O: AB

Here, seed O appears with A and B in the first document. It appears with seeds A and C in the second document. And so on. The entire set of co-occurrences represents a probability distribution of $(AB, 0.5)$, $(AC, 0.25)$, $(B, 0.25)$.

The probability of each pattern of co-occurrence, $P(x_i)$, for each of the $N = 3$ patterns, then goes into the normalized entropy equation:

$$H = -\frac{1}{log(N)} \sum_{i=1}^{N} P(x_i) log(P(x_i)).$$

Using this equation, we can compute the incoherence entropy for the various seeds in conspiracy and mainstream documents.

Figure 17.4 shows how seeds in the conspiracy subcorpora have higher incoherence entropy than seeds in the mainstream subcorpora. This suggests that though conspiracy seeds often co-occur with one another, they are more unpredictable than mainstream corpora. In addition, the panel on the right shows that seeds with higher betweenness are less predictable than seeds with lower betweenness. Those nodes with higher betweenness may be more representative conspiracies, and as such they therefore wind up being associated with a wider variety of other conspiracies even though there is no direct connection.

Does higher incoherence entropy speak to the question of costly signalling? It might, for at least two reasons. Higher incoherence entropy suggests the information is more

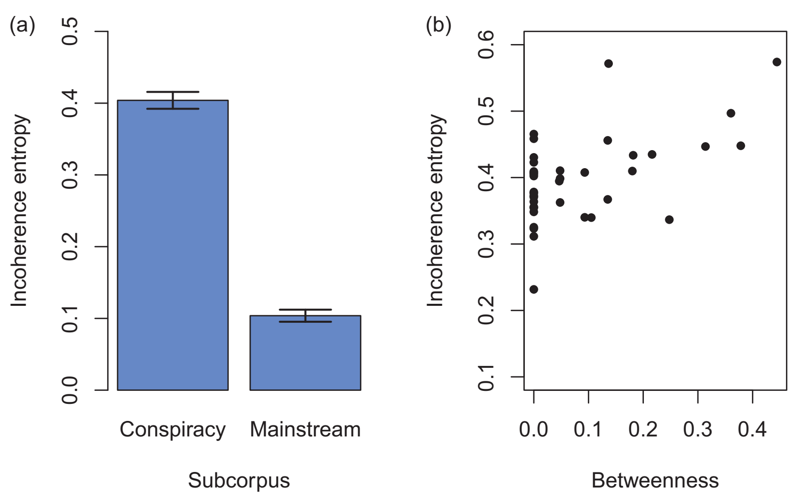

Figure 17.4 The incoherence entropy of the conspiracy and mainstream subcorpora. Incoherence entropy measures the extent to which documents containing specific seeds co-occur in unpredictable patterns with other seeds. Higher incoherence entropy indicates less predictable patterns of co-occurrence. (a) Across all seeds incoherence entropy is higher for conspiracy than mainstream subcorpora. (b) Across conspiracy seeds, seeds with higher betweenness also tend to have higher incoherence entropy. This may suggest that such seeds represent exemplars of conspiracies that can be called on as evidence to support arguments for other conspiracies.

complex and therefore less predictable. If we think predictable patterns are easier to fake and therefore less convincing to the ingroup, then high incoherence entropy is a good signal. Based on this idea, one could develop a measure of conspiracy expertise around people's ability to predict such a set of relationships. Second, unpredictability might be a costly signal and one valued by other conspiracy theorists. Indeed, the ability to make random but convincing associations between conspiracies may carry value that is respected by those high on conspiracist ideation. It is a hypothesis worth testing given that Oliver and Wood (2014) found that some individuals were willing to support novel conspiracies made up by the researchers, such as compact fluorescent light bulbs being promoted by the government because they make people more obedient.

17.10 Summary

Conspiracies represent a class of beliefs that involve explanations for anomalies based on the secret actions of self-interested elites. The network analysis shown here demonstrates how these beliefs are highly interconnected, supporting the notion that conspiracies are free-floating ideologies that find support through their interconnectedness. Basing edges on PMI further shows how focusing on more predictable patterns of co-occurrence reveals structural relationships that demonstrate cohesive groups (e.g., celebrity deaths and politics) and can facilitate theory building around specific nodes (e.g., JFK's assassination is a gateway conspiracy with high betweenness, connecting more disparate conspiracies). Finally, a new measure of degree entropy, incoherence entropy, was developed to investigate the predictability of associations between various conspiracies. Seeds in the conspiracy subcorpus had higher incoherence entropy than those in the mainstream subcorpus and, within the conspiracy subcorpus, seeds with higher betweenness had higher incoherence entropy. This analysis supports the notion that conspiracies tend to be connected with other conspiracies, but also suggests that conspiracy webpages tend to be highly unpredictable in terms of what conspiracies occur on those webpages.

17.11 Open Questions

17.11.1 Are All Worldviews Structured Like Conspiracies?

Miani et al. (2022a) focused their analysis on conspiracy seeds in conspiracy and mainstream corpora. Suppose they had focused instead on wine aficionados or movie buffs. Would they have found the same pattern of interconnectivity of ideas in these domains? On the face of it, the answer seems likely to be yes. A Pinot Grigio shares features with a Riesling, just as Akira Kurosawa shares features with Sergio Leone. Might the structure of online information on wine or movie buff websites differ from websites not devoted to those topics in a way similar to that found between conspiracy and mainstream websites? Presumably these various topics differ in structure and coherence. What motivates these differences?

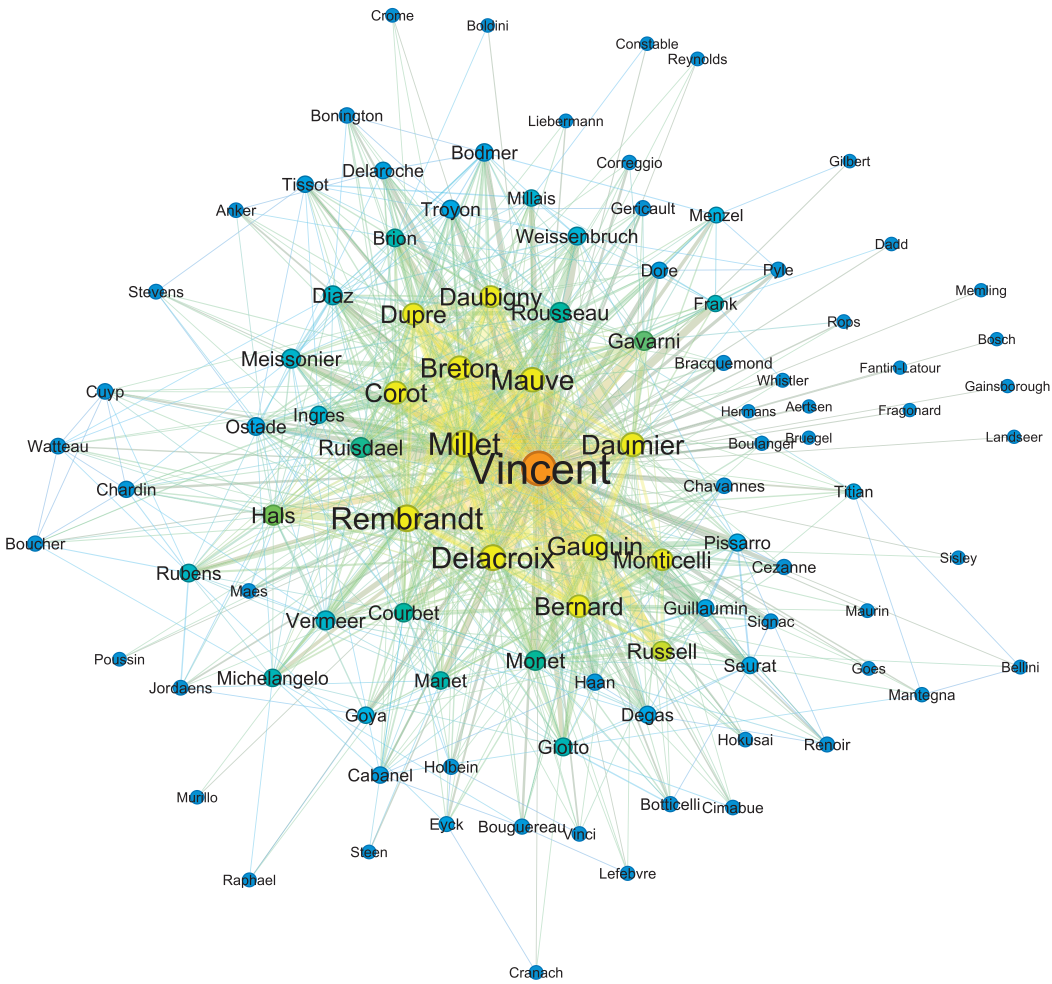

Figure 17.5 Colocations of artists' names within Van Gogh's letters. Weighted edges indicate how often artists were mentioned together in Van Gogh's letters. Node size is proportional to degree and nodes are colored with respect to their degree. Van Gogh, who signed most of his letters, is the highest degree node, followed by a close circle of his strongest influences, including Millet, Rembrandt, Breton, Daumier (shown in yellow). The force-directed network takes account of the weighted edges. One can see the relationships among Dutch artists (van Dyck, van Eyck, Peter Paul Rubens), and French artists (Watteau, Boucher, and Cuyp). Colors, places, and the valence of words associated with various artists, all present themselves as ways to investigate Van Gogh's understanding of the art world as he currently understood it. The image was created in Gephi, an application available online for visualizing networks.

17.11.2 Conspiracy Clusters

Figure 17.3 shows how a network built on PMI can reveal meaningful patterns between conspiracy co-occurrences. These patterns seem to reveal clusters of related conspiracies. Community detection algorithms could be applied to conspiracy networks to attempt to understand the various kinds (i.e., clusters) of conspiracies (e.g., Mahl et al., 2021; also see Williams et al., 2022 for a different approach). These could then be used to help hypothesis generation for future studies that ask about the validity and contagion of these various kinds.

17.11.3 Networks from Natural Language

The approach taken in this chapter is a simple method for building networks out of natural language. Using a dictionary of terms, one searches for terms in documents and builds edges where terms occur together.

Here I provide an example from the letters of Van Gogh. Using the Wikiart Vectors data set (Srinivasa Desikan et al., 2022), we can collect the names of artists and then search for co-occurrences of these names in the letters of Van Gogh. Van Gogh was a prolific writer, writing approximately a million words over 1000 letters to his brother Theo. Van Gogh also had experience as an art dealer and an inexhaustible interest in the work of fellow artists, often mentioning them in his letters. Van Gogh often painted the work of his heroes, and many of his most famous works are inspired by the work of others, such as Jean Millet, Anton Mauve, and Emile Bernard (Naifeh, 2021). By searching for co-occurrences of artists' names, we can visualize artist mentions and their likelihood of co-occurring with other artists in Van Gogh's letters. Figure 17.5 presents a network of these colocations.

More sophisticated methods for building networks from language are available as well, including the textual forma mentis approach (Stella, 2022; Stella, Citraro, et al., 2022). The textual forma mentis approach builds multiplex networks – having multiple edge types. In the case of forma mentis, the edge types usually involve syntactic and semantic information. The syntactic information encodes how words occur with one another in sentences based on their underlying grammar. The semantic information encodes shared meanings. Various data sources are combined to construct these networks, such as information about synonyms (i.e., WordNet) and valence information. The goal is to enrich the information available in the corpus to allow researchers to study smaller data sources, such as Tweets or short letters. Several recent studies have used methods like these to look at the structure of genuine suicide letters to evaluate their emotional complexity and to compare them with emotional distributions from other sources (Stella, Swanson, et al., 2022; Teixeira et al., 2021). Applying approaches such as these to matched conspiracy and mainstream documents could help answer questions about argument structure, emotional complexity, and other structural features of conspiracy ideation.

18 The Kennedy Paradox
Games of Conflict and Escalation

18.1 The Problem

What are the contexts that give rise to cooperation as opposed to conflict? When should we love thy neighbor, turn the other cheek, or escalate? Game theory is an effort to formalize this problem so we can ask what decisions we should make when the consequences depend on the decisions of others. In this formalism, games are systems where the outcomes for each individual depend on what other players do. By this accounting, war is a game, as is negotiation, rock-paper-scissors, and lovemaking. The difference is in the payoffs. That is, it is in the consequences, good and bad. Being rational is about making decisions that lead to the best outcomes. The hawks and doves of political foreign policy – who advocate for more or less aggressive military intervention – could be rational beings in this world because the right answer depends on how they think other people will respond. This mutual dependency underpins quotes like John F. Kennedy's "We can secure peace only by preparing for war." A sobering thought to be sure, but is Kennedy's statement rational by the logic of the game in which it is embedded? Is it rational when games of conflict are played repeatedly in a network of global interactions? When might the structure of those interactions catalyze cooperation or fuel pathways to escalating violence? By combining game theory with network science, we can make some progress toward understanding these issues.

18.2 From Cooperation to Conflict

Cooperation is often held up as a defining characteristic of life. This optimism is illustrated in a quote from Doebeli and Hauert (2005):

> The history of life on earth could not have unfolded without the repeated cooperative integration of lower level entities into higher level units … major evolutionary transitions, such as the evolution of chromosomes out of replicating DNA molecules, the transition from uni-cellular to multi-cellular organisms, or the origin of complex ecosystems, could not have occurred in the absence of cooperative interactions. Similarly, the emergence of complex animal and human societies requires cooperation (p. 748).

Despite this astonishing legacy, it remains worrisome that the capacity for violence in human populations would be compelling grounds for rewording this quote, not in favor of cooperation but of conflict. Perhaps mitochondria did not go willingly into the confines of the cell. And humans may cooperate, but how many human monuments were built not by volunteers but by people enslaved? No matter how often our better angels carry the day, it is hard not to find ourselves occasionally staring into the Hobbesian abyss – of lives "poor, nasty, brutish and short."

Nor is violence the only pathway to lives of desperation. Garret Hardin (1968) makes the observation that given free access to common resources, people often overuse them. This is the *tragedy of the commons*. If we can all graze our sheep in a common field, what is to keep me from adding one more of my own? I would benefit more if I did. But if we all make such a decision, the field becomes mostly sheep, and there is grass enough for none of them. Climate change is such a tragedy; we each benefit from our own carbon footprint and the benefit of minimizing it is hardly relevant if others are unwilling to make the same sacrifice. The tragedy of the commons is alarmingly common: traffic, doping in sports, over-fishing, deforestation, water scarcity, sharing misinformation, pollution . . . the list goes on. *Collective action problems* like these require our collective cooperation to solve, and yet the temptation to act selfishly is built into the systems that keep these problems alive.

Humans sometimes manage, though. And there are many fascinating accounts of how we have achieved it (Ostrom, 1999; Pinker, 2011). One of the ways we achieve this is through social structure. Social structure is important because it can keep a good thing going. When vampire bats share blood with their empty-stomached cave mates, they do so on the premise that the same individuals will repay the favor in the future (Wilkinson, 1984). This requires long-term relationships and the social structure to support them.

But it is important to remember that cooperation and conflict are two sides of a continuum. Whether we see one or the other more often depends on how we ask the question. In an early set of experiments using the dictator game, Kahneman et al. (1986) asked individuals to make a decision between splitting \$20 evenly between themselves and another partner, or splitting it \$18 and \$2, favoring themselves. The majority of individuals chose the equal split. This was long interpreted as suggesting that people are inherently fair. But in a clever adaptation, List (2007) asked people to play the same game with a minor variation: he allowed individuals to not only give money to their partner, but also to take money away. When the players could give as much as they could take, the majority became takers, not givers. So, it behooves us not only to examine the influence of structure on cooperation, but on conflict as well. In combining them, we can also ask when conflict might bring peace.

18.3 The Kennedy Paradox: Peace through Escalation

While running for the US presidency in 1960, Kennedy made the following controversial statement: "It is an unfortunate fact that we can secure peace only by preparing for war." Let's call this *the Kennedy paradox*. On hearing this, what should a potential enemy do?

Game theorists often advise us to resolve such problems by "looking forward and reasoning backwards" (Dixit & Nalebuff, 1993, p. 85). That is, to understand what to do, we must consider all available actions, work forward to their potential consequences, and then reason backward to the best course of action.

In Kennedy's paradox, countries have two choices: They can continue to arm themselves, or they can disarm. If they think Kennedy is lying and plans to disarm the USA, then they might choose to disarm themselves – a good outcome for both. However, if the USA does disarm, not disarming increases one's leverage in future conflict with an unarmed USA – a better outcome. If the USA does not disarm, then disarming would be catastrophic, leaving one at the US's mercy. Staying armed, on the other hand, would at worst maintain the status quo, making war costly to both sides, but potentially maintaining the peace.

The biggest potential losses go to the country that disarms. This is true even if Kennedy had made a claim like "disarmament is the path to peace." What Kennedy says is irrelevant – the payoffs in the game of war preparation are already set in place. Staying armed and, more importantly, making sure everyone knows you are, in the vision of nuclear strategists like Thomas Schelling and Daniel Ellsberg, is the only mechanism to maintain peace. Thus, Kennedy's statement is "an unfortunate fact" by the logic of the game.

18.4 The Prisoner's Dilemma

The Kennedy problem is what is called a prisoner's dilemma. In the prisoner's dilemma, each player must decide whether to *cooperate* or to *defect*. Like two prisoner's held separately for interrogation about a shared crime, to cooperate is to keep silent and to defect is to blame the crime on the other party. If both parties cooperate, this leads to the reward (R) outcome for both parties (shown in the payoff matrix in Table 18.1). But if you know the other party will cooperate, it is better to defect – leading to the temptation (T) payoff for the defector and the sucker (S) payoff for the cooperator. But reasoning further, if the

Table 18.1 The payoff matrix for the prisoner's dilemma. Each player can choose to cooperate or to defect. The values in each cell indicate the payoff for the strategy on the left played against the strategy on the top. *R* is the reward payoff. *T* is the temptation payoff. *P* is the punishment payoff. And *S* is the sucker's payoff. The exact numerical values assigned to these payoffs are less important than their position relative to one another. A prisoner's dilemma is any game with the structure $T > R > P > S$.

	Cooperate	Defect
Cooperate	$R = 3$	$S = 0$
Defect	$T = 5$	$P = 2$

other party thinks you will defect, their best option is to defect as well, leading to the punishment (P) payoff for both parties. Formally, a prisoner's dilemma occurs whenever the payoffs are ordered as follows: $T > R > P > S$. This applies to the Kennedy paradox as well.

Von Neumann and Morgenstern (1944) worked out a general mathematical theory for dealing with one-shot games of this kind. In a one-shot game, the game is over after one decision. The logic is as follows: determine which strategy is the best response for each of the other party's strategies. If the best response is always the same, this is the dominant strategy. In the prisoner's dilemma, if the other party cooperates, the highest payoff is to defect. If the other party defects, the highest payoff is still to defect. Therefore, defection *dominates* cooperation – it is the best response by the logic of the game.

When each player chooses their best strategy against the best strategy of the other party, this is the *Nash equilibrium* for the game. In the prisoner's dilemma, the Nash equilibrium is mutual defection. This is an example of how two parties, each acting in their own best interest, can create an outcome that is worse for both. In the Kennedy paradox, the Nash equilibrium is to stay armed. In the tragedy of the commons, the Nash equilibrium is to graze another sheep. More generally, these are all cases where maximizing self-interest is bad for the group. The economist Adam Smith's invisible hand is caught holding the knife in someone else's back.

18.4.1 Evolutionary Stable Strategies

In a world with no additional assumptions, defection is the *evolutionary stable strategy* (ESS) for the prisoner's dilemma: no other strategy can invade defection and defection can invade all other strategies. More technically, an ESS is a strategy such that if most of the individuals in a population have it, no other strategy can survive.

We can see this result using an agent-based simulation of the prisoner's dilemma on a fully connected network. Figure 18.1 shows the outcome of a series of simulations starting with various numbers of defectors and cooperators. On each round, every agent plays their strategy against each of their neighbors in turn, receiving the sum of their payoffs. The sum of each node's payoffs is indicated by their relative size in the figure. For the next round, agents *imitate-the-best*: they choose the strategy of the best-performing neighbor if it performs better than their current strategy. In each simulation, if there is even a single defector, the population immediately converges on defection.

Theorists often use the term *mean field approximation* to describe sufficiently large systems where it is assumed that agents interact at random. They play the field, experiencing the average behavior of the system. This makes for tidy mathematics, and game theorists have borrowed it to describe the behavior of groups of strategic agents in an unstructured population. Fully connected networks provide this approximation as well, because each agent interacts with all other agents. When this happens for the prisoner's dilemma, the ESS is defection.

18.4.2 So What's with All the Cooperation?

Despite what we might conclude from the above arguments, many people cooperate when given a one-shot prisoner's dilemma. They cooperate especially if they are not

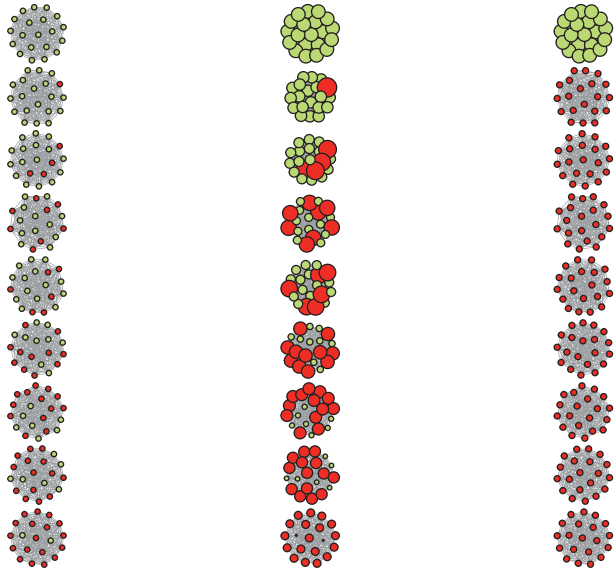

Figure 18.1 The prisoner's dilemma on a fully connected network. The probability of defectors in the first column increases by .1 in each row, moving from top to bottom. Defectors are indicated by the darker red dots. The first column is the initial population in a fully connected network before interaction. The second column shows the payoffs after each agent interacts with all their neighbors, with the size showing the relative payoffs. In the third column every agent that has a better-performing neighbor switches to that strategy and the agents play again. After the first column, the change in node size indicates the relative payoffs for that strategy. Thus, cooperators without defectors do best against one another. But defectors immediately invade and reduce the overall population performance.

economists. Frank et al. (1993) found that economics majors were about 60% likely to defect in a one-shot prisoner's dilemma, whereas noneconomists were about 60% likely to cooperate. My economist friends assure me this is not the result of some deep character flaw in economists (e.g., Yezer et al., 1996). The economist is taught that the payoffs in the game encompass everything – even the value of the future relationship. If the future relationship is not sufficiently valuable, it still pays to defect. But the high levels of cooperation among noneconomists, and even among the 40% of learned economists, deserves some explanation. Why do people cooperate if the rational outcome is to defect?

A variety of explanations have been offered to explain this. These include *kin selection* (cooperate with relatives; Hamilton, 1964), *direct reciprocity* (cooperate with those who will cooperate with you in the future; Trivers, 1971), *indirect reciprocity* (cooperate with groups of cooperators who will all cooperate with one another; Alexander, 1979), and *altruistic punishment* (in which cooperators pay an additional cost to punish defectors; Fehr & Fischbacher, 2003). Notice that these all require some mechanism by which the benefit of cooperating can be returned to oneself – either directly to one's kin (who carry the genetic predisposition to cooperate) or to others who will return the benefit in the future (reciprocity). Another way to achieve high levels of cooperation is through social structure.

18.5 Social Viscosity

Suppose we take the group of cooperators who fare so well at the top of Figure 18.1 and we take one of the groups of defectors below and we allow a single individual in one group to connect with a single individual in the other group. What happens now? Figure 18.2 shows that a single edge between the two networks does not lead to the invasion of defectors. It leads to the transformation of the connected defector into a cooperator. That sorry individual imitates his best neighbor (a cooperator among cooperators) and becomes a cooperator among (mostly) defectors. There is no kin selection or reciprocity – there is only imitate the best neighbor.

This approach to understanding the origins of cooperation is called *social viscosity* (Lion & Baalen, 2008). Social viscosity arises when communities of individuals practicing similar behaviors stick together. The importance of social viscosity was originally recognized by Hamilton in relation to *kin selection* (Hamilton, 1964). If cooperators give birth to cooperators, then cooperators will be surrounded by communities of cooperative partners. This follows *Hamilton's rule*: An altruistic behavior will be supported by natural selection if the cost, c, is less than the benefit to others, b, times the relatedness of the others, r: that is, $b \times r > c$. If related individuals are in close proximity, then cooperation with near neighbors can evolve when the benefit is sufficiently large relative to the cost. The idea of social viscosity was later expanded to include any form of strategic communities that create value through cooperative interactions, even when relatedness is zero.

From a network perspective, social viscosity is associated with behavioral assortativity. One way to see this assortativity in action is to simulate the prisoner's dilemma across networks with varying degrees of assortativity. We can do this as follows:

- Start with maximally assortative communities – agents only connect with individuals who share their strategy, either cooperation or defection.
- Transform one edge from within each community to two new edges between communities.

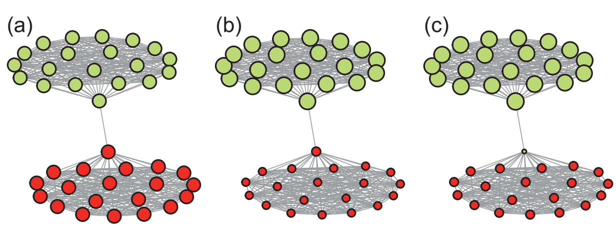

Figure 18.2 Prisoner's dilemma in metapopulations. A clique of cooperators (light) and defectors (dark) are connected by a single edge. The starting networks before competition are shown in panel (a). The network in panel (b) shows the payoffs (indicated by size) after the first round of play. This leads to the strategy change from defector to cooperator shown in the network in panel (c) – which is stable – and also shows the relative payoffs following an additional round of play for all nodes after the strategy change.

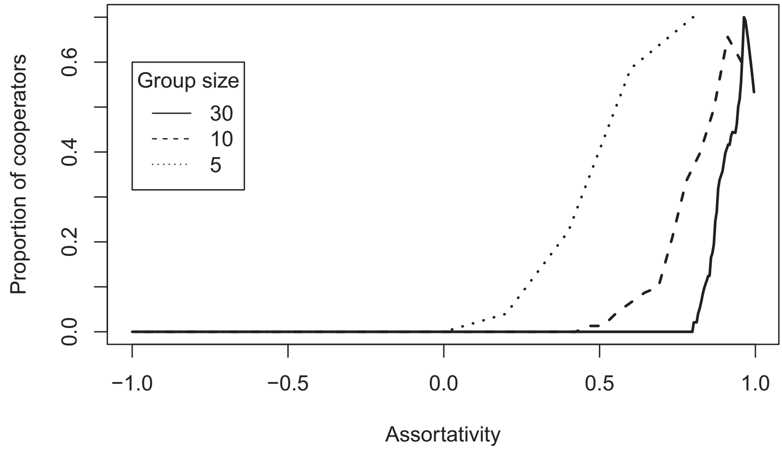

Figure 18.3 Social viscosity as strategic assortatitivity in the prisoner's dilemma and its influence on the stability of cooperators. The simulation is run 50 times for each network size at each assortativity value. For each simulation, the agents play the prisoner's dilemma with each of their neighbors. Then players choose the strategy associated with the best average performance. This is repeated until the strategy distribution is stable or for 30 rounds, whichever comes first. For each group size, the simulation starts with two separate but fully connected groups of cooperators and defectors. Then edges are swapped, two at a time, such that edges between similar strategies become edges between alternate strategies. Assortativity is recalculated after each swap. The proportion of cooperators are those remaining after the strategies stabilize. The lines indicate the average proportion across the 50 simulations for each group size.

- Allow the behaviors on the new network structure to interact and evolve through imitate-the-best over repeated rounds of interactions, until the strategies are stable. Compute the proportion of cooperators.
- Repeat steps b and c until the network is fully negatively assortative – all individuals only connect with individuals of the opposite strategy.

Figure 18.3 shows the results. When assortativity is high, cooperators proliferate. By supporting one another, cooperators do so well they support the spread of cooperation. However, as assortativity falls, so does the proportion of cooperators remaining after the network stabilizes.

By varying the size of the networks, we can also see that the size of the community matters (Boyd & Richerson, 1988b). Small communities are more resilient against defectors. This is handy as many researchers have suggested that *budding viscosity* – dispersal of communities, not individuals – can allow cooperators to invade defectors (Goodnight, 1992; West et al., 2007).

These results also tell us about where we should be more likely to find defectors. Larger and more fully connected networks should suffer greater defection. Extending the argument to cultural evolution, more mixing (such as that found in large cities, on highways, and online) should increase defection. Individuals who are not bound to local groups should also be more likely to defect. Compulsive defectors – such as people with antisocial personality disorder – have perfectly viable social strategies when they have large populations of fresh cooperators to prey upon (Dugatkin, 1992; Mealey, 1995). Even when cooperators can identify and avoid past defectors, free-floating defectors can outrun their reputation given a large enough population. In contrast, when there are clear ingroup and outgroup boundaries, cooperation should be more frequent within groups, and it often is (e.g., Balliet et al., 2014; Gross & De Dreu, 2019).

18.5.1 Learning to Cooperate in Different Worlds

We can further understand the role of structure in cooperation by examining networks designed to have different structural properties. For example, based on the intuitions developed earlier, we would expect networks with higher clustering coefficients to preserve cooperation more so than networks with lower clustering coefficients.

Mason and Watts (2012) developed an approach to generate networks that maximize or minimize various structural properties. I will explain this method in the next section and then we will use it to see how cooperation and defection fare in worlds with different structure.

18.6 Adaptive Network Search

An adaptive search process based on successive approximation is an algorithm that attempts to find an optimal solution to a hard problem by making repeated guesses. "Hard" means there are too many solutions to try them all. Adaptive search helps overcome this by making each new guess a variation of the best guess so far.

The adaptive network search process described here was developed by Mason and Watts (2012). It starts with random networks of N nodes in which all nodes have the same degree, k. This network becomes the benchmark network and the network metric of interest on this network becomes the current benchmark to beat. Then a random edge is chosen from the benchmark network and randomly rewired in a way that preserves the degree of all nodes, creating a new candidate network. If the network metric to optimize in the candidate network is better than the benchmark network, the candidate network becomes the new benchmark network. This is repeated many times to produce a final benchmark network. This entire process is then repeated many times again – starting from different initial random networks to avoid local minima created by different starting conditions – and the benchmark network that maximizes or minimizes the metric of interest across all final benchmarks becomes the network on which the study is run (see Barkoczi & Galesic, 2016).

We will produce networks optimized along the following criteria:

- Maximum mean clustering coefficient
- Minimum mean clustering coefficient[1]

[1] Minimum betweenness measures are nearly identical to the minimum mean clustering coefficient and they produce extremely similar results in the simulations. The R code provided can generate this for various metrics, sizes, and degree.

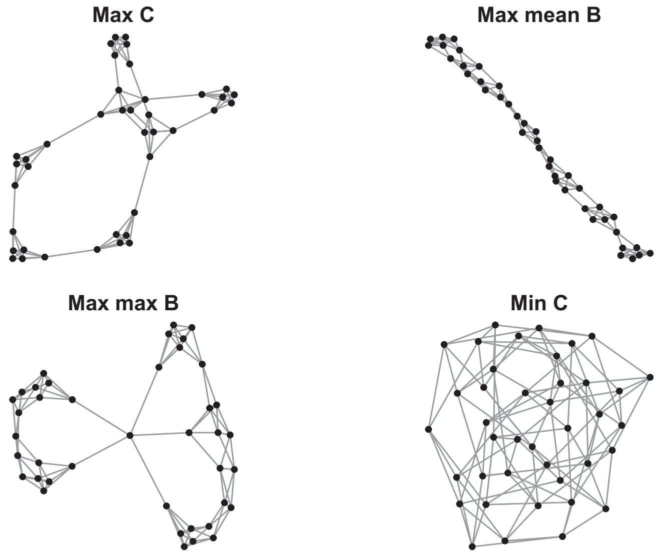

Figure 18.4 The networks produced using the adaptive network search optimization process from Mason and Watts, 2012. C means clustering coefficient and B means betweenness.

- Maximum mean betweenness
- Maximum maximum betweenness centrality[2]

We start the adaptive network search process with random networks each with $N = 50$ and $k = 5$. These are each rewired 5000 times and this is repeated for 20 different starting networks. Figure 18.4 shows the resulting best network for each of the metrics.

18.6.1 Simulating the Prisoner's Dilemma on Different Network Types

Figure 18.5 shows the results of simulating the prisoner's dilemma 1,000 times each on the different network types and with different starting probabilities of defectors. The networks that support cooperators manage to create communities that protect them. Networks that separate nodes into different clusters have the highest probability of sustaining cooperators. Maximum clustering coefficient and the two versions of maximum betweenness centrality maintain the highest levels of cooperation. Table 18.2 shows that those local communities are essentially absent in the network that is unable to maintain cooperation, even with the minimal number of defectors. This supports our intuitions based on social viscosity.

Also shown in Figure 18.5 is what happens if we redistribute strategies randomly over the network after strategy imitation – preserving strategy's relative proportions but removing their assortativity. The result (shown in the gray bar at the bottom of the figure) reveals that the persistence of cooperators in structured networks is not because they

[2] Maximize the largest betweenness value in the network.

Table 18.2 Network metrics resulting from adaptive network search.

Network type	C	b
Max CC	0.75	58.25
Max Mean Bet	0.62	91.12
Max Max Bet	0.50	48.30
Min CC	0.00	26.10

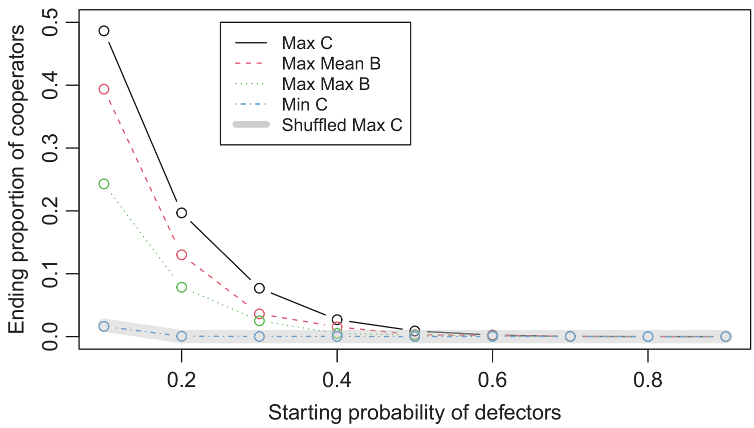

Figure 18.5 The proportion of cooperators sustained in each network type starting with various probabilities of defectors. Each network type and defector probability is run for 1,000 simulations. Strategies are randomly assigned to nodes and then nodes play against each of their neighbors. Each node then imitates the best strategy among its neighbors including itself. This is repeated until the strategies no longer change or for 30 iterations, whichever comes first. The thick gray line shows the effect of adding random dispersal (shuffling) after imitate-the-best on the number of cooperators. This same pattern is seen across network types. The relative disappearance of cooperation with dispersal tells us that structure is key to the persistence of cooperators.

are a Nash equilibrium, but because of structural barriers. In unshuffled networks, defectors must invade from poorly connected positions. When this constraint is removed, by allowing defectors to take up positions among clusters of cooperators, defectors rapidly invade. This is a consequence of *replicator dynamics*, which are the rules that determine how strategies change over time. The way we have allowed strategies to change so far is imitate-the-best neighbor. A cooperator surrounded by cooperators cannot suddenly become a defector. There are no neighboring defectors to emulate. The reason why cooperation is stable in communities of cooperators is not because cooperation is best. It is because strategies are learned locally and structure insulates cooperators from defectors. When the replicator dynamics change – for example, by letting strategies be learned globally – then other forces must come into play.

18.6.2 Selective Cooperation

Even when things are bad, cooperators can still manage to make a stand. They just need to take charge of their own network structure. If agents can form and dissolve edges based on the quality of their encounters, they can respond to defectors by ostracizing them (Bara et al., 2022; Rand et al., 2011). In communities where cooperators generate significant advantages, ostracized defectors run the risk of becoming evolutionary dead ends. One could fairly argue that it is the human capacity to alter its social structure that marks a particularly strong force supporting cooperation (Stevens & Hauser, 2004). Not only do we see the empirical consequences, but we have a long history of evidence for cultural practices of ostracism. Humans have developed ostracism into a high art, with various forms of blacklisting, boycotting, and formal religious sanctions such as excommunication among Catholics, shunning among Amish, and herem among Jews.

But social partner selection is not only edge-destroying, it can also be edge-forming. Social network theorists have long considered that edge formation can create value (see Figure 18.6). Burt's theory of *structural holes* argues that individuals who form edges between communities add value to that individual and the communities they bridge. They add value in the form of new information and other resources, including status (Burt, 2004; Redhead & Power, 2022). This was promoted by Burt most strongly in relation to business innovation, where the ability of individuals to bring insights from new communities creates a competitive advantage (Burt, 1995).

The value of these bridge builders is also seen in Granovetter's (1973) *strength of weak ties*. Here the observation is that it is our weak ties that are most likely to provide insights, job opportunities, and other novelties. Our weak ties are populated by individuals who share unique social information. Our strong ties, on the other hand, are likely to be connected to our other strong ties, making them informationally redundant. To

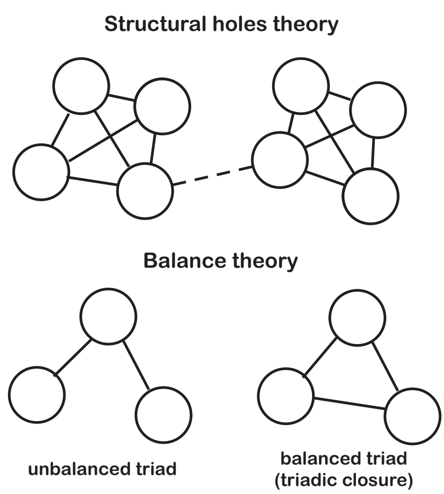

Figure 18.6 A visual depiction of structural holes and balance theory. Structural holes focuses on the value created by bridging communities. Balance theory focuses on the value created by aligning communities. This often involves signed edges, with a "−" edge indicating an enemy and "+" edge indicating a friend.

capture this quantitatively, Burt (1995) offered a measure called *effective network size*, which measures the extent to which an individual (ego) minimizes redundancy in their ego network. The measure ranges between 1 and n, where is n is the number of an ego's neighbors. If one's neighbors all know one another, the effective network size is 1 – you will learn as much from knowing one of them as knowing them all. If none of them are connected, the effective network size is n, as they all provide unique information. Specifically, effective network size is n minus the redundancy, or $n - \frac{2t}{n}$, where t is the number of edges between neighbors (Borgatti, 1997).

A counterforce to structural holes theory is the consolidation of ties within communities. These forces were originally outlined in Simmel's theory of *triadic closure* and Heider's *balance theory*. The basic intuition is that a friend of my friend is my friend and the enemy of my enemy is my friend. If Billy and Jim are friends, and Jim is friends with Peggy, then we can predict that Billy will become friends with Peggy. That is, we should expect triadic closure. Simmel (1950) gives an example of children acting as supporting mediators between their parents – helping to keep the triad together. Heider, on the other hand, initially developed the notion of balance theory to refer to cognitive concepts, proposing that there is cognitive pressure to align our attitudes (similar to *cognitive dissonance theory*). Cartwright and Harary (1956) applied this to interpersonal relationships.

Together, structural holes theory and balance theory provide opposing forces for edge formation and social network development. These will have different impacts on cooperation and defection. It seems straightforward to acknowledge that triadic closure is a mechanism for generating more cooperative communities. In contrast, attempting to close structural holes – by forming bridges between unconnected communities – represents a risk to cooperative communities. But then, cooperation with individuals outside one's community may generate the biggest payoffs.

18.7 Chaos and Cooperation

Cooperation and defection do not always stabilize. It is also possible to see waves of cooperation and defection in appropriately structured networks. One of the most colorful examples of this was discovered during the chaos era, when researchers were interested in understanding how deterministic (nonstochastic) systems – which followed simple and predictable rules – could generate complex and unpredictable behavior like that seen in natural systems. *Chaos theory* was formulated to describe how systems could appear chaotic and yet be predictable if one understood the underlying processes. A key feature of such systems is that they amplify small differences. *The butterfly effect* is a common example: The flapping of a butterfly's wings in Beijing will be amplified in deterministic ways to generate large effects later, like hurricanes in Havana.

Nowak and May (1992) showed that this *sensitivity to initial conditions* can be seen in the social evolution of the prisoner's dilemma. A specific regime of parameter values in the payoff structure of the prisoner's dilemma can generate spatial chaos when played out on a lattice network. Figure 18.7 shows the effect of introducing a single defector into

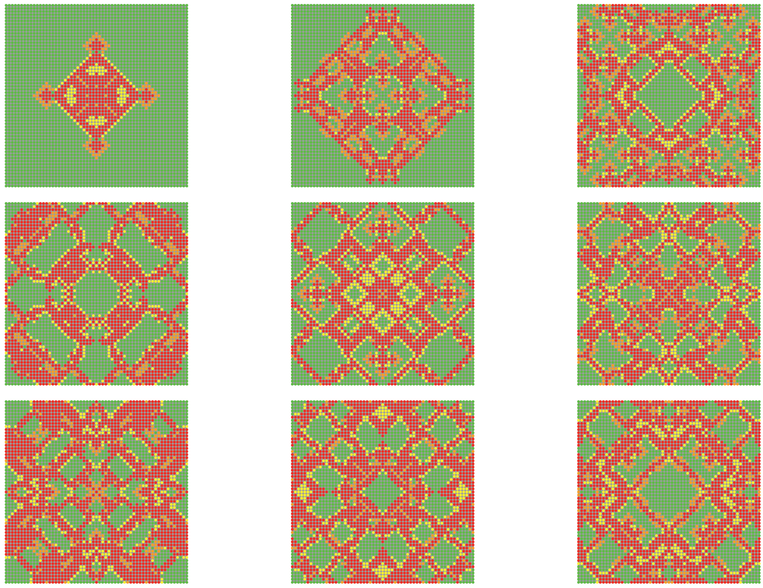

Figure 18.7 Chaos in the spatial prisoner's dilemma. All agents either cooperate or defect each round in a one-shot prisoner's dilemma. Each agent plays against all its neighbors and then imitates-the-best to determine its strategy for the next round. Payoffs are $P = S = 0$, $R = 1$, and $T = 1.56$. The networks are separated by 10 rounds between each panel. The progression starts with all cooperators but the center node, who is seeded with a single defector. Cooperators this turn and last are green. Defectors this turn and last are red. Cooperators who were defectors last turn are yellow. Defectors who were cooperators last turn are orange.

a sea of cooperators. After a round of play against its neighbors, each node then imitates-the-best to determine its strategy in the next round. Each successive panel shows the results after 10 additional rounds of play.

Like many chaotic systems, the parameter values required for chaos belong to a specific range and depend on other features of the network structure, such as how many neighbors any individual has. For example, with 8 neighbors, Nowak and May (1992) found that a ratio of T to R between 2 and 1.8 generated chaotic behavior. In Figure 18.7, each agent has 12 neighbors and the regime in the chaotic range is near 1.5. The patterns shown in the figure continue to evolve indefinitely. One can think of it as turbulence in the social viscosity of cooperation.

Oscillating patterns of strategic change have been observed in the real world, such as in the mating strategies of the side-blotched lizard in Merced County, California (Sinervo & Lively, 1996). This lizard plays an evolutionary game of rock-paper-scissors. The male lizard has three mating strategies, which are maintained through genetic variation: the "orange" aggressive large territory holder, the "yellow" less aggressive small territory holder, and the "sneaker" male who has no territory and mimics females. Each strategy can be invaded by another: the sneaker beats the large territory holder (who cannot defend his harem against sneakers), the small territory holder can beat the sneaker (as a smaller harem is easier to defend), and the large territory holder can beat the small territory holder (as he is more aggressive in open combat). This is a type of *frequency-dependent selection* – the rare strategy has the advantage, just as in rock-paper-scissors. This leads to a carousel of strategies, one rising up against another.

18.7.1 The Iterated Prisoner's Dilemma

It is important to juxtapose social viscosity against another common explanation for cooperation: the *iterated prisoner's dilemma*. The iterated prisoner's dilemma embeds structure not in space but in time. Players play repeatedly against one another without knowing when the game will end. Unlike a mean field one-shot prisoner's dilemma where the Nash equilibrium is to defect, in an iterated prisoner's dilemma the best strategy is often to cooperate. A player who defects is asking to be defected against in the future, harming their potential for future rewards. But two individuals who cooperate from the beginning can benefit for the duration of their relationship, far outstripping the benefits of a single temptation.

Because the iterated prisoner's dilemma lacks a formal solution like that proposed by Von Neumann and Morgenstern (1944), Axelrod (1980) invited game theorists to submit strategies for the iterated prisoner's dilemma and then played them against one another in a tournament. The best strategy was Rapaport's *tit-for-tat*: cooperate on the first play, and then copy what the other player does. Note that this allows short episodes of assortative cooperators to persist in time. In a summary of what he learned across many simulations, Axelrod (1984) describes the best strategies as being nice, retaliating, and forgiving: Never be the first to defect, defect against those who defect against you, and cooperate after mistakes from the other party.

Various other strategies can do reasonably well under a variety of circumstances. For example, *grim trigger* cooperates until it is defected against, and then defects thereafter with that partner. *Pavlov* is a win-stay, lose-shift strategy that changes its strategy whenever it receives the worst payoff for that strategy. In other words, Pavlov continues what it is doing until it is defected against, then it switches.

The iterated prisoner's dilemma pits the future against the present. Defecting is always better in the short run. For players to benefit in the long run, they must inhibit their short-term interests (Stevens & Hauser, 2004). This is a problem of *intertemporal discounting*, which arise when people are challenged with a trade-off between smaller-sooner versus larger-later rewards (Read et al., 2018). This is also sometimes studied under the concept of *melioration*. Melioration is another form of maximizing in the short term, by choosing larger rewards now that diminish future rewards later. Overcoming melioration can be particularly difficult for people to learn. When saddled with a choice between a large and a small reward, people favor the large reward – even when it shrinks over time – instead of a small reward that grows (Gureckis & Goldstone, 2009; Herrnstein & Prelec, 1991).

As the payoff for defection grows, at some point it exceeds even the collective benefit to cooperators. Then cooperation should be doomed. But cooperators can find a way to turn this to their advantage. I call this the *sumo wrestler's dilemma* after the work by Duggan and Levitt (2002) described in Levitt and Dubner's (2014) *Freakonomics*. The story told by the data is that sumo wrestler's sometimes trade wins – throwing key fights to the wrestler who will benefit most. This favor is returned by a similar favor later.

18.8 Wars of Attrition, Madmen, and Mutually Assured Destruction

Contests like the game of chicken and Russian roulette are games of escalation. Escalation games capture the risk of escalating by penalizing pairs that take things too far. The game carries the risk that the reward for winning will be smaller than the cost of attaining it. Escalation games are not uncommon. Two children fighting over a toy, bar room brawls, and the pile of dirty dishes by the sink are games of escalation.

Labor strikes impose a cost on both sides, which increases over time, until one side concedes or goes bankrupt. Large businesses can use this method when breaking contracts with smaller businesses, using threats of costly litigation that the smaller businesses cannot afford. Competing marketing campaigns drive up costs for both sides, with no appreciable benefit to either. Negotiation instructors often demonstrate this through a dollar auction: a dollar is auctioned and both the highest and second highest bidder must pay, but only the winner wins (Shubik, 1971). Two individuals inevitably get stuck bidding repeatedly against one another, until the cost of the dollar exceeds its value.

A key feature of escalation games in international conflict is what Ellsberg (1968) describes as a balance of terror. Both sides must believe that the other side will match annihilation with annihilation. Of course, national leaders often attempt to tilt the balance in their favor by signalling their commitment to disproportionate military responses, to the point of self-harm, a behavior summarized in *madman theory* (McManus, 2019).

In "The political uses of madness," Ellsberg (1959) explains how threats of massive retaliation can beat rational strategies. When both sides can respond to attacks with an assurance of mutual annihilation, we have a case where the capacity for escalation can in principle create peace. This is the curious logic underpinning mutually assured destruction, which involves both parties in a nuclear standoff recognizing that responses will be in kind and automatic (Schelling, 1980). Nikita Khrushchev said as much during the Cold War, responding to US threats: "If you send in tanks, they will burn and make no mistake about it. If you want war, you can have it, but remember, it will be your war. Our rockets will fly automatically" (quoted in Ellsberg, 1968).

Games of escalation are not limited to humans; they routinely play out in evolutionary contexts, over millions of years. One of my favorite examples is that of late succession trees, which postpone reproduction to gain access to the sunlight at the forest canopy. This requires other trees to do the same. Genetic variants that postpone reproduction for longer literally escalate their height, but they also escalate the costs of reaching reproductive maturity for all competitors. This incurs a collective risk that exposes species that play such games to catastrophe. The risk is that an event will occur that kills the trees before they can reproduce: for example, a fire or drought that wipes out an entire forest, leaving only the seeds to recover. If the next fire occurs before the trees reach reproductive maturity, the forest will vanish, as they have many times in the past (Sheil & Burslem, 2003).

There are two games that neatly capture the basic properties of escalation games. The first is the hawk–dove game. The second is the war of attrition. In what follows I will first show how the hawk–dove game is sensitive to network structure. Then I will introduce and adapt the war of attrition to a game of nuclear brinkmanship – in which both sides can escalate in the face of annihilation – and demonstrate why structure matters to Kennedy's paradox.

Table 18.3 The hawk–dove game. Payoffs are to the strategy on the left played against the strategy above.

	Hawk	Dove
Hawk	$(V - C)/2$	V
Dove	0	$V/2$

18.8.1 The Hawk–Dove Game

The *hawk–dove game* was developed by Smith (1978).[3] The question Smith was attempting to answer is why territory wars among animals often end without bloodshed. In the hawk–dove game each player chooses a hawk or dove strategy. Hawk is better than dove, but hawk is costly against another hawk. Table 18.3 shows the payoff matrix for the game. The cost of playing a hawk against another hawk is greater than the victory payoff, $C > V$. As a result, neither strategy dominates – the best strategy depends on what the other party does. Relative to the payoff matrix for the prisoner's dilemma ($T > R > P > S$), for the hawk-dove game (where hawk equals defect and dove equals cooperate) the payoff structure is correspondingly $T > R > S > P$ – that is, it is better to be a sucker (S) than suffer the penalty of escalation (P). Smith (1978) predicted that this is a coordination problem organisms should evolve to resolve without conflict, and evidence suggests they do (Pietraszewski & Shaw, 2015).

Smith (1982) also computed the optimal probability of playing hawk for games against the field and found it to be V/C. That is, the best strategy is a *mixed strategy* that sometimes plays hawk and sometimes dove. Mixed strategies are common to many games. Rock-paper-scissors is a clear example, with the optimal probability being $1/3$ for each strategy. The best mixed strategy is one the opponent cannot use to predict what you will do.

As the costs of hawk-on-hawk conflict rise, the proportion of players playing dove should increase. Here we already see inklings of a resolution to the Kennedy paradox. Even though a dove can never beat a hawk, hawks can punish one another so badly that a single dove can invade. On the flip side, a single hawk can invade a community of doves. This *frequency dependent selection* favors the rare species. And it does so up to a point determined by the relative costs and benefits of being a hawk. On networks, the hawk–dove game is also capable of the spatial chaos we saw earlier for the prisoner's dilemma (Killingback & Doebeli, 1996) – meaning that in some worlds we can expect to see phoenix-like waves of hawks and doves, rising up, one after another.

The relative prevalence of hawks and doves are also predictably influenced by network structure (e.g., Hauert & Doebeli, 2004). This can be seen by running the hawk–dove game on the Mason–Watts networks described earlier. We will use replicator dynamics

[3] The hawk–dove game is also sometimes called the chicken game or snowdrift game, motivated by the question of which of two individuals will shovel the snow blocking their mutual access to the road. See also the volunteer game (Krueger, 2019).

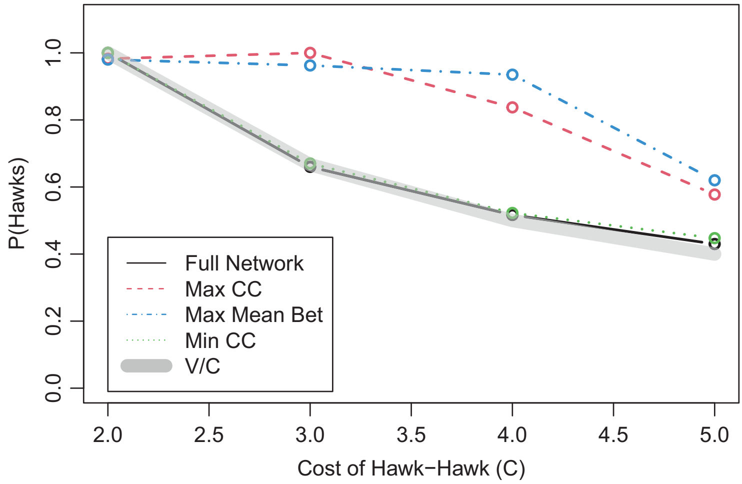

Figure 18.8 The hawk–dove game on different networks with different costs of hawk–hawk outcomes, C, and V = 2. Each network configuration is run for 100 games to stabilize, with 10 different randomized starting arrangements. Each starting arrangement randomly assigns hawk or dove to each node with equal probability. The thick gray line represents the predicted proportion of hawks for the mean field analysis, V/C.

such that individuals identify the best strategy but they only change to that strategy 10% of time. This prevents the model from converging on all doves or hawks on the full network, when one of the two strategies will always be the best performer and available to all. Figure 18.8 shows the proportion of hawks remaining for various costs, C, of hawk-on-hawk interactions. As the costs increase, the proportion of hawks falls. However, it falls the least for the more clustered networks. If we consider doves as the hawk–dove equivalent of cooperators, this is the opposite of what we saw for the prisoner's dilemma. For the prisoner's dilemma, more clustered networks had the highest numbers of cooperators. For the hawk–dove game, more clustered networks increase the number of hawks (Killingback & Doebeli, 1996).

18.8.2 The War of Attrition

Unlike the hawk–dove game, the war of attrition is a continuous strategy game. Two players stand off over a resource and the one willing to pay a higher cost to get it wins. Smith (1982) describes two wasps fighting over a hole in the ground: the winner wins the hole, in which it can lay its offspring. Either wasp could in principle beat the other if she were more committed to doing so. Both sides pay the cost of commitment associated with the least committed individual. That is, once one individual gives up, that individual loses the resource, but both sides pay the price up until that point.

Table 18.4 shows the payoffs for the two individuals competing in the war of attrition. No single strategy is an ESS in this game as once one side knows the other party's strategy, m_A, they can always play a strategy $m_A + \epsilon$, where ϵ is any value greater than 0, and they will win. However, when the costs exceed the reward of victory (V), then $m_A = 0$ can invade.

Table 18.4 The war of attrition

Comparison	Player A	Player B
$m_A > m_B$	$V - m_B$	$-m_B$
$m_A = m_B$	$V/2 - m_B$	$V/2 - m_B$
$m_A < m_B$	$-m_A$	$V - m_A$

To give an example of the game, suppose both sides play $m = 2$. Then they tie and split the victory, $V/2$. However, they also suffer a loss of 2, making their total reward -1.

Smith (1982) showed that the ESS in the war of attrition is a mixed strategy, in which individuals choose from the following distribution:

$$P(m) \propto \frac{1}{V} e^{-m/V}.$$

This equation tells us that the probability of escalating is a decreasing function of the level of escalation. Lower levels of escalation are more likely than higher levels. Higher rewards for victory increase the likelihood of greater escalation. Importantly, playing a mixed strategy prevents one's opponent from knowing your strategy ahead of time and choosing $m + \epsilon$. Smith points out that this distribution could be played by individuals – who each play all possible strategies – or it could be played out by different individuals, each choosing their own strategy from the distribution.

The war of attrition captures escalation, but what it lacks is catastrophe, especially of the kind alluded to by Kennedy in the era of nuclear brinkmanship. The hawk–dove game can have catastrophic loss, C, which as it grows should drive down the number of hawks. But a more continuous version of the kind seen in the war of attrition would make it easier to see how structure influences various levels of escalation. Moreover, it would also be nice if the catastrophe were more probabilistic, so that individuals take a risk of pushing things too far, but they never know exactly how far that it is. If escalation in the face of catastrophe is the domain of madmen and existential threats, it is an area worth understanding better. The models may not provide the answer, but they will refine our intuitions and help us ask better questions, and that is a start.

18.9 The Brinkmanship Game

Brinkmanship is a military strategy associated with bringing a state of affairs to the brink of war, with the intention of getting the other party to back down. Ellsberg (1959) describes this as "bargaining with the threat of mutual suicide." Whether we call the willingness to coerce through threats brinkmanship, coercion, extortion, blackmail, or deterrence, the evidence suggests these strategies work and the game theory underlying them suggests they *should* work to get the other party to back down.

If brinkmanship increases the probability of catastrophic events, then one may ask under what structural conditions catastrophes are most likely to occur. The brinkmanship game allows us to address that question in a controlled setting, by adding the probability of catastrophe to the war of attrition. As in the war of attrition, the only way to win is to

Table 18.5 The brinkmanship game.

Comparison	Player A	Player B
$s_A > s_B$	V	0
$s_A = s_B$	$V/2$	$V/2$
$s_A < s_B$	0	V
$P(s_{minimum})$	$-s_{minimum} \times C$	$-s_{minimum} \times C$

stay in the game. But in the brinkmanship game – which I introduce here – the longer one stays, the higher the probability of catastrophic loss.

The rules of the brinkmanship game are as follows:

- Players are assigned to nodes (or nations) on a network.
- At the beginning of the game, each player chooses a uniform random value between 0 and 0.1 to be their strategy, s_i. This is the maximum level of escalation they are willing to accept before withdrawing.
- Each round, each player plays against all their neighbors. If $s_A > s_B$, then player A wins and gets reward V. If $s_A = s_B$, then they tie and each get $V/2$.
- Catastrophic costs are paid with probability equivalent to the lower of the two player's strategies, $s_{minimum}$. The cost of a catastrophic loss is an amount $C \times s_{minimum}$. The greater the escalation before someone pulls out, the higher the cost of an accident. This reflects the costs of war scaled to the degree of arms escalation. If both parties are lobbing rocks, the costs will be lower than if they are lobbing bombs. The costs are suffered by both parties and subtracted from any rewards. If the minimum strategy $s_{minimum} \geq 1$, catastrophic loss is certain.
- After each round, each node looks to its neighbor's payoffs and moves toward the strategy with the highest payoff (including itself). We will assume that parties do not know the exact value of their neighbor's strategies, but do know their relative value. If another party wins or loses, it is because their strategy was higher or lower, respectively, and this is known through play. The approach factor, λ, is then added or subtracted from the player's current strategy based on their best neighbor's relative position.
- When an individual's summed payoffs across games with their neighbors are negative, the node suffers a total loss and is replaced. The replacement chooses a random uniform value between 0 and the current strategy. This can be likened to reproduction with mutation, revolution, or some other calamity that replaces the current leadership with a new set of minds.

Because catastrophes are costly, especially to pairs of nations with higher s, it is possible for low s nations to outperform high s nations. This mirrors the lessons of the hawk–dove game. If high s nations are surrounded by many similarly aggressive neighbors, who also have high s, the results are costly. Pairs of nations with high s will eventually suffer catastrophes. This can make more peaceful strategies preferable. Moreover, the more catastrophic encounters a nation endures the more likely it is to suffer a total loss.

18.9.1 Simulation: Resolving the Kennedy paradox

Let's run a simulation where $V = 2$ and C is varied over values ranging from 2 to 8. The approach factor is $\lambda = 0.01$. This means that in 100 rounds, a nation starting at $s = 0$ will catch up to another nation at $s = 1$.

Figure 18.9 shows some representative timelines for the various networks with $C = 8$, showing the simulations after 1,000 and 2,000 iterations. The visualizations are compelling and worth inspection. Across networks, the distribution of strategies tends to favor a few nodes with high escalation values and the rest with low values. And these high escalators change over the thousand iterations, showing us that escalators rise and fall over time.

Why do escalation levels rise above the starting distribution? This happens because the starting values are uniform random numbers to seven decimal places and $\lambda = 0.01$. This creates a situation in which nations leapfrog one another. A leading neighbor's value could be 0.8814363, which in turn causes its neighbor with strategy 0.8732493 to jump to 0.8832493. Through successive encounters, this can drive strategies up beyond their initial starting values. Ties will be vanishingly rare, which has a certain realism when escalators cannot inspect one another's stock piles and would prefer to keep them secret.

Escalating nations with high s can reach values greater than 1 if they pair off against another escalating neighbor. This is possible because such nations can often afford to pay the cost of a few catastrophes if they win sufficiently often against weaker neighbors. Two nations with similar s values can leapfrog one another repeatedly if they are both the best performing nations in their local neighborhoods. As the costs of catastrophes increase, this becomes less likely. Eventually, a less aggressive neighbor, with lower s, will do better – a Switzerland of sorts – and they will start following that neighbor and thus withdraw from their high stakes arms race.

18.9.2 The Impact of Catastrophic Costs

Figure 18.10 shows how the outcomes differ after 2,000 simulations from 20 different random allocations of strategies for each of the different network types. What is most apparent is that as the costs of catastrophe increase, the average level of escalation declines as does the frequency of catastrophes.

More costly wars limit the frequency of wars. Here we see results echoing the words of Kennedy. However, this is evident in more than just the decline in catastrophes. The total costs of conflict also decline. Kennedy's "unfortunate fact" is supported by the brinkmanship game, even if catastrophes still occur with disparaging frequency.

There are variations, but these each support what is evident here. War costs can be fixed, so that war is not scaled to the degree of brinkmanship. This produces a similar decline in escalation with increasing costs, but the categorical nature of catastrophes removes some of the sensitivity to network structure. One can also implement an additional cost of escalation, so that the winner and loser both pay the minimum cost of escalation $s_{minimum}$. This too produces the same general patterns described earlier.

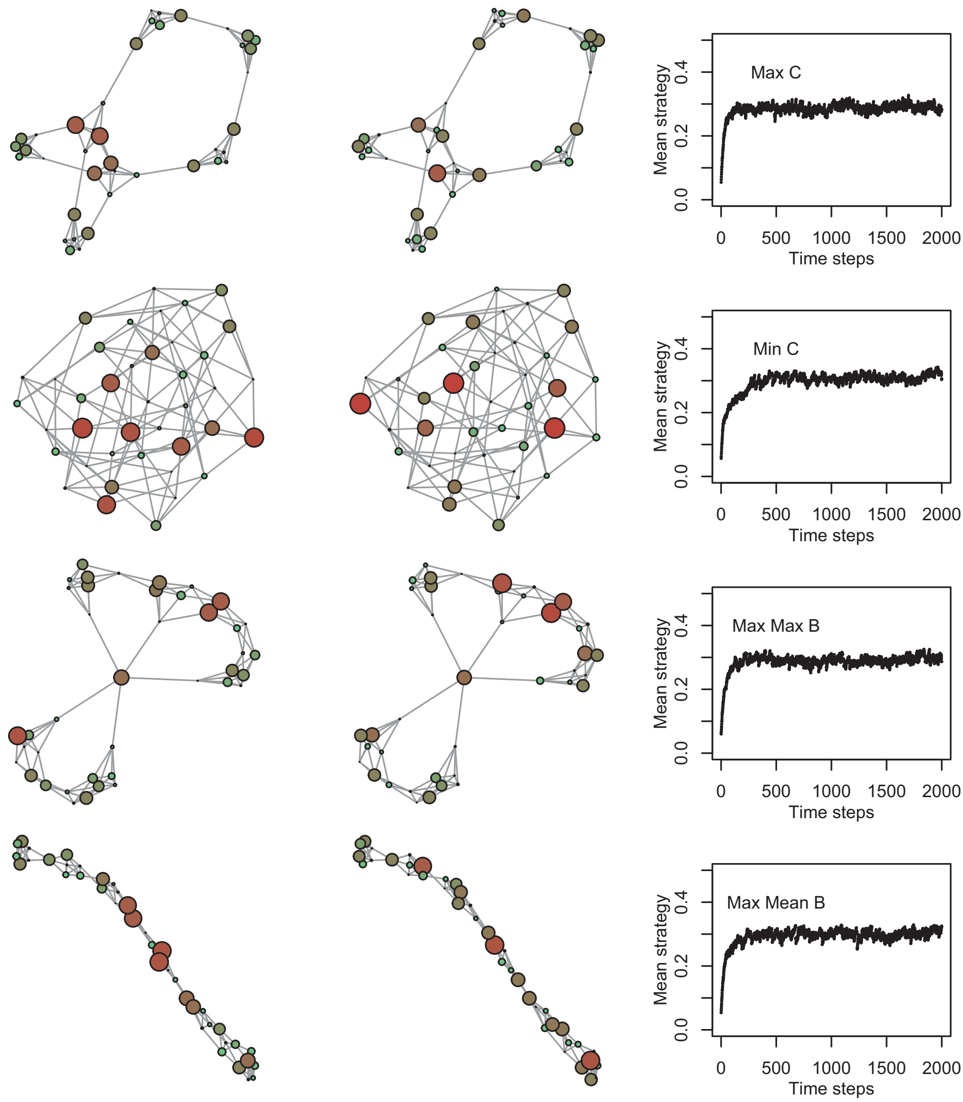

Figure 18.9 The brinkmanship game. The networks on the far left show the distribution after 1,000 generations and then, in the center, after 2,000 iterations. Starting nodes are assigned strategies uniformly between 0 and 0.1. In each round, they play each of their neighbors. The winner is the player with the higher escalation value, who wins V = 2. The loser's payoff is 0. Parties pay costs only in the event of a catastrophe. The probability of a catastrophe is proportional to the minimum escalation between the two parties. The cost of a catastrophe is C = 4. Strategies replicate by escalating upwards if the best neighboring strategy is higher than their own, or downwards if it is lower, by amount 0.01. If the total payoffs to a node at the end of play are less than 0, this is a total loss and their strategy in the next round is uniformly chosen between 0 and the current strategy.

The various network types also differ in their response to increasing costs. Networks with more clustering, such as maximum clustering coefficient (Max C) and maximum mean betweenness (Max Mean B), both have lower average escalation and higher numbers of catastrophes.

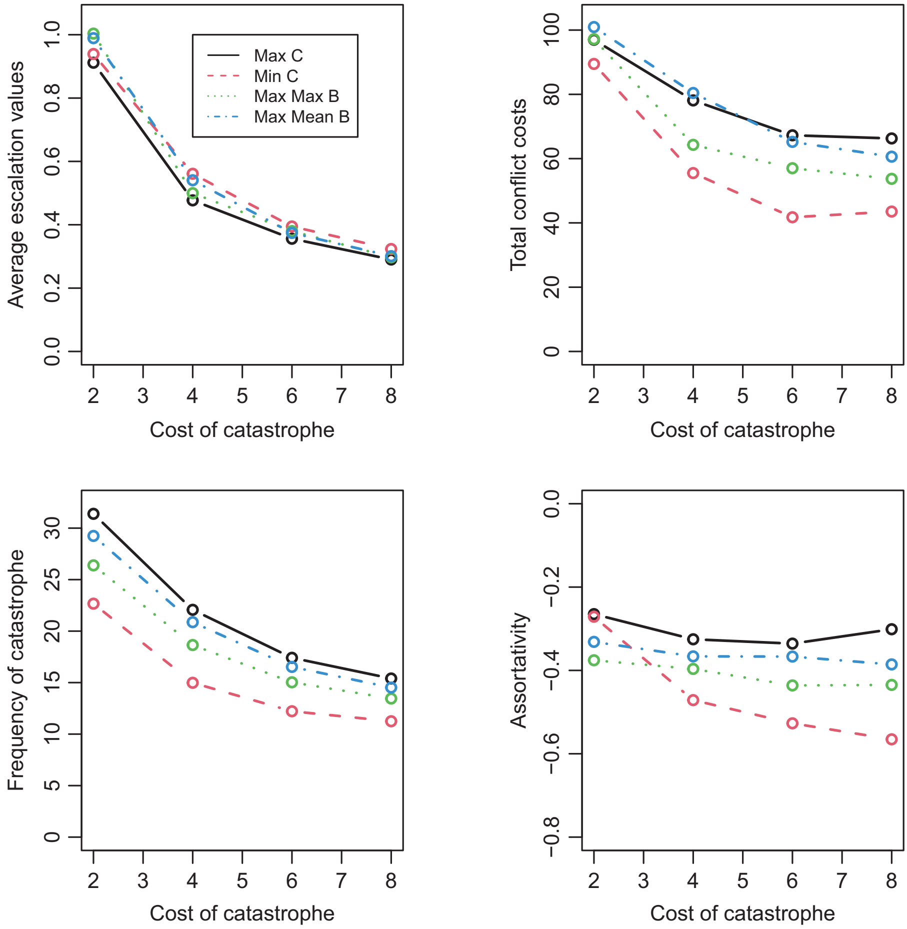

Figure 18.10 The influence of the costs of catastrophe on strategy, catastrophic losses, catastrophe frequency, and assortativity in the brinkmanship game for different network structures.

Why are the networks different in their average strategy and frequency of conflicts? If networks change over time and this brings escalators into closer contact (e.g., through alliances), we should expect more catastrophes. One way to investigate this further is by looking at assortativity. Recall that for the prisoner's dilemma, strategic assortativity helped support cooperation by placing cooperators among other cooperators. In the brinkmanship game, the opposite is true: assortativity drives up the cost and frequency of conflict. This is further facilitated by clustering, where groups with lower effective networks sizes redundantly learn from one another, creating outbreaks of escalation. Less clustered networks (like Min C) facilitate more negatively assortative networks. These experience fewer catastrophes and can therefore maintain higher levels of escalation.

Finally, we may ask about the mixed nature of the ESS, which in these contests remains mixed after 2,000 iterations. Figure 18.11 shows the relative distribution of strategies for the maximum clustering coefficient network across different C values (note: the y-axis is on a log scale). There are few high-escalation strategies and many low-escalation strategies, with the highest escalation strategies falling as C increases. The pattern is

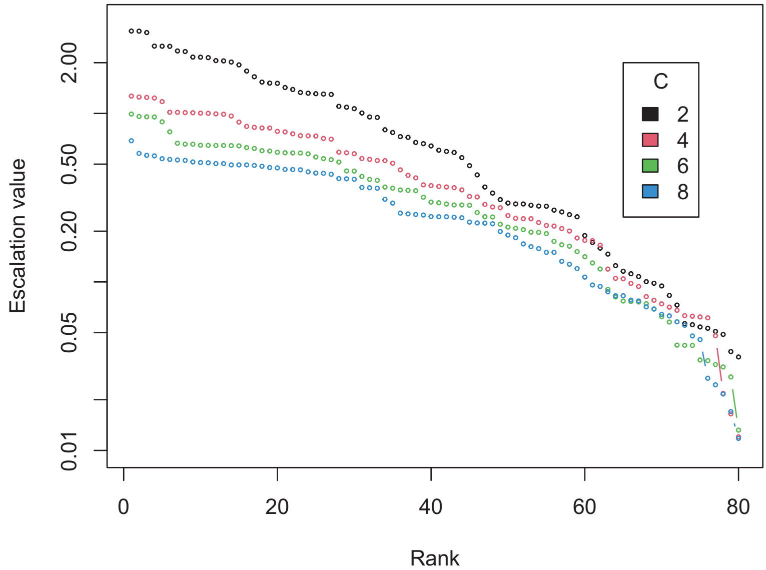

Figure 18.11 Strategy distribution in the brinkmanship game after 2,000 iterations for the maximum clustering coefficient network (Max C) from Figure 18.9.

roughly similar for the other networks. Further analyses could investigate the differences within the network. Where do escalators appear? Where are conflicts most frequent? Work by Aquino et al. (2019) taking a multiplex approach with real-world conflict data provides a useful guide.

18.10 Summary

Games of conflict and cooperation are influenced by the structure of the communities in which they reside. Simple frequency-dependent selection is one kind of structure, in which rare strategies become preferable. But spatial structure can also facilitate social viscosity, with pockets of cooperators thriving in more assortative networks. Whether structure drives cooperation or escalation can depend on the subtleties of the payoff structure. Clustering benefited cooperators in the prisoner's dilemma and hawks in the hawk–dove game. Structure can also facilitate chaotic mixtures of strategies that move through environments in waves, rising and falling as they invade and are then themselves invaded. When the stakes are raised through escalation, negative assortativity is desirable – keeping madmen apart. The brinkmanship game brought this to life but also allowed us to see that raising the cost of catastrophe reduced escalation in accordance with Kennedy's "unfortunate fact."

What these games and simulations teach us is partly how such systems might work and how to better think about these problems. Though the simulations described here offer some insight into how strategies might develop among agents that are learning from their neighbors, they fail to capture an agent's ability to anticipate costs. In these simulations, an agent's single cognitive ability is the ability to learn from others. For example, in the brinkmanship game, catastrophic costs "teach" populations to escalate less by decimating the catastrophic escalators. The limitation of this approach is that when catastrophic costs grow sufficiently large, no one will remain to learn from them.

18.11 Open Questions

18.11.1 Personality and Frequency-Dependent Games

Recent work has shown that behavioral specialization (i.e., personalities) can arise when agents learn in frequency-dependent environments, like those described here for mixed strategies. This was demonstrated using the hawk–dove game and producer–scrounger game (Leimar et al., 2022). The producer–scrounger game was developed by Barnard and Sibly (1981) and is based on strategic decisions between groups of individuals who can choose to search for food (produce) or steal the food from others (scrounge). The notion that individuals might learn various strategies based on the incentive structure of their environment is straightforward. But with frequency-dependent environments such as those seen in mixed-strategy games, a question arises about specialization and the degree to which individuals might benefit from it. Dall et al. (2004) have speculated that such environments can lead to different personalities, for example, in relation to aggression and trust. Others have found that individual differences in personality are associated with different social network structures (Kalish, 2006) and that these can influence friend/enemy relationships (Ruiz-García et al., 2023). Individual differences in intelligence also lead to different strategies (Proto et al., 2019). One of the interesting questions here is how the structure of different environments might lead to the development of certain personality features, or vice versa, such that personalities influence structure (as suggested by Ruiz-García et al., 2023). Leimar et al. (2022) studied frequency-dependent learning on lattices and small-world networks. But the adaptive network search described earlier allows this question to be extended to a variety of different network structures. Moreover, and much less often explored, is what happens when networks change over time, due to factors such as crowding or age (e.g., Wrzus et al., 2013).

18.11.2 Alliances

If the enemy of my enemy is my friend, then alliances are part of the natural order. The usual approach to game theory is to allow each player to play against their neighbors in a single dyad and then compare strategies over repeated iterations. But some kinds of conflict involve active conquest, defense of borders, spread of technology, and interactions in structured spaces that allow neighbors to leverage their collective strength (Guo et al., 2016; Turchin et al., 2013). The absorption of neighbors, through conquest, might be modeled as the replication of a neighboring strategy. But this fails to capture the power of strategic alliances. An alliance can potentially channel strength through network paths, or lead to the development of negative edges to shared enemies (e.g., Ruiz-García et al., 2023). How might these capacities be modeled using agent-based models and how would they influence network structure?

19 Fund People Not Projects
A Universal Basic Income for Research

19.1 The Problem

A universal basic income is widely endorsed as a critical feature of effective governance. It is also growing in popularity in an era of substantial collective wealth alongside growing inequality. But how could it work? Current economic policies necessarily influence wealth distributions, but they are often sufficiently complicated that they hide their inefficiencies. Simplifications based on network science can offer plausible solutions and even offer ways to base universal basic income on merit. Here we will examine a case study based on a universal basic income for researchers. This is an important case because numerous funding agencies currently require costly proposal processes. These are costly for the proposal writers, their evaluators, and the progress of science itself. Moreover, grant outcomes are known to be biased and inefficiently managed. Network science can help us redesign funding allocations in a less costly and potentially more equitable way.

19.2 Universal Basic Income

The concept of a universal basic income is simple: Everyone receives some minimal income, regardless of how they contribute to the economics or well-being of the communities to which they belong. Though this concept is much discussed, it has been around for a long time. Caesar gave Roman citizens a basic income. Many notable figures have called for it since, including Thomas More, Thomas Paine, C. H. Douglas, and Bertrand Russell. US president Lyndon B. Johnson and his successor Richard Nixon both supported a basic income such as the "negative income tax." More recently, the economist Thomas Piketty has argued for a universal inheritance to help restore equality and reduce the growing wealth gap (Piketty, 2022).

However one feels about such policies, the evidence from research is substantial, with many knock-on effects associated with better health, education, sustainable environmental impacts, economic productivity, and education (Baird et al., 2013; Bastagli et al., 2019; McGuire et al., 2022). There is also good evidence that the standard narrative around poverty is inverted – it is not that low intelligence leads to poverty, it is that poverty is a burden on the mind (Mani et al., 2013). Moreover, people in poverty also suffer a greater share of social injustices, which range across insurance, loans, profiling, and fraud (e.g., O'Neil, 2017). If we would like to see better outcomes for people and society at large, we need to invest more effectively in people to make it happen.

Though the funding of scientific inquiry is often distanced from more pressing social issues, it nonetheless shares similar challenges of effective wealth distribution. The question we will address here is how we might make research funding less costly and more equitable while still rewarding merit.

19.3 The Inefficiencies of Grant Writing

There are a wide range of reasons for being critical of existing grant funding schemes (Bollen et al., 2014, 2017; Gordon & Poulin, 2009; Ioannidis, 2011; Luebber et al., 2023; Vaesen & Katzav, 2017). Current grant review schemes reveal numerous selection biases, including biases based on ethnicity, gender, age, and affiliation. Existing grant allocations also tend to reveal long-tailed distributions, where a few researchers continue to win the lion's share of the grant funding and most others go without. This is unwarranted, given the evidence that scientific impact per dollar awarded is lower for large-grant holders (Fortin & Currie, 2013). The ability of peer review to identify grant quality is also questionable: An analysis of more than 100,000 grants found no relationship between the quality of research outputs and the review scores received during peer-review (Fang et al., 2016).

One study of time spent writing proposals for the Australian National Health and Medical Research Council estimated that researchers lost more than 400 working years of research time in 2012 writing unfunded proposals (Herbert et al., 2013). An additional 100 years was spent writing the successful grants. The cost of the unfunded time was estimated to be about Aus$66 million. A back-of-the-envelope order-of-magnitude estimation arrives at nearly the same amount of wasted effort in the UK. Assume approximately 100 universities with approximately 1,000 eligible faculty members at each and 10% of them each writing 1 grant per year at a cost of 100 hours each. With 48×40 working hours in a year, we get an equivalent of approximately 500 researcher years spent writing grants each year. If the award rate for grants is roughly 25%, this is 400 years with little to show for it, or roughly the entire research careers of 10 people. My favorite term for this kind of time unit is faculty-adjusted life year (pronounced like "folly") by which one can measure all things that consume the lifespan of a researcher.

An investigation by Gross and Bergstrom (2019) finds that the cost to science in time spent writing unfunded grants is likely to be on par with the benefits accrued for writing successful grants – effectively breaking even. As Gross and Bergstrom (2019) argue, these costs are hard to escape. If funding pots grow smaller, individuals will spend more time competing for them. With currently shrinking funding budgets, the time wasted may only increase. One effort to counter these costs by "streamlining" grant proposals only increased the time researchers spent preparing them (Barnett et al., 2015, p. 2).

Collectively, the cost of existing grant structures calls for novel approaches. So what are the alternatives?

19.4 What if Funds Were Allocated Equally?

According to Vaesen and Katzav (2017):

> Although the equal distribution of research money among qualified researchers would depart most from how funds are currently allocated, it could have numerous benefits: it could reduce the drop out of talented scientists, the effects of gender and other biases in grant peer review, the reinforcement of established views, the incentives to commit scientific fraud, the number of experienced researchers excluded from teaching, the oversupply of junior scientists, workplace stress and negative effects on personal and family life [. . .] Furthermore, a baseline grant system is much cheaper than schemes that involve peer assessment and, in light of our uncertainty about the future course of science, effectively spreads our risks. Finally, such a system accommodates the worry that awarding large grants only to a happy few might be ineffective; scientific impact per dollar appears lower for large grant-holders [. . .] and awardees do not perform extraordinarily well when compared with rejected applicants (see the review in Neufeld, 2016).

How much would each researcher receive with equal allocation? Vaesen and Katzav (2017) estimated the primary funding sources available to researchers in the UK, the USA, and the Netherlands and then divided the funding by the number of researchers eligible for that funding. They found that five-year budget allocations for individuals would exceed $300,000 in each of the three countries. For example, a Dutch professor would receive approximately $500k for a five-year budget. As the authors note, this money could be pooled of course to produce budgets on the order of $2.5 million for a five-faculty-member team.

Of course, some disciplines are more expensive than others, and Vaesen and Katzav (2017) also took this into account. Based on their estimates of relative need, researchers would receive between $290k and $715k depending on discipline. This is sufficient to employ the current number of PhDs and postdocs and leave a substantial portion remaining for travel, equipment, overheads, and so on.

One potential complaint is that many of these numbers are lower than existing grant awards, which are roughly between $1.5 million and $2.5 million depending on the country. However, only around 10% of faculty receive grants of this size. There are potentially other criticisms too, many of which are likely to flow either from worries that some good researchers will not be able to fund their research or that an equal distribution does not support merit.

19.5 What if Researchers Allocated Funding to People, Not Projects?

Bollen and colleagues proposed an alternative to grant proposal peer review that takes advantage of the wisdom of the crowds. Their approach assesses relative merit while also funding people instead of projects, overcoming the need for individuals to write costly proposals (Bollen, 2018; Bollen et al., 2019). Figure 19.1 provides a comparison of the old and new approaches. The old way involves multiple stages of oversight and review, with multiple tiers of administration.

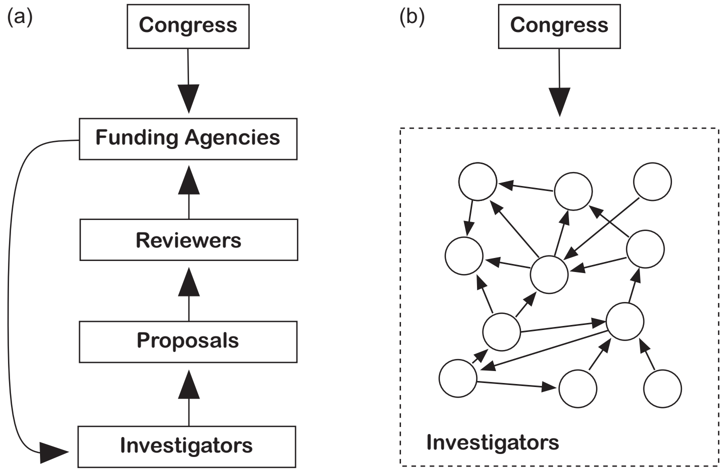

Figure 19.1 A depiction of how things presently are (panel a) versus how things would be under Bollen's system (panel b).

The new idea involves two steps:

1. Allocate to each researcher an equal portion of the share of available funding as a baseline allocation. This can differ by discipline such that more expensive sciences receive larger shares.
2. Each scientist then anonymously donates a share of their budget to other scientists.

The above procedure then repeats annually, with new allocations of baseline funding and new allocations of anonymous donations. Importantly, each individual allocates a portion of their previous year's budget. Because of this requirement, well-funded researchers also become the most generous donors.

A formal description of this allocation process is as follows. Each researcher, i, receives U as their baseline allocation in addition to $A_{j,i}$ from each of the other j contributors. A is a directed and weighted adjacency matrix or, in this case, the contribution matrix. This amounts to a budget of B at time t:

$$B_i^t = U + \sum\nolimits_{j\in[1,N]} A_{j\to i}^{t-1}.$$

The contribution of an individual researcher is a fraction, F, of their budget, B, in the previous year, which is then divided among those they contribute to. Thus, the total amount contributed by individual i to all others is as follows:

$$\sum\nolimits_{j\in[1,N]} A_{i\to j}^t = FB_i^t \quad \forall i \in N.$$

This function iterates over successive years. The budgets for this year influence the budgets for next year, and so on. This is sometimes referred to as a discrete dynamical system. It is discrete because the time is not continuous but considered at specific time points. It is dynamical because people's budgets change over time.

19.6 A Case Study Using Citation Graphs

Understanding how this would work in practice requires us to know how people are likely to contribute their funding. If each person contributed their funding to the same individual, this would create an outcome more lopsided than the present system. However, if each person gave their money to someone unique, then the funding distribution would be flat and budgets would be equal even after allocation (Vaesen & Katzav, 2017). Neither of these are necessarily bad, but it would be useful to understand how things might work under various values of F and more realistic distribution regimes. Pragmatically, this allows us to avoid complaints that the system will fail by design or that the system is unnecessary. With some data, we can start to understand how the design matters and why.

Bollen et al. (2017) explored the plausibility of the proposed system by using an agent-based model with funding distributions based on article citations. Citation networks offer numerous possibilities for creating edges (Börner et al., 2004; Newman, 2018). Here Bollen et al. (2017) allow that if an individual cites a set of scientists, then we can anticipate that they might also contribute some of their budget to those scientists. This is open to the immediate complaint that citation counts are long-tailed distributions that lead to rich-get-richer dynamics (Clauset et al., 2017), but we leave this aside for now.

Bollen et al. (2017) used Thomson-Reuters' 1992 to 2010 Web of Science (WoS) database to find articles and their associated citations. Using this, they extracted 37 million articles and 770 million references to produce a citation graph. This produced a set of 4,195,734 unique authors, of which they took only those authors who had a 5-year streak of at least 1 paper published per year. This narrowed the list to 867,872 names. To fill out the adjacency matrix, they then identified the number of times each author cited each other author. This produced the weighted and directed graph A.

With A, Bollen et al. (2017) then simulated the process by allocating each researcher a baseline allocation of $U = \$100,000$. Each individual's contribution to others was then a fraction F of their budget, allocated to each of the authors they cited in proportion to the number of times they cited them. The simulation was run for various values of F.

The results of the simulation showed two things. First, with lower values of F (less redistribution), budgets were more equal; with larger values of F (more redistribution), the distribution become more heavy-tailed and less equal. Second, a value of $F = 0.5$ most closely matched the heavy-tailed distribution of actual NSF and NIH funding. Bollen et al. (2017) were able to go one step further by matching the author names with their actual NSF and NIH funding. The simulation matched the observed funding with a correlation of $r = 0.27$.

19.7 Lottery and Ranked Preference Graphs

The simulation in Bollen et al. (2017) reflects one potential instantiation of the budget distribution. This is useful because it provides a sufficiency argument. We learn that resources distributed in relation to citation networks can produce a pattern similar to what we currently observe for costly grant funding procedures. The analysis above also reflects the long-term behavior of the system. We might ask how long it takes for a system to settle on this arrangement. Would other network architectures or redistribution schemes produce similar results? And how long would it take for such systems to settle down?

To address this we can produce simulations based on various topologies and with various additional constraints. We will use two approaches. One is a directed network similar to an ER random graph with the constraint that each agent has a fixed outdegree. We will call this the *lottery graph* as it reflects random contributions from each individual chosen with a uniform preference across all other individuals. Bollen et al. (2017) deals with many potential concerns about constraints on giving, including the creation of distribution cartels. Suppose we imagine that those constraints are so wretched that researchers are effectively forced to choose at random: then we have a lottery graph.[1]

The lottery graph is a good model to evaluate for three reasons: (1) It is simple and easy to understand – people are contributing at random. (2) It is sufficiently different from a weighted citation network that it provides some sense of boundary conditions. If it produces a similar budget distribution to other networks, then we know that the network structure may not be that important. And, perhaps most important, (3) we can easily imagine reasons why contributors or regulators might be averse to inequality and may therefore choose to provide funding in a more equitable way, which might be more akin to a lottery. Some have even argued that we effectively have a costly lottery system as things stand (Jerrim & Vries, 2023) and others have argued that acknowledging this and implementing it as such would be more equitable in any case (Fang & Casadevall, 2016; Heyard et al., 2022).

The other network we will investigate is a *ranked preference graph* (similar to that described in Chapter 12). Each agent has a fixed outdegree, but chooses their recipients in proportion to their rank, which is objectively shared across all agents:

$$P(i) = \frac{r_i^{-\gamma}}{\sum_k r_k^{-\gamma}}.$$

This is a fitness model, in which indegree is a function of rank. For our purposes, this is the opposite of a lottery graph. With a high preference parameter, γ, it can be made such that only a minimum number of nodes receive contributions and the rest are left with whatever fraction they can keep. Here we set $\gamma = 1$, which with 500 nodes make some recipients 500 times more preferred than others. This is certainly moving in the direction of a long-tailed citation graph.

Figure 19.2 shows the outcome of 20 budget iterations for the lottery and ranked preference graphs, with 500 nodes and each node choosing 4 recipients. The gray shading gets darker as the number of iterations increase. This reveals how the budget distributions settle down fairly rapidly. It is clear from the comparison that the ranked graph shows a much more linear budget distribution on a log-log plot. After the 20 iterations, the biggest budget ($\$378,000$) and the smallest budget ($\$100,000$) differ by a factor of approximately 4 in the lottery network. For the ranked preference graph, the largest budget is $\$6,126,587$ and the smallest $\$100,000$, differing by a factor of approximately 60.

As Bollen et al. (2017) note, we can also modulate the inequality by controlling the fraction of redistribution, F. Perhaps one lesson here is that the shape of the redistribution

[1] This is easy to envision: one cannot contribute to the same person twice, no past collaborators, researchers must search a randomized list forcing them to learn about others' research, unsearchable contribution profiles, caps on contributions, and so forth.

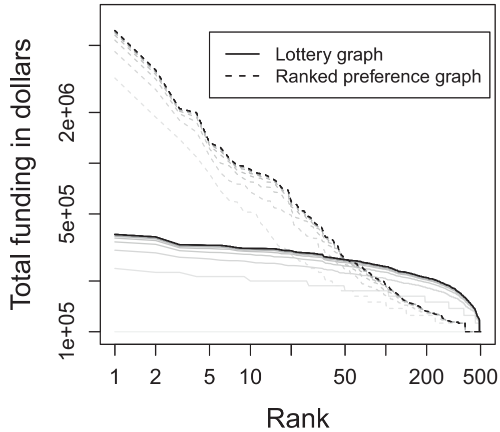

Figure 19.2 The model simulated on a uniform preference graph (lottery) and a ranked preference graph – with allocations in proportion to rank. The two directed networks have $N = 200$ agents with $e = 4$ targets for contribution from each agent (outdegree). Each agent contributes half its wealth each iteration, $F = 0.5$. The line darkens with each additional iteration.

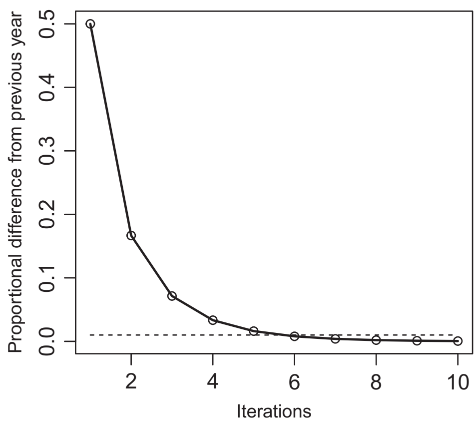

Figure 19.3 Convergence time over successive iterations. The proportion is measured as a percentage of the overall budgets in play. The dotted line represents budgets within 1% of their previous year's distribution. The lottery and ranked preference graphs show the same pattern.

can be modulated as well, to the extent that redistribution is based on shared objective rankings versus random preferences.

As shown in Figure 19.3, it takes approximately 6 iterations for the distribution to converge within 1% of the stable distribution. This is likely to be true for other formulations as well. An individual's budget is not just a function of how many contributors they have. It also depends on how much those contributors contribute. And that depends on what contributions they have themselves received from others in the past. This is not only a process based on popularity, but also on what Wasserman and Faust (1994) calls prestige.

19.7.1 Predicting Budgets: Eigenvector Centrality, PageRank, and Alpha Centrality

Could we have predicted these allocations ahead of time? For this system, individuals with larger budgets in one year will allocate more to fellow researchers in subsequent years. Thus we can imagine the money taking a random walk through the network. Nodes will have higher budgets in places where the money tends to collect. This should remind us of eigenvector centrality, which is a path-based measure of centrality that gives higher centrality to nodes that have more paths leading to them. Thus, budget prediction might be amenable to approaches like eigenvector centrality.

However, there is a problem with eigenvector centrality for asymmetric networks. As Bonacich (1987) notes, eigenvector centrality is not ideal for directed networks that have some nodes with zero indegree. Such nodes have zero status because they receive zero status from others. Because this makes their contributions to others' status zero as well, this leads to a lack of discrimination among low status nodes. Therefore, the contribution problem might be better represented by alpha centrality (Bonacich, 1987) or PageRank (Brin & Page, 1998).

Alpha centrality (or the similar Katz centrality, after Katz, 1953) overcomes this problem by providing each agent with an exogenous baseline contribution to centrality, b. The formulation also has a parameter, α, which weights endogenous factors (based on others' contributions) relative to the exogenous baseline. Alpha centrality is the solution to the following equation:

$$x_i = \alpha \sum_j A^T_{j,i} x_j + b.$$

For symmetric matrices, this essentially reduces to eigenvector centrality. For asymmetric matrices like the ones we have here, alpha centrality is more useful. It is even valid for acyclic graphs, like tree graphs, that have no cyclic paths – paths that can turn back on themselves.

The exogenous factor, b, is often set to 1 for every node. The value of α is more subtle: α is an attenuation factor that weights paths through the network, with larger values taking into account larger paths. When α is large (and < 1), then alpha centrality more closely resembles eigenvector centrality. When α is very near zero, alpha centrality approximates indegree. When $\alpha = 0$, all nodes have the same alpha centrality. Setting α may be more art than science and we will fit it to the data that follows.

PageRank is similar to alpha centrality (Katz centrality) but provides the exogenous baseline in a slightly different way. It does this by assuming that random walkers occasionally jump to a random node with some probability $1 - \beta$. PageRank also differs in another important way: it divides a node's centrality contribution by its outdegree. This gives us an equation for PageRank as follows:

$$R_i = \beta \sum_j A_{j,i} \frac{R_j}{k_{j,out}} + \frac{1-\beta}{n}$$

Thus, each node divides its PageRank contribution, R_j by the number of nodes it contributes to, $k_{j,out}$. In comparison, alpha centrality treats each edge as equivalent, whereas

PageRank reduces the impact of an edge in proportion to the outdegree of the node it is leaving. Here again, fitting β may be more art than science.

In the present case, these each work quite well as predictors and for both the lottery and ranked preference graphs the results are similar. Eigenvector centrality correlates with final budgets (after 20 iterations) for the lottery graph with a Pearson's correlation of $r = 0.97$ and for the ranked preference graph of $r = 0.89$. PageRank (with $\beta = 0.5$) and alpha centrality (with $\alpha = 0.1$) both have correlations with the final budget of $r = 1$.

This should reaffirm our understanding of how these allocations work. Funding agencies could get a reasonably good picture of how this system might behave simply by asking researchers who they would allocate funding to and doing the math. This would in turn allow them to compute F such that top budgets were constrained, or evaluate other allocation mechanisms.

19.8 Summary

Universal basic income is a concept with a great deal of evidence and theory to support it. How it might affect scientific productivity is an open question, but it could reduce a number of known costs and produce more equitable outcomes. This could enhance scientific exploration and potentially motivate many researchers to think bigger – for example, to explore projects that take longer than the standard grant allocation period (about three years) to complete. Furthermore, this could still produce funding based on merit, rewarding researchers for work well done. If we researchers make contributions similar to the way they make citations, this might not lead to funding allocations dramatically different from what we currently observe. However, they would be arrived at with potentially much lower administrative costs. Moreover, given certain assumptions, we can predict relative budget inequality in advance – using PageRank and alpha centrality – and take measures to wield it appropriately toward collective scientific goals. More pedagogically, the redistribution scheme nicely demonstrates an intuitive way to understand these metrics.

Like funding policies, a universal basic income – even with its merit-based adaptation – would no doubt need modification and fine tuning. It would also need adaptations to accommodate funder needs. But, given the costs of current approaches, it is a policy worth investigating.

19.9 Open Questions

19.9.1 Dynamic Architectures

One potential outcome of the previously mentioned merit-based funding allocations is that funding might itself lead to more researcher salience in the future. A well-funded individual this year will have more resources to hire researchers, publish in open access journals, and attend conferences to promote their work. If this leads to an enhancement of one's public profile, then this should lead to greater capture of next year's funding allocations. This alters the network structure over time in ways that should increase the prestige of well-funded individuals, creating a feed-forward process. This negative consequence may not be entirely present in the fuzzy merit-based funding allocations of our current system. Models of this process would help demonstrate how this might occur but also offer researchers a framework for asking how best to curb these effects.

19.9.2 Recursive Centrality in Identity

The principle of nodes allocating resources to other nodes can be applied in other domains. For example, we can ask what kinds of things contribute to our identity and self-understanding. Chen et al. (2016) used a recursive centrality approach to address self-identity. To do this, they adapted an approach from Sloman et al. (1998) which focused on causal centrality. This involved asking people what concepts underlying their identity were caused by other features underlying their identity. They also asked people which concepts were most important to their identity. They measured centrality similar to the recursive centrality method described here (e.g., eigenvector centrality) and found that more important concepts were also more causally central. Extending this approach to the self-concept, things we view as more critical to our identity are more highly correlated with their causal centrality in our identity network (Chen et al., 2016). One of the most central features of our identity is our morality – this is what Strohminger et al. (2017) describe as a component of our "true" self. Changing our morals changes the structural coherence of our identity and what others perceive as who we are. In this cognitive economics of identity, morality is a source of identity. Given that lottery networks and ranked preference networks produce such dramatic differences in centrality measures, we might ask if certain degree distributions in identity networks produce stronger identity. Moreover, how are these correlated with well-being, and what kinds of experiences or interventions produce stronger or more satisfying identities?

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